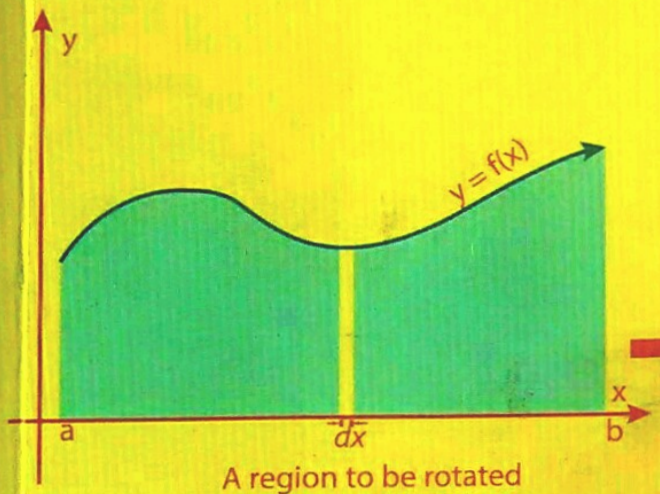
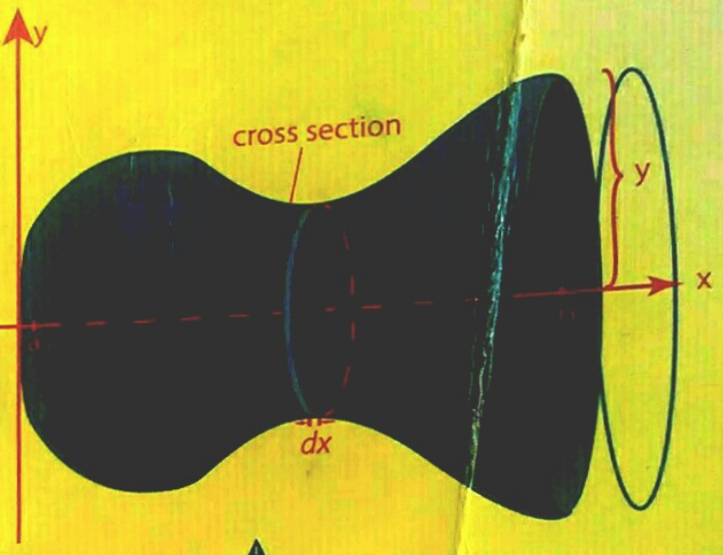
Extreme Series

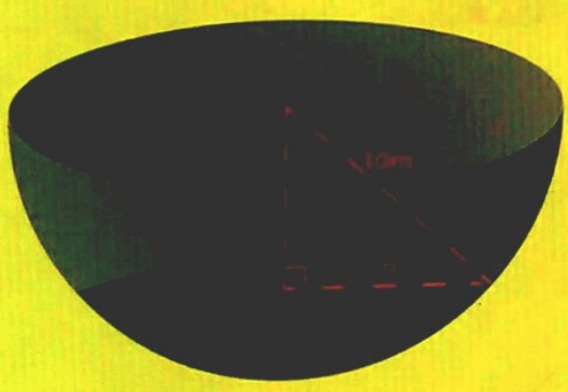
MATHEMATICS

GRADE 11

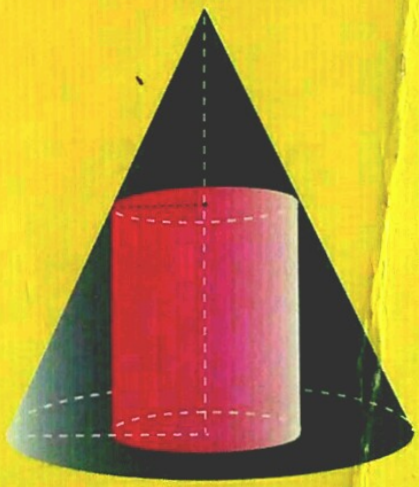


A region to be rotated





Hemispherical bowl



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UNIT ONE

Revision on Relations

Definition: A relation is a set whose elements are ordered pairs.

- Relation the way things are related.

Usually we use relating phrase like:

"*is smaller than*"
"*is greater than*"
"*is multiple of*"
"*is factor of*"
"*is father of*".....etc

Definition: If A and B are sets then a relation R from A to B is any subset of A × B if and only if $R \subseteq A \times B$

Definition: If R is relation from A to A, then R is relation on A.

i.e. $R \subseteq A \times A$

Let R be relation from A to B. Then

- Domain of R = $\{x \in R: (x, y) \in R$, for some $y \in B\}$
- Range of R = $\{y \in B: (x, y) \in R$, for some $x \in A\}$

Inverse of a relation

Let R be a relation from A to B. The inverse of R, denoted by R^{-1} is relation from B to A, given by: $R^{-1} = \{(y, x)(x, y) \in R\}$.

- **Note:** $(x, y) \in R \Rightarrow (y, x) \in R^{-1}$

Illustrative Example

1. Let A = {1, 2, 3, 4, 5, 6, 7, 8} B = {2, 4, 6, 8}
   If R is relation from A to B such that: $R = \{(x, y): y = 2x - 4\}$
   Then:
   i) a) List all elements of relation R
   b) Find the domain and range of relation R.
   c) List all elements of R^{-1}
   d) Domain and range of R^{-1}

   Solution: We know that $R \subseteq AxB \Rightarrow x \in A$ and $y \in B$ satisfies the relation $R = \{(x, y): y = 2x - 4\}$

   Let $x = 1 \in A \Rightarrow y = 2(1) - 4 = -2 \notin B$ $\therefore (1, 2) \notin R$
   Let $x = 2 \in A \Rightarrow y = 2(2) - 4 = 0 \notin B$ $\therefore (1, 2) \notin R$
   Let $x = 3 \in A \Rightarrow y = 2(3) - 4 = 2 \in B$ $\therefore (3, 2) \in R$

Let $x = 4 \in A \Rightarrow y = 2(4) - 4 = 4 \in B$ $\therefore (4, 4) \in R$
Let $x = 5 \in A \Rightarrow y = 2(5) - 4 = 6 \in B$ $\therefore (5, 6) \in R$
Let $x = 6 \in A \Rightarrow y = 2(6) - 4 = 8 \in B$ $\therefore (6, 8) \in R$
Let $x = 7 \in A \Rightarrow y = 2(7) - 4 = 10 \notin B$ $\therefore (7, 10) \notin R$
Let $x = 8 \in A \Rightarrow y = 2(8) - 4 = 12 \notin B$ $\therefore (8, 12) \notin R$

a) Thus, all elements of R from A to B is
$R = \{(3, 2), (4, 4), (5, 6), (6, 8)\}$
b) Domain of $R = \{3, 4, 5, 6\}$
Range of $R = \{2, 4, 6, 8\}$
c) All element of R from B to A i.e.
$R^{-1} = \{(2, 3), (4, 6), (6, 5), (8, 6)\}$
d) Domain of $R^{-1} = \{2, 4, 6, 8\} =$
Range of $R = \{2, 4, 6, 8\}$
Range of $R^{-1} = \{3, 4, 5, 6\} =$
Domain of $R = \{3, 4, 5, 6\}$

Explanation
To find the inverse of R
Interchange x and y
$x \rightarrow y$
$y \rightarrow x$ then solve for y
$\therefore R^{-1} = \{(y,x): y = 2x-4\}$
$= \{(x,y): x = 2y-4\}$
$= \{(x,y): y = \frac{1}{2}x + 2\}$
Solve for y

2. Let A = the set of natural number.
If R is relation from A to A such that $R = \{(x, y): 2x + y = 10\}$ then
a) find all elements of R
b) find the domain and range of R
c) find R^{-1}
d) find the domain and range of R^{-1}

Solution: We know $R \subseteq A \times A \Rightarrow x \in A$ and $y \in A$, satisfies the relation
$R = \{(x,y): 2x + y = 10\}$
a) Let $x = 1 \in A \Rightarrow y = 8 \in A \Rightarrow (1, 8) \in R$
Let $x = 2 \in A \Rightarrow y = 6 \in A \Rightarrow (2, 6) \in R$
Let $x = 3 \in A \Rightarrow y = 4 \in A \Rightarrow (3, 4) \in R$
Let $x = 4 \in A \Rightarrow y = 2 \in A \Rightarrow (4, 2) \in R$
Let $x = 5 \in A \Rightarrow y = 0 \notin A$
Because 0 is not natural number $\therefore (5, 0) \notin R$
Hence $R = \{(1, 8), (2, 6), (3, 4), (4, 2)\}$
b) domain $= \{1, 2, 3, 4\}$ range of $R = \{2, 4, 6, 8\}$
c) all elements of $R^{-1} = \{(2, 4),(4,3),(6,2),(8,1)\}$
d) domain of $R^{-1} = \{2, 4, 6, 8\}$ = Range of R
range of $R^{-1} = \{1, 2, 3, 4\}$ = Domain of R

Note: $R^{-1} = \{(y, x): 2x + y = 10\} = \{(x, y): x + 2y = 10\}$
Solving for $y = \{(x,y): y = 5 - \frac{1}{2}x\}$

3. If A = {0, 1, 2, 3, 4, 5}
B = {0, 1, 2, 3, 4} and R is a relation from A to B such that
$R = \{(x, y) : y = \sqrt{x-1}$ then what is the range of R^{-1}

A. {1, 2, 5}
B. {0, 1, 2}
C. $\{0, 1, \sqrt{2}, \sqrt{3}, 2\}$
D. {0, 1, 2, 3}

Solution: $R \subseteq AxB \Rightarrow x \in A$ and $y \in B$ satisfies the relation

$$R = \{(x, y) : y = \sqrt{x-1}$$

- Let $x = 0 \in A \Rightarrow y = \sqrt{0-1} = \sqrt{-1} \notin B$
- Let $x = 1 \in A \Rightarrow y = \sqrt{1-1} = 0 \in B$
  $\therefore (1, 0) \in R \Rightarrow (0, 1) \in R^{-1}$
- Let $x = 2 \in A \Rightarrow y = \sqrt{2-1} = 1 \in B \quad \therefore (2, 1) \in R$
  $\Rightarrow (1, 2) \in R^{-1}$
- Let $x = 3 \in A \Rightarrow y = \sqrt{3-1} = \sqrt{2} \notin B$
  $\therefore (3, \sqrt{2}) \notin R \Rightarrow (\sqrt{2}, 3) \notin R^{-1}$
- Let $x = 4 \in A \Rightarrow y = \sqrt{4-1} = \sqrt{3} \notin B$
  $\therefore (4, \sqrt{3}) \notin R \Rightarrow (\sqrt{3}, 4) \notin R^{-1}$
- Let $x = 5 \in A \Rightarrow y = \sqrt{5-1} = 2 \in B$
  $\therefore (5, 2) \in R \Rightarrow (2, 5) \in R^{-1}$

$\therefore$ Element of R = {(1, 0),(2, 1),(5, 2)}
$\therefore$ Element of R^{-1} = {(0, 1), (1, 2), (2, 5)}

- Domain of R^{-1} = {0, 1, 2} = Range of R
- Range of R^{-1} = {1, 2, 5} = Domain of R

Answer: A

Note: $R^{-1} = \{(y, x): y = \sqrt{x-1}\} = \{(x, y): x = \sqrt{y-1}\}$
$= \{(x, y): y = x^2 + 1\}$ solving for y.

4. Let A = {1, 2, 3, 4, 5, 6, 7}
If R is relation from A to A such that R = {(x,y):y = 2x + 1}
Then a) List element of R
b) List domain and range of R
c) List element of R^{-1}
d) List domain and range of R^{-1}

Solution: $R \subseteq A \times B \Rightarrow x \in A$ and $y \in A$ satisfies $y = 2x + 1$
Let $x = 1 \in A \Rightarrow y = 3 \in A$

$\therefore (1, 3) \in R \Rightarrow (3, 1) \in R^{-1}$

Let $x = 2 \in A \Rightarrow y = 5 \in A$

$\therefore (2, 5) \in R \Rightarrow (5, 2) \in R^{-1}$

Let $x = 3 \in A \Rightarrow y = 7 \in A$

$\therefore (3, 7) \in R \Rightarrow (7, 3) \in R^{-1}$

Let $x = 4 \in A \Rightarrow y = 9 \notin A$

$\therefore (4, 9) \notin R \Rightarrow (9, 4) \notin R^{-1}$ b/c $9 \notin A$

a) $\therefore R = \{(1, 3), (2, 5),$

b) domain $= \{1, 2, 3\}$

Range $= \{3, 5, 7\}$

c) $R^{-1} = \{(3, 1), (5, 2),$

d) domain of $R^{-1} = \{3,$

Range of $R^{-1} = \{1, 2,$

Explanation

$R^{-1} = \{(y,x): y = 2x+1\}$

$= \{(x,y): x = 2y+1\}$

$= \{(x,y): y = \frac{1}{2}x - \frac{1}{2}$

5. Let $R = \{(x, y): 2x - 3y > 3$ then which of the following points satisfied the relation R^{-1}. $y > 3x + 3$

A. (-2, 3) B. (1, 3) C. (2, -1) D. (-3, -2)

Solution: A) $(-2, 3) \in R^{-1} \Rightarrow (3, 2) \in R$

$\Rightarrow 2(3) - 3(-2) > 3 \Rightarrow 12 > 3 =$ True

$\therefore (-2, 3) \in R^{-1}$

Explanation

If $(a, b) \in R$ then $(b,a) \in R^{-1}$

B) $(1, 3) \in R^{-1} \Rightarrow (3, 1) \in R$

$\Rightarrow 2(3) - 3(1) > 3 \Rightarrow 3 > 3 =$ False

$\therefore (1, 3) \notin R^{-1}$

C) $(2, -1) \in R^{-1} \Rightarrow (-1, 2) \in R$

$\Rightarrow (2)(-1) - 3(2) > 3 \Rightarrow -8 > 3$, False

$\therefore (2, -1) \notin R^{-1}$

D) $(-3, -2) \in R^{-1} \Rightarrow (-2, -3) \in R$

$\Rightarrow -2(-1) - 3(-3) > 3 \Rightarrow 5 > 3$, True

$\therefore (-3, -2) \in R^{-1}$

$\therefore$ (-2, 3) and (-3, -2) satisfies the relation R^{-1} and (3, -2) and (-2, -3) satisfies the relation R.

6. Let $R = \{(x, y): y \geq x^2 - 4$ and $y \leq 5\}$

a) Sketch the graph of R

b) Find the domain and range of R

c) Find R^{-1}

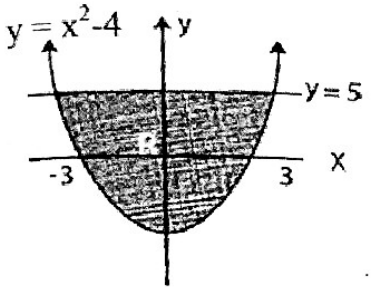
d) Find the domain and range of R^{-1}

Solution:

a) To sketch the graph, first we find the x-intercept set $y = 0$

$\Rightarrow x^2 - 4 = 0 \Rightarrow x = 2$, and $x = -2$

b) Domain of R = $\{x: x^2 - 4 \leq 5\}$
$= \{x: x^2 \leq 9\} = \{x: |x| \leq 3\}$
$= \{x: -3 \leq x \leq 3\}$



Range = $\{y: -4 \leq y \leq 5\}$

c) $R^{-1} = \{(y, x): y \geq x^2 - 4 \text{ and } y \leq 5\}$
$= \{(x, y): x \geq y^2 - 4 \text{ and } x \leq 5\}$
$= \{(x, y): y^2 \leq x + 4 \text{ and } x \leq 5\}$

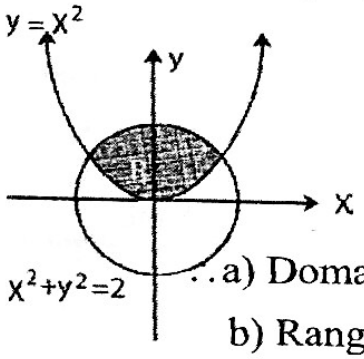
d) Domain of $R^{-1} = \{x: -4 \leq x \leq 5\}$
Range of $R^{-1} = \{y: -3 \leq y \leq 3\}$

7. Let $R = \{(x, y): x^2 + y^2 \leq 2 \text{ and } y \geq x^2\}$
Find a) Domain of R
b) Range of R
c) Domain of R^{-1}
d) Range of R^{-1}

Solution: The boundary of $x^2 + y^2 \leq 2$ and $y \geq x^2$ are $x^2 + y^2 = 2$ and $y = x^2$

Solving together $\begin{cases} x^2 + y^2 = 2 \\ y = x^2 \end{cases}$

$\Rightarrow x^4 + x^2 - 2 = 0$
$\Rightarrow (x^2 - 1)(x^2 + 2) = 0$
$\Rightarrow x^2 - 1 = 0 \text{ or } x^2 + 2 = 0$
$\Rightarrow x = 1, -1$



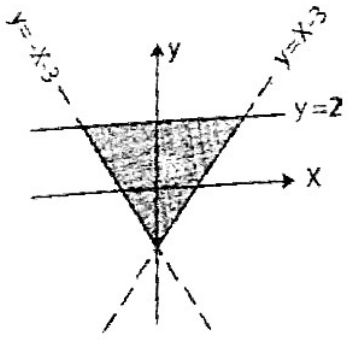
∴ a) Domain of $R = \{x: -1 \leq x < 1\}$
b) Range of $R = \{y: 0 \leq y \leq \sqrt{2}\}$
c) Domain of $R^{-1} = \{x: 0 \leq x \leq \sqrt{2}\}$
d) Range of $R^{-1} = \{y: -1 \leq y \leq 1\}$

8. Find the domain and range of the inverse of each of the following
a) $R = \{(x, y): y < 0\}$
b) $R = \{(x, y): y \leq 2, y > x - 3 \text{ and } y > -x - 3\}$
c) $R = \{(x, y): y = \sqrt{1 - |x|}\}$
d) $R = \{(x, y): y = 1 - 3^x\}$

Solution: a) $R^{-1} = \{(x, y): x < 0\} = \{(y, x): y < 0\}$
Domain of $R^{-1} = \{x: x < 0\}$

Range of $R^{-1} = \{y: y \in R\}$

b) To find the intersection point solving together $\begin{cases} y = 2 \\ y = x - 3 \\ y = -x - 3 \end{cases}$



$\Rightarrow$ when $y = 2$, $x = 5$ and $x = -5$
$\Rightarrow$ when $x = 0$, $y = -3$
$\therefore$ Domain of $R = \{x: -5 < x < 5\}$
Range of $R = \{y: -3 < y \leq 2\}$
$\therefore$ Domain of $R^{-1} = \{x: -3 < x \leq 2\}$
Range of $R^{-1} = \{y: -5 < y < 5\}$

c) $R = \{(x, y): y = \sqrt{1-|x|}$

$R^{-1} = \{(x,y): x = \sqrt{1-|y|}$ or $R^{-1} = |y| = 1 - x^2\}$

$\therefore$ Domain of $R^{-1} = \{x: 0 \leq x \leq 1\}$

Range of $R^{-1} = \{y: -1 \leq y \leq 1\}$

d) $R = \{(x, y): y = 1 - 3^x\}$

$R^{-1} = \{(x, y): x = 1 - 3^y\} = \{(x,y): y = \log_3^{1-x}\}$

Domain of $R^{-1} = \{x: x < 1\}$ and Range of $R^{-1} = \{y: y \in R\}$

Revision on Function

Definition: A function is a relation in which each element of the domain is paired with exactly one element in the range.

Note: Let A and B two non – empty sets.

- ✓ Then a **function** *f* from A to B denoted by *f*: A $\longrightarrow$ B is a rule for assigning to each element of **A** *exactly one element of* **B**.
- ✓ Domain of *f* = A and
- ✓ *f* is single valued (well – defined) i.e., if (a, b) $\in f$ and (a, c) $\in f$ imply that b = c, we write *f* (a) = b

Illustrative Example

9. Determine and justify which of the following sets are **function from A to B**: (*f*: A $\longrightarrow$ B) given by:

a) $f = \{(1, 2), (1, 3), (2, 3), (3, 4)\}$; where $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$

b) $f = \{(a, b), (b, c), (c, d), (d, a)\}$
Where, $A = \{a, b, c, d\}$, $B = \{a, b, c, d\}$

c) $f = \{(a, b), (b, c), (c, d)\}$;
Where, $A = \{a, b, c, d\}$; $B = \{a, b, c, d\}$

d) $f = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} | \ y = x + 4\}$; $A = B = \mathbb{Z}$

e) $f = \{(x, y) \in \mathbb{R} \times \mathbb{R}\} | \ y = \sqrt{x}\ \}$; $A = B = \mathbb{R}$

Solution:

a) Here f is not function because $f(1) = 2$ and $f(1) = 3$ thus $f(1)$ is not unique (not single valued).

b) f is function from A to B, because each element of A has exactly one image.

c) In this case, f is not a function from A to B, because, the domain of this relation f is $\{a, b, c\} \subset A$. But $A = \{a, b, c, d\}$

d) f is function from A to B.

e) f is not function, since it is not defined for negative real number.

10. What the domain and range of $f(x) = \sqrt{x - |x|}$?

Solution: Domain $= \{x: x - |x| \geq 0\} = \{x: |x| \leq x\} = \{x: x \geq 0\}$
Range $= \{y: y = 0\}$

Even and odd function

Definition: consider a function f: A $\longrightarrow$ B. Then f is called
i. **Even**, if and only if, for any $x \in A$; $f(-x) = f(x)$.
ii. Odd, if and only if for all $x \in A$, $f(-x) = -f(x)$.

Illustrative Example

11. Determine whether the following functions are even or odd or neither.

a) $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = x^3 - x$

b) $f: \mathbb{R} \longrightarrow [-3, \infty)$ defined by $f(x) = x^2 - 3$

c) $f: \mathbb{R} \longrightarrow [-3, \infty)$ defined by $f(x) = x^2 + 2x - 2$

d) $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = |x|x$

e) $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = |x| + x$

Solution:

a) $f(-x) = (-x)^3 - x = -(x^2 + x) = -f(x)$, therefore ***f* is odd.**

b) $f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x)$, therefore ***f* is even.**

c) $f(-x) = (-x)^2 + 2(-x) - 2 = x^2 - 2x - 2$, which is neither equal to $f(x)$ nor equal to $-f(x)$. Hence f is **neither** even nor an odd function.

d) $f(-x) = |-x|(-x) = -x|x| = -f(x)$, therefore f is odd

e) $f(-x) = |-x| + (-x) = |x| - x$, neither.

12. Which of the following function is an odd function?

A. $f(x) = |x|$ C. $f(x) = [x] - x$

B. $f(x) = |x - 2|$ D. $f(x) = \ell n\left(\frac{1-x}{1+x}\right)$

Solution: We need to show $f(-x) = -f(x)$

A. $f(-x) = |-x| = |x| = f(x)$, therefore f is even

B. $f(-x) = |-x - 2| = |x + 2|$ neither

C. $f(-x) = [-x] + x$, neither

D. $f(-x) = \ell n\left(\frac{1+x}{1-x}\right) = \ell n(1+x) - \ell n(1-x)$

$= [\ell n(1-x) - \ell n(1+x)]$

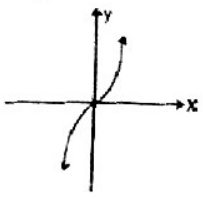
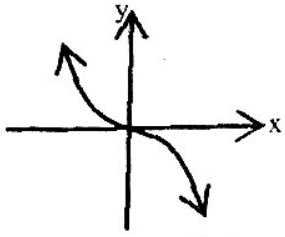
$= -\ell n\left(\frac{1-x}{1+x}\right) = -f(x)$ therefore, f is odd. **Answer: D**

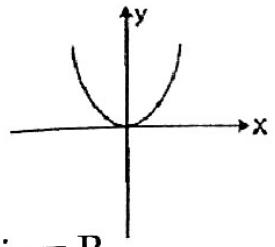
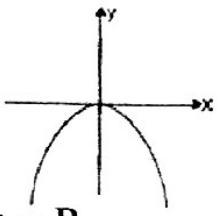
Some Types of Function

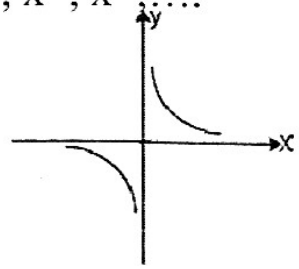
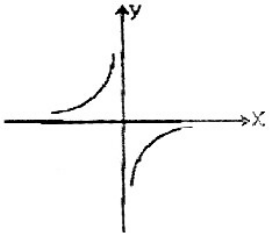
These are • Power function
- Absolute value (modulus) function
- Signum function
- Greatest integer function

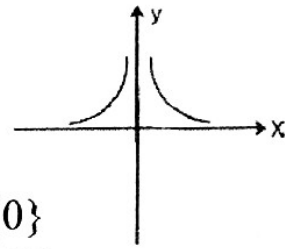
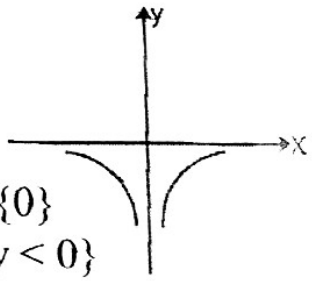
Power function

A function of the form $f(x) = ax^n$ where n is constant and $a \in R$

i) $f(x) = ax^n$	n is odd and $n > 0$
If $a > 0$  • domain R • Range R • odd function • Increasing Ex. $f(x) = x^3\ x^5, \ldots$	If $a < 0$  • Domain = $\mathbb{R}$ • Range = $\mathbb{R}$ • odd • decreasing Example, $f(x) = -x^3, -x^5, -x^7, \ldots$

ii) $f(x) = ax^n$,	n is an even and $n > 0$
If $a > 0$  • Domain = R • Range = $[0, \infty)$ • Even function	If $a < 0$  • Domain = R • Range = $(-\infty, 0]$ • Even function Example: $f(x) = -x^2, -x^4, \ldots$

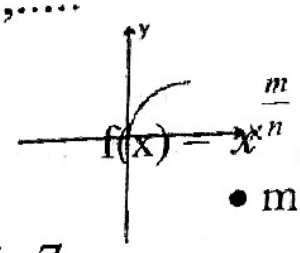
iii) $f(x) = ax^n$,	n is odd and n <
If $a > 0$ **Example:** $f(x)=x^{-3}, x^{-5}, x^{-7}, \ldots$  • Domain = R\{0} • Range = R\{0} • odd function • decreasing function	If $a < 0$ **Example:** $f(x)=-x^{-3}, -x^{-5}, -x^{-7}, \ldots$  • Domain = R\{0} • Range = R\{0} • odd function • Increasing function

iv) $f(x) = ax^{\frac{n}{m}}$ n is even and $n < 0$ m is odd	
If $a > 0$ **Example:** $f(x)= x^{\frac{-2}{3}}, x^{\frac{-4}{3}}, x^{\frac{-6}{5}}, \ldots$  • Domain = R\{0} • Range = {y:y>0} • Even function	If $a < 0$ **Example:** $f(x)= -x^{\frac{-2}{3}}, -x^{\frac{-4}{3}}, -x^{\frac{-6}{5}}, \ldots$  • Domain = R\{0} • Range = {y: y < 0} • even function

v) If $f(x) = x^{\frac{m}{n}}$, for m is odd, n is even the graph of f

- not symmetric
- neither even nor odd
- has range $[0, \infty)$
- has domain $[0, \infty)$
- increasing

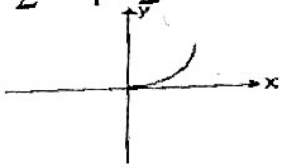
Example: $\frac{m}{n} = \frac{1}{2}, \frac{3}{4}, \ldots$



$f(x) = x^{\frac{m}{n}}$, for $m < n$

- m odd
- n even

Example: $\frac{m}{n} = \frac{3}{2}, \frac{5}{4}, \frac{7}{2}, \ldots$



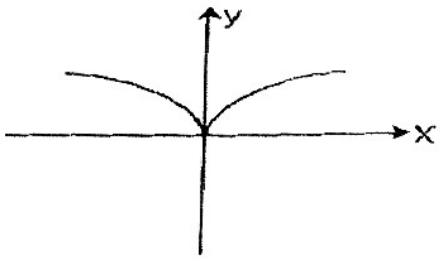
$f(x) = x^{\frac{m}{n}}$, for $m > n$

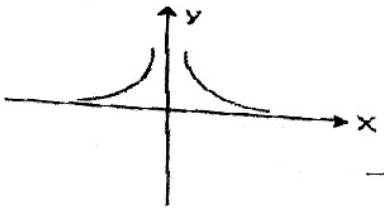
- m odd
- n even

vi) $f(x) = x^{\frac{m}{n}}$ for m is even, n is odd then the graph of f

- symmetric w.r.t y axis
- even function
- has domain real number
- has range: $0 \leq y < \infty$

Example: $f(x) = x^{\frac{2}{3}}, x^{\frac{4}{5}}, \ldots$





$\longrightarrow \frac{m}{n} < 0$

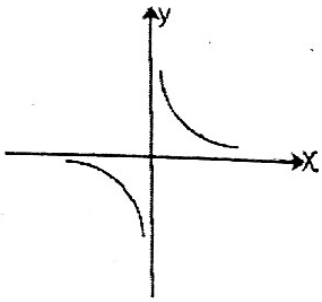
$f(x) = x^{\frac{m}{n}}$

Example, $f(x) = x^{\frac{-2}{3}}$ .....

vii) $f(x) = x^{\frac{-1}{n}}$, n is odd, then the graph of f

- symmetric
- odd function
- increasing
- Domain R\{0}

Example, $f(x) = x^{-1}, x^{-\frac{1}{3}}, x^{\frac{-1}{5}}, x^{\frac{-1}{7}}$ .....



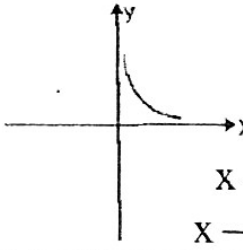
Explanation

- $x \longrightarrow 0^+$
  $y \longrightarrow \infty$
- $x \longrightarrow 0^-$
  $y \longrightarrow -\infty$
- $x \longrightarrow \infty$
  $y \longrightarrow 0$
- $x \longrightarrow -\infty$
  $y \longrightarrow 0^-$

viii) $f(x) = x^{\frac{-1}{n}}$, n is even, then the graph of f

- not symmetric
- neither even nor odd
- has domain $\{x: x > 0\}$
- Range = $\{y: y > 0\}$

Example, $f(x) = x^{\frac{-1}{2}} = \frac{1}{\sqrt{x}}$



Explanation

$x \longrightarrow 0^+, \ y \longrightarrow \infty$

$x \longrightarrow \infty, \ y \longrightarrow 0^+$

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Illustrative Example

13. Which of the following are power functions and which are not. (Justify)

a) $f(x)=\sqrt{\frac{1}{x}}$

b) $f(x)=\sqrt[3]{\frac{1}{2x}}$

c) $f(x)=\frac{x^2}{2}$

d) $f(x)=2x^{\frac{-4}{5}}$

e) $f(x)=2^x$

f) $f(x)=3x^2+4$

Solution: Power function is written in the form $f(x)=ax^r$

Thus, a) $f(x)=\sqrt{\frac{1}{x}}=\left(\frac{1}{x}\right)^{\frac{1}{2}}=x^{\frac{-1}{2}}$ ← power function

b) $f(x)=\sqrt[3]{\frac{1}{2x}}=\left(\frac{1}{2x}\right)^{\frac{1}{3}}=\left(\sqrt[3]{\frac{1}{2}}\right)x^{\frac{1}{3}}$ ← power function

with $a=\frac{1}{\sqrt[3]{2}}$

c) $f(x)=\frac{x^2}{2}=\frac{1}{2}x^2$ ← power function with $a=\frac{1}{2}$

d) $f(x)=2^x$ ← Not power function, it is exponential function

f) $f(x)=3x^2+4$ ← Not power function

Note: All power function of the form $f(x)=\mathbf{ax^r}$ with a = 1 satisfies multiplicative property of $f(xy)=f(x).f(y)$

Illustrative Example

14. Which of the following power function does not satisfies the condition $f(xy)=f(x)\cdot f(y)$?

a) $f(x)=\sqrt{\frac{1}{x}}$

c) $f(x)=x^3$

b) $f(x) = x^3$ d) $f(x) = 2x^4$

Solution: a) $f(xy) = \sqrt{\frac{1}{xy}} = \sqrt{\frac{1}{x}} \cdot \sqrt{\frac{1}{y}} = f(x).f(y)$

b) $f(xy) = (xy)^3 = x^3 \cdot y^3 = f(x) \cdot f(y)$

c) $f(xy) = (xy)^{\frac{2}{3}} = x^{\frac{2}{3}} \cdot y^{\frac{2}{3}} = f(x) \cdot f(y)$

d) $f(xy) = 2(xy)^4 = 2x^4 \cdot y^4$

but $f(x) \cdot f(y) = 2x^2 \cdot 2y^4 = 4x^4 y^4$

hence $f(xy) \neq f(x) \cdot f(y)$ **Answer: D**

Absolute value (Modulus) Function

Definition: The absolute value (Modulus) of x is defined by

$$f(x) = |x| = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$$

- Domain of $f = R$
- Range of $f = [0, \infty)$

Illustrative Example

15. Evaluate each of the following

a) $|-3|$ c) $\left|\sqrt{2} - 2\right|$ e) $\left|\sqrt{6} - 3\right|$

b) $|\pi - 4|$ d) $|x - y|$ f) $\left|3 - \sqrt{7}\right|$

Solution:

a) $|-3| = -(-3) = 3$, since $-3 < 0$, thus $|-3| = 3$

b) $|\pi - 4| = -(\pi - 4) = 4 - \pi$ since $\pi - 4 < 0$

c) $\left|\sqrt{2} - 2\right| = 2 - \sqrt{2}$

d) $|x - y| = \begin{cases} x - y \text{ if } x \geq y \\ y - x \text{ if } x < y \end{cases}$

e) $\left|\sqrt{6} - 3\right| = 3 - \sqrt{6}$

f) $\left|3 - \sqrt{7}\right| = 3 - \sqrt{7}$

Explanation

$\sqrt{9} = 3$

$\therefore \left|\sqrt{6} - \sqrt{9}\right| = \sqrt{9} - \sqrt{6}$

$= 3 - \sqrt{6}$

Note: $|x| = a$ iff $x = \pm a$, if $a < 0$ then $|x| = a$ has no ***Solution***.

16. Find the ***Solution*** set of each of the following equations.

a) $|x - 1| = \pi$ c) $|x| = -1$ e) $|4x + 3| = 5$

b) $|x| = \sqrt{2} - 1$ d) $|4x + 3| = 0$ f) $|3 - 5x| = 6$

g) $|x^2 - 2| = 1$ h) $|x^3 - 2| = 6$

Solution: Here

a) $|x - 1| = \pi \Leftrightarrow x - 1 = \pi$ or $x - 1 = -\pi \Rightarrow x = 1 + \pi$ or $x = 1 - \pi$

$\therefore$ S.S $= \{1 + \pi, 1 - \pi\}$

b) $|x| = \sqrt{2} - 1 \Leftrightarrow x = \sqrt{2} - 1$ *or* $x = 1 - \sqrt{2}$

c) $|x| = -1$ is $\varnothing$ because $|x| \geq 0$

d) $|4x + 3| = 0 \Leftrightarrow 4x + 3 = 0 \Rightarrow 4x = -3$

$$\therefore \text{S.S} = \left\{\frac{-3}{4}\right\}$$

e) $|4x + 3| = 5 \Leftrightarrow 4x + 3 = 5$ or $4x + 3 = -5$

$\Rightarrow 4x = 2$ or $4x = -8$

$$\therefore x = \frac{1}{2} \text{ or } x = -2$$

f) $|3 - 5x| = 6 \Leftrightarrow 3 - 5x = 6$ or $3 - 5x = -6$

$$\Rightarrow -5x = 3 \text{ or } -5x = -9 \Rightarrow x = \frac{-3}{5} \text{ or } x = \frac{9}{5}$$

$$\therefore \text{S.S} = \left\{\frac{-3}{5}, \frac{9}{5}\right\}$$

g) $|x^2 - 2| = 1 \Leftrightarrow x^2 - 2 = 1$ or $x^2 - 2 = -1$

$\Rightarrow x^2 = 3$ or $x^2 = -1 + 2 = 1$

$\Rightarrow x = \pm\sqrt{3}$ or $x = \pm 1$

$\therefore$ S.S $= \{\sqrt{3}, -\sqrt{3}, 1, -1\}$

h) $|x^3 - 2| = 6 \Leftrightarrow x^3 - 2 = 6$ or $x^3 - 2 = -6$

$\Rightarrow x^3 = 8$ or $x^3 = -4$

$\Rightarrow x = 2$ *or* $x = \sqrt[3]{-4}$

$\therefore$ S.S $= \{2, \sqrt[3]{-4}\}$

17. For what real number x true and solve each of the following.

a) $|2x + 3| = 4x + 8$

b) $|3x-4| = 4-x^2$

Solution: Universe $|2x+3| \geq 0 \Rightarrow 4x+8 \geq 0$

$\Rightarrow 4x \geq -8 \Rightarrow x \geq -2$

$\therefore$ Universe, $\{x: x \geq -2\} \longleftarrow$ domain

And $|2x+3| = 4x+8 \Leftrightarrow 2x+3 = 4x+8$ or $2x+3 = -4x-8$

$\Rightarrow -2x = 5$ or $6x = -11$

$\Rightarrow x = \dfrac{-5}{2}$ or $x = \dfrac{-11}{6} \in [-2, \infty)$

↑

Reject

The only solution set is $\left\{x : x = \dfrac{-11}{6}\right\}$ because $\dfrac{-5}{2} \notin [-2, \infty)$

b) $|3x-4| = 4-x^2 \Leftrightarrow 3x-4 = 4-x^2$ or $3x-4 = x^2-4$

$\Rightarrow x^2+3x-8 = 0$ or $x^2-3x = 0$

$\Rightarrow x(x-3) = 0$

$\therefore 0, x = 3$

Since the universe $4-x^2 \geq 0 \Leftrightarrow 4 \geq x^2$

$\Leftrightarrow x^2 \leq 4$

$\Leftrightarrow -2 \leq x \leq 2$

$\therefore$ The only S.S is $\{0\}$, but -4 and 3 are not ***Solution*** because $4-x^2$ is negative at $x = 3$.

Graph of Absolute value

Definition: The graph of absolute value is the pictorial representation of set of points (x, f(x)) in a plane of absolute value function.

Illustrative Example

18. Sketch the graph find the domain and range of each of the following.

a) $f(x) = |x|$

b) $f(x) = |x| + 2$

c) $f(x) = |x+2|$

d) $f(x) = \left|\dfrac{1}{x+2}\right|$

e) $f(x) = |x^2-4|$

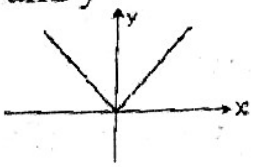
f) $f(x) = |x| - x + 1$

g) $f(x) = |x-3|$

h) $f(x) = |x-3| + 1$

Solution: a) We have $|x| = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$ and we know $y = f(x)$

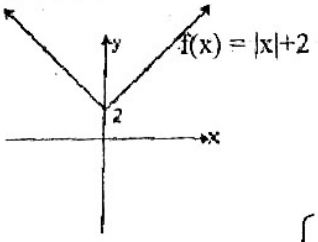
$\Rightarrow y = x$ if $x \geq 0$ and $y = -x$ if $x < 0$



f(x) = |x|

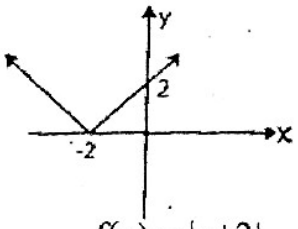
- Domain = R
- Range = $[0, \infty)$

b) Adding 2 from each value of $f(x) = |x|$



- domain = R
- range = $[2, \infty)$

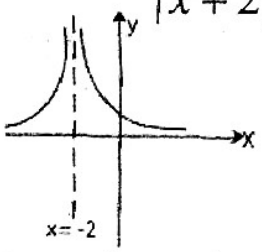
c) We have $f(x) = |x + 2| = \begin{cases} x+2 \text{ if } x+2 \geq 0 \\ -x-2 \text{ if } x+2 < 0 \end{cases}$



f(x) = |x+2|

- Domain = R
- Range = $[0, \infty)$
- Symmetry line x = -2

d) $f(x) = \left|\frac{1}{x+2}\right| = \begin{cases} \frac{1}{x+2} \text{ if } x \geq -2 \\ \frac{-1}{x+2} \text{ if } x < -2 \end{cases}$



- Domain = R\{-2}
- Range = {y: y > 0}
- Symmetry, the line x = -2

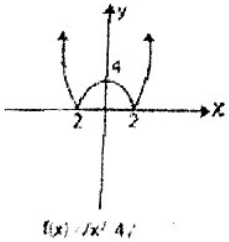
$f(x) = \left|\frac{1}{x+2}\right|$

$x^2 = 4$

e) $f(x) = |x^2 - 4| = \begin{cases} x^2 - 4 \text{ if } x^2 - 4 \geq 0 \text{ i.e } x \leq -2 \text{ or } x \geq 2 \\ 4 - x^2 \text{ if } -2 < x < 2 \end{cases}$

$x^2 \geq 4$

$-2 \geq x$



f(x) = |x² - 4|

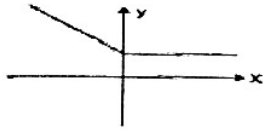
- Domain = R

- Range = $[0, \infty)$

$f(x) = |x^2 - 4|$

f) we have $|x| = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$

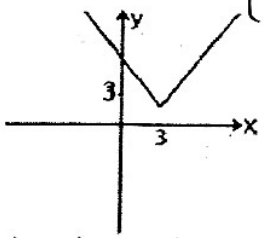
Hence $f(x) = |x| - x + 1 = x - x + 1 = 1$ if $x \geq 0$ and
$f(x) = -x - x + 1 = -2x + 1$ if $x < 0$



- Domain = R
- Range = $[1, \infty)$

$f(x) = |x| - x + 1$

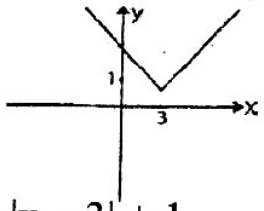
g) $f(x) = |x - 3| = \begin{cases} x - 3 \text{ if } x \geq 3 \\ -x + 3 \text{ if } x < 3 \end{cases}$



- Domain = R
- Range = $[0, \infty)$
- axis of symmetry line x=3

$f(x) = |x - 3|$

h) we shift 1 unit up the graph of $|x - 3|$



- Domain = R
- Range = $[1, \infty)$
- symmetry, x = 3

$f(x) = |x - 3| + 1$

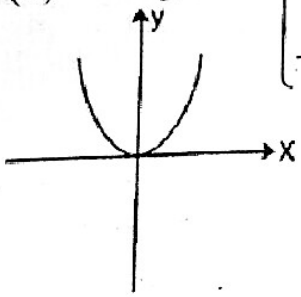
Signum function

The signum function, read as **signum (x)** is written as sign x and is

defined by $f(x) = \text{sgn}x = \begin{cases} 1 \text{ for } x > 0 \\ 0 \text{ for } x = 0 \\ -1 \text{ for } x < 0 \end{cases}$

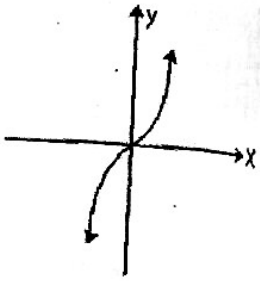
Or

c) $f(x) = x^n sgnx = \begin{cases} x^n & if\ x > 0 \\ 0 & if\ x = 0 \\ -x^n & if\ x < 0 \end{cases}$



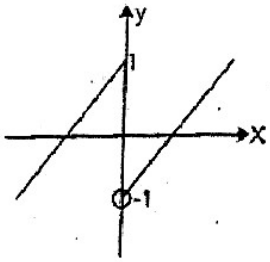
$f(x) = x^n sgnx$, if n is odd

- Domain = R
- Range = $[0, \infty)$
- even function



$f(x) = x^n sgnx$ if n is even

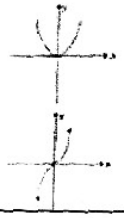
d) $f(x) = x - sgnx = \begin{cases} x-1 & if\ x > 0 \\ 0 & if\ x = 0 \\ x+1 & if\ x < 0 \end{cases}$



- Domain = R
- Range = R\{1,-1}
- f in neither even nor odd

$f(x) = x - sgnx$

Note: i) sgnx is an odd function



ii) xsgnx, x^3sgnx, x^5sgnx .... are even function of graph of the form

iii) x^2sgnx, x^4sgnx, ...are **odd function** of graph of the form

22. If $f(x) = 3(sgnx + 1)$ then find $f(-x) + f(x)$

Solution: Here

$\Rightarrow f(-x) = 3(sgn(-x) + 1) = 3[-sgnx + 1]$

$= -3sgnx + 3$

Thus $f(-x) + f(x) = -3sgnx + 3 + 3sgnx + 3$

$= 6$

$\therefore f(-x) + f(x) = 6$

Explanation
Sgn(-x) = -sgnx

Explanation
f(-2) + f(2) = 6
f(-1) + f(1) = 6
$\therefore$ f(-x) + f(x) = 6

23. If $h(x) = \frac{1}{2}(\operatorname{sgn} x + 2)$, then

Find a) h(x) + h(-x)

b) h (2) + h(-2)

c) $h\left(\frac{1}{3}\right) + h\left(-\frac{1}{3}\right)$

Solution: Here

a) $h(-x) = \frac{1}{2}(\operatorname{sgn}(-x) + 2)$

$= \frac{-1}{2}\operatorname{sgn} x + \left(\frac{1}{2}\right)(2) = \frac{-1}{2}\operatorname{sgn} x + 1$

Explanation
sgn(-x) = -sgnx
because
signum function is
odd function

And $h(x) = \frac{1}{2}(\operatorname{sgn} x + 2)$

$= \frac{1}{2}\operatorname{sgn} x + (2)\left(\frac{1}{2}\right) = \frac{1}{2}\operatorname{sgn} x + 1$

$\therefore h(x) + h(-x) = \frac{1}{2}\operatorname{sgn} x + 1 \ -\frac{1}{2}\operatorname{sgn} x + 1 = 2$

b) $h(2) + h(-2) = \left(\frac{1}{2}\operatorname{sgn}(2) + 1\right) + \left(-\frac{1}{2}\operatorname{sgn}(2) + 1\right)$

$= \frac{1}{2} + 1 - \frac{1}{2} + 1 = 2$

Explanation
sgn(2) = 1
sgn(-2) = -1

c) $h\left(\frac{1}{3}\right) + h\left(-\frac{1}{3}\right) = \left(\frac{1}{2}\operatorname{sgn}\left(\frac{1}{3}\right) + 1\right) + \left(-\frac{1}{2}\operatorname{sgn}\left(\frac{1}{3}\right) + 1\right)$

$= \frac{1}{2} + 1 - \frac{1}{2} + 1 = 2$

Greatest integer function

The greatest integer function is denoted by [x] and is defined by $f(x) = [x]$ = the greatest integer less than or equal to x. Note $[x] \leq x$

Illustrative Example

Evaluate each of the following

a) $[2\cdot64]$ d) $[-1\cdot24]$ g) $[-1\cdot5]$

b) $[1\cdot24]$ e) $[\pi]$ h) $\left[-\sqrt{3}\right]$

c) $[-2\cdot64]$ f) $[-2]$ o) $[0\cdot25]$

p) $[-0\cdot5]$ q) $[3]$ r) $[2\cdot47]$

s) $[-31\cdot004]$

Solution: Here

a) $[2\cdot64] = 2$ e) $[\pi] = 3$ o) $[0\cdot25] = 0$

b) $[1\cdot24] = 1$ f) $[-2] = -2$ p) $[-0\cdot5] = -1$

c) $[-2\cdot64] = -3$ g) $[-1\cdot5] = -2$ q) $[3] = 3$

d) $[-1\cdot24] = -2$ h) $\left[-\sqrt{3}\right] = -2$ r) $[2\cdot47] = 2$

Equation involving greatest integer

24. Find the value of x, for each part

a) $[x] = a$ d) $[x - 2] = 6$

b) $[x] = -3$ e) $[2x + 1] = 10$

c) $[x] = 3$

Solution: Here

a) $[x] = a$ if $a \leq x < a + 1$, where a is integer

b) $[x] = -3$ if $-3 \leq x < -3 + 1 \Rightarrow -3 \leq x < -2$

c) $[x] = 3$ if $3 \leq x < 3 + 1 = 3 \leq x < 4$

d) $[x - 2] = 6$ if $6 \leq x - 2 < 7$

adding both sides 2

e) $[2x + 1] = 10$ if $10 \leq 2x + 1 < 11$

subtracting (1) and dividing by $2 \Rightarrow \frac{9}{2} \leq x < 5$

Special properties of greatest integer function

If $f(x) = [x]$, then

i) $[x] \leq x$

ii) $f(x + k) = f(x) + k$, where k is integer and x is real number

iii) $f(x) + f(y) \leq f(x + y) \leq x + y$

iv) $f(x) \leq x < f(x) + 1$

Example: Let $x = 2.54$, $y = -3.46$, $k = 2$

i) $(2.54) \leq 2.54 \Rightarrow 2 \leq 2.54$

ii) $f(2.45 + 2) = f(2.54) + 2 = [2.54] + 2 = 2 + 2 = 4$

iii) $f(2.54) + f(-3.46) \leq f(2.54 + -3.46) \leq 2.54 + -3.46$

$\Rightarrow [2.54] + [-3.46] \leq [-0.92] \leq -0.92$
$\Rightarrow 2 + -4 \leq -1 \leq -0.92$
$\Rightarrow -2 \leq -1 \leq -0.92$
$\therefore f(x) + f(y) \leq f(x + y) \leq x + y$ since $f(x) = [x]$
$\therefore [x] + [y] \leq [x + y] \leq x + y$ and $f(y) = [y]$

iv) $[2.54] \leq 2.54 < [2.54] + 1$
$\Rightarrow 2 \leq 2.54 < 2 + 1$
$\therefore f(x) \leq x < f(x) + 1$, if $f(x) = [x]$
i.e. $[x] \leq x < [x] + 1$

25. Let $f(x) = x - [x]$
Find a) the domain
b) the possible value of f(x) will be $0 \leq f(x) < 1$
i.e. all the out put will be lie between 0 and 1

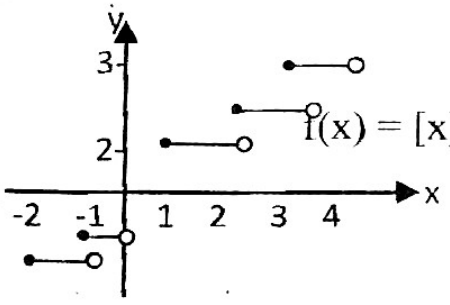
Example: $f(2.9) = 2.9 - [2.9] = 0.9$
$f(-2.9) = -2.9 - [-2.9] = -2.9 + 3 = 0.1$
$f(-2.1) = -2.1 + [-2.1] = -2.1 + 3 = 0.9$
$f(-2) = -2 - [-2] = -2 + 2 = 0$

Thus, conclude that
$0 \leq x - [x] < 1$ for every value of x

Graph of greatest integer function f(x) = [x]

- Domain = R
- Range = the set of integer (z)

Hence f: R $\longrightarrow$ z given by $f(x) = [x]$



$f(x) = [x]$

$$f(x) = [x] = \begin{cases} -2 \text{ if } -2 \leq x < -1 \\ -1 \text{ if } -1 \leq x < 0 \\ 0 \text{ if } 0 \leq x < 1 \\ 1 \text{ if } 0 \leq x < 2 \\ 2 \text{ if } 2 \leq x < 3 \end{cases}$$

Note: The graph of the greatest integer function y = [x] lies on below the line y = x, so it provides an integer **floor for x.**

26. Let $y = 2[x] + 3$ and $y = 3[x - 2] + 5$
Then find the value of $[x + y]$

Solution: we have

$\Rightarrow 2[x] + 3 = 3[x - 2] + 5$

$\Rightarrow 2[x] + 3 = 3[x] - 6 + 5$

$\Rightarrow 3[x] - 2[x] = 3 + 1 = 4 \Rightarrow [x] = 4$ if $4 \le x < 5$

Therefore, $y = 2[x] + 3 = 2(4) + 3$, since $[x] = 4$

$\therefore y = 11$

And $x = 4 +$ fractional part $= 4 + f$

Hence $[x + y] = [4 + f + 11] = [15 + f] = 15$

27. Indicate whether the statement is true for all real number x, y, and z. If false give a counter example

a) $[x + y] = [x] + [y]$

b) $[x] + [y] \le [x + y]$

c) $[x] - [y] = [x - y]$

d) $[x]([y] + [z]) = [xy] + [xz]$

e) $[x]^2 = [x^2]$

f) $[x]^2 \le [x^2]$

g) $\dfrac{[x]}{2} = \left[\dfrac{x}{2}\right]$

h) $[-x] = -[x]$

i) $[x] + \dfrac{1}{2} =$

j) $[xy] = [x][y]$

k) $[xy] \ge [x][y]$

Solution: Here

a) False: counter example
Let $x = 3.6$, and $y = 2.4$
$\Rightarrow [x] + [y] \ne [x + y] \Rightarrow [3.6] + [2.4] \ne [3.6 + 2.4]$
$\Rightarrow 3 + 2 \ne 6$
$\Rightarrow 5 \ne 6 \quad \therefore [x] + [y] \ne [x + y]$

b) True, $[x] + [y] \le [x + y]$

c) False, counter example, Let $x = -3.8$, $y = -5.2$
then $[x] = [-3.8] = -4$ and $[y] = [-5.2] = -6$
thus, $[x] - [y] = -4 - 6 = -10$
and $[x - y] = [-3.8 - (-5.2)] = [-3.8 + 5.2] = [1.4] = 1$
$\therefore [x] - [y] \ne [x - y]$

d) False

e) False, counter example, Let $x = 2.4$
$[2.4]^2 \ne [(2.4)^2] \Rightarrow 2^2 \ne [5.76]$
$4 \ne 5$
$\therefore [x]^2 \ne [x^2]$

And $[x]^2 \leq [x^2]$

f) True

g) False, counter example, Let x = 7.6 then $\frac{[7.6]}{2} = \frac{7}{2} = 3.5$

and $\left[\frac{7.6}{2}\right] = [3.8] = 3$

$\therefore \frac{[x]}{2} \neq \left[\frac{x}{2}\right]$

h) False, counter example, Let x = 3.7 then [-3.7] = -4 and –[3.7] = -3

$\therefore [-x] \neq -[x]$ note $[-x] \leq -[x]$

i) False, counter example, Let x = 3.6

then $[3.6] + \frac{1}{2} = 3 + \frac{1}{2} = 3.5$ and

$\left[3.6 + \frac{1}{2}\right] = [3.6 + 0.5] = [4.1] = 4$

$\therefore [x] + \frac{1}{2} \neq \left[x + \frac{1}{2}\right]$

j) False, counter example, Let x = 2.4, and y = 3.4
then, $[(2.4)\cdot(3.4)] = [8.16) = 8$
and $[2.4][3.4] = (2)(3) = 6$ $\therefore [xy] \neq [x][y]$

k) $[xy] \geq [x][y]$ is true

Classification of Function

Based on some relating situation between the domain and its value 'range' function can classify as:

i. One – to -- one (injective)
ii. On to (surjective)
iii. One – to – one correspondence (bijective)

One – to – one function

Definition: A function f: A $\longrightarrow$ B is said to be **one – to – one** (injective) if and only if, each element of the range is paired with exactly one element of the domain. That means;

a) if $f(a_1) = f(a_2)$, then $a_1 = a_2$

b) if $f(a_1) \neq f(a_2)$, then $a_1 \neq a_2$ for all $a_1, a_2 \in A$.

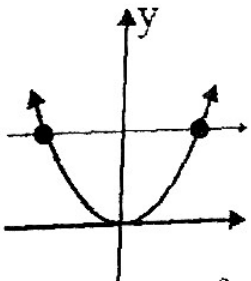
Horizontal – Line Test

If the graph of a function f is known, there is a simple test, called the **horizontal – line test**, to determine whether f is one – to – one.

Theorem: A function f: A $\longrightarrow$ B is one – to – one if and only if every horizontal line crosses the graph of a function f in at most one point.

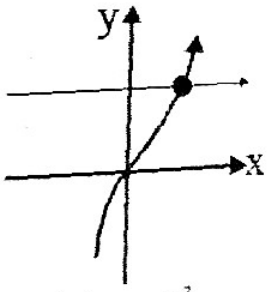
Consider the following figure

a) A horizontal line crosses the graph of f twice; f is not – one – to – one.



$\rightarrow f(x) = ax^2$

b) Every horizontal line crosses the graph of g exactly once. Therefore, g is one – to – one.



$g(x) = ax^3$

Theorem:

A function that is increasing over its domain is one – to – one function.

A function that is decreasing over its domain is one – to – one.

Study the following Example

✓ Quadratic function with domain on set of real number is not **one – to – one.**

- f: $\mathbb{R} \longrightarrow [0, \infty)$ given by $f(x) = x^2$ is not one – to – one but:

i) $f: [0, \infty) \longrightarrow [0, \infty)$ given by $f(x) = x^2$ is one – to – one

ii) $f: (-\infty, 0] \longrightarrow [0, \infty)$ given by $f(x) = x^2$ is one – to – one.

✓ Absolute value function with domain set of real number is **not one – to – one.**

- $f(x) = |x|$ is not one – to – one when $x \in (-\infty, \infty)$, but
- $f(x) = |x|$ is one – to – one when $x \in (-\infty, 0]$ or $x \in [0, \infty)$

✓ Linear function $f(x) = mx + c$, for $m \neq 0$ are one – to – one.

✓ Logarithm function and exponential function are one – to – one.

- $f(x) = \log_a^x$ is one – to – one when $x \in (0, \infty)$
- $f(x) = a^x$ is one – to – one $x \in (-\infty, \infty)$

Illustrative Example

28. Determine which of the following functions are one – to – one

A. $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = 2x - 3$

B. $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = |x| + x$

C. $f: \mathbb{R} \longrightarrow \mathbb{Z}$ defined by $f(x) = [x]$

D. $f: (2, \infty) \longrightarrow \mathbb{R}$ defined by $f(x) = \log_5^{x-2}$

E. $f: \mathbb{R} \longrightarrow (0, \infty)$ defined by $f(x) = 3^x$

Solution:

A. Linear function are one – to – one function.

B. Not – one to one function, because $f(-2) = |-2| + (-2) = 2 + (-2) = 0$ and $f(-3) = |-3| + (-3) = 3 - 3 = 0$

$\Rightarrow f(-2) = f(-3) = 0$, but $-2 \neq -3$

C. Not one – to – one, let $a_1 = 2.1$ and $a_2 = 2.7$

$\Rightarrow f(2.1) = [2.1] = 2$ and $f(2.7) = [2.7] = 2$

$\Rightarrow f(2.1) = f(2.7) = 2$, but $2.1 \neq 2.7$

D. Logarizm function of the form $f(x) = \log_a^x$ are one – to – one function, because a horizontal line crosses its graph one time only.

E. Exponential of the form $f(x) = a^x$ are one to one function, because a horizontal line crosses its graph at most once.

9. Determine which of the following functions are one – to – one

A. $f: \mathbb{R} \longrightarrow [3, \infty)$, defined by $f(x) = (x + 2)^2 + 3$

B. $f: [-2, \infty] \longrightarrow [3, \infty)$ defined by $f(x) = (x + 2)^2 + 3$

C. $f:\mathbb{R}\longrightarrow[0,\infty)$ defined by $f(x)=|x-4|$

D. $f:[4,\infty)\longrightarrow(0,\infty)$ defined by $f(x)=|x-4|$

Solution:

A. Let $a_1=0$ and $a_2=-4$

$f(0)=(0+2)^2+3=7$ and $f(-4)=(-4+2)^2+3=7$

$\Rightarrow f(0)=f(-4)$, but $0\neq -4$

For different domain, with the same y – value.

Therefore f is not – one – to one.

B. For all, $a_1, a_2\in[-2,\infty)$, $f(a_1)\neq f(a_2)$ and $a_1\neq a_2$ for all a_1, a_2. Therefore, f is one – to – one.

C. Let $a_1=5$ and $a_2=3$, thus $f(5)=|5-4|=1$, and

$f(3)=|3-4|=1$

$\Rightarrow f(5)=f(3)=1$, but $5\neq 3$

$\Rightarrow$ Different domain with the same y – value.

$\therefore f$ is not – one – to one.

D. f is one – to – one for all $x\in[4,\infty)$

30. Find the set A, for which f is one – to – one function.

a) $f:A\longrightarrow\mathbb{R}$, given by $f(x)=\log_5^{2x-3}$

b) $f:A\longrightarrow[-6,\infty)$ given by $f(x)=x^2+8x+10$

c) $f:A\longrightarrow[2,\infty)$ given by $f(x)=|4x+3|+2$

Solution:

a) f is one – to – one on the domain:

$$A=\{x:2x-3>0\}=\left\{x:x>\frac{3}{2}\right\}$$

b) f is one – to – one on the interval.

$$A=\left\{x:x\geq\frac{-b}{2a}\text{ or }x\leq\frac{-b}{2a}\right\}$$

$$\text{Then, } A=\left\{x:x\geq\frac{-8}{2}\text{ or }x\leq\frac{-8}{2}\right\}$$

$\therefore A=(-\infty,-4]$ or $[-4,\infty)$

c) f is one – to – one on the interval

$A=\{x:4x+3\geq 0$ or $4x+3\leq 0\}$

$$\therefore A=\left(-\infty,\frac{-3}{4}\right)\text{or}\left[\frac{-3}{4},\infty\right)$$

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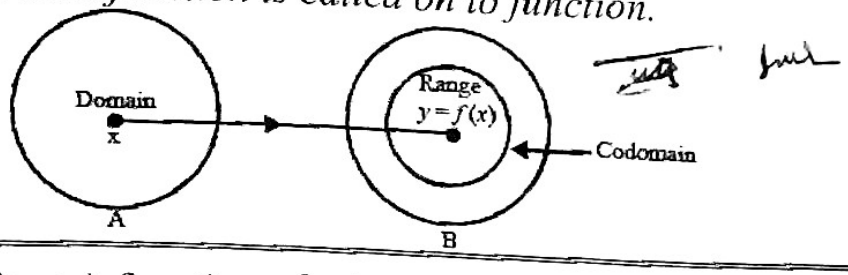
On to (Surjective function)

Function is mapping say A to B denoted by f: A $\longrightarrow$ B,
for each $x \in$ A paired with exactly one element $y \in$ B, such that $f(x) = y$.

- The element y is called **the image of x** under f.
- The set A is called the **domain of the function f.**
- The set B is called the **codomain of f.**
- All the image of the element in A under f is a subset of B, which is called the range of the function f.

In some cases, range of a function may be equal to the whole codomain, in which case such function is called on to function.

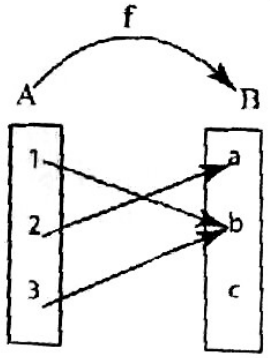


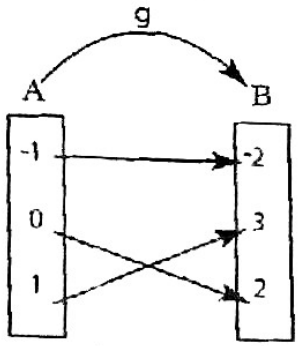
Definition: A function f : A $\longrightarrow$ B is on to if and only if:

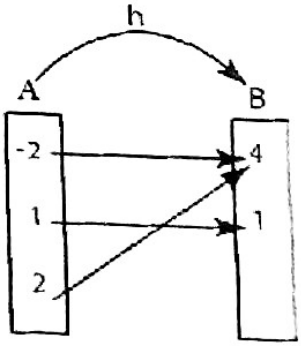
- Range if f = B, i.e., f(A) = B
- For every b $\in$ B, the exist a $\in$ A such that f(a) = b
- Every element of B has pre-image in A.

Illustrative Example

31. Which of the following Venn diagram are onto

a.


b.


c.


a. f: A $\rightarrow$ B is not onto, because c $\in$ B has no pre- image in A, i. e., image of $f \neq$ B,
and image of f = {a, b} but B = {a, b, c}

b. g: A $\rightarrow$ B is onto, because image of g = B = {– 2, 3, 2}

c. h : A $\rightarrow$ B is onto, since image of h = B = {4, 1}

32. Determine which of the following sets are onto function from A to B,

a. $f = \{(-1,2), (0,1) (1,2), (2, 0)\}$,
where $A = \{-1, 0, 1, 2\}$ and $B = \{0, 1, 2, 5\}$

b. $g = \{(-2, 4) (-1,1) (1, 3), (2, 4)\}$
where $A = \{-2, -1, 1, 2\}$ and $B = \{1, 3, 4\}$

Solution: f is onto, if im $(f) = B$.

a. f is not onto, because $f(A) \neq B$,
$\Rightarrow$ image of $f = \{0, 1, 2\}$, but $B = \{0, 1, 2, 5\}$
i.e $5 \in B$, has no pre- image in A.

b. g is onto, because $g(A) = B = \{1, 3, 4\}$

Note: *If $f: A \to B$ is onto iff the range of $f = B$*

Illustrative Example

33. Determine which of the following functions are onto:

a. $f: \mathbb{R} \to \{-1, 0, 1\}$, given by $f(x) = (-1)^x$

b. $f: \mathbb{R}^+ \to (0, \infty)$, given by $f(x) = x^2$, for all $x \in \mathbb{R}$

c. $f: \mathbb{R} \to \mathbb{Q}$, given by $f(x) = 2x$, for all $x \in \mathbb{R}$

d. $f: \mathbb{R} \to \mathbb{R}$, given by $f(x) = |x| + x$, for all $x \in \mathbb{R}$

e. $f: \mathbb{R} \to [3, \infty]$ given by $f(x) = x^2 - 4x + 7$

f. $f: [1, \infty) \to \mathbb{R}$, given by $f(x) = 4 - \sqrt{x-1}$

g. $f: \mathbb{R} \to [1, \infty)$, given by $f(x) = |x - 3| + 1$

Solution: Let $f: A \to B$, then

a. $B = \{-1,0,1\}$ and range of $f = \{-1, 1\}$
$\Rightarrow 0 \in B$, has no pre – image in A, such that $f(x) = 0$
$\Rightarrow$ Range of $f \neq B \Leftrightarrow \{-1, 1\} \neq \{-1, 0, 1\}$
$\therefore f$ is not onto.

b. $B = (0, \infty)$ and range of $f = (0, \infty)$
$\Rightarrow$ Range of $f = B, \Rightarrow (0, \infty) = (0, \infty)$ $\therefore f$ is onto

c. $B = \mathbb{Q}$ and range of $f = (0, \infty)$ since $-1 \in Q$, but there does not exist any $x \in \mathbb{R}$ such that $f(x) = 2^x = -1$
$\therefore f$ is not onto

Method II, Range of $f \neq B \Rightarrow \mathbb{Q} \neq (0, \infty)$
$\therefore f$ is not onto.

d. $B = \mathbb{R}$, and range of $f = (0, \infty)$ since $-2 \in \mathbb{R}$, but there does not exist any $x \in \mathbb{R}$, such that $f(x) = |x| + x = -2$
$\therefore f$ is not onto

Method II, Range of $f \neq B$
$\therefore f$ is not on to

e. $B = [3, \infty)$ and range of $f(x) = (x-2)^2 + 3$ is $[3, \infty)$

$\Rightarrow$ Range of f = B $\Rightarrow [3, \infty) = [3, \infty)$

$\therefore f$ is onto

f. B = $\mathbb{R}$ and range of $f(x) = 4 - \sqrt{x-1}$ is $(-\infty, 4]$ since $5 \in \mathbb{R}$, but there does not exist any $x \in [1, \infty)$. Such that

$f(x) = 4 - \sqrt{x-1} = 5 \Rightarrow \sqrt{x-1} = -1$

$\therefore f$ is not onto.

Method II. Range of $f \neq$ B $\Rightarrow \mathbb{R} \neq (-\infty, 4]$

$\therefore f$ is not onto

g. B = $[1, \infty)$ and range of $f(x) = |x - 3| + 1$ is $[1, \infty)$.

$\Rightarrow$ Range of f = B = $[1, \infty)$

$\therefore f$ is onto.

34. For each of the following function, find the set B for which $f: \mathbb{R} \to$ B is onto

a. $f(x) = x^2 + 4$
b. $f(x) = \text{s gn}x$
c. $f(x) = [x]$
d. $f(x) = 2 - 5|x|$
e. $f(x) = |x| + 3$
f. $f(x) = x^2 + 6x + 10$

Solution:

Remember, $f: \mathbb{R} \to$ B is onto if range of f = B.

a. Range of $f(x) = x^2 + 4$ is $[4, \infty]$ for all $x \in \mathbb{R}$

$\therefore$ B = $[4, \infty)$

b. Range of $f(x) = \text{sgn}x$ is $\{-1, 0, 1\}$ for all $x \in \mathbb{R}$

$\therefore$ B = $\{-1, 0, 1\}$

c. Range of $f(x) = [x]$ is z for all $x \in \mathbb{R}$

$\therefore$ B = z = the set of integer

d. Range of $f(x) = 2 - 5|x|$ is $(-\infty, 2)$ for all $x \in \mathbb{R}$

$\therefore$ B = $(-\infty, 2]$

e. Range of $f(x) = |x| + 3$ is $[3, \infty)$, for all $x \in \mathbb{R}$

$\therefore$ B = $[3, \infty)$

f. Range of $f(x) = x^2 + 6x + 5 = (x + 3)^2 - 4$ is $[-4, \infty)$

$\therefore$ B = $[-4, \infty)$

One – to – one Correspondence

Two sets are said to be equivalent set if there exists one- to- one correspondence (bijective) between them.

Definition: A function $f: A \to B$ is a one – to – one correspondence (bijective) if and only if f is both onto and one -- to one

Illustrative Example

35. Determine and justify which of the following sets are one – to – one, onto or one – to – one correspondence from A to B.

a. $f = \{(x, 1), (y, 1) (z, 2), (w, 3)\}$, $A = \{x, y, z, w\}$ and $B = \{1,2,3\}$

b. $g = \{(x, 1), (y, 2), (z, 3)\}$, $A = \{x, y, z\}$ and $B = \{1, 2, 3,4\}$

c. $h = \{(x, 1), (y, 2) (z, 3)\}$, $A = \{x, y, z\}$ and $B = \{1, 2, 3\}$

d. $k = \{(x, 1), (y, 2), (z, 3), (w, 3)$, $A = \{x, y, z\}$ and $B = \{1, 2, 3, 4\}$,

Solution:

a. f is onto function but not one- to – one

b. g is one- to- one, but not onto, because range of $g \neq B$

c. h is one – to – one correspondence, because h is both onto and one – to –one

d. k is neither onto nor one – to – one

36. Determine which of the following functions are onto, one – to – one or one – to – one correspondence .

a. $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|\, x$ for all $x \in \mathbb{R}$

b. $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = \dfrac{4x+3}{7}$ for all $x \in \mathbb{R}$

c. $f: [2, \infty) \to [0, \infty)$ defined by $f(x) = \sqrt{x-2}$, for all $x \geq 2$

d. $f: \mathbb{R} \to \mathbb{Q}$, defined by $f(x) = 3^x$, for all $x > \mathbb{R}$

e. $f: (1, \infty) \to (0, \infty)$ defined by $f(x) = \log_5^{x-1}$, for all $x > 1$

f. $f: [3, \infty) \to [1, \infty)$ defined by $f(x) = (x-3)^2 + 1$ for all $x \geq 3$

g. $f: \mathbb{R} \to \{-1, 0, 1\}$ defined by $f(x) = \text{sgn}x$.

Solution: Let $f: A \longrightarrow B$

a) i) Let $a_1, a_2 \in (-\infty, \infty) \Rightarrow f(a_1) \neq f(a_2)$ then $a_1 \neq a_2$

$\therefore f$ is one – to – one

ii) $B = \mathbb{R}$ and range of $f = \mathbb{R}$

$\Rightarrow$Range of $f = B = \mathbb{R}$, then f is on to

iii) Since f is both one – to – one and on to, then

f is one to one correspondence

b) Linear functions are one – to – one and on to.

$\Rightarrow f$ is one to one correspondence

c) i) For all $a_1, a_2 \in [2, \infty)$, then $f(a_1) \neq f(a_2)$

$\Rightarrow \sqrt{a_1 - 2} \neq \sqrt{a_2 - 2}$ then $a_1 \neq a_2$

$\Rightarrow f$ is one to one

ii) B $= [0, \infty)$ and range of $f = [0, \infty)$ for all $x \geq 2$

$\Rightarrow$Range of f = B $= [0, \infty) \Rightarrow f$ is on to

iii)Since f is both one to one and onto

$\Rightarrow f$ is one to one correspondence

d) i) For all $a_1, a_2 \in Q \Rightarrow f(a_1) \neq f(a_2) \Rightarrow 3^{a_1} \neq 3^{a_2}$ then $a_1 \neq a_2, \Rightarrow f$ is one to one

ii) B = $\mathbb{Q}$ and range of $f = (0, \infty)$ for all $x \in \mathbb{R}$

$\Rightarrow$Range of $f \neq$ B, therefore f is not on to

e) i) For all $x > 1$, $f(x) = \log_5^{x-1}$ is one to one

ii) B $= (0, \infty)$ and range of $f = (0, \infty)$

$\Rightarrow$Range of f = B $= (0, \infty)$ thus f is on to

iii) Since f is both one – to – one and on to

Therefore f is one – to – one correspondence.

f) i) For all $a_1, a_2 \in [3, \infty) \Rightarrow f(a_1) \neq f(a_2)$

$\Rightarrow (a_1 - 3)^2 + 1 \neq (a_2 - 3)^2 \Rightarrow a_1 \neq a_2$

Therefore f is one – to – one correspondence

ii) B $= [1, \infty)$ and range of $f = [1, \infty)$, for all $x \geq 3$

$\Rightarrow$Range of f = B $= [1, \infty)$ thus f is on to

iii) Since f is both one to one and on to therefore f is one to one correspondence

g) i) Let $a_1 = 3$ and $a_2 = 4 \Rightarrow f(3) = 1$ and $f(4) = 1$,

so that $f(3) = f(4)$ but $3 \neq 4$, therefore f is no one to one

ii) B $= \{-1, 0, 1\}$ and range of $f = \{-1, 0, 1\} \Rightarrow f$ is on to

7. Which one of the following is a one – to – one correspondence function from A $= [0, 1]$ to B $= [2, 3]$?

A. $f(x) = x + 1$ C. $f(x) = 2x + 3$

B. $f(x) = x^3 + 1$ D. $f(x) = x^2 + 2$

Solution: f : A $\longrightarrow$ B

A. B $= [2, 3]$ and range of $f = [1, 2]$ for all $x \in [0, 1]$

$\Rightarrow$ Range of $f \neq$ B, thus f not on to

B. B $= [2, 3]$ and range of $f = [1, 2]$ for all $x \in [0, 1]$

$\Rightarrow$Range of $f \neq$ B Therefore, f is not on to.

C. B $= [2, 3]$ and range of $f = [3, 5]$ for all $x \in [0, 1]$

$\Rightarrow$Range of f $\neq$ B Therefore, f is not on to.

D. i) B $= [2, 3]$ and range of $= [2, 3]$ for all $x \in [0, 1]$

$\Rightarrow$Range of f = B $= [2, 3]$ Therefore, f is on to

ii) For all $a_1, a_2 \in [0, 1]$ $f(a_1) \neq f(a_2)$

$\Rightarrow a_1^2 + 2 \neq a_2^2 + 2 \Rightarrow a_1 \neq a_2$

$\therefore f$ is one – to – one

iii) since f is both one – to – one and on to

$\Rightarrow f$ is one – to – one correspondence

Combination function

(Sum, difference, and quotient) of function

Term	Functional value	Domain
i) $f+g$	$(f+g)(x) = f(x) + g(x)$	Domain of $(f \cap g)$
ii) $f-g$	$(f-g)(x) = f(x) - g(x)$	
iii) $f.g$	$(f.g)(x) = f(x).g(x)$	
iv) $\frac{f}{g}$	$\frac{f}{g}(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$	

Illustrative Example

32. Let $f(x) = \sqrt{9 - x^2}$ and $g(x) = 4x + 5$

Find a) $(f+g)(x)$ b) $(f-g)(x)$ c) $(fg)(x)$

d) $\left(\frac{f}{g}\right)(x)$ e) $\left(\frac{g}{f}\right)(x)$

Solution: i) The domain of $f = \{x: 9 - x^2\} = \{x: -3 \leq x \leq 3\}$

ii) The domain of $g = \{x: x \in R\} = (-\infty, \infty)$

The domain of $f+g$, $f-g$ and $f.g$ are

{the domain of f} $\cap$ {the domain of g} $= \{x: -3 \leq x \leq 3\}$

Therefore,

	Value	Domain
a)	$(f+g)(x) = \sqrt{9 - x^2} + 4x + 5$	$\{x: -3 \leq x \leq 3\}$
b)	$(f-g)(x) = \sqrt{9 - x^2} - (4x + 5)$	$\{x: -3 \leq x \leq 3\}$
c)	$(fg)(x) = \sqrt{(9 - x^2)}(4x + 5)$	$\{x: -3 \leq x \leq 3\}$

d) $\left(\frac{f}{g}\right)(x) = \frac{\sqrt{9-x^2}}{4x+5}$ $\left\{x : -3 \le x \le 3 \text{ and } x \ne \frac{-5}{4}\right\}$

e) $\left(\frac{g}{f}\right)(x) = \frac{4x+5}{\sqrt{9-x^2}}$ $\{x: -3 < x < 3\}$

Composition of Function

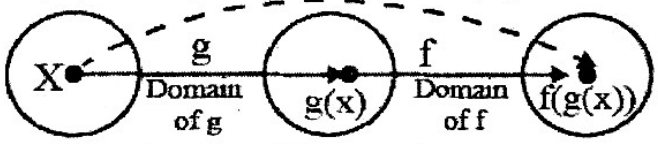
The composition of f and g is denoted by $(fog)(x) = f(g(x))$ provided that:

- g (x) is in the domain of f.
- Domain of fog ⊑ Domain of g.
- Range of fog ⊑ Range of f.

The domain of fog is the set of all real number

x in the domain of g such that g(x) in the domain of f. i.e.

x ∈ Domain of g and g(x) ∈ Domain of f.



Illustrative Example

38. Let $f(x) = 3x - 4$ and $g(x) = x^2 + 3x$. **Find**

a) (fog)(1) c) (fog)(x) e) (fof)(2)

b) (gof)(1) d) (gof)(x) f) (fof)(x)

Solution:

a) $(fog)(1) = f(g(1)) = f(4) = 12 - 4 = 8$

b) $(gof)(1) = g(f(1)) = g(-1) = 1 - 3 = -2$

c) $(fog)(x) = f(g(x)) = f(x^2 + 3x) = 3(x^2 + 3x) - 4$

d) $(gof)(x) = g(f(x)) = g(3x - 4) = (3x - 4)^2 + 3(3x - 4)$
$= 9x^2 - 24x + 16 + 9x - 12 = 9x^2 - 5x + 4$

39. Find f and g such that $(fog)(x) = h(x)$ if:

a) $h(x) = \sqrt{x^2 + 3x}$ b) 3^{2x^2-1}

Solution: $(fog)(x) = f(g(x)) = h(x)$

a) Let $g(x) = x^2 + 3x$ and $f(x) = \sqrt{x}$

$\Rightarrow f(g(x)) = f(x^2+3x) = \sqrt{x^2+3x} = h(x)$

b) Let $g(x)=2x^2-1$ and $f(x)=3^x$

$\Rightarrow f(g(x)) = f(2x^2-1) = 3^{2x^2-1} = h(x)$

40. If $f(x) = 3x^2 - 7$ and $g(x) = 4x + k$, for what value of k, so that the graph of (fog)(x) crosses the y-axis at 8?

A. $k = 0$ B. $k = \pm\sqrt{3}$ C. $k = \pm\sqrt{5}$ D. $k = \pm 5$

Solution:

(fog)(x) crosses, y-axis when $x = 0 \Rightarrow f(g(0)) = f(k) = 3k^2 - 7 = 8$

$$k^2 = \frac{15}{3} = 5 \Leftrightarrow k = \pm\sqrt{5}$$

Answer: C

41. Let $f(x) = 3x - 1$. Then find g(x):

a) if $f(g(x)) = 9x^2 + 12x + 8$

b) if $g(f(x)) = x^2 - 1$

Solution: $f(x) = 3x - 1$

a) $f(g(x)) = 3g(x) - 1 = 9x^2 + 12x + 8 \Rightarrow 3g(x) = 9x^2 + 12x + 9$

$\therefore g(x) = 3x^2 + 4x + 1$

b) Let $g(x) = ax^2 + bx + c$, then $g(g(x)) = g(3x - 1)$

$\Rightarrow g(3x-1) = a(3x-1)^2 + b(3x-1) + c = x^2 - 1$

$\Rightarrow 9ax^2 - 6ax + a + 3bx - b + c = x^2 - 0x - 1$

Comparing like coefficient

$$\Rightarrow 9ax^2 = x^2 \Rightarrow a = \frac{1}{9} \text{ and } -6ax + 3bx = 0$$

$$\Rightarrow 3b = 6a = 6\left(\frac{1}{9}\right) \Rightarrow b = \frac{2}{9}$$

$$\text{Also, } a - b + c = -1 \Rightarrow \frac{1}{9} - \frac{2}{9} + c = -1 \Rightarrow c = \frac{-8}{9}$$

$$\therefore g(x) = \frac{1}{9}x^2 + \frac{2}{9}x - \frac{8}{9}$$

42. The function $f(x)$ is defined in [0, 1].
What are the domains of the function $f(2x^2)$.

A. $\mathbb{R}$ B. $-\frac{1}{2} \le x \le \frac{1}{2}$ C. $-\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}$ D. $[0, \infty)$

Explanation: Let $u = 2x^2$

$\Rightarrow f(u)$ is defined for $0 \le u \le 1$

$\Rightarrow 0 \le 2x^2 \le 1$, since $2x^2 \ge 0$ for all x

$$\Rightarrow x^2 \le \frac{1}{2} \Rightarrow -\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}}$$ **Answer: C**

43. If $f(x) = \log(4 - x^2)$ and $g(x) = 3x + 7$ then what is the domain of $(fog)(x)$?

Solution: $(fog)(x) = f(g(x))$

$x \in D_g$ and $g(x) \in D_f$

The function $f(x)$ is defined in $4 - x^2 > 0 = \{x: -2 < x < 2\}$

$\Rightarrow f(g(x))$ is defined for: $-2 < g(x) < 2$

$\Rightarrow -2 < 3x + 7 < 2$

$$\Rightarrow -9 < 3x < -5 \Rightarrow -3x < x < -\frac{5}{3}$$

$$\therefore \text{Domain of } (fog)(x) = \{x: -3 < x < -\frac{5}{3}\}$$

44. If $f(x) = \dfrac{1}{x-2}$ and $g(x) = \dfrac{1}{x+4}$

a) What is the domain of $(fog)(x)$?

b) What is the domain of $(gof)(x)$?

Solution:

a) Domain of $g(x) = \{x: x \neq -4\}$ and domain of $f(x) = \{x: x \neq 2\}$

$$\Rightarrow f(g(x)) = \frac{1}{g(x) - 2}, g(x) - 2 \neq 0$$

Domain of $f(g(x)) = \{x \in D_g \text{ and } g(x) \in D_f\}$

$= \{x: x \neq -4 \text{ and } g(x) \neq 2\}$

$$= \{x: x \neq -4 \text{ and } \frac{1}{x+4} \neq 2\}$$

$= \{x: x \neq -4 \text{ and } 2x + 8 \neq 1\}$

$$= \{x: x \neq -4 \text{ and } x \neq \frac{-7}{2}\}$$

$$\therefore \text{Domain of } (fog)(x) \text{ is } \mathbb{R} \setminus \left\{-4, \frac{-7}{2}\right\}$$

b) Exercise left for you. (Ans: $\mathbb{R} \setminus \left\{2, \frac{7}{4}\right\}$.

45. If $f(x) = \ell n\,(x + 3)$ and $g(x) = x^3 - 4$, then what is the domain of $(fog)(x)$?

A. $(-1, \infty)$ B. $(1, \infty)$ C. $[1, \infty]$ D. $[-1, \infty)$

Solution: $(fog)(x) = f(g(x))$

Domain of $f(x) = \{x: x + 3 > 0\} = \{x > -3\}$

$\Rightarrow f(g(x))$ is defined for $-3 < g(x) < \infty \Rightarrow g(x) > -3$

$\therefore$ Domain of $(fog)(x) = \{x: g(x) > -3\} = \{x: x^3 - 4 > -3$

$\therefore$ Domain of $(fog)(x) = \{x: x^3 > 1\} = (1, \infty)$ **Answer: B**

INVERSE FUNCTIONS

The inverse of a one to one function f, denoted f^{-1}, is a function whose ordered pairs are obtained from f by interchanging the x and y – coordinates.

That means:

If $(a, b) \in f$, then $(b, a) \in f^{-1}$ or if $(a, b) \in f^{-1}$, then $(b, a) \in f$.

Note:

1. A function is invertible if and only if one – to – one.
2. If f is strictly monotonic on its entire domain, then it is invertible.

Illustrative Example

46. Determine whether each function is invertible. If it is invertible, then find the inverse function.

a) $f = \{(-2,1), (-1,0), (1,2), (3, 2)\}$ b) $g = \{(3, 1), (0, 2), (2, 3)\}$

Solution:

a) Since (1, 2) and (3, 2) have the same y coordinate, this function is not one – to – one, and f is not invertible.

b) This function is one – to – one, and so it is invertible $g^{-1} = \{(1, 3), (2, 0), (3, 2)\}$.

To find the inverse of $y = f(x)$ (when it exists):

1. Replace $f(x)$ by y
2. Interchange x and y
3. Solve for y then replace y by $f^{-1}(x)$

Note: Domain of f^{-1} = Range of f.
Range of f^{-1} = Domain of f.

HORIZONTAL – LINE TEST FOR INVERS FUNCTION

A function f has an inverse if and only if no horizontal line crosses its graph more than once.

Illustrative Example

47. Determine whether each function is invertible. If it is invertible, then find the inverse function.

a) $f = \{(x, y): y = 3x - 2\}$ b) $g = \{(x, y): y \geq 3x - 2\}$

c) $h = \{(x, y): y = \log_3^{4x}\}$ d) $k = \{(x, y): y = e^{2x}\}$

Solution:

a) $f = \{(x, y): y = 3x - 2\}$ is linear function and linear functions are one-to-one and so f is invertible.

To find the inverse:

Let $y = f(x) = 3x - 2$, interchange x and y

$\Rightarrow x = 3y - 2 \Leftrightarrow 3y = x + 2$

$\Rightarrow y = \dfrac{x+2}{3}$, therefore, $f^{-1}(x) = \dfrac{x+2}{3}$

b) $g = \{(x, y): y \geq 3x - 2\}$. This function is not invertible because a horizontal line crosses the graph at two point $(-2, 1)$ and $(0, 1)$.

c) $h = \{(x, y): y = \log_3^{4x}\}$. This function is invertible because every horizontal line that crosses the graph only once.

To find the inverse: h^{-1}, let $y = h(x) = \log_3^{4x}$ interchange x and y.

$$\Rightarrow x = \log_3^{4y} \Leftrightarrow 4y = 3^x \Rightarrow y = \frac{3^x}{4}$$

$$\therefore h^{-1}(x) = \frac{3^x}{4}$$

d) $k = \{(x, y): y = e^{2x}\}$. This function is one-to-one and so that k has an inverse.

To find k^{-1}, let $y = k(x) = e^{2x}$, $x \longleftrightarrow y$

$$\Rightarrow x = e^{2y} \Leftrightarrow 2y = \ln x \Rightarrow y = \frac{\ln x}{2}$$

$$\therefore k^{-1}(x) = \frac{\ln x}{2}$$

Note:
If the point (a, b) lies on the graph of f, then the point (b, a) must lie on the graph of f^{-1}, and vice versa. This means that the graph of f^{-1} is a reflection of the graph of f in the line $y = x$.

48. If the point (4, -5) is on the graph of $y = f(x)$, which point is on the graph of $y = f^{-1}(x)$.

A. $\left(\frac{1}{4}, -5\right)$ B. $(-4, 5)$ C. $(-5, 4)$ D. $\left(4, -\frac{1}{5}\right)$

Answer: C

Identify Inverse Functions

> Functions ***f*** and **g** are inverses of each other if and only if
> i) $(fog)(x) = x$ for every value of x in the domain of g.
> ii) $(gof)(x) = x$ for every value of x in the domain of *f*.

Illustrative Example

49. Determine whether each pair of functions ***f*** and **g** are inverses of each other.

a) $f(x) = 3x + 7$ and $g(x) = \frac{x-7}{3}$

b) $f(x) = x^2$ and $g(x) = \sqrt{x}$

c) $f(x) = \sqrt[3]{2x+1}$ and $g(x) = \frac{x^3 - 1}{2}$

d) $f(x) = \sqrt{x+2}$ and $g(x) = x^2 - 2$

e) $f(x) = \sqrt{x+2}$ and $g(x) = x^2 - 2$, where $x \geq 0$

Solution: We must verify that $f(g(x)) = x$ and $g(f(x)) = x$.

a) $f(g(x)) = 3g(x) + 7 = 3\left(\frac{x-7}{3}\right) + 7 = x$

$$g(f(x)) = \frac{f(x) - 7}{3} = \frac{3x + 7 - 7}{3} = x, \text{ for all } x.$$

$\therefore f$ and g are inverses of each other.

b) If $x \in (-\infty, \infty)$, we can write:

$$f(g(x)) = (g(x))^2 = \left(\sqrt{x}\right)^2 = \sqrt{x^2} = |x|$$

The domain of f is $(-\infty, \infty)$, and $|x| \neq x$ if x is negative.
$\therefore f$ and g are not inverse of each other.

DIGITAL LIBRARY FOR ETHIOPIAN STUDENTS
Digital Library for Ethiopian students
Link: https://telegram.me//digitallibrary4ethiopianstudents or digitallibrary4ethiopianstudents
YouTube Digital[0,1] Ethiopia

c) $f(g(x)) = \sqrt[3]{2g(x)+1} = \sqrt[3]{\frac{2(x^3-1)}{2}+1} = x$

$\therefore f$ and g are inverse of each other.

d) If $x \in (-\infty, \infty)$, then:

$f(g(x)) = \sqrt{g(x)+2} = \sqrt{(x^2-2)+2} = \sqrt{x^2} = |x|$

Since $x \in (-\infty, \infty)$, therefore, $|x| \neq x$

$\therefore f$ and g are not inverses of each other.

e) If $x \in [0, \infty)$ then:

$f(g(x)) = \sqrt{g(x)+2} = \sqrt{x^2-2+2} = \sqrt{x^2} = |x| = x$

For all $x \geq 0$, $f(g(x)) = g(f(x)) = x$

$\therefore f$ and g are inverse of each other.

50. Find the inverse of the following function and determine the domain and range of f^{-1}.

a) $f(x) = \frac{1}{3x-2}$ b) $f(x) = \frac{4x}{x+3}$

c) $f(x) = \frac{3x+2}{4x-1}$ d) $f(x) = 2x^3 + 5$

Solution: Replace $y = f(x)$

a) $y = \frac{1}{3x-2} \Leftrightarrow 3xy - 2y = 1 \Rightarrow 3xy = 2y+1 \Rightarrow x = \frac{2y+1}{3y} = f^{-1}(y)$

$\therefore f^{-1}(x) = \frac{2x+1}{3x}$

Domain of $f^{-1} = \mathbb{R} \setminus \{0\}$ and

Range of $f^{-1} = \mathbb{R} \setminus \left\{\frac{2}{3}\right\}$

b) $y = \frac{3x+2}{4x-1}$, interchange x to y

$x = \frac{3y+2}{4y-1} \Leftrightarrow 4xy - x = 3y + 2$

$\Rightarrow 4xy - 3y = x + 2 \Leftrightarrow y(4x-3) = x + 2$

$\Rightarrow y = \frac{x+2}{4x-3} \Rightarrow f^{-1}(x) = \frac{x+2}{4x-3}$

$\therefore$Domain of $f^{-1} = \mathbb{R} \setminus \left\{\frac{3}{4}\right\}$ and

Range of $f^{-1} = \mathbb{R} \setminus \left\{\frac{1}{4}\right\}$

c) $y = \frac{4x}{x+3}$ ⟵ interchange x and y

$$x = \frac{4y}{y+3} \Leftrightarrow xy + 3x = 4y$$

$$\Rightarrow xy - 4y = -3x \Leftrightarrow y(x-4) = -3x$$

$$y = \frac{-3x}{x-4} \Rightarrow f^{-1}(x) = \frac{3x}{4-x}$$

Domain of $f^{-1} = \mathbb{R}\setminus\{4\}$ and

Range of $f^{-1} = \mathbb{R}\setminus\{-3\}$

d) $y = 2x^3 + 5$ ⟵ interchange x to y

$\Rightarrow x = 2y^3 + 5 \Rightarrow 2y^3 = x - 5$

$\Rightarrow y = \sqrt[3]{\frac{x-5}{2}}$, therefore $f^{-1}(x) = \sqrt[3]{\frac{x-5}{2}}$

Domain of $f^{-1} = \mathbb{R}$ and Range of $f^{-1} = \mathbb{R}$

51. Find the inverse of each of the following function.

a) $f(x) = \ell n(2x + 1)$ c) $f(x) = \log_2^{\sqrt[3]{4x-1}}$

b) $f(x) = 1 + \frac{1}{2}\ell n(x-3)$ d) $f(x) = \ell n\left(\frac{x}{x-1} + 2\right)$

Solution:

a) $y = \ell n\,(2x + 1)$ ⟵ interchange x to y

$$\Rightarrow x = \ell n(2y+1) = \log_e^{(2y+1)}$$

$$\Rightarrow 2y + 1 = e^x \Rightarrow y = \frac{e^x - 1}{2}$$

$$\therefore f^{-1}(x) = \frac{1}{2}(e^x - 1)$$

b) $y = 1 + \frac{1}{2}\ell n(x-3)$ ⟵ interchange x to y

$\Rightarrow x = 1 + \frac{1}{2}\ell n(y-3) \Leftrightarrow x - 1 = \frac{1}{2}\ell n(y-3)$

$\Rightarrow 2x - 2 = \ln(y-3) = \log_e^{(y-3)} \Rightarrow e^{2x-2} = y - 3$

$\therefore y = e^{2x-2} + 3 \Rightarrow f^{-1}(x) = e^{2x-2} + 3$

c) $y = \log_2^{\sqrt[3]{4y-1}}$, interchange x to y

$\Rightarrow x = \log_2^{\sqrt[3]{4y-1}} \Rightarrow \sqrt[3]{4y-1} = 2^x$

$\Rightarrow 4y - 1 = (2^x)^3 \Leftrightarrow 4y = 2^{3x} + 1$

$\therefore y = \frac{2^{3x}+1}{4} \Rightarrow f^{-1}(x) = \frac{1}{4}(2^{3x}+1)$

d) $y = \ell n\left(\frac{x}{x-1} + 2\right)$ ⟵ interchange x to y

$x = \ell n\left(\frac{y}{y-1} + 2\right) = \ln\left(\frac{3y-2}{y-1}\right) \Rightarrow e^x = \frac{3y-2}{y-1} \Leftrightarrow e^x y - e^x = 3y - 2$

$\Rightarrow e^x y - 3y = e^x - 2 \Leftrightarrow y(e^x - 3) = e^x - 2$

$\therefore y = \frac{e^x - 2}{e^x - 3} \Rightarrow f^{-1}(x) = \frac{e^x - 2}{e^x - 3}$

52. Find the inverse of the following function:

a) $f(x) = \frac{1}{e^x - 1}$ b) $f(x) = \frac{1}{e^x + 1}$

c) $f(x) = \sqrt[3]{1 + e^{-x}}$ d) $f(x) = \frac{1}{e^{-x} + 1}$

Solution:

a) $y = \frac{1}{e^x - 1}$ then $x = \frac{1}{e^y - 1}$

$\Rightarrow xe^y - x = 1 \Rightarrow xe^y = x + 1 \Rightarrow e^y = \frac{x+1}{x} \Rightarrow y = \ell n\left(\frac{x+1}{x}\right)$

$\therefore f^{-1}(x) = \ln\left(\frac{x+1}{x}\right) = \ell n(x+1) - \ln x$

b) Exercise left for you. **Ans:** $\ell n\left(\frac{1-x}{x}\right)$.(UEE)

c) Exercise left for you. **Ans:** $\ell n\left(\frac{1}{x^3 - 1}\right)$.(UEE)

Let $y = f(x) = |x + 2| + 3 \Rightarrow y = x + 2 + 3 = x + 5$, then
$x = y + 5 \Rightarrow y = x - 5$
Therefore $f^{-1}(x) = x - 5$

57. Which of the following functions are invertible?
A. $f: \mathbb{R} \longrightarrow$ be defined by $f(x) = [x]$
B. $f: \mathbb{R} \longrightarrow \{-1, 0, 1\}$ be defined by $f(x) = \text{sgn}(x)$
C. $f: \mathbb{R} \longrightarrow [-9, \infty)$ be defined by $f(x) = x^2 + 2x - 8$
D. $f: [-2, 0) \longrightarrow [-4, 0]$ be defined by $f(x) = x^2 - 4$

Solution:
When $x \in [-2, 0]$, $f(x) = x^2 - 4$ is one to one so that invertible.
Answer:

58. What is the set of value of B, for which $f: \mathbb{R} \longrightarrow$ B defined by $f(x) = x^2 + 6x + 7$ is on to?
A. $[-4, \infty)$ B. $[-2, \infty)$ C. $[-3, \infty)$ D. $(-\infty, -1]$

Solution:
$f: \mathbb{R} \longrightarrow$ B is said to be on to if and only if range of f = B
Hence $f(x) = x^2 + 6x + 7 = x^2 + 6x + 9 + 7 - 9$
$\Rightarrow f(x) = (x + 3)^2 - 2$
$\therefore$The range of $f = [-2, \infty)$ **Answer:**

59. For what values of A, $f: A \longrightarrow [5, \infty)$ be defined by $f(x) = x^2 + 2x + 6$ is $1 - 1$.
A. $\mathbb{R}$ B. $[-1, \infty)$ C. $(-\infty, -1]$ D. B and C

Solution:
$f: A \longrightarrow [5, \infty)$ is $1 - 1$ correspondence if and only if one – to – one and on to.
A. $f: \mathbb{R} \longrightarrow [5, \infty)$ given by $f(x) = x^2 + 2x + 6$ is on to because image of $f(x)$ $x^2 + 2x + 6$ is $[5, \infty)$. But not one to one since $(-2, 6)$ and $(0, 6)$.
B. $f: [-1, \infty) \longrightarrow [5, \infty)$, given by $f(x) = x^2 + 2x + 6$ is on to and one – to – one
C. $f: (-\infty, -1] \longrightarrow [5, \infty)$, given by $f(x) = x^2 + 2x + 6$ onto and one – to – one.

60. Let $F(x) = f(2g(x))$ where $f(x) = x^4 + x^3 + 1$ for $0 \le x \le 2$ and $g(x) = f^{-1}(x)$. Find F (3)
Solution: We have $f(x) = x^4 + x^3 + 1$, for $0 \le x \le 2$.
$f(0) = 0^4 + 0^3 + 1 = 1 \Rightarrow f^{-1}(1) = 0$ g(1)

$f(1) = 1^4 + 1^3 + 1 = 3 \Rightarrow f^{-1}(3) = 1 = g(3)$
$f(2) = 2^4 + 2^3 + 1 = 25$
$\therefore F(3) = f(2g(3)) = f(2) = 25$

61. Let $f(x) = 2x^3 + 5x + 8$ and $f^{-1}(2x) = -1$. Find the value of x.

Solution: If $f^{-1}(a) = b \Rightarrow f(b) = a$

Then, $f^{-1}(2x) = -1 \Rightarrow f(-1) = 2x \Rightarrow -2 + -5 + 8 = 2x \Leftrightarrow x = \frac{1}{2}$

62. Let $f^{-1}(3x - 1) = -2$ and $f(-2) = 11$. Find x

Solution: $f^{-1}(3x - 1) = -2 \Rightarrow f(-2) = 3x - 1$

$\Rightarrow 3x - 1 = 11 \Rightarrow 3x = 12 \qquad \therefore x = \frac{12}{3} = 4$

63. If $y = f(x)$ and $f(x) = \frac{1}{1-x}$ and $z = f(y)$, then $f(z)$ is equal to:

A. $-x$ B. x C. $x-1$ D. 2

Solution:

Now since $f(x) = \frac{1}{1-x}$, we have,

$$f(z) = \frac{1}{1-z} = \frac{1}{1-f(y)} = \frac{1}{1-\frac{1}{1-y}}$$

$$= \frac{1-y}{1-y-1} = \frac{y-1}{y} = \frac{\frac{1}{1-x}-1}{\frac{1}{1-x}} = \frac{1-(1-x)}{1-x}\cdot\left(\frac{1-x}{1}\right)$$

$$= \frac{1-1+x}{1} = x$$

Answer: B

64. Let $f(x) = \frac{x}{1+x^2}$, what is the range of the function;

A. $\left[-\frac{1}{2}, \frac{1}{2}\right]$ B. [-2, 2] C. (-1, 1] D. $\mathbb{R}$

Solution: Let $f(x)=y$ then $y = \frac{x}{1+x^2} \Rightarrow x = y + yx^2 \Leftrightarrow yx^2 - x + y = 0$

We observe that the expression is a quadratic in x. So that

$\Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, Where b = 1, a = y and c = y

$$\therefore x = \frac{-1 \pm \sqrt{1-4y^2}}{2y}$$

Since x is real numbers, so $y \neq 0$ and $1 - 4y^2 \geq 0$.

$\therefore$ Range of $f = \{y: 1 - 4y^2 \geq 0\} = \{y: (1 + 2y)(1 - 2y) \geq 0\}$

$$= \{y: \left(y + \frac{1}{2}\right)\left(y - \frac{1}{2}\right) \leq 0\}$$

Hence, range $\left\{y: -\frac{1}{2} \leq y \leq \frac{1}{2}\right\}$ **Answer: A**

65. Let $f(x) = (x - 3)^2 + 2$ for $x \leq 3$. Find $f^{-1}(x)$

Solution: Let $y = f(x)$, then $y = (x - 3)^2 + 2$, $x \leq 3$

$\Rightarrow x = (y - 3)^2 + 2$ for $y \leq 3$ ⟵ interchange x and y

$\Rightarrow (y - 3)^2 = x - 2$, for $y \leq 3$ ⟵ solve for y

$\Rightarrow y = 3 \pm \sqrt{x-2}$, for $y \leq 3$ ⟵ $y = f^{-1}(x)$

$\therefore f^{-1}(x) = 3 - \sqrt{x-2}$

$\therefore$Domain of $f^{-1} = \{x: x \geq 2\}$ and range of $f^{-1} = \{y: y \leq 3\}$

66. Which one of the following is a one – to – one correspondence function from $A = \{-1, 0, 1\}$ to $B = \{1, 2, 3\}$?

A. $f(x) = 2x + 1$
B. $f(x) = x^3 + 2$
C. $f(x) = |x| + 2$
D. $f(x) = x^2 + 2$

Solution:

A function f: A ⟶ B is one – to – one correspondence if f is both one-to-one and on to.

A. For $2 \in B$, there does not exist any $x \in \{-1, 0, 1\}$ such that

$$f(x) = 2 \Rightarrow 2x + 1 = 2 \Rightarrow 2x = 1 \Rightarrow x = \frac{1}{2} \notin \{-1, 0, 1\}$$

B. $f(x) = x^3 + 2$ is on to because for every $y \in B$, there is $x \in A$ such that $f(x) = y$. That means image of f is B. Also f is one – to – one. Therefore, f is a one – to – one correspondence.

C. $f(x) = |x| + 2$ is not on to because for $1 \in B$, there does not exist any $x \in \{-1, 0, 1\}$ such that $f(x) = 1$ for which $|x| + 2 = 1$
$\Rightarrow |x| = 1 - 2 = -1$

D. $f(x) = x^2 + 2$ is not on to, because for $1 \in B$, there does not exist any $x \in \{-1, 0, 1\}$ such that $f(x) = 1$ for which $x^2 + 2 = 1$

Answer: B

67. If $f(x) = x + \sqrt{1+x^2}$, then what will be $f^{-1}(x)$?

Solution: Let $y = f(x) \Rightarrow y = x + \sqrt{1+x^2}$

$\Rightarrow y - x = \sqrt{1+x^2} \Leftrightarrow (y - x^2) = 1 + x^2$

$\Rightarrow y^2 - 2xy + x^2 = 1 + x^2 \Leftrightarrow y^2 - 2xy = 1 \Rightarrow 2xy = y^2 - 1$

$\Rightarrow x = \frac{y^2-1}{2y} = f^{-1}(y)$ $\quad \therefore f^{-1}(x) = \frac{1}{2}\frac{x^2-1}{x} = \frac{1}{2}\left(x - \frac{1}{x}\right)$

Review Exercise on Unit – 1

1. Determine which of the following sets are functions from A to B;
   a. $f = \{(a, b), (a,c), (b,d), (c,a)$; $A = \{a, b, c\}$, $B = \{a, b, c, d\}\}$
   b. $f = \{(1, 3), (2, 2), (3, 4)\}$; $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$
   c. $f = \{(a, b) \in \mathbb{R} \times \mathbb{R} / b = \sqrt{a}$, $A = \mathbb{R}$, $B = \mathbb{R}\}$
   d. $f = \{(1,2), (2,3), (3,4)$, where , $A=\{1,2,3,4\}$, $B = \{1,2,3,4\}$
2. Determine whether the given function is even, odd or neither.

   a) $f(x) = x^3 + 5x$

   b) $f(x) = \frac{x}{1+x^2}$

   c. $f(x) = \frac{1+x^2}{x^3}$

   d. $f(x) = x^2 - 4|x| + 3$
3. Determine which of the following are **onto, one – to – one or one – to – one correspondence** function.
   a. $f: \mathbb{R} \to \mathbb{R}$, defined by $f(x) = 4x + 1$
   b. $f: \mathbb{R} \to [-3, \infty)$ defined by $f(x) = 2(x - 1)^2 - 3$
   c. $f: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = f(x) = |x| - x$
   d. $f: [1, \infty) \to [5, \infty)$ defined by $f(x) = 3(x - 1)^2 + 5$
   e. $f: [0, \infty) \to [-1, \infty)$ defined by $f(x) = \sqrt{x} - 1$
   f. $f: \mathbb{R} \to \mathbb{R}^+$, defined by $f(x) = 4^x$, $x \in \mathbb{R}$
   g. $f: \mathbb{Z} \to \{-1, 1\}$ defined by $f(x) = (-1)^{x+}$, $x \in \mathbb{Z}$
   h. $f: \mathbb{R} \to [-4, \infty)$ defined by $f(x) = |x| - 4$
4. Determine which of the following are **onto, one – to – one or one – to – one correspondence** function from $A = [0, 1]$ to $B = [1, 2]$
   a. $f(x) = x$ b) $f(x) = 2x^3 + 1$ c. $f(x) = x^3 + 1$
5. Find the domain of the composite function fog.
   a. $f(x) = \sqrt{x-4}$ and $g(x) = 2x + 3$

d) Exercise left for you. **Ans:** $\ell nx - \ell n(1-x)$

53. Find the inverse of $f(x) = \log_5^{3x-4}$.

Solution: Let $y = f(x) \Rightarrow y = \log_5^{3x-4}$ ⟵ Interchange x to y

$$\Rightarrow x = \log_5^{3y-4} \Leftrightarrow 5^x = 3y - 4 \Rightarrow 3y = 5^x + 4 \Leftrightarrow y = \frac{1}{3}(5^x + 4)$$

$$\therefore f^{-1}(x) = \frac{1}{3}(5^x + 4)$$

54. Find the inverse of $f(x) = 3 + \sqrt{x}$.

Solution: Let $y = f(x)$

$y = 3 + \sqrt{x}$ then $x = 3 + \sqrt{y} \Rightarrow \sqrt{y} = x - 3 \Rightarrow y = (x-3)^2$

$\therefore f^{-1}(x) = (x-3)^2$ for $x \geq 3$

Domain of $f^{-1} = \{x: x \geq 3\}$ and range of $f^{-1} = \{y: y \geq 0\}$

55. Find the inverse of $f(x) = \dfrac{e^x - e^{-x}}{e^x + e^{-x}}$.

Solution: Let $y = f(x)$

$$y = \frac{e^x - e^{-x}}{e^x + e^{-x}} \text{ then } x = \frac{e^y - e^{-y}}{e^y + e^{-y}}$$

$$\Rightarrow xe^y + xe^{-y} = e^y - e^{-y} \Leftrightarrow xe^{-y} + e^{-y} = e^y - xe^y$$

$$\Rightarrow e^{-y}(x+1) = e^y(1-x) \Leftrightarrow \frac{e^y}{e^{-y}} = \frac{x+1}{1-x}$$

$$\Rightarrow e^{2y} = \frac{x+1}{1-x} \Rightarrow y = \frac{1}{2}\ell n\left(\frac{x+1}{1-x}\right)$$

$$\therefore f^{-1}(x) = \frac{1}{2}\ell n\left(\frac{x+1}{1-x}\right) = \frac{1}{2}[\ell n(x+1) - \ell n(1-x)]$$

Inverse of Even power and Quadratic function

As such quadratic function and even power has no inverse becau horizontal line test shows this function is not one to one, and hen cannot have inverse.

However, by restricting their domains to some "suitable" intervals, one is attained and the function becomes invertible.

b. $f(x) = x^2 + 4$ and $g(x) = \sqrt{x-2}$

c. $f(x) = \ln(x-3)$ and $g(x) = x^3 - 5$

6. If f: A $\rightarrow$ B and g: B $\rightarrow$ C are functions, then which of the following statement are **True or false** about the composite function.

a. Domain of (gof) $\sqsubseteq$ Domain of f.

b. Domain of (gof) $\sqsubseteq$ Domain of g.

c. Range of (gof) $\sqsubseteq$ Range of f.

d. Range of (gof) $\sqsubseteq$ Range of g.

7. Find the inverse of the following function.

a. $f(x) = 3 + \frac{1}{2}\ell n(x-1)$ c. $f(x) = 2(x-3)^2 + 1$, for $x \leq 3$

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$\frac{1}{e^{-x}+1}$

8. If $f(x) = 2x^2 + 5$ and $g(x) = 3x + k$, find k so that the graph of **fog** crosses the y-axis at 45.

9. If $f(x+2) = 2x^2 - 3x - 1$, then find $f(x+1)$.

10. If the domain of $f = \{x \in \mathbb{R}: 0 \leq x \leq 8\}$ and the domain of $g = \{x \in \mathbb{R}: -3 \leq x \leq 4\}$ then what is the domain of a) $f+g$, b) $f-g$.

11. Find the inverse of each function:

a) $f(x) = (x+5)^2$ for $x \geq -5$

b) $f(x) = x^2 + 3$ for $x \geq 0$

c) $f(x) = \frac{x+1}{3x-4}$ e) $f(x) = \sqrt{x+2}$

d) $f(x) = \sqrt[3]{x-4}$ f) $f(x) = x^3 - 2$

Unit Two

Rational function

Definition: A rational function of x is a function which can be defined by $f(x)=\frac{N(x)}{D(x)}$, where N(x) and D(x) are polynomials and $D(x)\neq 0$

Domain of Rational function

Domain of rational function is the set of all real numbers x for which $D(x)\neq 0$.

Illustrative example

1. In each part, find the domain and the range

a) $f(x)=\frac{x+2}{x-3}$ b) $f(x)=\frac{3}{x^2+1}$

c) $f(x)=\frac{x^2+2x}{x^2-1}$ d) $f(x)=\frac{1}{x^2+3x+2}$

Solution: a) domain = $\{x: x-3\neq 0\}=\{x: x\neq 3\}$ or $R\backslash\{3\}$
range = $\{y\in R: y\neq 1\}$ or $R\backslash\{1\}$

b) domain = R, range = $\{y: 0<y\leq 3\}$

c) domain = $\{x: x^2-1\neq 0\}=\{x: x\neq 1, -1\}$
range = R

d) domain = $\{x: x^2+3x+2\neq 0\}=\{x: x\neq -1, -2$
range = $R\backslash\{0\}$

2. Let $f(x)=\frac{x+1}{x-2}$ then, Find domain and range of

a) f(x) b) f(x – 1) c) $f(x^2-2)$

Solution:

a) domain = $\{x: x-2\neq 0\}=\{x: x\neq 2\}$ Range = $\{y: y\neq 1\}$

b) we have $f(x)=\frac{x+1}{x-2}$ then

$f(x-1)=\frac{x-1+1}{x-1-2}=\frac{x}{x-3}$. Therefore, domain of $\frac{x}{x-3}$ is R\{x = 3} range = {y: y ≠ 1

c) $f(x)=\frac{x+1}{x-2}$ then $f(x^2-2)=\frac{x^2-2+1}{x^2-2-2}=\frac{x^2-1}{x^2-4}$

then domain of $f(x^2-2)$ = {x: $x^2-4\neq 0$ = {x: x ≠ 2, -2}

Range = {R\ y ≠ 1}

Simplification of rational expression

Definition:

A rational expression is said to be reduced to lowest terms (or simplified). If the numerator and denominator have no common factor except -1 and 1.

Illustrative example

3. Simplify the following

a) $\frac{x}{x^2-1}+\frac{2}{x^2+2x+1}$

b) $\frac{\frac{x}{x^2-1}-\frac{2}{x^2-1}}{x-\frac{2}{x-1}}$

c) $\frac{y-1}{1-\frac{1}{1-\frac{1}{y}}}$

d) $\frac{\frac{1}{x-3}+\frac{3}{x-1}}{\frac{2}{x^2-4x+3}+\frac{4}{x-3}}$

Solution:

a) $\frac{x}{x^2-1}+\frac{2}{x^2+2x+1}=\frac{(x+1)x}{(x-1)(x+1)}+\frac{2(x-1)}{(x+1)(x-1)}$

$$= \frac{x^2+3x-2}{(x+1)^2(x-1)}, x \neq -1,1$$

b) $$\frac{x-\frac{2}{x^2-1}}{x-\frac{2}{x-1}} = \frac{x-\frac{2}{x^2-1}}{\frac{x(x-1)}{x-1}-\frac{2}{x-1}} = \frac{\frac{x-2}{x^2-1}}{\frac{x^2-x-2}{x-1}} = \frac{x-2}{x^2-1}\cdot\frac{x-1}{x^2-x-2}$$

$$= \frac{x-2}{(x+1)(x-1)}\cdot\frac{x-1}{(x-2)(x+1)} = \frac{1}{(x+1)^2}, x \neq -1$$

c) $$\frac{y-1}{1-\frac{1}{1-\frac{1}{y}}} = \frac{y-1}{1-\frac{1}{\frac{y-1}{y}}} = \frac{y-1}{1-\frac{y}{y-1}} = \frac{y-1}{\frac{y-1-y}{y-1}} = \frac{y-1}{\frac{1}{y-1}} = -(y-1)^2$$

d) $$\frac{\frac{1}{x-3}+\frac{3}{x-1}}{\frac{2}{x^2-4x+3}+\frac{4}{x-3}} = \frac{\frac{1(x-1)+3(x-3)}{(x-3)(x-1)}}{\frac{2}{(x-1)(x-3)}+\frac{4}{(x-3)}\cdot\frac{(x-1)}{(x-1)}}$$

$$= \frac{\frac{x-1+3x-9}{(x-3)(x-1)}}{\frac{2+4(x-1)}{(x-3)(x-1)}} = \frac{4x-10}{(x-3)(x-1)}\cdot\frac{(x-3)(x-1)}{(4x-2)} = \frac{2x-5}{2x-1}, x \neq \frac{1}{2}$$

Rational Equations

The easiest way to solve most equations involving rational expressions is to **multiply all the terms in the equations** by the least common denominator.

Illustrative Example

1. Solve the equation

a) $\frac{2}{3x+1} = \frac{1}{x} - \frac{6x}{3x+1}$

b) $\frac{3}{x-2} + \frac{21}{x^2-4} = \frac{14}{x+2}$

c) $\frac{2}{m-3} - \frac{3}{3+m} = \frac{12}{m^2-9}$

d) $\frac{2}{x-4} - \frac{3}{x+1} = \frac{6}{x-1}$

Solution: To solve $\frac{2}{3x+1}=\frac{1}{x}-\frac{6x}{3x+1}$

Multiply both sides by x(3x + 1) the resulting equation is

$\Rightarrow 2x = 3x + 1 - 6x^2$

$\Rightarrow 6x^2 + 2x - 3x - 1 = 0$ ..... Standard form

$\Rightarrow (3x + 1)(2x - 1) = 0$

$\Rightarrow 3x + 1 = 0$ or $2x - 1 = 0 \Rightarrow x = -\frac{1}{3}$ or $x = \frac{1}{2}$

Using $-\frac{1}{3}$ in the original equation causes the denominator 3x + 1 to equal zero. So it is not a **Solution**.

$\therefore$ The **Solution** set is $\{\frac{1}{2}\}$

b) $\frac{3}{x-3}+\frac{21}{x^2-4}=\frac{14}{x+2}$

Multiply both sides by (x – 2)(x + 2)

$\Rightarrow 3(x + 2) + 21 = 14(x - 2) \Rightarrow 3x + 6 + 21 = 14x - 28$

$\Rightarrow 27 + 28 = 14x - 3x \Rightarrow 55 = 11x \Rightarrow x = 5$

c) $\frac{2}{m-3}-\frac{3}{3+m}=\frac{12}{m^2-9}$

Multiply both sides by (m – 3)(m + 3)

$\Rightarrow 2(m + 3) - 3(m - 3) = 12$

$\Rightarrow 2m + 6 - 3m + 9 = 12 \Rightarrow -m = 12 - 15 = -3$

$\Rightarrow m = 3 \longleftarrow$ not in the domain

$\therefore$ m = 3 is not **Solution** set $\quad\quad$ $\therefore$ Solution set is $\varnothing$

d) $\frac{2}{x-4}-\frac{3}{x+1}=\frac{6}{x-1}$

Multiply both sides by (x + 1)(x – 1)(x – 4)

$\Rightarrow 2(x + 1)(x - 1) - 3(x - 1)(x - 4) = 6(x + 1)(x - 4)$

$\Rightarrow 2(x^2 - 1) - 3(x^2 - 5x + 4) = 6(x^2 - 3x - 4)$

$\Rightarrow 2x^2 - 2 - 3x^2 + 15x - 12 = 6x^2 - 18x - 24$

$\Rightarrow -x^2 + 15x - 14 = 6x^2 - 18x - 24$

$\Rightarrow 7x^2 - 33x - 10 = 0 \Rightarrow (7x + 2)(x - 5) = 0$

$\Rightarrow 7x + 2 = 0$ or $x - 5 = 0$

$\therefore x = \frac{-2}{7}$ or $x = 5$ $\quad\quad$ $\therefore S \cdot S = \left\{\frac{-2}{7}, 5\right\}$

Decomposition of rational expressions into partial fractions

Decomposition of rational expressions in to sums of simpler expressions is called **Partial fractions decomposition**

We may verify that $\frac{2}{x^2-1} = \frac{1}{x-1} + \frac{-1}{x+1}$

Rule of Finding partial fraction

Decomposition of $\frac{P(x)}{Q(x)}$

1. **Linear factor Rule (Degree of P(x) < degree of Q(x).** For each factor of the form (ax +b), the partial fraction decomposition contains the following sum of k, partial fraction.

$$\frac{1}{(ax+b)^k} = \frac{A}{ax+b} + \frac{B}{(ax+b)^2} + \frac{C}{(ax+b)^3} \ldots\ldots + \frac{D}{(ax+b)^k}$$

Illustrative Example

1. Find the partial fraction decomposition of the following:

a) $\frac{1}{x^2-9}$ $\quad\quad$ c) $\frac{x^2+10x-36}{x(x-3)^2}$

b) $\frac{4x^2+13x-9}{x^3+2x^2-3x}$ $\quad\quad$ d) $\frac{2x-3}{(x-1)^2}$

***Solution*:** Here

$$\frac{1}{x^2-9} = \frac{1}{(x-3(x+3)} = \frac{A}{x-3} + \frac{B}{x+3} = \frac{A(x+3)+B(x-3)}{(x-3)(x+3)}$$

Now, 1 = A(x+3) + B (x-3)

To find for A, Let x =3 $\Rightarrow$ 1= A (3+3) + B (3-3)

$\Rightarrow$ 1=6A $\therefore$ A = $\frac{1}{6}$

To find for B, Let x = -3$\Rightarrow$=A(-3+3) + B(-3-3)$\Rightarrow$1=-6B

$$\therefore B = -\frac{1}{6}$$

$$\therefore \frac{1}{x^2-9} = \frac{1}{6(x-3)} + \frac{-1}{6(x+3)}$$

b) $\frac{4x^2+13x-9}{x^3+2x^2-3x}$, Since the degree of numerator less than degree of denominator so, long division is not required.

Thus, the partial fraction decomposition has the form:

$$\frac{4x^2+13x-9}{x^3+2x^2-3x} = \frac{4x^2+13x-9}{x(x+3)(x-1)} = \frac{A}{x} + \frac{B}{x+3} + \frac{c}{x-1}$$

$\Rightarrow 4x^2 + 13x-9 = A\,(x+3)(x-1) + Bx(x-1) + Cx(x+3)$

To find for A, Let X=0

$\Rightarrow 4(2)^2 + 13(0) - 9 = A\,(0+3)\,(0-1) + 0 + 0 \Rightarrow -9 = -3A \quad \therefore A = 3$

To find for B, Let x =-3

$\Rightarrow 4(-3)^2 + 13(-3)-9 = A\,(-3+3) + B(-3)(-3-1) + C\,(-3+3)$

$\Rightarrow 36-39-9 = 0 + 12B + 0 \Rightarrow -12 = 12B \quad \therefore B = -1$

To find for C, Let x =1

$\Rightarrow 4(1)^2 + 13(1) -9 = A\,(1+3)(1-1) + B(1)(1-1) + C(1)(1+3)$

$\Rightarrow 8 = 4C \therefore C = 2$

$$\Rightarrow \frac{4x^2+13x-9}{x(x+3)(x-1)} = \frac{3}{x} + \frac{-1}{x+3} + \frac{2}{x-1}$$

C) $$\frac{x^2+10x-36}{x(x-3)^2} = \frac{A}{x} + \frac{B}{x-3} + \frac{C}{(x-3)^2}$$

$\Rightarrow x^2 + 10x-36 = A\,(x-3)^2 + Bx\,(x-3) + Cx$

To find A, Let x = 0

$\Rightarrow 0^2 + (10)(0) -36 = A\,(0-3)^2 + 0 + 0 \Rightarrow -36 = 9A \quad \therefore A = -4$

To find C, Let x=3

$\Rightarrow 3^2 + 10(3) -36 = A\,(3-3)^2 + B\,(3)(3-3) + 3C \Rightarrow 3 = 3C \therefore C = 1$

To find B, Let x = 1

$\Rightarrow 1^2 + 10(1) -36 = A\,(1-3)^2 + B(1)(1-3) + C\,(1)$

$\Rightarrow -25 = 4A -2B + C$

But A + -4 and C = 1

Hence, $\Rightarrow -25 + 4(-4) + -2B + 1$

$\Rightarrow -25 + 15 = -2B \Rightarrow -10 = -2B \quad \therefore B = 5$

Thus, $\frac{x^2+10x-36}{x(x-3)^2} = \frac{-4}{x} + \frac{5}{x-3} + \frac{1}{(x-3)^2}$

d) $\frac{2x-3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} \Rightarrow 2x-3 = A(x-1)^2 + B$

$\Rightarrow 2x-3 = Ax^2 - 2Ax + A + B \ldots \Rightarrow -2A = 2 \therefore A = -1$

And $A + B = -3$; $-1 + B = -3 \Rightarrow B = -3 + 1 = -2$

Explanation: Squaring $(x-1)^2 = x^2 - 2x + 1$

$\therefore A(x-1)^2 = Ax^2 - 2Ax + A$

Thus, $\frac{2x-3}{(x-1)^2} = \frac{-1}{x-1} + \frac{-2}{(x-1)^2}$

Rule2, when the degree of numerator is greater than or equal to the degree of denominator, we first perform along division, then decompose into partial fraction.

Illustrative example

2. Find partial decomposition.

a) $\frac{x^2-x-2}{x^2-9}$ b) $\frac{x^3}{x^2-3x+2}$

Solution: Here

a) The degree of numerator is equal to the degree of denominator, hence by long division

$\frac{x^2-x-2}{x^2-9} = 1 + \frac{7-x}{x^2-9}$ *and* $\frac{7-x}{x^2-9} = \frac{A}{x-3} + \frac{B}{x+3}$

$\Rightarrow 7-x = A(x+3) + B(x-3)$

To find A, Let $x = 3 \Rightarrow 7-3 = A(3+3) + B(3-3)$

$\Rightarrow 4 = 6A \quad \therefore A = \frac{4}{6} = \frac{2}{3}$

To find B, Let $x = -3$

$\Rightarrow 7-(-3) = A(-3+3) + B(-3-3) \Rightarrow 10 = -6B$

$\therefore B = \frac{-10}{6} = \frac{-5}{3}$

$$\Rightarrow \frac{x^2-x-2}{x^2-9} = 1 + \frac{2}{3(x-3)} + \frac{-5}{3(x+3)}$$

b) $\frac{x^3}{x^2-3x+2}$, The degree of numerator is greater than the degree of denominator. We perform long division.

$$\frac{x^3}{x^2-3x+2} = x+3+\frac{7x-6}{x^2-3x+2} = x+3+\frac{7x-6}{(x-2)(x-1)}$$

Again: $\frac{7x-6}{(x-2)(x-1)} = \frac{A}{x-2} + \frac{B}{x-1}$

To find B, Let x = 1 $\Rightarrow$7-6 = B (1-2) $\Rightarrow$ 1 = -B $\therefore$ B = -1

To find A, Let x = 2 $\Rightarrow$ 7(2)-6 = A (2-1) + B (2-2) $\Rightarrow$ 8 = A

$$\therefore \frac{x^3}{x^2-3x+2} = x+3+\frac{8}{x-2}+\frac{-1}{x-1}$$

Rule 3 for prime quadratic factor: ax^2+bx+C, (with b^2-4ac)

A prime quadratic factor of the form ax^2+bx+c (with $b^2-4ac<0$) in the denominator require partial fraction $\frac{Bx+c}{ax^2+bx+c}$

Illustrative example

3. Find partial fraction decomposition.

a) $\frac{8x^3+13x}{(x^2+2)^2}$ b) $\frac{2x^3-4x-8}{(x^2-x)(x^2+4)}$

Solution:

a) $\frac{8x^3+13x}{(x^2+2)^2} = \frac{Ax+B}{x^2+2} + \frac{Cx+D}{(x^2+2)^2}$

$8x^3 + 13x = (Ax + B)(x^2 + 2) + Cx + D$

Expanding the basic equation and collecting like terms on opposite sides of the equation.

$\Rightarrow 8x^3+13x = Ax^3 + 2Ax + Bx^2 + 2B + Cx + D$

$\Rightarrow 8x^3 + 13x = Ax^3 + Bx^2 + (2A + C)x + 2B + D$

$\therefore A = 8, B = 0, 2A+C = 13$ and $2B + D = 0$

In $2A + C = 13$

$\therefore 2(8) + C = 13, \therefore C = 13-16 = -3$ $\therefore D = 0$

Thus, $\frac{8x^3+13x}{(x^2+2)^2} = \frac{8x}{x^2+2} + \frac{-3x}{(x^2+2)^2}$

Graph of rational function

A graph of a rational function is pictorical representation of rational function.

When sketching the graph of rational function f, it is important to investigating

- The intercepts, x and y
- The asymptote
- The behavior of function near the vertical asymptote (functional value of f(x) when x is close to a zero of the denominator)
- The behavior of function as x goes to very large number
- The party i.e. f is odd or even or neither.

Asymptote

There are three:

- Vertical asymptote
- Horizontal asymptote
- Oblique asymptote

Definition: The line x = a is a vertical asymptote for the graph of a function f if $f(x) \to \infty$ or $f(x) \to -\infty$ as x approaches **a** form either the left or the right.

Notation	Terminology
$\to a^+$	x approaches a from the right
$\to a^-$	x approaches a from the left
$f(x) \to \infty$	f(x) increase with out bound
$f(x) \to -\infty$	f(x) decrease with out bound

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Illustrative example

8. Find all vertical asymptotes of each rational function

a) $f(x)=\dfrac{2x-1}{x^2+3x}$

b) $f(x)=\dfrac{3x^2+5}{x^2-4}$

c) $f(x)=\dfrac{x-2}{x^2+2x-8}$

d) $f(x)=\dfrac{4}{x-3}$

e) $f(x)=\dfrac{x^2-x-6}{x^2-9}$

Solution: To find vertical asymptote simplify the common factor then set denominator zero

a) $x^2+3x=0 \Rightarrow x(x+3)=0 \Rightarrow x=0$ and $x=-3$

$\therefore$ V.A, $x=0$ and $x=-3$

b) $x^2-4=0 \Rightarrow x=2$ and $x=-2$

$\therefore$ V.A, $x=-2$ and $x=2$

c) $\dfrac{x-2}{x^2+2x-8}=\dfrac{(x-2)}{(x-2)(x+4)}=\dfrac{1}{x+4}$ simplifying common factor

$\therefore$ V.A is only $x=-4$, but not $x=2$

$\therefore$ the function has hole at $x=2$

d) $x-3=0 \Rightarrow x=3 \longleftarrow$ V.A

e) $\dfrac{(x-3)(x+2)}{(x-3)(x+3)}=\dfrac{x+2}{x+3}$, simplifying common factor

$\therefore$ V.A is only $x=-3$ but not $x=3$

$\therefore$ The graph of function has hole at $x=3$.

Definition of Horizontal asymptote

The line $y=b$ is a horizontal asymptote of the graph of function f if $f(x) \to b$ as $x \to \infty$ or as $x \to -\infty$

Rule of Horizontal asymptote

Let $f(x)=\dfrac{N(x)}{D(x)}=\dfrac{ax^n+\ldots\ldots+a_0}{bx^m+\ldots\ldots+b_0}$ be rational function

1. If $n < m$, then x-axis ($y=0$) is the horizontal asymptote

2. If n = m, then the line $y = \frac{a}{b}$ is the horizontal asymptote
3. If n > m, then the graph has no horizontal asymptote. It has oblique asymptote if n = m + 1
4.

Illustrative Example

9. Find the horizontal asymptote for the graph of f if it exist

a) $f(x) = \frac{2x+3}{x^2+1}$ c) $f(x) = \frac{2x-4}{5x+6}$

b) $f(x) = \frac{4x^2+1}{3x^2+4}$ d) $f(x) = \frac{x^3+1}{x-2}$

Solution:

a) $f(x) = \frac{2x+3}{x^2+1}$, here, the degree of the numerator $2x + 3$ is less than the degree of the denominator $x^2 + 1$

$\therefore y = 0 \leftarrow$ H.A

b) $f(x) = \frac{4x^2+1}{3x^2+4}$, since the degree of numerator equal to the degree of denominator

$\therefore$ The line $y = \frac{4}{3} \leftarrow$ H.A

c) since n = m $\therefore$ The line $y = \frac{2}{5} \leftarrow$ H.A

d) since n > m $\therefore$ It has no horizontal asymptote

The zeros of rational function

Definition: Let $f(x) = \frac{N(x)}{D(x)}$ be rational function an element **a** in the domain of f is called a zero of f, if and only if N(a)=0 and $D(a) \neq 0$

Illustrative Example

Find the zeros of each of the following rational function

a) $f(x)=\dfrac{x^2-4}{x^2-9}$

b) $f(x)=\dfrac{x^2+5x+6}{x^2+6x+8}$

c) $f(x)=\dfrac{x+4}{x-3}$

d) $f(x)=\dfrac{x-3}{x^2-9}$

Solution:

a) $\dfrac{x^2-4}{x^2-9}\Rightarrow x^2-4=0\Rightarrow (x-2)(x+2)=0$

$\therefore$ The zeros are x = 2 and x = -2

b) First we factorize both numerator and denominator

$f(x)=\dfrac{(x+2)(x+3)}{(x+2)(x+4)}=\dfrac{x+3}{x+4}$, domain = R\{-2, -4}

$\therefore$ The zeros of f are only x = -3, but not x = -2

c) $f(x)=0 \Leftrightarrow \dfrac{x+4}{x+3}=0 \Rightarrow x=-4 \leftarrow$ the zeros of f

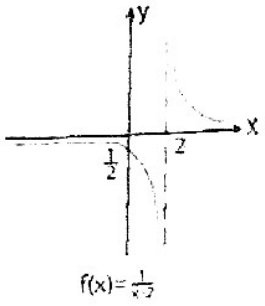
d) $\dfrac{x+3}{(x-3)(x+3)}=\dfrac{1}{x-3}$

$\therefore$ No zeros, that makes f(x) = 0.

Sketching the graph of rational function

Sketch the graph of the following function

A) Let $f(x)=\dfrac{1}{x-2}$, then



Find a) x and y-intercept
b) domain and range
c) asymptote
d) how behaves f as $x \to 2^+$ and $x \to 2^-$
e) how behaves f(x) as $x \to \infty$ and $x \to -\infty$
f) for what values of x
i) $f(x) > 0$ ii) $f(x) < 0$ iii) $f(x) > 3$

Solution: a) No, x-intercept

y –intercept $\left(0,-\frac{1}{2}\right)$

b) Domain = R{2}
Range = R\{0}
c) V.A, x = 2, H.A, y = 0
d) as $x \to 2^+$, $f(x) \to \infty \leftarrow$ f(x) increase with out bound
as $x \to 2^-$, $f(x) \to -\infty \leftarrow$ f(x) decrease with out bound
e) as $x \to \infty$, $f(x) \to 0^+$
as $x \to -\infty$, $f(x) \to 0^-$
f) i) $f(x) < 0$ when $x < 2$
ii) $f(x) > 0$ when $x > 2$
iii) $f(x) \geq 3$ when $2 < x \leq \frac{7}{3}$

- f is symmetric with respect to (2, 0)
- parity, neither even nor odd.

B) Let $f(x) = \frac{x+1}{x+3}$, Investigate the behavior. Sketch the graph

Solution: i) The x-intercept, (-1, 0)

The y-intercept $\left(0,\frac{1}{3}\right)$

ii) Domain = R\{-3}
Range = R\{1}
iii) V.A, x = -3
H.A, y = 1

Explanation

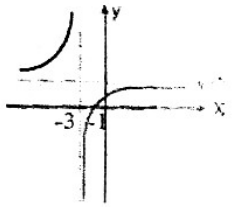
as $x \to -3^+$

$$\frac{-3^+ + 1}{-3^+ + 3} = \frac{-2}{0^+} \to -\infty$$

as $x \to -3^-$

$$\frac{-3^- + 1}{-3^- + 3} = \frac{-2}{0^-} \to \infty$$

iv) as $x \to -3^+$, $f(x) \to -\infty \longleftarrow$ f(x) decrease with out bound
as $x \to -3^-$, $f(x) \to \infty \longleftarrow$ f(x) increase with out bound
v) as $x \to \infty$, $f(x) \to 1 \longleftarrow$ f(x) approaches to 1 as x goes
as $x \to -\infty$, $f(x) \to 1$ very large number.
From the graph, we can observe

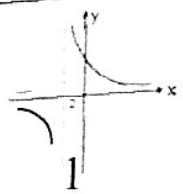
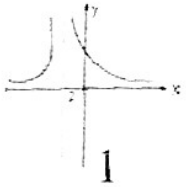


vi) • $f(x) < 1 \Rightarrow \frac{x+1}{x+3} < 1$, when x > -3

• $f(x) > 1 \Rightarrow \frac{x+1}{x+3} > 1$, when x < -3

- $f(x) < 0 \Rightarrow \dfrac{x+1}{x+3} \leq 0$ when $-3 < x \leq -1$
- $f(x) \geq 0 \Rightarrow \dfrac{x+1}{x+3} \geq 0$ when $x \in (-\infty, -3] \cup [-1, \infty)$

c) Quick sketch of the following table

i)	ii)	iii)
		
$f(x) = \dfrac{1}{x+2}$	$f(x) = \dfrac{1}{(x+2)^{2n}}$	$f(x) = \dfrac{x}{x+2}$
as $x \to -2^+$, $f(x) \to \infty$	as $x \to -2$, $f(x) \to \infty$	as $x \to -2^+$, $f(x) \to -\infty$
as $x \to -2^-$, $f(x) \to -\infty$	as $x \to \infty$, $f(x) \to 0$	as $x \to -2^-$, $f(x) \to \infty$
as $x \to \infty$, $f(x) \to 0$		as $x \to \infty$, $f(x) \to 1$

D) Let $f(x) = \dfrac{x^2 - 3x - 4}{x^2 - 6x + 8}$

i) Investigate all the behavior

ii) sketch the graph iii) for what value of x

a) $f(x) \geq 0$ b) $f(x) < 0$ c) $f(x) \geq 2$ d) $f(x) < 1$

Solution: First we factorize the numerator and denominator and simplify $f(x) = \dfrac{x^2 - 3x - 4}{x^2 - 6x + 8} = \dfrac{(x-4)(x+1)}{(x-4)(x-2)} = \dfrac{x+1}{x-2}, x \neq 4,2$

Hence,

i) • the x-intercept by setting $f(x) = 0$, we get $(-1, 0)$

- the y-intercept by setting, $x = 0$, we get

$$\left(0, -\frac{1}{2}\right) \to f(0) = -\frac{1}{2}$$

- the vertical asymptote, $x = 2 \leftarrow$ the zero of denominator
- the graph has hole at $x = 4$
- the horizontal asymptote, $y = 1$
- the behavior near the V.A, $x = 2$

as $x \to -2^+, f(x) \to \infty$

as $x \to -2^-, f(x) \to -\infty$

- what happens $f(x)$ as x goes to very large number

as $x \to \infty, f(x) \to 1^+$

as $x \to -\infty, f(x) \to 1^-$

iii) a) $f(x) \geq 0$ when $x \in (\infty, -1] \cup (2, 4) \cup (4, \infty)$

b) $f(x) < 0$ when $x \in (-1, 2)$

c) $f(x) \geq 2$ when $x \in (2, 4) \cup (4, 5]$ because 4 is not in domain

d) $f(x) < 1$ when $x \in (-\infty, 2)$

iv) f is neither even nor odd

E) Let $f(x) = \dfrac{x-1}{x^2-x-6}$

i) Discuss all the behavior.

ii) sketch the graph

iii) For what value of x

a) $\dfrac{x-1}{x^2-x-6} > 0$ b) $\dfrac{x-1}{x^2-x-6} < 0$

Solution: we have $\dfrac{x-1}{x^2-x-6} = \dfrac{x-1}{(x-3)(x+2)}$

i) * The x-int, $(1, 0) \longrightarrow f(x) = 0$

* The y-int $\left(0, \dfrac{1}{6}\right) \longleftarrow f(0) = \dfrac{1}{6}$

* V.A, $x = -2$ and $x = 3 \longleftarrow$ the zero of denominator

* H.A, $y = 0$

The behavior near the vertical asymptote

as $x \to -2^+$, $f(x) \to \infty$, the graph goes up

as $x \to -2^-$, $f(x) \to -\infty$, the graph goes down

as $x \to 3^+$, $f(x) \to \infty$

as $x \to 3^-$, $f(x) \to -\infty$

- The behavior as x goes to very large number

as $x \to \infty$, $f(x) \to 0^+$

as $x \to -\infty$, $f(x) \to 0^-$

From the graph we observe that

iii) $\dfrac{x-1}{x^2-x-6} > 0$, when $x \in (-2, 1) \cup (3, \infty) \leftarrow$ The graph lies above the x-axis

b) $\frac{x-1}{x^2-x-6} < 0$, when $x \in (-\infty, -2) \cup (1, 3)$ ← The graph lies below the x-axis

iv) parity, neither even nor odd

F) Let $f(x) = \frac{x^2+1}{x^2+x-2}$

i) Discuss all the behavior
ii) sketch the graph iii) for what values of x

a) $\frac{x^2+1}{x^2+x-2} < 0$ b) $\frac{x^2+1}{x^2+x-2} \geq 0$

Solution: Here, $\frac{x^2+1}{x^2+x-2} = \frac{x^2+1}{(x+2)(x-1)}$

i) No, x-intercept

- y-intercept $\left(0, -\frac{1}{2}\right)$
- vertical asymptote, x = -2 and x = 1 ← the zero of denominator
- Horizontal asymptote, y = 1

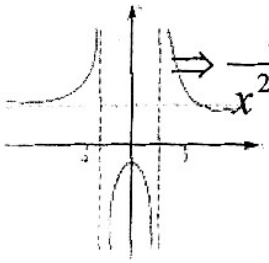
The behavior near the V.A

- as $x \to -2^+, f(x) \to -\infty$, the graph goes down
- as $x \to -2^-, f(x) \to \infty$, the graph goes up
- as $x \to 1^+, f(x) \to \infty$ • as $x \to 1^-, f(x) \to -\infty$

The behavior as x goes to very large number

- $x \to \infty, f(x) \to 1$ ← the graph approaches to 1
- $x \to -\infty, f(x) \to 1$

The graph crosses the horizontal asymptote at y = 1
This can be determined by solving $f(x) = 1$



$$\Rightarrow \frac{x^2+1}{x^2+x-2} = 1 \Rightarrow x^2+1 = x^2+x-2$$

$$\Rightarrow x - 2 = 1$$

$$\therefore x = 3$$

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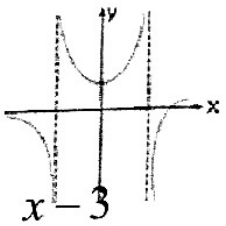
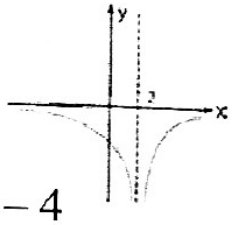
iii) $\frac{x^2+1}{x^2+x-2} < 0$, when $x \in (-2, 1)$ ← The interval on which the graph lies below the x-axis

b) $\frac{x^2+1}{x^2+x-2} \geq 1$ When $x \in (-\infty, 2) \cup (1,3]$

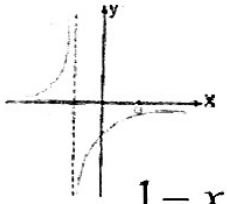
G. Quick sketch of the graph



$\frac{x-3}{x^2-1}$



$\frac{-4}{(x-1)^2}$



$f(x) = \frac{1-x}{x^2-1}$

Oblique Asymptote

Let $f(x) = \frac{a_1 x^n + \ldots + a_0}{bx^m + \ldots + b_0}$ f has oblique asymptote

If and only if n = m +1 and we can find by long division.

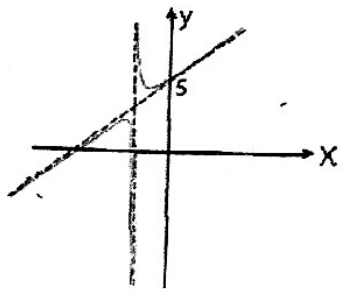
Illustrative Example

Find the oblique asymptote and sketch the graph of f.

a) $f(x) = \frac{x^2+6x+8}{x+1}$ b) $f(x) = \frac{x^2+2x-3}{(x-2)}$

Solution: a) First write f(x) in two different ways. Factoring the numerator

$$f(x) = \frac{x^2+6x+8}{x+1} = \frac{(x+2)(x+4)}{x+1}$$



* The x-intercepts: (-2, 0), (-4, 0) ← The graph meets the x-axis
* The y-intercepts: (0, 8) ← $f(0) = 8$
* V.A, x = -1 ⟶ the zeros of denominator
* $f(x) = \frac{x^2+6x+8}{x+1} = x+5+\frac{3}{x+1}$

∴ y = x + 5 ⟵ oblique asymptote

The behavior near V.A, x = -1

$$\text{As } x \to -1^+ = \frac{(-1+2)(-1+4)}{-1^+ + 1} = \frac{3}{0^+} = \infty$$

$\therefore$ as $x \to -1^+, f(x) \to \infty$

$x \to -1^-, f(x) \to -\infty$

Note: From the graph we can observe that

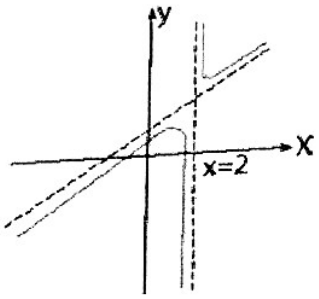
* $\frac{x^2+6x+8}{x+1} \geq 0$ when $x \in (-1, \infty) \cup [-4, -2)$

* $\frac{x^2+6x+8}{x+1} \leq 0$ when $x \in [-2, -1) \cup (-\infty, -4]$

Parity, neither even nor odd

b) $f(x) = \frac{x^2+2x-3}{x-2} = \frac{(x-1)(x+3)}{x-2}$

* x-intercepts: (1, 0) and (-3, 0)



* y-intercepts: $\left(0, \frac{3}{2}\right) \to f(0) = \frac{3}{2}$

* V.A, $x = 2 \longleftarrow$ the zeros of denominator

* $\frac{x^2+2x-3}{x-2} = x + 4 + \frac{5}{x-2}$

$\therefore$ $y = x + 4 \longleftarrow$ oblique asymptote

The behavior near V.A, x = 2

as $x \to 2^+, f(x) \to \infty$

$x \to 2^-, f(x) \to -\infty$

Note: From the graph we can observe that

* $\frac{x^2+2x-3}{x-2} \leq 0$ when $x \in (-\infty, -3] \cup [1, 2)$

* $\frac{x^2+2x-3}{x-2} \geq 0$ when $x \in [-3, 1) \cup [2, \infty)$

Parity, neither even nor odd

Solved Problem

1. Perform the indicated operation and simplify

a) $\dfrac{x^2-6x+9}{x^2-1}\cdot\dfrac{2x-2}{x-3}$

c) $\dfrac{\frac{x}{y}-\frac{y}{x}}{\frac{1}{x}-\frac{1}{y}}$

b) $\dfrac{x+2}{2x-3}\div\dfrac{x^2-4}{2x^2-3x}$

Solution:

a) $\dfrac{x^2-6+9}{x^2-1}\cdot\dfrac{2x-2}{x-3}=\dfrac{(x-3)^2\cdot 2(x-1)}{(x-1)(x+1)(x-3)}=\dfrac{2(x-3)}{x+1}$,

if $x\neq 3$, $x\neq -1$

b) $\dfrac{x+2}{2x-3}\div\dfrac{x^2-4}{2x^2-3x}=$

$$\frac{x+2}{2x-3}\cdot\frac{2x^2-3x}{x^2-4}=\frac{x+2}{2x-3}\cdot\frac{(x)(2x-3)}{(x+2)(x-2)}$$

$$=\frac{x}{x-2},\ x\neq -2,\frac{3}{2}$$

c) $\dfrac{\frac{x}{y}-\frac{y}{x}}{\frac{1}{x}-\frac{1}{y}}=\dfrac{\frac{x^2-y^2}{xy}}{\frac{y-x}{xy}}=\dfrac{(x-y)(x+y)}{xy}\cdot\dfrac{xy}{y-x}=-(x+y)$

2. The value (S) of x where the graph of $y=\dfrac{x^2-4x+3}{x^3+1}$ crosses its horizontal asymptote, is (are) .... EHEECE

A. $x=-2$ C. $x=0$

B. $x=3$ and $x=1$ D. $x=\sqrt{2}$ *and* $x=1$

Solution: The degree of numerator is less than the degree of denominator by one

$\therefore$ H.A, $y=0$

$\Rightarrow y = \dfrac{x^2-4x+3}{x^3+1} = 0 \Rightarrow x^2-4x+3=0 \Rightarrow (x-3)(x-1)=0$

$\Rightarrow x = 3$ and $x = 1$

3. Which one of the following represent the *Solution* set for the equation $\dfrac{1}{x-2} = \dfrac{3}{x+2} - \dfrac{6x}{x^2-4}$ ..... EHEECE

Answer: B

A. $\left\{-\dfrac{1}{2}, \dfrac{1}{2}\right\}$ B. {-4, 2} C. {-2} D. $\varnothing$

Solution: Restriction, $x \neq 2$, $x \neq -2$

LCM of $(x-2)(x+2)(x^2-4) = (x-2)(x+2)$

Multiply both side by LCM$(x-2)(x+2)$, we get

$\Rightarrow x+2 = 3(x-2) - 6x$

$\Rightarrow x+2 = 3x-6-6x \Rightarrow 4x = -6-2 \Rightarrow 4x = -8$

$x = -2 \longleftarrow$ not in the domain

$\therefore$ S.S $= \varnothing$,

Answer: D

4. For what values of x, the following inequality true

a) $\dfrac{-1}{x^2+1} < 0$ b) $\dfrac{-1}{x^2+1} > 0$

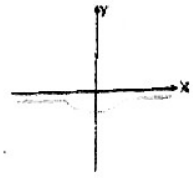
c) $\dfrac{4x^2}{(x-3)(x+2)} < 0$ d) $\dfrac{4x^2}{(x-3)(x+2)} > 0$

e) $\dfrac{x-3}{x^2-9} > 0$ f) $\dfrac{x-3}{x^2-9} < 0$

g) $\dfrac{-x^2-3}{x(x+2)} < 0$

Solution: a) $\dfrac{-1}{x^2+1} < 0$, for all real number the graph lies below the x-axis

$\therefore \dfrac{-1}{x^2+1} < 0$ on $(-\infty, \infty)$



b) $\dfrac{-1}{x^2+1} > 0$, is $\varnothing$.

No ***Solution***, for all x, the graph lies below the x-axis

c) $\frac{4x^2}{(x-3)(x+2)} < 0 \Leftrightarrow (x-3)(x+2) < 0 \Rightarrow \{-2 < x < 3\}$.
because $4x^2 > 0$ for all x.

d) $\frac{4x^2}{(x-3)(x+2)} > 0 \Leftrightarrow (x-3)(x+2) > 0 \Rightarrow x < -2$ or $x > 3$
because $4x^2 > 0$ for all x.

e) $\frac{x-3}{x^2-9} > 0 \Leftrightarrow \frac{1}{x+3} > 0$ on $(-3, 3) \cup (3, \infty)$

f) $\frac{x-3}{x^2-9} < 0 \Leftrightarrow \frac{1}{x+3} < 0$ on $(-\infty, -3)$

g) $\frac{-x^2-3}{x(x+2)} < 0 \Leftrightarrow \frac{x^2+3}{x(x+2)} > 0 \Leftrightarrow x(x+2) > 0$ because $x^2 + 3 > 0$ for all x

$\therefore \frac{-x^2-3}{x(x+2)} < 0$ on $(-\infty, -2) \cup (0\ \infty)$

$\therefore \frac{-x^2-3}{x(x+2)} > 0$ on $(-2, 0)$

5. Simplified form of $\frac{4-(a-2)^2}{a^4+2a^2+1} \div \frac{1-\frac{1}{a^2+1}}{a}$ ..... UEE 2004/12

A. $\frac{4-a}{a}$ C. $-\frac{1}{a^2(a^2+1)}$

B. $\frac{a}{a^2+1}$ D. $\frac{4-a}{a^2+1}$ **Answer: D**

6. If $f(x) = \frac{x-1}{(x-2)^2(x+1)}$ which of the following is true about f ...UEE 2004/12.

A. It has an oblique asymptote
B. The graph does not meet its asymptote
C. As $\rightarrow -1^+$, $f(x) \rightarrow -\infty$
D. As $\rightarrow 2^-$, $f(x) \rightarrow -\infty$

Solution: • It has no oblique asymptote

• The graph meets the horizontal asymptote at x = 1

• As $x \to -1^+$, $f(x) = \dfrac{-1^+ - 1}{(-1^+ - 2)^2(-1^+ + 1)} = \dfrac{-}{(+)(0^+)} \to -\infty$

• As $x \to 2^-$, $f(x) \to \infty$ ∴ Answer C

7. If $P(x) = 4x^3$ and $q(x) = 3x^2 - 3x$, then what is the solution set of $\dfrac{P(x)}{4xq(x)} + \dfrac{1}{x} = \dfrac{1}{q(x)}$?

A. {1, 4} B. {1} C. {4} D. {-1, 4}

Explanation:

Multiplying both side by q (x), we get

$$\Rightarrow \frac{P(x)}{4x} + \frac{q(x)}{x} = 1 \Leftrightarrow \frac{4x^3}{4x} + \frac{3x^2 - 3x}{x} = 1$$

$$\Rightarrow x^2 + 3x - 3 = 1 \Leftrightarrow x^2 + 3x - 4 = 0$$

$$\Rightarrow (x - 1)(x + 4) = 0 \Rightarrow x = 1 \text{ and } x = -4$$

But, q(1) = 0, so reject $x = 1$

∴ The solution set is {4}.

Answer: C

8. What is the solution set of $\dfrac{2 - \frac{1}{x}}{1 - \frac{1}{x^2}} = 4x^2 - \dfrac{2x}{2 + \frac{1}{x}}$?

A. $\left\{1, -\frac{1}{4}\right\}$ B. $\left\{\frac{1}{4}\right\}$ C. $\left\{-1, \frac{1}{4}\right\}$ D. $\left\{-\frac{1}{4}\right\}$

Solution:

Simplify both side;

$$\frac{2 - \frac{1}{x}}{4 - \frac{1}{x^2}} = \frac{2 - \frac{1}{x}}{\left(2 - \frac{1}{x}\right)\left(2 + \frac{1}{x}\right)} = \frac{1}{2 + \frac{1}{x}} = \frac{x}{2x + 1} \quad \text{.....(*)}$$

$$\text{and } 4x^2 - \frac{2x}{2 + \frac{1}{x}} = 4x^2 - \frac{2x^2}{2x + 1} = x^2\left(4 - \frac{2}{2x + 1}\right) \quad \text{.....(**)}$$

From (*) and (**) we have $\dfrac{x}{2x + 1} = x^2\left(4 - \dfrac{2}{2x + 1}\right)$

$$\Rightarrow \frac{1}{2x+1} = 4x - \frac{2x}{2x+1} \Leftrightarrow 4x = \frac{1}{2x+1} + \frac{2x}{2x+1}$$

$$\Rightarrow 4x = \frac{2x+1}{2x+1} = 1 \Rightarrow 4x = 1 \Leftrightarrow x = \frac{1}{4}$$

Answer: B

9. What are the respective values of A, B and C, so that $\frac{x-1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$? ..... UEE 2003/11

A. -1, 1, 1 B. -1, 1, 2 C. 1, -2, 0 D. 3, 1, 2

Solution: $\frac{x-1}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$?

$\Leftrightarrow x - 1 = A(x^2 + 1) + (Bx + C)x$

$\Rightarrow x - 1 = Ax^2 + A + Bx^2 + Cx \Rightarrow x - 1 = (A + B)x^2 + Cx + A$

$\Rightarrow 0x^2 + x - 1 = (A + B)x^2 + Cx + A \Rightarrow A + B = 0$

$\Rightarrow Cx = x \Rightarrow C = 1 \Rightarrow A = -1 \quad \therefore B = 1$

Hence $\frac{x-1}{x^3+x} = \frac{-1}{x} + \frac{x+1}{x^2+1}$

Answer: A

10. What is the solution set of $1 - \frac{5}{x^2+4} = \frac{x-1}{x^2-x} - \frac{1}{x}$ ....UEE 2003/11

A. $\{1, -1\}$ B. $\{2\}$ C. $\{-1\}$ D. $\{1, 2, -1\}$

Solution: Multiply both sides by $x(x - 1)(x^2 + 4)$

$\Rightarrow x(x - 1)(x^2 + 4) - 5x(x - 1) = (x-1)(x^2 + 4) - (x - 1)(x^2 + 4)$

$\Rightarrow (x^2 - x)(x^2 + 4) - 5x^2 + 5x = 0$

$\Rightarrow x^2(x^2 + 4) - x(x^2 + 4) - 5x^2 + 5x = 0$

$\Rightarrow x^4 + 4x^2 - x^3 - 4x - 5x^2 + 5x = 0$

$\Rightarrow x^4 - x^3 - x^2 - 4x + 5x = 0$

$\Rightarrow x^4 - x^3 - x^2 + x = 0 \Rightarrow x(x^3 - x^2 - x + 1) = 0$

$\Rightarrow x(x + 1)(x - 1)^2 = 0$

$\therefore x = 0, x = -1$, and $x = 1$

But, $x = 0$, and $x = 1$ are not in the domain

$\therefore x = 0$ and $x = 1$ are

$$= \frac{x^2 + 2x - x^2 - 8}{(x+2)(x-4)} = \frac{2(x-4)}{(x+2)(x-4)} = \frac{2}{x+2}$$ not in

the solution set

$\therefore$ Solution set is $\{-1\}$ only

Answer: C

11. Which of the following is not true about the graph of $g(x) = \frac{x^2-1}{x^2+1}$ .....UEE 2003/11

A. The range of g is $(-\infty, 1)$
B. g is even
C. $y = 1$ is a horizontal asymptote
D. As $x \to -\infty$, $g(x) \to 1$

Solution: To find the range, solve for x

$$\Rightarrow y = \frac{x^2-1}{x^2+1} \Leftrightarrow yx^2 + y = x^2 - 1 \Rightarrow yx^2 - x^2 = -1 - y$$

$$\Rightarrow x^2(y-1) = -1 - y \qquad \therefore x = \pm\sqrt{\frac{-1-y}{y-1}}$$

$$\therefore \text{range} = \left\{y : \frac{-1-y}{y-1} > 0\right\} = \{y : -1 \le y < 1\}$$

B. $g(x) = g(-x)$ $\therefore$ g is even
C. $y = 1$ is a horizontal asymptote
D. As $x \to -\infty$, $g(x) \to 1$ $\therefore$ Answer A

Review – Exercise on Unit 2

I. Find a. the vertical
b. the horizontal if any
c. oblique asymptotes if any
d. the intercepts if any of each rational function

1. $R(x) = \frac{3x+3}{2x+4}$ 2. $R(x) = \frac{3x+1}{x^2-4}$ 3. $R(x) = \frac{x^2}{x^2+x-6}$

4. $R(x) = \frac{x^3+2}{x^2-4}$ 5. $R(x) = \frac{2x^3-3}{x^2-9}$ 6. $R(x) = \frac{x^2+x-6}{x^2+7x+12}$

II. State the domain and solve each pf the rational equation

7. $\frac{x^2}{x+3} + \frac{3x}{x^2+5x+6} = \frac{2x}{x+3}$ 8. $\frac{2}{x-3} + \frac{3}{x+3} = \frac{5x-3}{x^2-9}$

9. $\frac{5x}{x-2} = 7 + \frac{10}{x-2}$ 10. $\frac{7}{x-4} - \frac{3}{x+4} = \frac{13-5x}{x^2-16}$

III. Simplify

11. $x-\dfrac{x}{1-\dfrac{x}{1-x}}$ 12. $\left(\dfrac{y}{y+2}-\dfrac{y}{y-2}\right)\div\left(\dfrac{y}{y+2}+\dfrac{y}{y-2}\right)$

ANSWER TO REVIEW EXERCISE

1. a. V.A: $x = -2$ b. H.A: $y=\dfrac{3}{2}$; not intersected
   d. x – intercept: - 1; no y – intercept.
2. a. vertical asymptote; $x = 2, x = -2$
   b. horizontal asymptote: $y = 0$, intersected at $\left(0,-\dfrac{1}{3}\right)$
   d. x – intercept: $-\dfrac{1}{3}$, y - intercept, $-\dfrac{1}{4}$
3. a. vertical asymptote: $x = 2, x = -3$
   b. horizontal asymptote: $y = 1$ intersected at (6,1)
   c. intercept: (0, 0)
4. a. vertical asymptote: $x = 2, x = -2$ b. No. H. Asy
   b. oblique asymptote: $y = x$, intersected at $\left(-\dfrac{1}{2},-\dfrac{1}{2}\right)$
   d. x – intercept, $-\sqrt[3]{2}$, y- intercept: $-\dfrac{1}{2}$
5. a. vertical asymptote: $x = 3, x = -3$ b. No, H.A
   b. oblique asymptote: $y = 2x$, intersected at $\left(\dfrac{1}{6},\dfrac{1}{3}\right)$
6. a. vertical asymptote: $x = -4$,
   b. horizontal asymptote: $y = 1$, intersected at (-3, 1)
7. $\{-1, 0, 1\}$ 8. $\mathbb{R}\setminus\{3, -3\}$ 9. $\emptyset$ 10. {-3}

3. Find the equation of the line with slope m and y-intercept b

a) m = -2, b = 1 b) $m = \frac{\pi}{2}$, b = 4, c) m = -3, b = 0

Solution: use y = mx + b

Thus a) y = -2x + 1 b) $y = \frac{\pi}{2}x + 4$, c) y = -3x

4. a) Find the equation of straight line parallel to the x-axis at distance

i) 4 unit above it ii) 5 unit below it

Solution: i) y = 4 ii) y = -5

5. Find the equation of straight line with x-intercept 3 y-intercept 2

Solution: use $\frac{x}{a}+\frac{y}{b}=1 \Rightarrow \frac{x}{3}+\frac{y}{2}=1 \Leftrightarrow 2x+3y=6$

6. Find the equation of straight line through (2, 5) and making equal intercept of opposite sign on x-axis

Solution: use $\frac{x}{-a}+\frac{y}{a}=1$

The line passes through (2, 5) $\Rightarrow \frac{2}{-a}+\frac{5}{a}=1 \Leftrightarrow 2-5=-a \Rightarrow a=3$

$\therefore$ The equation of straight line $\frac{x}{-3}+\frac{y}{3}=1 \Leftrightarrow x-y=-3$

Let ℓ_1 and ℓ_2 be two non-vertical line with respective slope m_1 and m_2 then

i) If $\ell_1 \perp \ell_2$ then $m_1 \cdot m_2 = -1$

ii) If $\ell_1 \parallel \ell_2$ then $m_1 = m_2$

Illustrative Example

7. Determine which of the following pairs if lines are perpendicular parallel, intersecting (but not perpendicular)

a) ℓ_1: 3x – y + 5 = 0 and ℓ_2: x + 3y – 1 = 0

b) ℓ_1: 3x – 4y + 1 = 0 and ℓ_2: 4x - 3y + 1 = 0

c) ℓ_1: 3x + 2y = 6 and ℓ_2: 6x + 4y = 10

d) ℓ_1: 2x + 3y = 5 and ℓ_2: 4x + 6y = 10

Solution: Here

a) slope of ℓ_1: $3x - y + 5 = 0$ is $\frac{-3}{-1} = 3 = m_1$

slope of ℓ_2: $x + 3y - 1 = 0$ is $\frac{-1}{3} = m_2$

$\Rightarrow m_1 \cdot m_2 = (3)\left(\frac{-1}{3}\right) = -1$ ⟵ intersecting perpendicular

$\therefore \ell_1$ is perpendicular to ℓ_2

b) slope of ℓ_1: $3x - 4y + 1 = 0$ is $m_1 = \frac{3}{4}$

slope of ℓ_2: $4x - 3y + 1 = 0$ is $m_2 = \frac{4}{3}$

$\therefore$ neither parallel nor perpendicular

$\therefore$ It is intersecting line

c) slope of ℓ_1: $3x + 2y = 6$ is $m_1 = \frac{-3}{2}$

slope of ℓ_2: $6x + 4y = 10$ is $m_2 = \frac{-6}{4} = \frac{-3}{2}$

$\therefore m_1 = m_2$

$\therefore \ell_1$ is parallel to ℓ_2

d) exercise left for student.

3. Find an equation of the line through p(5, -7) that is parallel to the line $6x + 3y = 4$

Solution: slope of $6x + 3y = 4$ is $m_1 = \frac{-6}{3} = -2$

Since parallel lines have the same slop $m_1 = m_2$

$\therefore$ slope of the required line has slope $m_2 = -2$

Now using $y - y_1 = m_2(x - x_1)$ in $(x_1, y_1) = p(5, -7)$

$\Rightarrow y + 7 = -2(x - 5) \Rightarrow y = -2x + 3$ ⟵ the required parallel line

$\Leftrightarrow y + 2x = 3$

Method II

t should be noted that any line parallel to $6x + 3y = 4$ is of the form
$x + 3y = k$ where k is constant

If $6x + 3y = k$ passes through (5, -7)

$\Rightarrow 6(5) + 3(-7) = k \Rightarrow k = 9$

$\therefore$ the required equation $6x + 3y = 9 \Leftrightarrow 2x$ [illegible]

9. Find an equation of the line through (2, -6) parall[illegible]

Solution: Any line parallel to $5x + y = 8$ is of the form $5x + y = k$

If $5x + y = k$ passes through (2, -6)

$\Rightarrow 5(2) + -6 = k \Rightarrow k = 4$

$\therefore 5x + y = 4 \longleftarrow$ the equation of the required line

10. Find the equation of the line that passes through the point (2, [illegible] perpendicular to the line $2x - 3y - 4 = 0$

Solution: slope of $2x - 3y = 4$ is $m_1 = \frac{2}{3}$

Slope of perpendicular line $m_2 = \frac{-3}{2}$

$\therefore y + 1 = \frac{-3}{2}(x - 2) \Leftrightarrow 2y + 3x = 4 \leftarrow$ the equation of required line

11. Let ℓ be the line with equation $4x - 3y = 8$. Find the equation of the line that passes through the point p(1, 2) that is

a) parallel to ℓ b) perpendicular to ℓ

Solution: Exercise left for the student

Answer: a) $4x - 3y = -2$ b) $3x + 4y = 11$

12. Find k so that the line $4x + ky - 12 = 0$ will be

a) parallel to the line with equation $x - 3y = 8$

b) perpendicular to the line with equation $x - 3y = 8$

Solution: we have

a) slope of $4x + ky - 12 = 0$ is $\frac{-4}{k}$

slope of $x - 3y = 8$ is $\frac{1}{3}$

since they are parallel $\Rightarrow \frac{-4}{k} = \frac{1}{3} \Leftrightarrow k = -12$

b) since they are perpendicular

$\Rightarrow \left(\frac{-4}{k}\right)\left(\frac{1}{3}\right) = -1 \Rightarrow \frac{-4}{3k} = 1 \Leftrightarrow 3k = 4 \quad \therefore k = \frac{4}{3}$

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13. Given lines ℓ_1: $(6-k)x + y = -1$ and ℓ_2: $8x + ky = 7$
Find the value of k so that
a) $\ell_1 \parallel \ell_2$ b) $\ell_1 \perp \ell_2$

Solution:

a) slope of ℓ_1 is $\frac{-(6-k)}{1} = k-6$ and slope of ℓ_2 is $\frac{-8}{k}$

since $\ell_1 \parallel \ell_2 \Rightarrow k - 6 = \frac{-8}{k} \Leftrightarrow k^2 - 6k + 8 = 0$

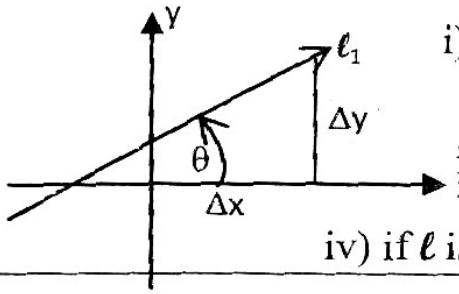
$\Rightarrow (k-2)(k-4) = 0 \Rightarrow k = 2$ or $k = 4$

b) $\ell_1 \perp \ell_2 \Rightarrow (k-6)\left(\frac{-8}{k}\right) = -1 \Leftrightarrow -8k + 48 = -k$

$\Rightarrow 7k = 48 \Rightarrow k = \frac{48}{7}$

Angle of inclination and slope

The angle measured from positive x-axis to a line in counter clock wise direction is called angle of inclination



i) $\tan\theta = \frac{\Delta y}{\Delta x}$ ii) slope $m = \frac{\Delta y}{\Delta x}$

iii) $\tan\theta = \text{slope of } \ell = \frac{\Delta y}{\Delta x}$

iv) if ℓ is horizontal, then $\theta = 0$

Illustrative Example

14. Find the slope of a line ℓ if its inclination is
a) 45° b) 150° c) 60°

Solution: a) slope of $\ell = \tan 45^0 = 1$

b) slope of $\ell = \tan 150^0 = -\tan 30^0 = \frac{-1}{\sqrt{3}}$

c) slope of $\ell = \tan 60^0 = \sqrt{3}$

15. Find the equation of straight line passing through (2, 3) and making with the x-axis an angle of
a) 45° b) 60° c) 90°

Solution: a) $m = \tan 45^0 = 1$, using $y - y_1 = m(x - x_1)$

$\Rightarrow y - 3 = 1(x - 2) \Leftrightarrow y = x + 1$

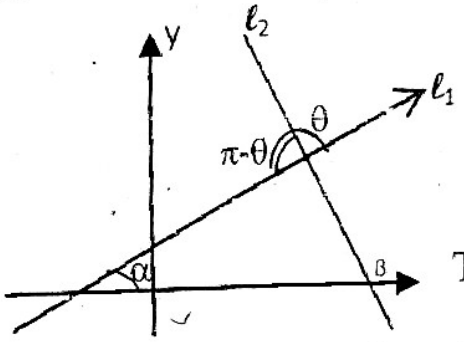
b) $m = \tan 60^0 = \sqrt{3} \Leftrightarrow y - 3 = \sqrt{3}(x - 2)$

$\Rightarrow y = \sqrt{3}x - 2\sqrt{3} + 3$

c) when $\theta = 90^0$, the line is vertical $\quad \therefore x = 2$

Angle Between two lines

The angle between two intersecting lines ℓ_1 and ℓ_2 is angle θ which is measured counter clock wise from ℓ_1 to ℓ_2 about the point of intersection in order to concide with ℓ_2.



- slope of $\ell_1 = m_1 = \tan\alpha$
- slope of $\ell_2 = m_2 = \tan\beta$

You know that $\alpha + \theta = \beta$

$\Rightarrow \theta = \beta - \alpha$

Thus, $\tan\theta = \tan(\beta - \alpha) = \dfrac{\tan\beta - \tan\alpha}{1 + \tan\beta\tan\alpha}$

Hence angle between ℓ_1 and ℓ_2 is given by

$$\tan\theta = \frac{m_2 - m_1}{1 + m_2 m_1} \text{ if } m_1 m_2 \neq -1$$

i) If $1 + m_2 m_1 = 0 \Rightarrow m_1 \cdot m_2 = -1$, then $\theta = 90^0$

ii) If $m_2 - m_1 = 0 \Rightarrow m_1 = m_2$, then $\theta = 0^0$

Illustrative Example

16. Find the tangent of the angle of intersection of the lines with slope.

a) $m_1 = 2$ and $m_2 = 3$

b) $m_1 = \dfrac{-2}{3}$ and $m_2 = -5$

c) $m_1 = \dfrac{-1}{3}$ and $m_2 = -2$

d) $m_1 = 0$ and $m_2 = 1$

e) $m_1 = 1$ and $m_2 = -1$

Solution: we use $\tan\theta = \dfrac{m_2 - m_1}{1 + m_2 m_1}$

a) $\tan\theta = \dfrac{3 - -2}{1 + (3)(2)} = \dfrac{1}{7}$ ⟵ the acute angle between the lines

$\tan\theta = \frac{-1}{7}$ ⟵ the obtuse angle between the line

b) $\tan\theta = \frac{-5+\frac{2}{3}}{1+(5)\frac{2}{3}} = \frac{\frac{-13}{3}}{\frac{13}{3}} = -1$

$\tan\theta = -1$ ⟵ the obtuse angle between the line

$\Rightarrow \theta = 135^0$

$\tan\theta = 1$ ⟵ the acute angle between the line

$\Rightarrow \theta = 45^0$ e) $\theta = 90^0$

c and d exercise

17. Find the acute angle of intersection of the lines with slopes $\frac{1-m}{1+m}$ and –m, where $m \neq -1$

Solution: we have $m_1 = \frac{1-m}{1+m}$ and $m_2 = -m$

Use $\tan\theta = \left|\frac{m_2 - m_1}{1+m_2 m_1}\right|$ for acute angle between thee two lines

$$\therefore \tan\theta = \left|\frac{-m-\left(\frac{1-m}{1+m}\right)}{1+(-m)\left(\frac{1-m}{1+m}\right)}\right| = \left|\frac{m^2+1}{m^2+1}\right| = 1 \Rightarrow \theta = 45^0$$

18. Find the acute angle of intersection of the lines with slope and -3

Solution: $\tan\theta = \left|\frac{-\frac{1}{2}+3}{1+\left(\frac{-1}{2}\right)(-3)}\right| = \frac{\frac{5}{2}}{\frac{5}{2}} = 1 \Rightarrow \theta = 45^0$

19. If a line L_1 passes through (1, -1) and (3, 1) and a line L_2 contains points (4, 5) and (6, 9) then find the tangent of the angle between ℓ_1 and ℓ_2

Solution: First find the slope

Slope of ℓ_1: $m_1 = \dfrac{1-(-1)}{3-1} = 1$

Slope of ℓ_1: $m_2 = \dfrac{9-5}{6-4} = \dfrac{4}{2} = 2$

If θ is the angle between ℓ_1 and ℓ_2 then

$$\tan\theta = \frac{m_2 - m_1}{1 + m_2 m_1} = \frac{2-1}{1+(2)(1)} = \frac{1}{3}$$

20. The angle between two lines is 45^0 and the slope of one of the line is $\dfrac{3}{5}$. Find the slope of the other line

Solution: **case 1**: Let $m_1 = \dfrac{3}{5}$ and $m_2 = ?$

$$\Rightarrow \tan 45^0 = \frac{m_2 - \frac{3}{5}}{1 + \frac{3}{5}m_2} = 1 \Rightarrow m_2 - \frac{3}{5} = 1 + \frac{3}{5}m_2$$

$$\Rightarrow m_2 - \frac{3}{5}m_2 = 1 + \frac{3}{5} \Rightarrow \frac{2m_2}{5} = \frac{8}{5} \Leftrightarrow 2m_2 = 8 \Rightarrow m_2 = 4$$

Case II Let $m_2 = \dfrac{3}{5}$ *and* $m_1 = ?$

$$\Rightarrow \tan 45^0 = \frac{\frac{3}{5} - m_1}{1 + \frac{3}{5}m_1} = 1 \Leftrightarrow \frac{3}{5} - m_1 = 1 + \frac{3}{5}m_1$$

$$\Rightarrow \frac{3}{5} - 1 = \frac{3}{5}m_1 + m_1 \Rightarrow \frac{-2}{5} = \frac{8m_1}{5} \Leftrightarrow m_1 = -\frac{1}{4}$$

21. What is the angle between the line through (-1, 3) and (2, 3) and the line with equation $\sqrt{3}y - x + 7 = 0$

A. 120° B. 60° C. 30° D. 15°

Solution: slope of ℓ_1: $m_1 = \frac{3-3}{2+1} = 0$

Slope of ℓ_2: $m_2 = \frac{1}{\sqrt{3}}$

$$\Rightarrow \tan\theta = \frac{\frac{1}{\sqrt{3}} - 0}{1+0} = \frac{1}{\sqrt{3}}$$

$\therefore \theta = 30^0$

22. Find the equation of the line that

a) passes through (2,-1) and the angle from the line $6x+5y-1=0$ to the line 45^0

b) have y-intercept 4 and is inclined at an angle of 135^0 to the line $2x + 3y - 7 = 0$

Solution: a) Let the slope of $\ell_1 = 6x + 5y - 1 = 0$ is $m_1 = \frac{-6}{5}$

Let the slope of required line ℓ_2 is m_2 the angle between the two line can be obtained by

$$\Rightarrow \tan\theta = \frac{m_2 - m_1}{1 + m_2 m_1} \Rightarrow \tan 45^0 = \frac{m_2 + \frac{6}{5}}{1 - \frac{6}{5}m_2}$$

$$\Rightarrow m_2 + \frac{6}{5}m_2 = 1 - \frac{6}{5} \Rightarrow \frac{11m_2}{5} = \frac{-1}{5}$$

$$\therefore m_2 = \frac{-1}{11}$$

Thus, the equation of the line passing through (2, -1) have slope

$$m = \frac{-1}{11}$$

Given by $y+1=\frac{-1}{11}(x-2)$

$\Rightarrow 11y + 11 = -x + 2$

$\Rightarrow x + 11y = -9$

Case II the angle between the given straight line and the required straight line is 135^0

Hence, $\tan 135^0 = \frac{m+\frac{6}{5}}{1-\frac{6}{5}m}$

$\Rightarrow -1=\frac{5m+6}{5-6m} \Leftrightarrow -5+6m=5m+6 \Rightarrow$ m=11 $\Leftrightarrow$ y + 1 = 11(x - 2)

$\therefore$ the required straight lines are $x + 11y + 9 = 0$ or $11x - y - 23 = 0$

b) Let m be the slope of the required lines.

Slope of $2x + 3y - 7 = 0$ is $\frac{-2}{3}$

Now the angle between the given straight line and the required straight line is 45^0 or 135^0

$\therefore \tan 45^0 = \frac{m+\frac{2}{3}}{1-\frac{2}{3}m}$

$\Rightarrow 1=\frac{m+\frac{2}{3}}{1-\frac{2}{3}m}$

$\Rightarrow 1-\frac{2}{3}m=m+\frac{2}{3}$

$\Rightarrow 3 - 2m = 3m + 2$

$\Rightarrow 1 = 5m$

$\therefore m = \frac{1}{5}$

$\therefore \tan 135^0 = \frac{m+\frac{2}{3}}{1-\frac{2}{3}m}$

$\Rightarrow -1=\frac{m+\frac{2}{3}}{1-\frac{2}{3}m}$

$\Rightarrow -1+\frac{2}{3}m=m+\frac{2}{3}$

$\Rightarrow -3 + 2m = 3m + 2$

$\Rightarrow -5 = m$

$\Rightarrow$ Equation of ℓ is $y = \frac{1}{5}x + 4$ or $y = -5x + 4$

Thus the equation of the straight lines are, for b = 4

$$y = \frac{1}{5}x + 4 \quad or \quad y = -5x + 4$$

23. Find the tangent of the angle between the given lines

a) $\ell_1: y = -4x + 2; \ell_2: y = -2x$

b) $\ell_1: y = 3x - 2; \ell_2: y = 4x - 6$

c) $\ell_1: -3x + 4y - 15 = 0; \ell_2: 4x + 3y - 30 = 0$

d) $\ell_1: 5x - 3y = 15; \ell_2: x + 2y - 4 = 0$

e) $\ell_1: y = 2; \ell_2: x + 2y = 8$

Solution: Here we use $\tan\theta = \frac{m_2 - m_1}{1 + m_2 m_1}$

a) the slope of ℓ_1: $m_1 = -4$

the slope of ℓ_2: $m_2 = -2$

$$\therefore \tan\theta = \frac{m_2 - m_1}{1 + m_2 m_1} = \frac{-2 - (-4)}{1 + (-2)(-4)} = \frac{2}{9} \longleftarrow \text{the angle between from } \ell_1 \text{ to } \ell_2$$

b) $m_1 = 3, m_2 = 4$

$$\Rightarrow \tan\theta = \frac{4 - 3}{1 + (4)(3)} = \frac{1}{13} \longleftarrow \text{the angle between from } \ell_1 \text{ to } \ell_2$$

c) $m_1 = \frac{3}{4}, m_2 = -\frac{4}{3}$

$$\Rightarrow \tan\theta = \frac{\frac{-4}{3} - \frac{3}{4}}{1 + \left(\frac{-4}{3}\right)\left(\frac{3}{4}\right)} = \frac{\frac{-25}{12}}{\frac{12 - 12}{12}} = \frac{-25}{0}$$

Since ℓ_1 is perpendicular to ℓ_2

$\therefore$ The angle between the two line is $\theta = 90^0$

d) $m_1 = \frac{5}{3}, m_2 = -\frac{1}{2}$

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$$\Rightarrow \tan\theta = \frac{-\frac{1}{2}-\frac{5}{3}}{1+\left(-\frac{1}{2}\right)\left(\frac{5}{3}\right)} = \frac{\frac{-3-10}{6}}{\frac{6-5}{6}} = -13 \leftarrow \text{the angle from } \ell_1 \text{ to } \ell_2$$

e) $m_1 = 0, m_2 = -\frac{1}{2} \Rightarrow \tan\theta = \frac{-\frac{1}{2}}{1} = -\frac{1}{2} \leftarrow$ the angle from ℓ_1 to ℓ_2

24. Find the tangent of the acute angle between the line

a) $3x - y + 5 = 0$ and $2x + y + 7 = 0$

b) $\sqrt{3}x + \sqrt{2}y - 2 = 0$ and $\sqrt{6}x - 3y + 3 = 0$

c) $6x + 5y - 1 = 0$ and $11x - y - 23 = 0$

Solution: To find the acute angle between the line we use

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_2 m_1}\right|$$

Note, slope of $Ax + By + C = 0$ is $\left(\frac{-A}{B}\right)$

a) $m_1 = 3, m_2 = -2$

$$\tan\theta = \left|\frac{-2-3}{1-6}\right| = \left|\frac{-5}{-5}\right| = 1$$

$\therefore \theta = 45^0$

b) $m_1 = \frac{-\sqrt{3}}{\sqrt{2}}, \; m_2 = \frac{\sqrt{6}}{3}, \quad Note, \; \frac{-\sqrt{3}}{\sqrt{2}} = \frac{-\sqrt{6}}{2} = \frac{-3}{\sqrt{6}}$

$$\tan\theta = \left|\frac{\frac{\sqrt{6}}{3}+\frac{\sqrt{3}}{\sqrt{2}}}{1+\left(\frac{\sqrt{6}}{3}\right)\left(\frac{-\sqrt{3}}{\sqrt{2}}\right)}\right| = \left|\frac{\frac{\sqrt{6}}{3}+\frac{3}{\sqrt{6}}}{1-1}\right| \Rightarrow \theta = 90^0$$

c) $m_1 = \frac{-6}{5}$, $m_2 = 11$

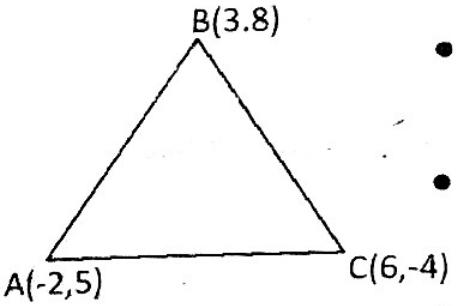
$$\tan\theta = \left|\frac{11+\frac{6}{5}}{1-\frac{66}{5}}\right| = \left|\frac{\frac{61}{5}}{\frac{-61}{5}}\right| \Rightarrow |-1| = 1$$

$\therefore \theta = 45^0$

25. Let A(-2, 5), B(3, 8) and C(6, -4) and vertices of a triangle (ΔABC) then, Find the tangent angle

a) at A b) at B c) at C

Solution: Consider the triangle, then find pair of slopes.



- slope of $\overline{AB} = \frac{8-5}{3-(-2)} = \frac{3}{5}$
- slope of $\overline{AC} = \frac{-4-5}{6-(-2)} = \frac{-9}{8}$
- slope of $\overline{BC} = \frac{8-(-4)}{3-6} = -4$

Therefore, tangent angle at

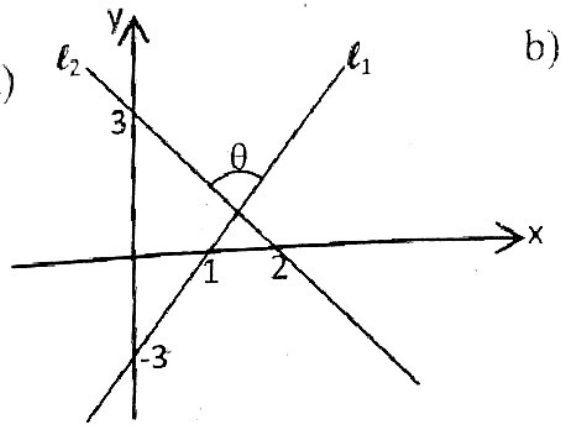
a) A is $\tan A = \frac{\frac{3}{5}-\left(\frac{-9}{8}\right)}{1+\left(\frac{3}{5}\right)\left(\frac{-9}{8}\right)} = \frac{69}{13}$

b) at B, $\tan\hat{B} = \frac{-4-\frac{3}{5}}{1+(-4)\left(\frac{3}{5}\right)} = \frac{23}{7}$

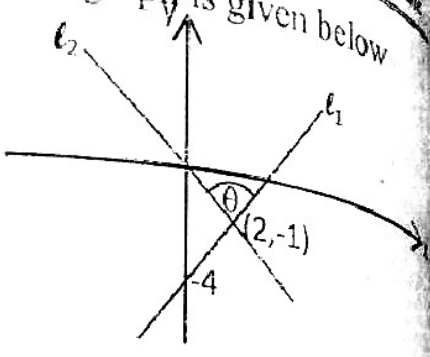
c) at C, $\tan\hat{C} = \frac{\frac{-9}{8}-(-4)}{1+\left(\frac{-9}{8}\right)(-4)} = \frac{23}{44}$

26. Find the angle between ℓ_1 and ℓ_2 whose graph is given below

a)



b)



Solution: To find the angle between ℓ_1 and ℓ_2, first determine slope,

a) slope of ℓ_1: $m_1 = 3$ and slope of $\ell_2 = -\frac{3}{2} = m_2$

$$\therefore \tan\theta = \frac{\frac{-3}{2} - 3}{1 + \left(\frac{-3}{2}\right)(3)} = \frac{9}{7} \leftarrow \text{angle from } \ell_1 \text{ to } \ell_2$$

b) since the point (0, -4) and (2, -1) on ℓ_1, and (0, 0) and (2, -1) on ℓ_2, so the angle between ℓ_1 and ℓ_2 will be

$$\therefore m_1 = \frac{3}{2} \text{ and } m_2 = -\frac{1}{2}$$

$$\therefore \tan\theta = \frac{\frac{-1}{2} - \frac{3}{2}}{1 + \left(\frac{3}{2}\right)\left(\frac{-1}{2}\right)} = -8$$

27. Find the tangent angle between

a) the x-axis and the line: $y = x - 4$

b) the x-axis and the line $y + x = 6$

c) the line $y = 3$ and the line $3x - 2y = 6$

Solution: (a) Now, ℓ_1 as x-axis (i.e. $y = 0$)

ℓ_2 as $y = x - 4 \therefore m_1 = 0$ and $m_2 = 1$

$$\therefore \tan\theta = \frac{1-0}{1+(1)(0)} = 1 \Rightarrow \theta = 45^0$$

(b) $m_1 = 0 \ and \ m_2 = -1$

$$\therefore \tan\theta = \frac{-1-0}{1+0} = -1 \ \therefore \theta = 135^0$$

(c) since the slope of horizontal line $m_1 = 0$

$$\Rightarrow m_1 = 0 \ and \ m_2 = \frac{3}{2} \ \therefore \tan\theta = \frac{\frac{3}{2}}{1+0} = \frac{3}{2}$$

Distance between two point

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ are coordinate points in a plane, then the distance d between P and Q is given by

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_2)^2}$$

Illustrative Example

28. Find the distance between the point of each of the following
    a) (2, 5) and (5, 9) c) (0, a) and (a, 0)
    b) (a, -b) and (-a, b) d) (x, -y) and (-y, x)

Solution: Let distance between be d

a) $d = \sqrt{(5-2)^2 + (9-5)^2} = \sqrt{9+16} = 5unit$

b) $d = \sqrt{(a-(-a))^2 + (-b-b)^2} = \sqrt{4a^2 + 4b^2} = 2\sqrt{a^2 + b^2}$

c) $d = \sqrt{(0-a)^2 + (a-0)^2} = \sqrt{a^2 + a^2} = \sqrt{2}\,|a|$

d) $d = \sqrt{(x+y)^2 + (x+y)^2} = \sqrt{2(x+y)^2} = \sqrt{2}\,|x+y|$

29. Find all points on the x-axis that are 10 units from (4, 6)

Solution: all points on the x-axis are (x, 0)

$$(x,0) \xrightarrow{d = 10 \ unit} (4,6)$$

$$\Leftrightarrow \sqrt{(x-4)^2 + (0-6)^2} = 10 \Rightarrow \sqrt{x^2 - 8x + 52} = 10$$

$$\Rightarrow x^2 - 8x + 52 = 100 \longleftarrow \text{squaring both side}$$

$$\Rightarrow x^2 - 8x - 48 = 0 \Rightarrow (x-12)(x+4) = 0$$

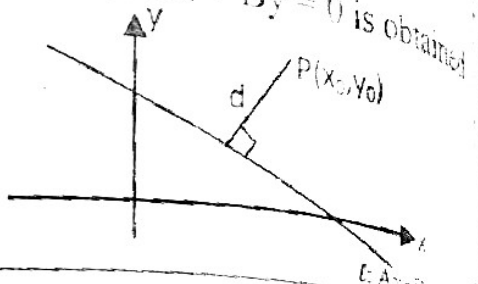
$\therefore$ x = 12 or x = -4

$\therefore$ The possible points are (12, 0) or (-4, 0)

Distance between a point $P(x_0, y_0)$ and a line ℓ: $Ax + By + C = 0$ on the coordinate plane.

The distance from a point $P(x_0, y_0)$ to the line ℓ: $Ax + By = 0$ is obtained by the formula

$$d = \left| \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \right|$$



Illustrative Example

30. Find the distance from each line to the indicated point.
   a) $x - 2y + 5 = 0$, P(0, 0) c) $3x - 5y = 0$, P(4, -2)
   b) $2x - 3y + 6 = 0$, P(4, 5) d) $5x + 2y - 1 = 0$, P(4, 1)

Solution: a) The distance from the origin to the line $x - 2y + 5 = 0$

$$\Rightarrow d = \left| \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \right| \Leftrightarrow d = \left| \frac{1 \cdot 0 - 2 \cdot 0 + 5}{\sqrt{1^2 + 2^2}} \right| = \frac{5}{\sqrt{5}} = \sqrt{5}$$

b) $$d = \left| \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \right| \Leftrightarrow \left| \frac{2(3) - 3(5) + 6}{\sqrt{2^2 + 3^2}} \right| = \frac{1}{\sqrt{13}} = \frac{\sqrt{13}}{13}$$

c) $$d = \left| \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \right| \Leftrightarrow \left| \frac{3(4) - 5(-2) + 0}{\sqrt{3^2 + 5^2}} \right| = \frac{22}{\sqrt{34}}$$

d) $$d = \left| \frac{Ax_0 + By_0 + C}{\sqrt{A^2 + B^2}} \right| \Leftrightarrow \left| \frac{5(4) + 2(1) - 1}{\sqrt{5^2 + 2^2}} \right| = \frac{21}{\sqrt{29}}$$

31. Find the point on the line $y = x - 2$ at a distance of $3\sqrt{2}$ units from (1, -1)

Solution: If $y = x - 2$, then $(x, y) = (x, x - 2)$

$$\therefore (x, x-2) \xrightarrow{d = 3\sqrt{2}} (1, -1)$$

Distance between point

$$\Rightarrow \sqrt{(x-1)^2 + (x-2+1)^2} = 3\sqrt{2} \longleftarrow \text{squaring both side}$$

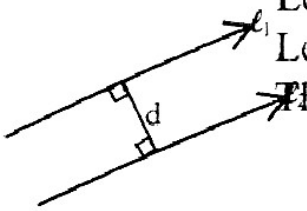
$\Rightarrow 2(x - 1)^2 = 18 \Leftrightarrow (x - 1)^2 = 9$

$\Rightarrow x - 1 = \pm 3 \Rightarrow x - 1 = 3$ or $x - 1 = -3$

$\therefore x = 4$ or $x = -3 + 1 = -2$

$\therefore$ Point on the line $y = x - 2$ are (4, 2) or (-2, -4)

The distance between two parallel lines



If ℓ_1 is parallel to ℓ_2

Let take point P(h, k) on ℓ_1.

Let the equation of ℓ_2: $Ax + By + C = 0$

Then the distance between ℓ_1 and ℓ_2 is obtained by $d = \dfrac{|Ah + Bk + C|}{\sqrt{A^2 + B^2}}$

Illustrative Example

32. Find the distance between the pair of parallel lines whose equations are given below.

a) ℓ_1: $3x + 4y - 10 = 0$ and ℓ_2: $6x + 8y + 3 = 0$

b) ℓ_1: $2x - 5y + 1 = 0$ and ℓ_2: $2x - 5y - 3 = 0$

c) ℓ_1: $y = -2x + 3$ and ℓ_2: $3y = -6x + 1$

Solution: a) First take point on ℓ_1(h, k) = (2, 1) on ℓ_1: $3x + 4y - 10$ then the distance between the point P(2, 1) and the line ℓ_2, $6x + 8y + 3 = 0$

$$\Rightarrow d = \frac{|6(2) + 4(1) + 3|}{\sqrt{6^2 + 8^2}} = \frac{23}{\sqrt{100}} = \frac{23}{10} = 2.3 unit$$

b) Let P(h, k) = $P\left(3, \frac{7}{5}\right)$ on ℓ_1 i.e. when $x = 3$, $y = \frac{7}{5}$ then distance between the point $P\left(3, \frac{7}{5}\right)$ and the line $2x - 5y - 3 = 0$ will be

$$\Rightarrow d = \frac{\left|2(3) + 5\left(\frac{7}{5}\right) - 3\right|}{\sqrt{2^2 + 5^2}} = \frac{4}{\sqrt{29}} unit \leftarrow \text{Distance between } \ell_1 \text{ and } \ell_2$$

c) Let P(h, k) = (-1, 5) on $y = -2x + 3$

$$\Rightarrow d = \frac{|6(-1) + 3(5) - 1|}{\sqrt{36 + 9}} = \frac{8}{\sqrt{45}} = \frac{8}{3\sqrt{5}} \leftarrow \text{Distance between } \ell_1 \text{ and } \ell_2$$

Locus

The locus of a point is the path traced out by the point as it moves under certain condition.

- All points satisfying the given condition lie on the locus.
- We define a locus as a set of points satisfying a given condition.

Circle

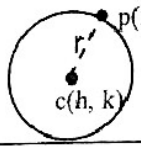
Definition: A circle is the locus of a point P which moves in plane with fixed distance from fixed point.

- The fixed distance is called radius
- The fixed point is called centre of circle.

Equation of a circle

Let P(x, y) be a point on circle with center C(h, k) and the distance PC = r, now using distance between two point is given by

$(x-h)^2 + (y-k)^2 = r^2$



Illustrative example

33. Find the equation of a circle with the following centers and radii

a) C(0, 0), r = 4 c) C(2, -1), r = $\sqrt{5}$

b) C(-1, -1), r = 3 d) C(3, 2), r = 2

Solution: use $(x-h)^2 + (y-k)^2 = r^2$, for c(h, k)

a) $(x-0)^2 + (y-0)^2 = 4^2 \Leftrightarrow x^2 + y^2 = 16$

b) $(x+1)^2 + (y+1)^2 = 3^2 \Rightarrow (x+1)^2 + (y+1)^2 = 9$

c) $(x-2)^2 + (y+1)^2 = 5$

d) $(x-3)^2 + (y-2)^2 = 4$

34. Find the equation of the circle of center C(2, -1) which passes

i) through the point (3, 4)

ii) through the point (4, 1)

Solution: Center (h, k) = (2, -1)

i) The equation of a circle is $(x-2)^2 + (y+1)^2 = r^2$ since the point (3, 4) on the circle

$\therefore (3-2)^2 + (4+1)^2 = r^2$

$\therefore r^2 = 26$

$\therefore$ The equation of a circle $(x-2)^2+(y+1)^2=26$

ii) Since the point (4, 1) lies on the circle

$\therefore (4-2)^2+(1+1)^2=r^2$

$\therefore r^2=8$

$\therefore$ The equation of circle is $(x-2)^2+(y+1)^2=8$

35. In each of the following find the equation of the circle with

a) center (-1, -3) passes through (-4, -2)

b) center (-4, 5) tangent to the x-axis

c) center (1, -7) tangent to the y-axis

d) end points of diameter are (-2, -3) and (4, 5)

Solution: use $(x-h)^2+(y-k)^2=r^2$, for center (h, k)

a) $(x+1)^2+(y+3)^2=r^2$, since the point (-4, -2) on the circle

$\therefore (-4+1)^2+(-2+3)^2=r^2$

$\therefore r^2=10$ and $r=\sqrt{10}$

$\therefore$ The equation of circle is $(x+1)^2+(y+3)^2=10$

Note: for center (h, k), if the circle is tangent to x-axis, then the radius, r = k for center (h, k) if te circle is tangent to y-axis, then the radius, r = h

$(x+4)^2+(y-5)^2=25$

c) $r=1$

$\therefore (x-1)^2+(y+7)^2=1$

d) center $=\left(\frac{-4+-2}{2},\frac{5+-3}{2}\right)=(-3,1)$

$\therefore$ the equation of circle $(x+3)^2+(y-1)^2=r^2$

Since (-2, -3) on the circle

$\therefore r^2=(-2+3)^2+(-3-1)^2=17$

$\therefore$ the equation of circle $(x+3)^2+(y-1)^2=17$

36. Find the equation of a circle containing the point P(0, -3) and Q(4, 0) and whose center is on the line $x+y=0$

Solution: Let C(h, k) the center, $\rightarrow \overline{PC}=\overline{QC}=r$

$\Rightarrow (h-0)^2+(k+3)^2=(h-4)^2+(k-0)^2$

$\Rightarrow h^2+k^2+6k+9=h^2-8h+16+k^2$

$\Rightarrow 6k+8h=7$, since (h, k) on the line $x+y=0 \Rightarrow h+k=0$

$\Rightarrow h=-k \Rightarrow 6k+8(-k)=7 \Rightarrow k=\frac{-7}{2},\quad h=\frac{7}{2}$

$\therefore$ center (h, k) $= \left(\frac{7}{2}, \frac{-7}{2}\right)$

Hence the equation of the circle $\left(x-\frac{7}{2}\right)^2+\left(y+\frac{7}{2}\right)^2=r^2$

Since (0, -3) on the circle

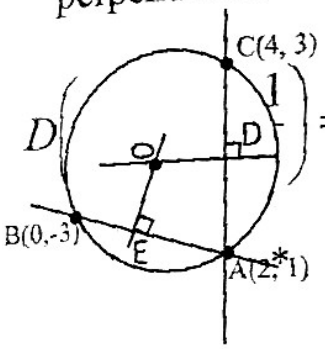
$$r^2=\left(0-\frac{7}{2}\right)^2+\left(-3+\frac{7}{2}\right)^2=\frac{49}{4}+\frac{1}{4}=\frac{50}{4}=\frac{25}{2}$$

$\therefore$ the equation of circle $\left(x-\frac{7}{2}\right)^2+\left(y+\frac{7}{2}\right)^2=\frac{25}{2}$

37. Find the equation of the circle which passes through the points A(2, 1), B(0, -3) and C(4, 3)

Solution; The center is the point of intersection of the perpendicular bisectors of any two chords

* mid-point of AC $= \left(\frac{4+2}{2}, \frac{3+1}{2}\right) = (3,2)$



mid-point of AB $= E\left(\frac{0+2}{2}, \frac{-3+1}{2}\right) = (1,-1)$

Since slope of AC $= \frac{3-1}{4-2} = 1$, therefore the slope of

OD = -1, because OD $\perp$ AC

$\therefore$ equation of OD, is $y - 2 = -1(x - 3) \Rightarrow y + x = 5 = 5$

And slope of AB $= \frac{1+3}{2-0} = 2$

$\Rightarrow$ slope of OE $= -\frac{1}{2}$, because $\overline{AB} \perp \overline{OE}$

$\therefore$ the equation of OE is $y + 1 = -\frac{1}{2}(x - 1) \Rightarrow 2y + x = -1$

From (*) and (**) solving together

$$\begin{cases} y + x = 5 \\ 2y + x = -1 \end{cases} \Rightarrow (11,-6) \longleftarrow \text{the center}$$

radius $= \sqrt{(11-0)^2 + (-6+3)^2} = \sqrt{130}$

Hence the equation of circle is $(x - 11)^2 + (y + 6)^2 = 130$

Note: The equation of circle with center (h, k) and tangent to the line ℓ: $Ax + By + C = 0$ is given by

$$(x - h)^2 + (y - k)^2 = r^2 \text{ where } r = \frac{|Ah + Bk + C|}{\sqrt{A^2 + B^2}}$$

Illustrative Example

38. In each of the following, find the equation of a circle with
   a) center (4, 3) and tangent to $2x - 3y + 6 = 0$
   b) center (-2, 1) and tangent to $3x - 4y - 1 = 0$
   c) center (0, 0) and tangent to $3x + 4y - 10 = 0$

Solution:

a) $(x - 4)^2 + (y - 3)^2 = r^2$ and $r = \frac{|2(4) - 3(3) + 6|}{\sqrt{2^2 + 3^2}} = \frac{5}{\sqrt{13}}$

$$\therefore (x-4)^2 + (y-3)^2 = \frac{25}{13}$$

b) $(x - 2)^2 + (y - 1)^2 = r^2$ where $r = \frac{|3(-2) - 4(1) - 1|}{\sqrt{3^2 + 4^2}} = \frac{11}{5}$

$$\therefore (x+2)^2 + (y-1)^2 = \frac{121}{25}$$

c) $(x - 0)^2 + (y - 0)^2 = r^2$, where

$$r = \frac{|0 + 4(0) - 10|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2$$

$\therefore x^2 + y^2 = 2^2 \Leftrightarrow x^2 + y^2 = 4$

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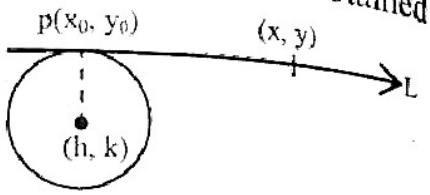
Circle and tangent line

Note:
Let a line L tangent to a circle $(x-h)^2+(y-k)^2=r^2$ at a point $P(x_0, y_0)$ then the equation of tangent line L can be obtained by

$$\frac{y-y_0}{x-x_0}=-\left(\frac{x_0-h}{y_0-k}\right)$$



Illustrative Example

39. Find the equation of the tangent line to each circle at the indicated point

a) $(x-1)^2+(y+2)^2=40$, P(3, 4)

b) $(x+3)^2+(y-1)^2=29$, P(2, 3)

c) $x^2+y^2=2$; +(1, -1)

d) $(x-3)^2+y^2=8$, P(1, 2)

Solution: a) $P(x_0, y_0)=(3, 4)$, $C(h, k)=(1, -2)$

$$\Rightarrow \frac{y-4}{x-3}=-\left(\frac{3-1}{4+2}\right)\Leftrightarrow y-4=-\frac{1}{3}(x-3)$$

$\therefore$ the equation of the tangent line L: $x+3y-15=0$

b) $$\frac{y-3}{x-2}=-\left(\frac{2+3}{3-1}\right)\Leftrightarrow y-3=-\frac{5}{2}(x-2)$$

$\therefore$ the equation of the tangent line L: $2y+5x=16$

c) $$\frac{y+1}{x-1}=-\left(\frac{1-0}{-1-0}\right)\Leftrightarrow y+1=x-1$$

$\therefore$ The equation of the tangent line L: $y-x+2=0$

Line and Circle

Definition: If the perpendicular distance from center of circle to the line ℓ: $Ax+By+C=0$ is

a) greater than radius of r, then line does not intersect the circle

i.e. $$\left|\frac{Ah+Bk+C}{\sqrt{A^2+B^2}}\right|>r$$

b) Smaller than radius or r, then the line intersect **at two point** which is called secant line.

c) equal to the radius of r, then the line intersects **at one point** only, the line called tangent line

$$\text{i.e. } r = \left| \frac{Ah + Bk + C}{\sqrt{A^2 + B^2}} \right|$$

Illustrative Example

40. Find the perpendicular distance from center of the circle with equation $(x - 1)^2 + (y + 4)^2 = 25$ and identify the line is intersect:

* at one point * at two point

* does not intersect to each of the following line with equations

a) $3x + 4y - 12 = 0$ b) $3x + 4y + 5 = 0$

b) $3x + 4y - 13 = 0$ d) $3x + 4y + 4 = 0$

Solution: Let the perpendicular distance from center of circle (1, -4) to line = D

Let be r radius, so that $r^2 = 25$ $\therefore r = 5$

a) $D = \dfrac{|3(1) + 4(-4) - 12|}{\sqrt{3^2 + 4^2}} = \dfrac{25}{5} = 5$ ⟵ perpendicular distance

Since D = r = 5 $\therefore$ It intersect at one point

$\therefore$ the line $3x + 4y - 12 = 0$ is tangent to $(x-1)^2+(y+4)^2= 25$

b) $D = \dfrac{|3(1) + 4(-4) + 5|}{\sqrt{3^2 + 4^2}} = \dfrac{8}{5}$ ⟵ perpendicular distance

since $D < r \Rightarrow \dfrac{8}{5} < 5$ $\therefore$ It intersect at two point

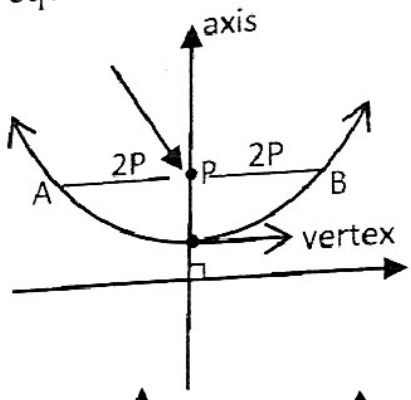
$\therefore$ $3x + 4y - 12$ is secant line to circle $(x-1)^2+(y+4)^2 = 25$

c) $D = \dfrac{|3(1) + 4(-4) + -13|}{\sqrt{3^2 + 4^2}} = \dfrac{26}{5}$ ⟵ perpendicular distance since $D > r \Rightarrow \dfrac{26}{5} > 5$

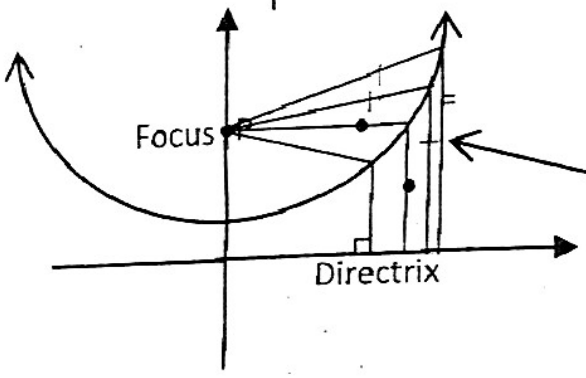
$\therefore$ The line does not intersect the circle.

Parabola

Definition: A parabola is the set of points in the plane that are equidistant from fixed line (directrix) and fixed point (the focus).



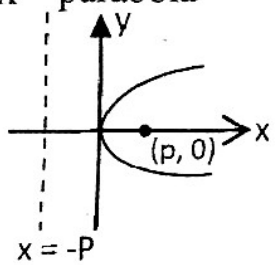
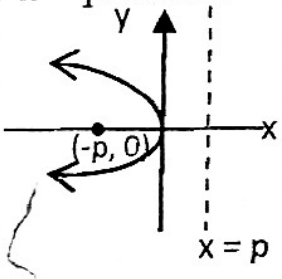
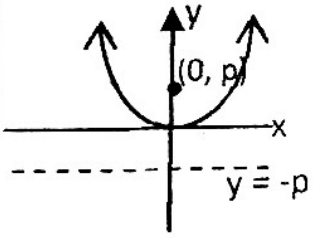
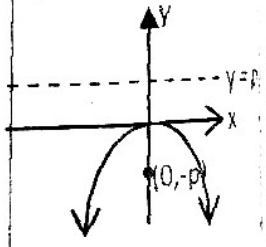
- The distance between the focus and the vertex is P
- The vertex is equidistant from thee focus and the directrix
- The distance between the focus and directrix is 2P
- The length $\overline{AB}$ that pass through focus (P) is |4P| (i.e. $\overline{AB}$ = |4P|) It is called Latus rectum.



All points on the parabola are equidistant from the focus and directrix

The equation of a parabola is simplest if the vertex is the origin and the axis of symmetry is along the x-axis or y-axis. These are called the standard equation.

Parabola in standard Equation

x – parabola	x – parabola	y– parabola	y– parabola
			
-Equation: $y^2=4Px$	$y^2 = -4Px$	$x^2 = 4Py$	$x^2 = -4Py$
- Focus: (P, O)	(-P, 0)	(0, P)	(0, -P)
- Vertex: (0, 0)	(0, 0)	(0, 0)	(0, 0)
- latus rectum:(4P)	|4P|	|4P|	|4P|
- directrix x = -P	x = p	y = -p	y = p

Illustrative example

8. Find the focus, vertex, latus rectum, directrix.
a) $y^2 = -8x$, b) $x^2 = 12y$ c) $y = x^2$ d) $y = -\frac{1}{2}x^2$

Solution: a) i) latus rectum $4P = |-8| \Leftrightarrow 4P = 8 \Rightarrow P = 2$
ii) Focus, (-2, 0) iii) direct rix, $x = 2$

b) i) Latusrectum, $4P = 12 \Leftrightarrow P = \frac{12}{4} = 3$

ii) focus, (0, 3) iii) directrix, $y = 3$ iv) vertex, (0, 0)

c) i) latus rectum $4P = 1 \Leftrightarrow P = \frac{1}{4}$ ii) focus, $(0, \frac{1}{4})$
iii) directrix, $y = -\frac{1}{4}$ iv) vertex, (0, 0)

d) i) latus rectum $4P = 2 \Leftrightarrow P = \frac{1}{2}$

ii) focus $\left(0, -\frac{1}{2}\right)$ iii) directrix, $y = \frac{1}{2}$

9. Find the equation of the parabola with the given properties
a) Focus (-2, 0) and directrix, $x = 2$
b) Focus (0, -6), and directrix, is $y = 6$
c) Symmertry, y, axis, vertex (0, 0), and passes through (5, 2)

Solution: a) since It is x – parabola, and $P = -2$
$\therefore 4P \Rightarrow 4(2) = 8$ $\therefore y^2 = 4Px \Leftrightarrow y^2 = -8x$

b) It is y-parabola and $P = -6$
$\therefore x^2 = 4Py \Leftrightarrow x^2 = 4(-6)y \Rightarrow x^2 = -24y$

c) since the parabola is symmetric about the y-axis
$\therefore x^2 = 4Py$ or $x^2 = -4Py$, since (5, 2) on parabola

$\therefore 25 = 4P(2) \Leftrightarrow 4P = \frac{25}{2}$ $\therefore x^2 = \frac{25}{2}y$

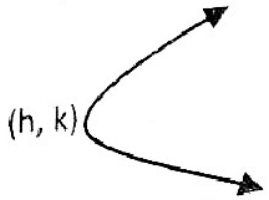
Parabolas with vertex at (h, k)

Equation of parabola that are translated from their standard position can be obtained by replacing for $x \longrightarrow x - h$ and for $y \longrightarrow y - k$ in their standard equations. So that

Vertex $(0, 0) \longrightarrow (h, k)$

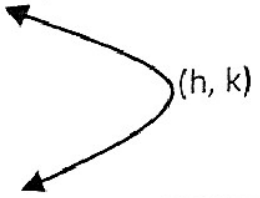
i) Parabola with vertex (h, k) and axis parallel to x-axis

a) $(y-k)^2 = 4P(x-h)$, [opens to right]



- latus rectum 4P
- Focus, $x - h = P \Leftrightarrow x = h + P$ and $y = k$
- Directrix, $x - h = -P \Leftrightarrow x = h - P$
- axis, $y - k = 0 \Leftrightarrow y = k$

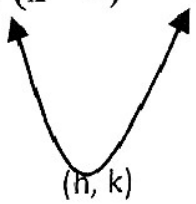
b) $(y-k)^2 = -4P(x-h)$, (opens to left)



- Latus rectum |4P|
- Focus, $x - h = -P$ and $y - k = 0$ i.e. $(h - p, k)$
- directrix, $x - h = P \Leftrightarrow x = h + P$
- axis, $y - k = 0 \Leftrightarrow y = k$

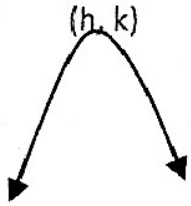
ii) Parabola with vertex (h, k) and axis parallel to y - axis

c) $(x-h)^2 = 4P(y-k)$ (opens up)



- Latus rectum |4P|
- Focus, $y - k = P$ and $x - h = 0$ $(h, p + k)$
- directrix, $y - k = -p \Rightarrow y = k - p$
- axis, $x - h = 0 \Rightarrow x = h$

d) $(x-h)^2 = -4P(y-k)$, opens down ward



- Latus rectum, |4P|
- Focus, $y - k = -P$ and $x - h = 0$ i.e. $(h, k - p)$
- directrix $y - k = P \Rightarrow y = k + P$
- axis $x - h = 0 \Leftrightarrow x = h$

10. Find the latus rectum, focus, directrix, axis and vertex of parabola given by each of the following

a) $y = 4(x-1)^2 + 2$ c) $3x - 2 = (y+3)^2$

b) $3x + 2y^2 + 8y - 4 = 0$ d) $2x^2 - 8x - 4y + 3 = 0$

Solution: $y = 4(x-1)^2 + 2 \Leftrightarrow y - 2 = 4(x-1)^2$

$\Rightarrow (x-1)^2 = \frac{1}{4}(y-2)$

i) latus rectum $4P = \frac{1}{4}$. Hence $P = \frac{1}{16}$

ii) Focus, $y - 2 = \frac{1}{16}$ and $x - 1 = 0$ $\therefore \left(1, \frac{33}{16}\right)$

iii) directrix, $y - 2 = -\frac{1}{16} \Leftrightarrow y = 2 - \frac{1}{16} = \frac{31}{16}$

iv) axis, $x - 1 = 0 \Leftrightarrow x = 1$

v) vertex (1, 2)

b) $2y^2 + 8y + 3x - 4 = 0 \Leftrightarrow 2y^2 + 8y = -3x + 4$

$\Rightarrow 2(y^2 + 4y + 4) = -3x + 4 + 8$

$\Rightarrow 2(y + 2)^2 = -3(x - 4)$

$\therefore (y + 2)^2 = \frac{-3}{2}(x - 4)$

i) latus rectum $4P = \frac{3}{2}$, hence $P = \frac{3}{8}$

ii) focus, $x - 4 = \frac{-3}{8}$ and $y + 2 = 0$

$\therefore$ Focus $\left(\frac{29}{8}, -2\right)$

iii) directrix, $x - 4 = \frac{3}{8} \Rightarrow x = \frac{35}{8}$

iv) axis, $y + 2 = 0 \Rightarrow y = -2$

v) vertex (4, -2)

c) $3x - 2 = (y + 3)^2 \Leftrightarrow (y + 3)^2 = 3(x - \frac{2}{3})$

i) latus rectum $4P = 3$, Hence focal length $P = \frac{3}{4}$

ii) Focus, $x - \frac{2}{3} = \frac{3}{4}$, and $y + 3 = 0 \therefore$ Focus $\left(\frac{17}{12}, -3\right)$

iii) Directrix, $x - \frac{2}{3} = -\frac{3}{4} \Leftrightarrow x = \frac{2}{3} - \frac{3}{4} = \frac{-1}{12}$

iv) axis, $y + 3 = 0 \Leftrightarrow y = -3$

v) vertex, $\left(\frac{2}{3}, -3\right)$

d) *Exercise left for you.*

11. Find an equation of parabola with the given properties

a) Focus (-5, -2); directrix, $x = 3$

b) Focus (3, 3), vertex (3, 2)

c) Vertex (6, 3), directrix, $x = 4$

d) Vertex (3, -2), passes through (1, 6), axis parallel to y axis

Solution: a) the distance between the focus and directrix is 2P

$\therefore 2P = 3 + 5 = 8 \Rightarrow P = 4$

$\therefore$ vertex $= (-5 + 4, -2) = (-1, -2)$

$\therefore$ the equation $(y + 2)^2 = -16(x + 1)$

b) The distance between focus and vertex is P

$\therefore P = 3 - 2 = 1 \quad \Rightarrow 4P = 4$

$\therefore$ the equation of parabola: $(x - 3)^2 = 4(y + 2)$

c) The distance between directrix and vertex is P

$P = 6 - 4 = 2 \quad \therefore 4P = 8$

$\therefore$ the equation of parabola: $(y - 3)^2 = 8(x - 6)$

d) The equation of parabola for the axis parallel to y axis is given by

$(x - h)^2 = 4P(y - k)$ or $(x - h)^2 = -4P(y - k)$

$\therefore (x - 3)^2 = 4P(y + 2)$

$$\Leftrightarrow (1 - 3)^2 = 4P(6 + 2) \Leftrightarrow 4P = \frac{4}{8} = \frac{1}{2}$$

$\therefore (x - 3)^2 = \frac{1}{2}(y + 2)$

12. Let $y = ax^2 + bx + c$

Find a) Latus rectum b) focal length c) vertex d) focus

Solution: $y = ax^2 + bx + c$

$$\Leftrightarrow y = a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right]$$

$$\Leftrightarrow y = a\left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} + \frac{c}{a} - \frac{b^2}{4a^2}\right]$$

$$\Leftrightarrow y = a\left[\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a^2}\right]$$

$$\Leftrightarrow y = a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac - b^2}{4a}$$

$$\Rightarrow \left(x + \frac{b}{2a}\right)^2 = \frac{1}{a}\left(y + \frac{b^2 - 4ac}{4a}\right)$$

a) Latus rectum, $4P = \frac{1}{a}$ c) Vertex $\left(-\frac{b}{2a}, \frac{4ac-b^2}{4a}\right)$

b) Focal length, $P = \frac{1}{4a}$

13) a) A parabolic reflector is to be constructed with diameter 12 meter and its depth 2m

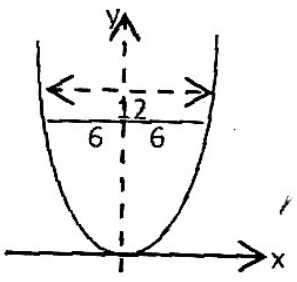
Find a) focus b) the equation

c) how wide is at the focus

d) how wide is at 4m from the vertex.

Solution: Assume the vertex origin (0,0)

$\Rightarrow X^2 = 4Py$



- Diameter 12m $\Rightarrow$ radius 6m

$\Rightarrow$ (6,2) on parabola

$\Rightarrow 36 = 4p(2) \Rightarrow P = \frac{36}{8} = \frac{9}{2} \leftarrow$ focal length

- Focus $\left(0, \frac{9}{2}\right)$

b) $x^2 = 4\left(\frac{9}{2}\right)y \Rightarrow x^2 = 18y$

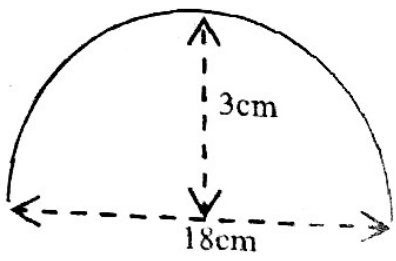
c) $4p = 18m \leftarrow$ The opening at the focus (length of lotus rectum)

d) when $y = 4 \Rightarrow x^2 = (18)(4) = 72 \Rightarrow x = 6\sqrt{2} \Rightarrow 12\sqrt{2} \leftarrow$ is wider.

II) The dimension of a parabolic head light lamp reflector are 18cm wide and 3cm deep

a) Find a) focal length

b) The equation c) How wide is at the focus



Solution: vertex origin (0,0)

- Use $x^2 = 4py$
- Diameter = 18 $\Rightarrow$ radius = 9
- (9, – 3) is point on parabola

$\Rightarrow 9^2 = 4P(-3) \Rightarrow -12P = 81 \Rightarrow P = \frac{-27}{4}$

$\Rightarrow$ Focus $\left(0, \frac{-27}{4}\right)$

b) $x^2 = 4Py \Rightarrow x^2 = 4\left(\frac{-27}{4}\right)y$

$\therefore x^2 = -27y \leftarrow$ the equation.

c) $|4P| = 27 \leftarrow$ the width at the focus (length of latus rectum)

C) Find the equation of parabola:

i) vertex (5,7); focus (2,7) ii) vertex (-4 2); focus (1,2)

iii) vertex (1,3); focus (1,5) iv) vertex (– 2, 9); focus (– 2, 7)

Solution: If y component value is the same then it is horizontal parabola, use $(y-k)^2 = 4P(x-h)$, and P = F – V of x components.

- If X component value is the same then it is vertical parabola, use $(x-h)^2 = 4P(y-k)$ and P = F – V of y, components.

i) h = 5, k = 7 and P = 2 – 5 = – 3,

$\therefore (y-7)^2 = -12(x-5) \leftarrow$ the same y component; horizontal parabola

ii) h = – 4, k = 2, p = 1 – (– 4) = 5, and the same y components $\therefore (y-2)^2 = 20(x+4)$

iii) h = 1, k = 3, P = F – v = 5 – 3 = 2, the same x components

$\therefore (x-1)^2 = 8(y-3)$

iv) h = – 2, k = 9, P = 7 – 9 = – 2

$\therefore (x+2)^2 = -8(y-9)$

d) Find the equation of parabola

i) vertex is at (– 3,5), axis parallel to the x – axis and passing through the point (5,9).

ii) vertex is at (2,3); passes through (4,6), vertical axis.

iii) vertex (– 5, –3); axis, y = – 3, P = 2

iv) focus (3,1); P = 5, vertical axis

Solution: Here

i) $(y-5)^2 = 4P(x+3)$, since (5,9) on the parabola

thus $(9-5)^2 = 4P(5+3) \Rightarrow 16 = 32P \Rightarrow P = \frac{1}{2}$

$\therefore (y-5)^2 = 2 \quad (x+3)$ ← equation of parabola

ii) $(x-2)^2 = 4P(y-3)$, since (4,6) on parabola

thus, $(4-2)^2 = 4P(6-3) \Rightarrow 4 = 12P = \Rightarrow P = \frac{1}{3}$

$\therefore (x-2)^2 = \frac{4}{3}(y-3)$ ← Equation of vertical of vertical parabola.

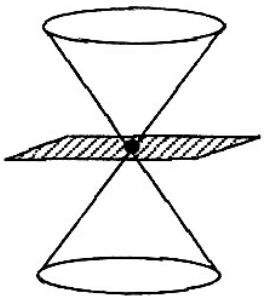
iii) $(y+3)^2 = 4P(x+5)$, since $P = 2$

$\therefore (y+3)^2 = 8(x+5)$

iv) Since it is vertical parabola, thus $V(3,1-5) = (3,-4)$

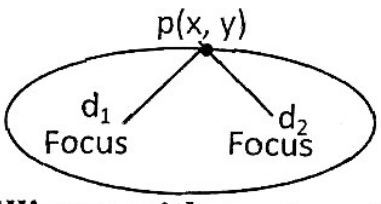
$\therefore (x-3)^2 = 20\,(y+4)$

The Ellipse



An ellipse is another of conic sections formed when plane intersects aright circular cone.

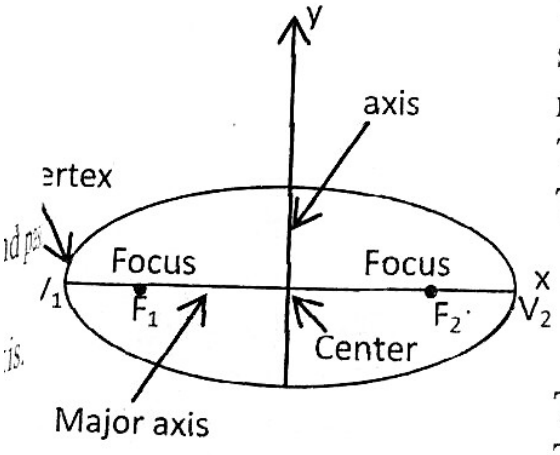
If the plane intersects the cone at the vertex of the cone so that the resulting figure is a point, the point is degenerate ellipse.



Definition

An ellipse is the set of all points p(x, y) in the plane the sum of whose distances from two fixed point (foci) is a positive constant.

Ellipses with center at (0, 0)



The graph of an ellipse has two axes of symmetry. The longer axis is called the major axis.

The foci of the ellipse are on major axis

The shorter axis is called the minor axis.

The length of major axis = 2a

The length of minor axis = 2b

The distance between foci = 2c

The mid point of major axis is the centre

The end point of major axis is the vertices

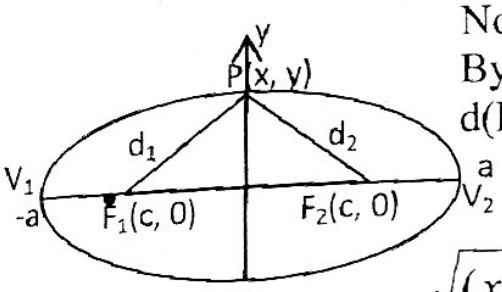
We not that the sum of distance from a point on the ellipse to each focus s constant

Now by hypothesis

$d_1 + d_2 = v_2F_1 + v_2F_2$

$$\sqrt{c^2+b^2}+\sqrt{c^2+b^2}=(a+c)+(a-c)$$

$2\sqrt{c^2+b^2}=2a \Rightarrow a^2=b^2+c^2 \Rightarrow c^2=a^2-b^2$



Now let P(x, y) be any point on the ellipse. By using definition of ellipse we have $d(PF_1) + d(PF_2) = 2a$

$$\sqrt{(x+c)^2+y^2}+\sqrt{(x-c)^2+y^2}=2a$$

$\Rightarrow \left[\sqrt{(x+c)^2+y^2}\right]=\left[2a-\sqrt{(x-c)^2+y^2}\right]^2$ ..... subtracting and squaring both side.

$\Rightarrow (x+c)^2+y^2=4a^2=4a\sqrt{(x-c)^2+y^2}+(x-c)^2+y^2$

$\Rightarrow 4cx-4a^2=-4a\sqrt{(x-c)^2+y^2}$

$\Rightarrow [-cx+a^2]^2=\left[a\sqrt{(x-c)^2+y^2}\right]^2$ ..... Divide by (-4) and then square each side.

$\Rightarrow c^2x^2-2cxa^2+a^4=a^2x^2-2cxa^2+a^2c^2+a^2y^2$

$\Rightarrow -a^2x^2+c^2x^2-a^2y^2=-a^4+a^2c^2$ ..... Rewrite with x and y terms on the left side. $\Rightarrow -(a^2-c^2)x^2-a^2y^2=-a^2(a^2-c^2)$

From hypothesis, $b^2=a^2-c^2$

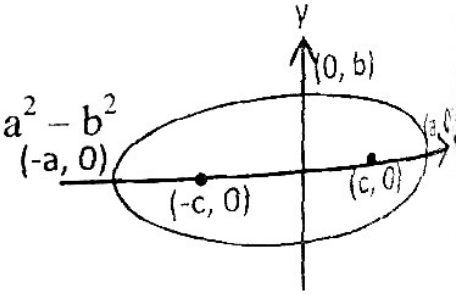
$\Rightarrow -b^2x^2-a^2y^2=-a^2b^2$ ..... Divide each side by $-a^2b^2$

$\Rightarrow \frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ ..... An equation of an ellipse with center at (0, 0)

Standard equation of an ellipse with unter (0, 0)

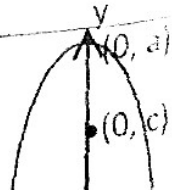
i) Major axis on the x-axis

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, for $a \geq b > 0$, where $c^2=a^2-b^2$



- Focas (c, 0) and (-c, 0)
- vertices are (-a, 0) and (a, 0)
- The length of latus rectum $\frac{2b^2}{a}$

ii) Major Axis on the y-axis (vertical ellipse)



$\frac{x^2}{b^2}+\frac{y^2}{a^2}=1$ for $a \geq b > 0$

- Foci are (0, c) and (0, -c)
- vertices are (0, a) and (0, -a)
- length of latus rectum $\frac{2b^2}{a}$

Illustrative example

13. Find the vertices and foci of the ellipse given by

a) $\frac{x^2}{25}+\frac{y^2}{49}=1$ b) $x^2 + 4y^2 = 8$ c) $3x^2 + y^2 = 3$

Solution: a) Because the y^2 term has the larger denominator, the major axis is on the y-axis

$a^2 = 49, b^2 = 25,$ $c^2 = a^2 - b^2 = 49 - 25 = 24$

$a = 7$ $b = 5$ $c = \sqrt{24} = 2\sqrt{6}$

- The vertices are (0, 7) and (0, -7)
- the foci are $(0, 2\sqrt{6})$ and $(0, -2\sqrt{6})$

b) Divide both sides of the equation by 8, we obtain

$$x^2 + 4y^2 = 8 \Leftrightarrow \frac{x^2}{8}+\frac{y^2}{2}=1$$

Because the x^2 term has the larger denominator the major axis is on the x-axis.

Thus $a^2 = 8 \Rightarrow a = \sqrt{8} = 2\sqrt{2}$

$b^2 = 2 \Rightarrow b = \sqrt{2}$

Therefore the foci lie on the x-axis and

$c = \sqrt{a^2 - b^2} = \sqrt{8-2} = \sqrt{6}$

- Consequently the foci are the points $(-\sqrt{6}, 0)$ and $(\sqrt{6}, 0)$
- The vertices are $(-2\sqrt{2}, 0)$ and $(2\sqrt{2}, 0)$

c) $3x^2 + y^2 = 3$ write as $\frac{3x^2}{3}+\frac{y^2}{3}=\frac{3}{3} \Rightarrow x^2 + \frac{y^2}{3} = 1$

Since y^2 term has larger denominator the major axis is y-axis.

Therefore, $a^2 = 3 \Rightarrow a = \sqrt{3}$

$b^2 = 1 \Rightarrow b = 1$

Also, $c = \sqrt{a^2 - b^2} = \sqrt{3-1} = \sqrt{2}$

Thus foci $(0, \sqrt{2})$ and $(0, -\sqrt{2})$

Vertices $(0, \sqrt{3})$ and $(0, -\sqrt{3})$

14. Find the equation of the ellipse having center at the origin

a) a focus at (2, 0) and the vertex (-3, 0)

b) a focus at (4, 0) and the length of major axis is 10

c) a focus at (0, 4) and the length of minor axis is 6

Solution

a) $c = 2$, and $a = 3$ and $b^2 = a^2 - c^2 = b^2 = 9 - 4 = 5$

Thus $\frac{x^2}{9} + \frac{y^2}{5} = 1$

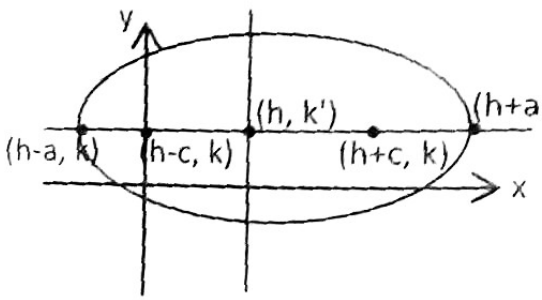
b) $c = 4$, $2a = 10 \Rightarrow a = 5$ and $b^2 = a^2 - c^2 = 25 - 16 = 9$

Thus $\frac{x^2}{25} + \frac{y^2}{9} = 1$

c) $c = 4$, $2b = 6 \Rightarrow b = 3$ and $b^2 = a^2 - c^2 = 9 = a^2 - 16 = a$

Thus $\frac{x^2}{9} + \frac{y^2}{25} = 1$

Standard equation of the ellipse with center at (h, k)



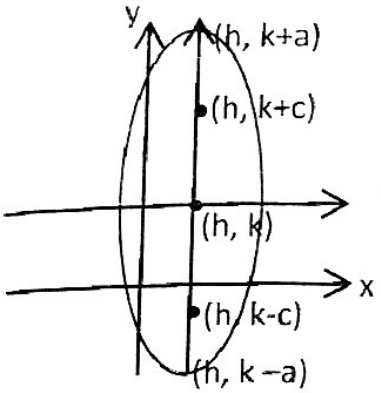
i) Major axis is parallel to the x-axis and center (h, k)

$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$

$c^2 = a^2 - b^2$ and $a > b$

* vertices are (h – a, k) and (h+a, k)
* Foci are (h – c, k) and (h +c, k)
* latus rectum $\frac{2b^2}{a}$

ii) Major axis parallel to the y-axis and center (h, k)



$\frac{(x-h)^2}{b^2}+\frac{(y-k)^2}{a^2}=1$, a > b, $c^2 = a^2 - b^2$

* vertices are (h, k+a) and (h, k-a)
* foci are (h, k+c) and (h, k-c)
* length of latus rectom $\frac{2b^2}{a}$
* eccentercity, $e = \frac{c}{a} = \frac{\sqrt{a^2-b^2}}{a}$

15. Find the equation of the ellipse having the given properties
   a) The foci are (2, 3) and (2, -3) and the length of the minor axis is 10
   b) The foci are (2, 1) and (2, -1) and the length of the major axis is 4.
   c) Center at (1, 2), one focus at (-4, 2) and passing through the point (4, 6)

Solution: a) center $= \left(\frac{2+2}{2}, \frac{3+(-3)}{2}\right) = (2,0)$

$b = 5, 2c = 6 \Rightarrow c = 3$

Thus $b^2 = a^2 - c^2 = 25 = a^2 - 9 \Rightarrow a^2 = 34$

Therefore, $\frac{(x-2)^2}{25}+\frac{(y-0)^2}{34}=1$

b) $2c = 2 \Rightarrow c = 1$ and $2a = 4 \Rightarrow a = 2$

Therefore, $b^2 = a^2 - c^2 \Rightarrow b^2 = 4 - 1 = 3$

Center (h, k) = (2, 0)

Thus, $\frac{(x-2)^2}{3}+\frac{y^2}{4}=1$

c) Let (m, n) be other side of the focus. Consequently follows;

$\frac{-4+m}{2}=1$, and $\frac{2+n}{2}=2$

m = 6 and n = 2

Thus $2c = 6 - (-4) = 10 \Rightarrow c = 5$

$$\Rightarrow \frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$$

$$\frac{(4-1)^2}{a^2}+\frac{(6-2)^2}{b^2}=1 \Rightarrow \frac{9}{a^2}+\frac{16}{b^2}=1 \text{ and } b^2=a^2-c^2$$

$$\Rightarrow b^2=a^2-25$$

$$\Rightarrow \frac{9}{a^2}+\frac{16}{a^2-25}=1 \Rightarrow 9a^2-225+16a^2=a^2(a^2-25)$$

$\Rightarrow 25a^2-225=a^4-25a^2 \Rightarrow 50a^2-225=a^4 \Rightarrow a^4-50a^2+$
$=0$

$\Rightarrow (a^2-45)(a^2+5)=0 \Rightarrow a^2=45$ and $a^2=-5$

We take $a^2=45$

Thus $\dfrac{(x-1)^2}{45}+\dfrac{(y-2)^2}{20}=1$

16. Find the vertices, foci and latus rectum of the ellipse

a) $3x^2+y^2-6x+6y+13=0$

b) $4x^2+9y^2-8x+36y+4=0$

Solution: $3x^2+y^2-6x+6y+13=0$

$3x^2-6x+y^2+6y+13=0$

$3(x^2-2x+1)+y^2+6y+9+13-9-3=0$

$3(x-1)^2+(y+3)^2=-1$

Therefore, the equation does not represent an ellipse.

b) $4x^2+9y^2-8x+36y+4=0$

$4x^2-8x+9y^2+36y+4=0$

$4(x^2-2x+1)+9(y^2+4y+4)+4-4-36=0$

$$4(x-1)^2+9(y+2)^2=36 \Rightarrow \frac{(x-1)^2}{9}+\frac{(y+2)^2}{4}=1$$

- center (1, -2), $a^2=9$. Thus $a=3$, thus the vertices are
  3, -2) and (1-3, -2) = (4, -2) and (-2, -2)
- foci are $(1+\sqrt{5}, -2)$ and $(1-\sqrt{5}, -3)$
- latus rectum $\dfrac{2b^2}{a}=\dfrac{2(4)}{3}=\dfrac{8}{3}$

17. Find the coordinates for the center, vertices, end point of
minor axis, length of latus rectum and foci of the ellipse given
$2x^2+3y^2+8x-18y-1=0$

$$\Rightarrow \frac{(x-1)^2}{a^2}+\frac{(y-2)^2}{b^2}=1$$

$$\frac{(4-1)^2}{a^2}+\frac{(6-2)^2}{b^2}=1 \Rightarrow \frac{9}{a^2}+\frac{16}{b^2}=1 \text{ and } b^2=a^2-c^2$$

$$\Rightarrow b^2=a^2-25$$

$$\Rightarrow \frac{9}{a^2}+\frac{16}{a^2-25}=1 \Rightarrow 9a^2-225+16a^2=a^2(a^2-25)$$

$\Rightarrow 25a^2-225=a^4-25a^2 \Rightarrow 50a^2-225=a^4 \Rightarrow a^4-50a^2+225=0$

$\Rightarrow (a^2-45)(a^2+5)=0 \Rightarrow a^2=45$ and $a^2=-5$

We take $a^2=45$

Thus $\dfrac{(x-1)^2}{45}+\dfrac{(y-2)^2}{20}=1$

16. Find the vertices, foci and latus rectum of the ellipse

a) $3x^2+y^2-6x+6y+13=0$

b) $4x^2+9y^2-8x+36y+4=0$

Solution: $3x^2+y^2-6x+6y+13=0$

$3x^2-6x+y^2+6y+13=0$

$3(x^2-2x+1)+y^2+6y+9+13-9-3=0$

$3(x-1)^2+(y+3)^2=-1$

Therefore, the equation does not represent an ellipse.

b) $4x^2+9y^2-8x+36y+4=0$

$4x^2-8x+9y^2+36y+4=0$

$4(x^2-2x+1)+9(y^2+4y+4)+4-4-36=0$

$$4(x-1)^2+9(y+2)^2=36 \Rightarrow \frac{(x-1)^2}{9}+\frac{(y+2)^2}{4}=1$$

- center (1, -2), $a^2=9$. Thus $a=3$, thus the vertices are (1+3, -2) and (1-3, -2) = (4, -2) and (-2, -2)
- foci are $(1+\sqrt{5}, -2)$ and $(1-\sqrt{5}, -3)$
- latus rectum $\dfrac{2b^2}{a}=\dfrac{2(4)}{3}=\dfrac{8}{3}$

17. Find the coordinates for the center, vertices, end point of the minor axis, length of latus rectum and foci of the ellipse given by $2x^2+3y^2+8x-18y-1=0$

Solution: First, we begin by arranging the equation in standard form and completing the square.

$(2x^2 + 8x) + (3y^2 - 18y) = 1$

$2(x^2 + 4x + 4) + 3(y^2 - 6y + 9) = 1 + 8 + 27$

$2(x + 2)^2 + 3(y - 3)^2 = 36$

$$\frac{(x+2)^2}{18} + \frac{(y-3)^2}{12} = 1$$

Now, $a^2 = 18$ and $b^2 = 12$

$a = \sqrt{18}$ and $b = \sqrt{12}$

$a = 3\sqrt{2}$ and $b = 2\sqrt{3}$

also, $c = \sqrt{18-12} = \sqrt{6}$

Thus center (-2, 3)

- vertices $(-2 - 3\sqrt{2}, -3 + 3\sqrt{2})$
- endpoint of minor axis $(-2, 3 + 2\sqrt{3})$ and $(-2, 3-2\sqrt{3})$
- foci: $(-2 - \sqrt{6}, 3)$ and $(-2 + \sqrt{6}, 3)$
- length of latus rectum $\frac{2b^2}{a} = \frac{2(12)}{3\sqrt{2}} = \frac{8}{\sqrt{2}}$
- eccentercity, $e = \frac{c}{a} = \frac{\sqrt{a^2 - b^2}}{a} = \frac{\sqrt{18-12}}{\sqrt{18}} = \sqrt{\frac{1}{3}}$

Eccentricity of an Ellipse

The graph of an ellipse can be very long and thin or it can be much like a circle.

The eccentricity of an ellipse is a measure of its "roundness."

Definition: The eccentricity of an ellipse is denoted bye and is defined as

$$e = \frac{distance\ \ between\ the foci}{length\ of\ \ the\ major\ axis}$$

That is $e = \frac{2c}{2a} = \frac{c}{a}$

Note, $0 < e < 1$

When $e \approx 0$, the graph is circle

When e ≈ 1, the graph is long and thin.

Example

18. Find the eccentricity of the ellipse given by $8x^2 + 9y^2 = 18$

Solution: First, write the equation of the ellipse in standard form

Divide each side of the equation by 18

$$\frac{8x^2}{18}+\frac{9y^2}{18}=1 \Leftrightarrow \frac{4x^2}{9}+\frac{y^2}{2}=1 \Rightarrow \frac{\frac{x^2}{9}}{\frac{4}{}}+\frac{y^2}{2}=1$$

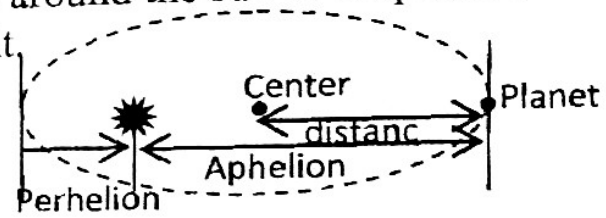
$$\Rightarrow a^2=\frac{9}{4} \text{ and } a=\frac{3}{2}$$

Use the definition $c^2 = a^2 - b^2$ to find c

$$c^2=\frac{9}{4}-2=¼ \text{ and } c=\sqrt{\frac{1}{4}}=\frac{1}{2} \Rightarrow e=\frac{c}{a}=\frac{\frac{1}{2}}{\frac{3}{2}}=\frac{1}{3}$$

Application

The planets travel around the sun in elliptical orbit. The sun is located a focus of the orbit.



The terms per in e lion and a phelion are used denote the position of planet in ts orbit around the sun.

Note, the length of semi major axis of a planet's elliptical orbit is called the mean distance of the planet from the sum

Illustrative example

19. Earth has mean distance of 93 million miles and perihelion distance of 91.5 million miles. Find an equation of Earth's orbit

Solution: Semi major axis = a = 93

Aphelion distance = 2(93) – 91.5 = 94.5

c = aphelion distance – a

c = aphelion distance -93 = 94.5 – 93 = 1.5 million mil

The length of b of the semi minor axis of the orbit is

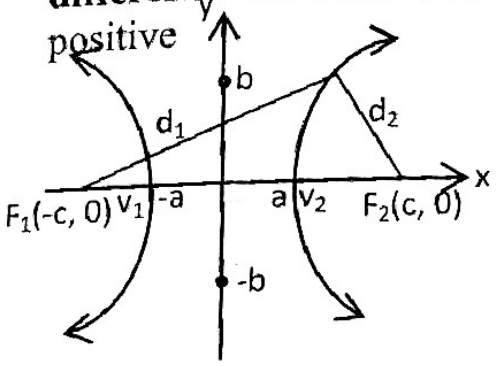
$$b = \sqrt{a^2 - c^2} = \sqrt{93^2 - 1.5^2} = \sqrt{8646.75}$$

Therefore an equation of Earth's orbit is $\frac{x^2}{93^2} + \frac{y^2}{8646.75} = 1$

3. Hyperbolas

Hyperbola is formed when plane intersects right circular cone parallel to the axis of the cone.

Definition: A hyperbola is the set of all points P(x, y) in the plane, the **difference between** whose distance from two fixed points (foci) is positive



- $|d_1 - d_2| = 2a$
- $\overline{v_1 v_2}$. *The transverse axis* $= 2a$
- *length of conjugate axis* $= 2b$
- $\overline{F_1 F_2}$, *distance between foci* $= 2c$

Find $|v_2F_1 - v_2F_2|$

Solution: $\overline{v_2 F_1} = a + c$ and $v_2F_2 = c - a$

Thus, $|v_2F_1 - v_2F_2| = |a + c - (c - a)| = 2a$

- The line through the two foci is an axis of symmetry

Note: i) If the axis of the hyperbola is parallel to the x-axis, then it is called an x-hyperbola or horizontal hyperbola.

ii) If the axis of the hyperbola is parallel to the y-axis, then it is called y-hyperbola or vertical hypherbola.

To find the standard equation using distance formula

For any point P(x, y) on the hyperbola, let

$d_1 = pF_1$, and $PF_2 = d_2$

$d_1 = \sqrt{(x+c)^2 + y^2}$ and $d_2 = \sqrt{(x-c)^2 + y^2}$

since $|PF_1 - PF_2| = |d_1 - d_2| = 2a$, we deduce that

$$\left|\sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2}\right| = 2a$$

$$\Leftrightarrow \sqrt{(x+c)^2 + y^2} - \sqrt{(x-c)^2 + y^2} = \pm 2a$$

$$\Leftrightarrow \sqrt{(x+c)^2+y^2} \pm 2a + \sqrt{(x-c)^2+y^2}$$

Squaring both side, simplifying, we get

$$cx^2 - a^2 = \pm a\sqrt{(x-c)^2+y^2} \Leftrightarrow (cx-a^2)^2 = a^2((x-c)^2+y^2)$$

Expanding and collecting like terms, we have

$(c^2-a^2)x^2 - a^2y^2 = a^2)c^2 - a^2)$

Since $c > a$ and letting $c^2 - a^2 = b^2$, $b > 0 \Rightarrow b^2x^2 - a^2y^2 = a^2b^2$

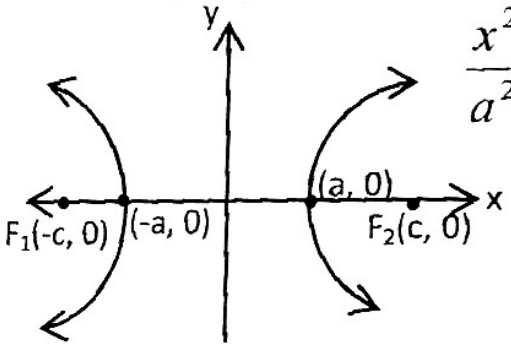
Dividing both side by a^2b^2, we get the coordinate of every point on the hyperbola satisfies the equation

$$\boxed{\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1}$$

Which is called standard equation of x-hyperbola with center origin.

Standard forms of the equation of hyperbola with center at the origin

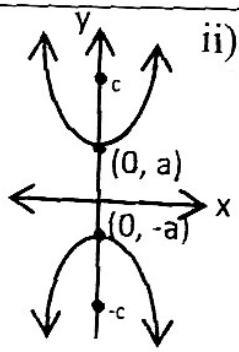
i) Transverse axis on the x-axis



$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ - vertice are (a, 0) and (-a, 0)

- Foci are (c, 0) and (-c, 0)
- $c^2 = a^2 + b^2$

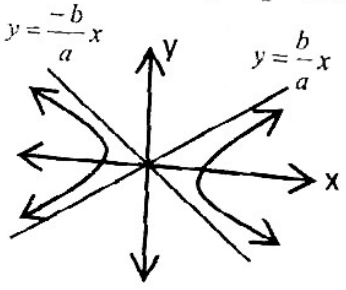
ii) Transverse axis on the y-axis



$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

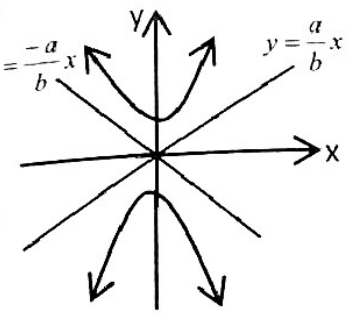
- vertices are (0, a) and (0, -a)
- foci are (0, c) and (0, -c)
- $c^2 = a^2 + b^2$

Asymptote of a hyperbola with center at origin



- Each hyperbola has two asymptote that pass through the center of hyperbola
- *The asymptotes of the hyperbola defined by* $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are given by the equation

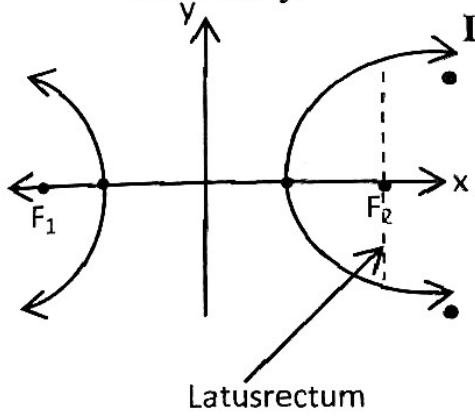
$$y = \frac{b}{a}x \quad and \quad y = -\frac{b}{a}x$$



- *The asymptotes of the hyperbola defined by* $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$ are given by the equation

$$y = \frac{a}{b}x \quad and \quad y = -\frac{a}{b}x$$

Note:- To find asymptote equation let 0 by 1 or replace 1 by 0 and then solve for y.



Latus rectum

- The line segment that pass through a focus of a hyperbola, is perpendicular to the principal axis, and has its end points on the hyperbola is called a latus rectum.
- Length of latus rectum for hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $\frac{2b^2}{a}$

Illustrative Example

20. Find the vertices, foci, and asymptote, and latus rectum of each equation

a) $\frac{x^2}{1} - \frac{y^2}{4} = 1$ b) $\frac{y^2}{9} - \frac{x^2}{4} = 1$

Solution: a) Because the x^2 term is positive, the trans verse axis on x –axis, we know

$a^2 = 1$; thus $a = 1$

$b^2 = 4$, thus, $b = 2$, and $c^2 = a^2 + b^2 = 1 + 4 = 5$

vertices are (1, 0) and (-1, 0)

$\Rightarrow$ foci: $(\sqrt{5}, 0)$ and $(-\sqrt{5}, 0)$

More over, the asymptote are the line $y = 2x$ and $y = -2x$

- length of latus rectum = $\frac{2(4)}{1} = 8$

b) Because the y^2 terms is positive, the transverse axis on y-axis, and $a^2=9$, thus $a = 3$, and $b^2 = 4$ thus $b=2$

* $c^2 = a^2 + b^2 = 9 + 4 = 13 \Rightarrow c = \sqrt{13}$

Vertice (0, 3) and (0, -3)

Foci are $(0, \sqrt{13})$ and $(0, -\sqrt{13})$,

Asymptote $y = \frac{a}{b}x$ *and* $y = \frac{-a}{b}x$

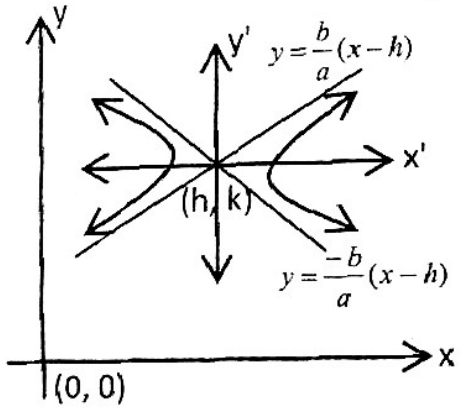
$\Rightarrow y = \frac{3}{2}x$ *and* $y = \frac{-3}{2}x$

Length of latus rectum is $\frac{2b^2}{a} = \frac{2(4)}{3} = \frac{8}{3}$

Note: Ayperbola for which a = b is called equilateral
Ayperbola is equilateral if and only if its asymptote are perpendicular.

Hyperbola with center at (h, k)

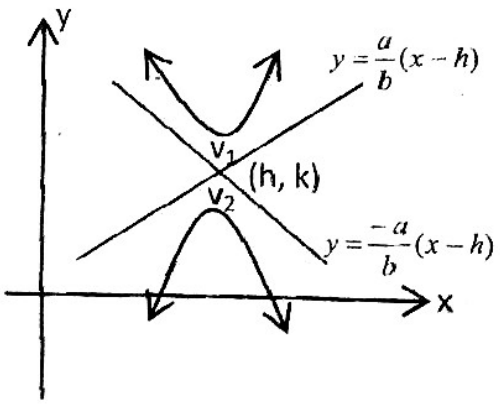
- Standard forms of the equation of hyperbola with center at (h, k)



i) Transverse axis parallel to the x-axis

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

- vertices are $v_1(h + a, k)$ and $v_2(h-a, k)$
- foci are $F_1(h - c, k)$ and $F_2(h - c, k)$
- $c^2 = a^2 + b^2$
- asymptote are $y - k = \pm\frac{b}{a}(x-h)$
- length of latus rectum $\frac{2b^2}{a}$
- eccentricity $e = \frac{c}{a} = \frac{\sqrt{a^2+b^2}}{a}$



ii) transverse axis parallel to y-axis

$$\frac{(y-k)^2}{a^2} - \frac{(x-h)^2}{b^2} = 1$$

Solution: $m_1 = \frac{4}{3}$, $m_2 = \frac{3}{2}$, angle between L_1 and L_2 from l_1 to l_2

$$\tan\alpha = \frac{m_2 - m_1}{1 + m_2 m_1} = \frac{\frac{3}{2} - \frac{4}{3}}{1 + \frac{3}{2}\cdot\frac{4}{3}} = \frac{1}{18}$$

65) The floor of conference hall is made in a shape of ellipse. If the largest chord has length 20 meters and the smallest is 16m what is the length between the foci?

A. 24m B. 12m C. $9\sqrt{2}$ m D. 6m

Solution: $2a = 20 \Rightarrow a = 10 \Rightarrow a^2 = 100$

$2b = 16 \Rightarrow b = 8 \Rightarrow b^2 = 64$

- $c^2 = a^2 - b^2 = 100 - 64 = 36 \Rightarrow c = 6 \Rightarrow 2c = 12$ **Answer: B**

66) What is an asymptote of the hyperbola $4x^2 - y^2 + 2y = 5$ ..... EHEECE.

A. $y = -2x + 1$ C. $y = \frac{-1}{2}x + 1$

B. $y = 2x - 1$ D. $y = \frac{1}{2}x - 1$

Solution: $4x^2 - y^2 + 2y = 5 \Rightarrow 4x^2 - (y^2 - 2y + 1)$

$$\Rightarrow 4x^2 - (y-1)^2 = 4 \Rightarrow x^2 - \frac{(y-1)^2}{4} = 1$$

$\Rightarrow$ Asymptote: $y - 1 = \pm 2x \Rightarrow y = 2x+1$ and $y = -2x+1$

Answer:

67) For what value of K is $x^2 + y^2 + 6x - 4y + k = 0$ represent a circle ...EHEECE.

A. For all $k \geq 0$ C. For all $K < 13$

B. For all $K \geq 10$ D. for $K = 16$

Solution: $x^2 + 6x + y^2 - 4y = -K$

$\Rightarrow (x+3)^2 + (y-2)^2 = -k + 9 + 4 \Rightarrow$ circle if $13 - k > 0 \Rightarrow$ for $k < 13$,

Answer:

68) What is the distance between the two foci of the hyperbola $4(y-1)^2 - 16(x+3)^2 = 1$....(EHEEC

A. $3\sqrt{2}$ B. $\frac{\sqrt{5}}{2}$ C. $\frac{1}{4}$ D. $\frac{1}{\sqrt{10}}$

Solution: $\Leftrightarrow \frac{(y-1)^2}{\frac{1}{4}} - \frac{(x+3)^2}{\frac{1}{16}} = 1 \Rightarrow a^2 = \frac{1}{4}, b^2 = \frac{1}{16}$

$\Rightarrow c^2 = a^2 + b^2 = \frac{1}{4} + \frac{1}{16} = \frac{5}{16} \Rightarrow c = \frac{\sqrt{5}}{4}$

$\Rightarrow 2c = (2)\left(\frac{\sqrt{5}}{4}\right) = \frac{\sqrt{5}}{2}$ .... **Answer: B**

69) Let l be line containing (1,5) and its angle of inclination be $\theta = 135^0$. What is the distance from the line ℓ to (– 1,1) … EHEECE

A. $2\sqrt{2}$ B. $3\sqrt{2}$ C. $4\sqrt{2}$ D. $6\sqrt{2}$

Solution: Slope of the line, $\tan\theta = \tan 135^0 = -1$

- equation of the line ℓ : $\frac{y-5}{x-1} = -1 \Rightarrow y - 5 = -x + 1$

  $\Leftrightarrow y - 5 = -x + 1 \Leftrightarrow x + y = 0$
- Distance, from (– 1,1) + 0 x + y – 6 = 0 is

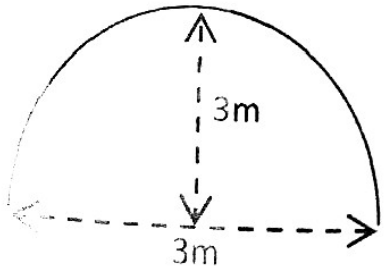
$$\frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|(-1)(1) + (1)(1) - 6|}{\sqrt{1+1}} = \frac{6}{\sqrt{2}} = 3\sqrt{2}$$

70) Consider a gate whose shape is a parabolic arch of height 3m and base width 3m If the origin of the coordinate system is at the vertex, the equation of the edge of the gate is ….EHEECE 2002/2010

A. $3x^2 + 3y = 0$ C. $y = -3x^2 + 1$

B. $4x^2 + 3y = 0$ D. $3y^2 - 4x = 0$

Solution: Vertex (0,0), the point $\left(\frac{3}{2}, -3\right)$ on parabola.



$\Rightarrow x^2 = 4py \Rightarrow \frac{9}{4} = 4P(-3)$

$\Rightarrow P = \frac{-3}{16}$

$$\Rightarrow x^2 - 4py = 0 \Rightarrow x^2 - 4\left(\frac{-3}{16}\right)y = 0$$

$$\Rightarrow x^2 + \frac{3}{4}y \Leftrightarrow 4x^2 + 3y = 0$$

Answer:

71) The equation $x^2 - 4y^2 = -1$ represents... EHEECE 2002/10

Solution: $x^2 - 4y^2 = -1 \Rightarrow -x^2 + 4y^2 = 1$

$\Rightarrow 4y^2 - x^2 = 1 \leftarrow$ hyperbola. $\Rightarrow a^2 \frac{1}{4} \Rightarrow a = \frac{1}{2}, b^2 = 1 \Rightarrow b =$

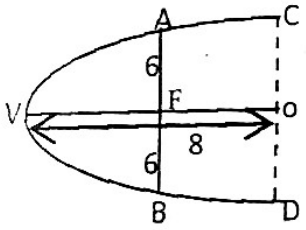
$$\Rightarrow c^2 = a^2 + b^2 = \frac{1}{4} + 1 = \frac{5}{4} \Rightarrow c = \frac{\sqrt{5}}{2}$$

$\therefore$ Vertice $\left(0, \pm\frac{1}{2}\right)$ *and foci* $\left(0, \pm\frac{\sqrt{5}}{2}\right)$

72) A cross – section of parabolic reflector is shown below. The opening at the focus $\overline{AB}$ is 12cm. What is the diameter of the opening $\overline{CD}$, 8cm from the vertex. . .UEE 2003/11.

A. $8\sqrt{6}$ cm B. 12cm C. 9cm D. $6\sqrt{8}$ cm

Solution:



- $4P = 12 \Rightarrow P = 3$, $\overline{VF} = P = 3$
- Assume the vertex origin.

$\Rightarrow y^2 = 4Px \Rightarrow y^2 = 12x$

When $x = 8 \Rightarrow y^2 = (12)(8) = 96$

$\Rightarrow y = \overline{OC} = 8\sqrt{6}$

Answer:

Supplementary exercise

1. Find the distance between the points whose coordinates are given.
   a) $(\sqrt{3}, \sqrt{8})$, $(\sqrt{12}, \sqrt{27})$ b) (a, b), (-a, -b) c) (x, 4x), (-2x,
2. Find all points on the x-axis that are 10 units from (4, 6)
3. Find all points on the y-axis that are 12 units from (5, -3)
4. Find the equation of a line which passes through the point (and makes the intercepts on the x-axis equal in magnitude opposite in sign.

5. The vertices of a triangle are (2, 0), (0, 2) and (4, 6). Find the median through the first vertex.
6. Find the ratio in which the join of (-1, 0) and (-2, 3) is divided by the line $x + 2y = 3$
7. Find the acute angle between the lines:
   a) $3x + 2y = 11$ and $2x + y = -12$
   b) $\frac{x}{a} + \frac{y}{b} = 1$ *and* $\frac{x}{a'} + \frac{y}{b'} = 1$
   c) $2x + y = 3$ and $3x - 2y = 5$
8. A circle passes through the points (-3, 1) and (-1, 5). Its centre lies on the x-axis

Find a) the coordinate of the center b) the equation of the circle

9. Find the coordinates of the circum center of a triangle whose coordinates are (7, -1),(5, 1) and (-3, -7). Find the circum center
10. If (-3, 2); (1, -2) and (5, 6) are the mid-points of the sides of a triangle, find the coordinates of the vertices of the triangle.
11. The points $\left(2, \frac{3}{2}\right), \left(-3, \frac{-7}{2}\right), \left(k, \frac{9}{2}\right)$ are collinear, find k
12. Find the equation of the two lines through the point (3, -2) and inclined at 60^0 to the straight line $\sqrt{3}x + y = 1$
13. If $x + y^2 + kxy + 8x - 6y + 9 = 0$ represents a circle state the value of k. substituting this value of k in the equation, find the center and radius of the circle.
14. Find the equation of the circle
    a) whose center lies on the x-axis and which passes through the points (-1, 0) and (5, 0)
    b) whose center lies on the y-axis and which passes through the points (0, 3) and (0, -7)
15. Find the equations of the circles passing though the following sets of points A(-2, 2), B(2, 2) and C(-2, -4)
16. Which of the points A(2, 5), B(1, -3) and C(1, 0) lies inside or outside the circle $x^2 + y^2 - 3x + 2y - 12 = 0$
17. Find the length of a tangent from (-3, 2) to the circle $x^2+y^2 = 3+2x -2y$
18. Find how the circle $x^2 + y^2 = 1$ and $x^2 + y^2 - 6x + 5 = 0$ touch each other, and find the coordinates of the point of contact.
19. Find the equation of the tangents to the circles

a) $x^2 + y^2 + 4x - 10y - 12 = 0$ at (3, 1)
b) $x^2 + y^2 - 6x - 3y - 16 = 0$ at (-2, 0)

20. Find the equation of the circle whose center lies on the line $x = 2y$, passes through the point (2, 5), and for which the length of the tangent from the origin is $\sqrt{7}$

21. Find the equation of the parabola with vertex at the origin having its axis along the x-axis and passing through the point (2, 3)

22. Find latus rectum, focus, vertex, and directrix of each of the following

a) $y^2 = 1x$
b) $x^2 = -9y$
c) $(y-3)^2 = 6x - 12$
d) $(x+2)^2 = -(y+2)$
e) $x^2 = 4x + 2y - 1 = 0$
f) $x = y^2 - 4y + 2$

23. Find an equation for the parabola that satisfies the given condition

A. vertex (0, 0); focus (3, 0)
B. vertex (0, 0), directrix $x = 7$
C. focus (0, -3), directerix, $y = 3$
D. vertex (1, 1), directerix, $y = -2$
E. vertex (-1, 2), focus, (-1, 3)
F. vertex (-4, 1) , has its axis of symmetry parallel to the y-axis, and passes through the point (-2, 2)
G. focus (3, -3) and directrix $y = -5$

24. Find the vertices and foci of the ellipse given

a) $\frac{x^2}{16} + \frac{y^2}{25} = 1$
b) $4x^2 + 9y^2 = 36$
c) $9x^2 + y^2 = 9$
d) $\frac{(x-3)^2}{9} + \frac{(y+2)^2}{16} = 1$
e) $\frac{(x+2)^2}{9} + \frac{y^2}{25} = 1$
f) $\frac{(x-1)^2}{21} + \frac{(y-3)^2}{4} = 1$
g) $\frac{9(x-1)^2}{16} + \frac{(y+1)^2}{9} = 1$
h) $3x^2 + 4y^2 = 12$
i) $4x^2 + y^2 - 24x - 8y + 48 = 0$
j) $x^2 + 9y^2 + 2x - 18y + 1 = 0$
k) $4x^2 + y^2 + 8x - 10y + 13 = 0$

25. Find an equation for the ellipse that satisfies the given condition.

a) Ends of major axis $(\pm 3, 0)$; end of minor axis $(0, \pm 2)$
b) length of major axis 26; foci $(\pm 5, 0)$

c) Foci $(\pm 1, 0)$; $b = \sqrt{2}$

d) vertices (6, 0), (-6, 0); ellipse passes through (0, -4) and (0, 4)

e) major axis of length 12 on the x-axis center (0, 0), ellipse passes through (2, -3)

f) center (-2, 4), vertices (-6, 4) and (2, 4), foci at (-5, 4) and (1, 4)

g) center (2, 4) major axis parallel to the y-axis and length 10; ellipse passes through the point (3, 3)

h) Eccentricity $\frac{2}{5}$ major axis on the x-axis and of length 10 center at (0,0)

i) eccentricity $\frac{2}{3}$, foci (0, -4) and (0, 4)

j) eccentricity $\frac{2}{5}$, foci at (-1, 3) and (3, 3)

26. Find an equation for a hyperbola that satisfies the given conditions.

a) vertices $(\pm 2, 0)$; foci $(\pm 3, 0)$

b) vertices $(\pm 1, 0)$; asymptote $y = \pm 2x$

c) asymptote $y = \pm \frac{3}{2}x$; $b = 4$

d) foci $(0, \pm 5)$; asymptote $y = \pm 2x$

e) vertices (2, 4) and (10, 4); foci 10 units apart

f) asymptote $y = 2x+1$ and $y = -2x+3$; passes through the origin.

g) asymptote $y = ½x$ and $y = -\frac{1}{2}$d, vertices (0, 4) and (0, -4)

h) foci (1, -2) and (7, -2), slope of an asymptote $\frac{5}{4}$

i) Passing through (9, 4), slope of asymptote $\frac{1}{2}$, center (7, 2), transverse axis parallel to the y-axis

27. Use the eccentricity to find the equation in standard form of each hyperbola

a) vertices (1, 6) and (1, 8) hyperbola.
b) Eccentricity 2, foci (4, 0) and (-4, 0)
c) center (4, 1), conjugate axis of length 4 eccentricity $\frac{4}{3}$ (there are two answer)

Use the definition of a hyperbola to find the equation of the hyperbola in standard form.

a) foci (2, 0) and (-2, 0); passes through the point (2, 3)

b) foci (0, 4) and (0, -4); passes through the point $\left(\frac{7}{3}, 4\right)$

Find the center, vertices, foci and asymptotes for the hyperbola given by each equation.

a) $\frac{x^2}{16} - \frac{y^2}{25} = 1$

b) $\frac{y^2}{4} - \frac{x^2}{25} = 1$

c) $\frac{4x^2}{9} - \frac{y^2}{16} = 1$

d) $\frac{(x-3)^2}{16} - \frac{(y+4)^2}{9} = 1$

e) $\frac{(y+2)^2}{4} - \frac{(x-1)^2}{16} = 1$

f) $\frac{(x+2)^2}{9} - \frac{y^2}{25} = 1$

g) $\frac{9(x-1)^2}{16} - \frac{(y+1)^2}{9} = 1$

h) $x^2 - y^2 = 9$

i) $x^2 - y^2 - 6x + 8x - 3 = 0$

j) $9x^2 - 4y + 36x - 8y + 68$

- $c^2 = a^2 + b^2 = 16 + 9 = 25 \Rightarrow c = 5$
- Distance $c = 5$

Answer: A

9) One of the asymptote of the hyperbola: $y^2 - x^2 + 4x - 5 = 0$

A. $y = x + 2$ B. $y = -x + 2$ C. $y = -x - 2$ D. $y = x - 4$

Solution: $y^2 - (x^2 - 4x + 4) = 5 - 4 = 1$

$$\Rightarrow \frac{y^2}{1} - \frac{(x-2)^2}{1} = 1 \Rightarrow a^2 = 1, b^2 = 1$$

- Asymptote: $y = \pm(x - 2) \Rightarrow y = -x + 2$ and $y = x - 2$

Answer: B

11) The equation of the circle with center at $(-3,1)$ and tangent to the circle $(x-1)^2 + (y+2)^2 = 1$ is . . . EHEECE

A. $(x + 3)^2 + (y - 1)^2 = 4$ C. $(x - 3)^2 + (y + 1)^2 = 16$

B. $(x+3)^2 + (y - 1)^2 = 25$ D. $(x + 3)^2 + (y - 1)^2 = 16$

Solution: Distance between $(-3,1)$ and $(1, -2)$

$$\Rightarrow r_1 + r_2 = \sqrt{(1+3)^2 + (-2-1)^2} = 5$$

$r_2 = 5 - r_1 = 5 - 1 = 4$

$\therefore$ the equation of circle $(x + 3) + (y - 1)^2 = 16$

12) An iron wire bent in shape of parabola has latus rectum of length 60cm. what is its focal length.

A. 60cm B. 30cm C. 15cm D. 7.5cm

Solution: length of latus rectum, $4P = 60 \Rightarrow P = 15$

13) Length of the major axis of the ellipse $x^2+9y^2-2x+18y + 1 = 0$ is

A. 3 B. $\sqrt{10}$ C. 6 D. 9

Solution:

$x^2 - 2x + 9y^2 + 18y = -1 \Rightarrow (x - 1)^2 + 9(y+1)^2 = -1 + 1 + 9$

$$\Rightarrow \frac{(x-1)^2}{9} + \frac{(y+1)^2}{1} = 1 \Rightarrow a^2 = 9 \Rightarrow a = 3 \Rightarrow 2a = 6$$

Answer: C

14) Let L_1,: $3y - 9 = 4x$ and L_2: $2y - 3x + 2 = 0$ represent two intersecting lines If α is the angle between L_1 and L_2. then what is $\tan\alpha$? . . . (EHEECE).

A. $\frac{3}{2}$ B. $\frac{4}{3}$ C. $\frac{1}{6}$ D. $\frac{1}{18}$

54) Find all values of F such that the graph of the equation $x^2 + y^2 - 6x - 10y + F = 0$ represent.

a) circle b) point circle c) no locus

Solution: $x^2 + y^2 - 6x - 10y + 25 = -F + 9 + 25$

$\Rightarrow x^2 - 6x + 9 + y^2 - 10y + 25 = -F + 9 + 25$

$\Rightarrow (x-3)^2 + (y-5)^2 = 34 - F$

a) circle, if $34 - F > 0 \Rightarrow F < 34$

b) point circle if $34 - F = 0 \Rightarrow F = 34$

c) no circle if $34 - F < 0 \Rightarrow F > 34$

55) Find the equation of circle whose center is (4,5) and which goes through the center of the circle $x^2 + y^2 + 4x - 6y - 12 = 0$

Solution: $x^2 + y^2 + 4x - 6y - 12 = 0 \Rightarrow (x+2)^2 + (y-3)^2 = 25$

$\Rightarrow$ center $(-2,3) \Rightarrow (x-4)^2 + (y-5)^2 = r^2$, since $(-2, 3)$ on it

$\Rightarrow (-2-4)^2 + (3-5)^2 = r^2 \Rightarrow r^2 = 100 \Rightarrow (x-4)^2 + (y-5)^2 = 100$

56) What is the equation of circle passing through (3,1), its center on the x – axis and radius, $r = \sqrt{5}$

Solution: Let (a,0) be center on x – axis $(x-a)^2 + (y-0)^2 = r^2$

$\Rightarrow (3-a)^2 + 1 = 5 \Rightarrow 3 - a = \pm 2 \Rightarrow a = 1$ or $a = 5$

$\Rightarrow (x-1)^2 + y^2 = 5$ or $(x-5)^2 + y^2 = 5$

57) The foci of the conic section defined by the equation $4x^2 + 9y^2 - 24x + 18y + 9 = 0$

A. $(3-\sqrt{5},-1)$ *and* $(3+\sqrt{5},-1)$ C. $(3,-1)$

B. $(3,-1-\sqrt{5})$ *and* $(3,-1+\sqrt{5})$ D) $(3-\sqrt{13},-1)$ *and* $(3+\sqrt{13},-1)$

Solution: $4x^2 + 9y^2 - 24x + 18y + 9 = 0 \Rightarrow 4x^2 - 24x + 9y^2 + 18y + 9 = 0$

$\Rightarrow (4x^2 - 6x + 9) + 9(y^2 + 2y + 1) = -9 + 36 + 9 = 36$

$$\Rightarrow \frac{(x-3)^2}{9} + \frac{(y+1)^2}{4} = 1, \text{ center } (3,-1)$$

$\Rightarrow a^2 = 9$, and $b^2 = 4 \Rightarrow c^2 = a^2 - b^2 = 9 - 4 = 5$

It is x – ellipse.

$\therefore$ foci $(3-\sqrt{5},-1)$ *and* $(3+\sqrt{5},-1)$

58) What is the distance from center to one of the foci of hyperbola $\dfrac{(y-3)^2}{16} - \dfrac{(x+1)^2}{9} = 1$

A. 5 B. 10 C. 6 D.

Solution: center $(-1,3)$, $a^2 = 16$, $b^2 = 9$

- Slope of the line y = x is 1
- Slope of the line passing through h(2,2) and (a, 0) is -1

$$\Rightarrow \frac{2-0}{2-a} = -1 \Rightarrow a-2=2 \Rightarrow a=4$$

$\therefore$ the center of circle is (4,0)

- the radius, r = $\sqrt{(2-0)^2+(2-4)^2} = \sqrt{8} \Rightarrow r^2 = 8$

$\Rightarrow$ Equation of circle $(x-4)^2 + y^2 = 8$

Answer: C

52) What is the vertex and the equation of the directrix, resp. of the parabola $x + y^2 + 2y + 1 = 0$

A. $(0, -1), x = \frac{-1}{4}$ C. $(0, -1), x = \frac{1}{4}$

B. $(-1,0), y = \frac{-1}{4}$ D. $(-1, 0), y = \frac{1}{4}$

Solution: $x + y^2 + 2y + 1 = 0 \Rightarrow (y+1)^2 = -x$

- vertex: $(0, -1)$, $4P = 1 \Rightarrow p = \frac{1}{4}$
- directrix, $x = \frac{-1}{4}$, **Answer: A**

3) The or bit of Mercury around the sun forms an ellipse with eccentricity, e = 0.206, length of the major axis 1.16×10^8km and sun at one focus. What is the best approximation of the maximum distance from Mercury to the sun? . . . UEE 2004/2012.

A. 7.596×10^7 km C. 8.695×10^7 km

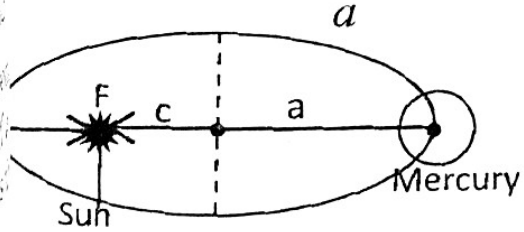
B. 5.695×10^7 km D. 6.995×10^7 k

Solution: $2a = 1.16 \times 10^8$km $\Rightarrow a = 5.8 \times 10^7$ km

$$e = \frac{c}{a} \Rightarrow c = ea = (0.206)(5.8 \times 10^7)$$

$\Rightarrow c = 1.195 \times 10^7$

maximum distance;



$\Rightarrow a+c=(1.195+5.8)\times 10^7 \simeq 6.995\times 10^7$ km **Answer: D**

Minimum distance, $a - c = (5.8 - 1.195) \times 10^7 = 4.61 \times 10^7$ km.

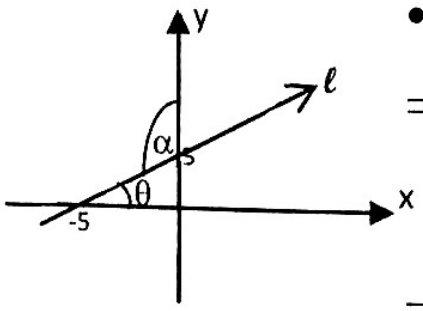
- foci, $F\left(1,-3+\frac{5\sqrt{13}}{6}\right), F^1\left(1,-3-\frac{5\sqrt{13}}{6}\right)$
- vertice $\left(1,-3+\frac{5}{2}\right) and \left(-3-\frac{5}{2}\right)=\left(1,\frac{-1}{2}\right),\left(1,\frac{-11}{2}\right)$
- $B\left(1,-\frac{5}{3},-3\right) and\ B^1\left(1+\frac{5}{3},-3\right)=\left(\frac{-2}{3},-3\right),\left(\frac{8}{3},-3\right)$
- Eccentricity, $e=\frac{c}{a}=\frac{5\sqrt{13}}{6}\cdot\frac{2}{5}=\frac{\sqrt{13}}{3}$
- asymptote, $y+3=\pm\frac{2}{3}(x-1)\Rightarrow y=\pm\frac{2}{3}(x-1)-3$

Solved Problem

50) Line l passes through (0,5) and (-5, 0) what is the angle between the y – axis and l in radian measure . . . UEE 2004/2012.

A) $\frac{\pi}{4}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{2}$ D) $\frac{3}{4}\pi$

Solution: Slope of $\ell=\frac{5-0}{0-(-5)}=\frac{5}{5}=1$



- Angle of inclination, $\tan\theta = 1$

$\Rightarrow\theta=\frac{\pi}{4}\leftarrow$ the angle measured from positive x – axis to line ℓ

$\Rightarrow\alpha=\pi-\frac{\pi}{4}=\frac{3\pi}{4}\leftarrow$ the angle between y – axis and ℓ measured C.C.W from y to l

51) Consider a circle whose center is on the x – axis. If a line given by y = x is tangent to the circle at point (2,2), what is the equation of the circle . . . UEE 2004/2012.

A) $x^2+y^2=8$ C) $(x-4)^2+y^2=8$

B) $(x-2)^2+y^2=4$ D) $(x-1)^2+y^2=5$

Solution: Let the center (a,0)

- eccentricity, $e = \frac{c}{a} = \frac{\sqrt{10}}{\sqrt{2}} = \sqrt{5}$
- asymptote, $y = \pm 2x$

c) $8y^2 - 2x^2 = 16 \Rightarrow \frac{y^2}{2} - \frac{x^2}{8} = 1$

- $a^2 = 2 \Rightarrow a = \sqrt{2}, b^2 = 8 \Rightarrow b = \sqrt{8}$
- $c^2 = a^2 + b^2 = 2 + 8 = 10 \Rightarrow c = \sqrt{10}$
- center (0,0), foci $(0, \pm\sqrt{10})$, $V(0, \pm\sqrt{2})$
  $B(0, \pm 2\sqrt{2})$ ← conjugate vertex.
- eccentricity, $e = \frac{c}{a} = \frac{\sqrt{10}}{\sqrt{2}} = \sqrt{5}$
- asymptote, $y = \pm \frac{1}{2}x$

d) $\frac{(x-2)^2}{4} - \frac{(y+5)^2}{9} = 1 \Rightarrow a^2 = 4, b^2 = 9$

- $c^2 = a^2 + b^2 \Rightarrow c^2 = 4 + 9 = 13 \Rightarrow c = \sqrt{13}$
- center (2, – 5), foci $F(2 - \sqrt{13}, -5)$ and $F^1(2 + \sqrt{13}, -5)$
- vertice, V(4, –5), V^1(0, –5) and B(2, –2),B^1(2, –8)
- eccentricity, $e = \frac{c}{a} = \frac{\sqrt{13}}{2}$
- asymptote, $y + 5 = \pm \frac{3}{2}(x-2) \Rightarrow y = \pm\frac{3}{2}(x-2) - 5$

e) $4(y+3)^2 - 9(x-1) = 25 \Rightarrow \frac{4}{25}(y+3)^2 - \frac{9}{25}(x-1)^2 = 1$

- $a^2 = \frac{25}{4} \Rightarrow a = \frac{5}{2}$ and $b^2 = \frac{25}{9}, b = \frac{5}{3}$
- $c^2 = a^2 + b^2 \Rightarrow c^2 = \frac{25}{4} + \frac{25}{9} \Rightarrow c = \frac{5}{6}\sqrt{13}$
- center, (1, - 3),

$$\Rightarrow \frac{y^2}{36} - \frac{x^2}{64} = 1$$

j) V(4,1) and $V^1(-2,1)$, $\Rightarrow 2a = 4 - (-2) = 6 \Rightarrow a = 3, a^2 = 9$

- $e = \frac{c}{a} \Rightarrow \frac{4}{3} = \frac{c}{3} \Rightarrow c = 4$
- $c^2 = a^2 + b^2 \Rightarrow 16 = 9 + b^2 \Rightarrow b^2 = 16 - 9 = 7$
- It is horizontal hyperbola with center (1,1)

$$\Rightarrow \frac{(x-1)^2}{9} - \frac{(y-1)^2}{7} = 1$$

49) Find center, foci, vertice, eccentricity and equations of the asymptote.

a) $4x^2 - y^2 = 16$

b) $8x^2 - 2y^2 = 16$

c) $8y^2 - 2x^2 = 16$

d) $\frac{(x-2)^2}{4} - \frac{(y+5)^2}{9} = 1$

e) $4(y+3)^2 - 9(x-1)^2 = 25$

Solution: a) $4x^2 - y^2 = 16 \Rightarrow \frac{4x^2}{4} - \frac{y^2}{16} = 1 \Rightarrow \frac{x^2}{1} - \frac{y^2}{16} = 1$

- $a^2 = 1 \Rightarrow a = 1$, $b^2 = 16 \Rightarrow b = 4$
- $c^2 = a^2 + b^2 = 1 + 16 = 17 \Rightarrow c = \sqrt{17}$
- center (0,0), foci, $F(-\sqrt{17}, 0)$ and $F^1(-\sqrt{17}, 0)$
- vertice, V(– 1, 0) and $V^1(1,0)$, B $(0, \pm 4)$
- eccentricity, $e = \frac{c}{a} = \frac{\sqrt{17}}{1} = \sqrt{17}$
- asymptote, $y = \pm 4x$

b) $8x^2 - 2y^2 = 16 \Rightarrow \frac{x^2}{2} - \frac{y^2}{8} = 1 \Rightarrow a^2 = 2, b^2 = 8$

- $c^2 = a^2 + b^2 = 2 + 8 = 10 \Rightarrow c = \sqrt{10}$
- center (0,0), foci $(\left(\pm\sqrt{10},0\right), V\left(\pm\sqrt{2},0\right), B\left(\pm\sqrt{8},0\right))$

f) Center (– 3, 6), V(0,6), $\Rightarrow$ a = 0 – (– 3) = 3 $\Rightarrow a^2 = 9$

- It is horizontal (x – hyperbola)

$$\Rightarrow \Rightarrow \frac{(x+3)^2}{a^2} - \frac{(y-6)^2}{b^2} = 1, \text{ since (1,1) on graph}$$

$$\Rightarrow \frac{(1+3)^2}{9} - \frac{(1-6)^2}{b^2} = 1 \Rightarrow \frac{16}{9} - \frac{25}{b^2} = 1$$

$$\Rightarrow \Rightarrow b^2 = \frac{225}{7} \Rightarrow \frac{(x+3)^2}{9} - \frac{7(y-6)^2}{225} = 1$$

g) V(– 4, 2), V^1(8,2), F(– 7,2), $\Rightarrow$ 2a = 8 – (– 4) = 12 $\Rightarrow$ a = 6

- Center $\left(\frac{8-4}{2}, \frac{2+2}{2}\right) = (2,2), \Rightarrow C = 2 - (-7) = 9$
- $c^2 = a^2 + b^2 \Rightarrow 81 = 36 + b^2 \Rightarrow b^2 = 81 - 36 = 45$
- it is horizontal (x – hyperbola)$\Rightarrow \frac{(x-2)^2}{36} - \frac{(y-2)^2}{45} = 1$

h) V(3,0) and V^1(–3,0), 2a 3 – (–3) = 6 $\Rightarrow$ a = 3, $a^2 = 9$

- It is horizontal hyperbola with center

$$\left(\frac{3+-3}{2}, \frac{0+0}{2}\right) = (0,0)$$

$$\Rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \text{asymptote, } y = \pm 3x$$

$$\Rightarrow \frac{b}{a} = \frac{b}{3} = 3 \Rightarrow b = 9 = b^2 = 81 \Rightarrow \frac{x^2}{9} - \frac{y^2}{81} = 1$$

i) F(0,10) and F^1 (0, - 10) $\Rightarrow$ 2c = 10 – (–10) = 20, c = 10

- It is vertical hyperbola with center (0,0)

$$\Rightarrow \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1 \Rightarrow \text{asymptote, } y = \pm \frac{a}{b}x = \pm \frac{3}{4}x$$

$$\Rightarrow \frac{a}{b} = \frac{3}{4} \Rightarrow 4a = 3b \Rightarrow b = \frac{4a}{3} \Rightarrow b = 9 = b^2 = 81$$

- $c^2 = a^2 + b^2 \Rightarrow 10^2 = a^2 + \frac{(16a^2)}{9} \Rightarrow a^2 = 36$ and b^2 $= 8^2 = 64$

axis, $\Rightarrow \frac{(x-2)^2}{a^2} - \frac{(y-1)^2}{b^2} = 1, a = 6, b = 4$

$\Rightarrow \frac{(x-2)^2}{36} \frac{-(y-1)^2}{16} = 1$

b) F(-2,-1),F^1(-2,9),2a = 6 $\Rightarrow$ a = 3

$\Rightarrow$ 2c = 9 – (– 1) = 10 $\Rightarrow$ C = 5

- $C^2 = a^2 + b^2 \Rightarrow 5^2 = 3^2 + b^2 \Rightarrow b^2 = 25 - 9 = 16$
- It is y – hyperbola (vertical) with center

$\left(\frac{-2+-2}{2}, \frac{-1+9}{2}\right) = (-2,4) \Rightarrow \frac{(y-4)^2}{9} - \frac{(x+2)^2}{16} = 1$

c) V(1,2) and V^1(-1,2), 2b = 8 $\Rightarrow$ b = 4, $b^2 = 16$

2a = 1 – (- 1) = 2 $\Rightarrow$ a = 1 $\Rightarrow a^2 = 1$

- it is x – hyperbola with center (0,2)

$\Rightarrow \frac{(x-0)^2}{1} - \frac{(y-2)^2}{16} = 1$

d) center (3,– 1), F(5, –1),V (4, –1)

- Distance between center (3, –1) and V(4, –1) is a = 4 – 3 = 1 $\Rightarrow a^2 = 1$
- Distance between center (3, –1) and G(5, –1): C = 5 – 3 = 2 $\Rightarrow C^2 = 4$
- $C^2 = a^2 + b^2 \Rightarrow 4 = 1 + b^2 \Rightarrow b^2 = 4 - 1 = 3$
- It is horizontal (x – hyperbola)

$\Rightarrow \frac{(x-3)^2}{1} - \frac{(y+1)^2}{3} = 1$

e) Center (2,3), V(2,8); a = 8 – 3 = 5 $\Rightarrow a^2 = 25$

- Asymptote, 5y – 4x = 7 $\Rightarrow y = \frac{4}{5}x + \frac{7}{5}$
- It is vertical (y – hyperbola)
- From asymptote, we have $\frac{a}{b} = \frac{4}{5} \Rightarrow \frac{5}{b} = \frac{4}{5}$

$\Rightarrow b = \frac{25}{4} \Rightarrow b^2 = \frac{625}{16} \Rightarrow \frac{(y-3)^2}{25} - \frac{4(x-2)^2}{625} = 1$

- vertices $v_1(h, k + a)$ and $v_2(h, k - a)$
- Foci are $F_1(h, k + c)$ and $F_2(h, k - c)$
- asymptote $y - k = \pm \frac{a}{b}(x - h)$
- eccentricity, $e = \frac{c}{a} = \frac{\sqrt{a^2 + b^2}}{a}$

Eccentricity of Hyperbola

The graph of a hyperbola can be very wide or very narrow. The eccentricity of a hyperbola measure of its "wideness."

$$eccentricity(e) = \frac{distance\ between\ foci}{distance\ between\ vertices} = \frac{2c}{2a} = \frac{c}{a}$$

Note, i) $e > 1$ for hyperbola ii) if $e = 0$ circle iii) if $0 < e < 1$, Ellipse

48. Find the equation of each hyperbola that satisfies the states condition:
   a) center (2,1), a = 6, b = 4 with horizontal transverse axis
   b) foci, F(– 2, – 1) and F^1(– 2, 9) and transverse axis has length 6
   c) vertice V(1,2) and V^1 (– 1, 2), length of conjugate axes are equal to 8
   d) center (3, – 1), focus F(5, – 1); vertex V(4, – 1).
   e) center C(2,3), vertex V(2,8); equation of the asymptote, 5y – 4x = 7.
   f) center (– 3,6), vertex V(0,6) passing through P(1,1)
   g) vertices at (– 4, 2) and V1(8,2) one focus at F(-7 , 2)
   h) vertices V(3,0) and V1(-3,0), asymptote: $y = \pm 3x$
   i) focus, F(0,10) and F1(0,– 10), asymptote: $y = \pm \frac{3}{4}x$
   j) vertices, V(4,1) and V^1(– 2, 1), eccentricity, $e = \frac{4}{3}$

Solution:

Note: • Distance between vertices = 2a, (length of transverse)
- Distance between foci = 2C
- Center is mid – point of vertices or foci.
- Distance between center and vertice = a

a) center (2,1) with horizontal transverse

Note: Quadratic function $f(x) = ax^2 + bx + c$ is one to one and invertible in $x \geq -\frac{b}{2a}$ or $x \leq -\frac{b}{2a}$.

Illustrative Example

56. Determine whether the following functions are invertible if so, find their respective inverses.

a) $f: \mathbb{R} \longrightarrow [1, \infty)$ be defined by $f(x) = (x-2)^2 + 1$

b) $f: [2, \infty) \longrightarrow [1, \infty)$ be defined by $f(x) = (x-2)^2 + 1$

c) $f: (-\infty, -3] \longrightarrow [0, \infty)$ be defined by $f(x) = (x+3)^2$

d) $f: [-\infty, \infty) \longrightarrow [3, \infty)$ be defined by $f(x) = |x+2| + 3$

e) $f: [-2, \infty) \longrightarrow [3, \infty)$ be defined by $f(x) = |x+2| + 3$

Solution:

a) This function is not one to one in the given domain $(-\infty, \infty)$. For example, $(-1, 10)$ and $(5, 10)$ have the same y – coordinate. $\therefore f$ is not invertible.

b) This function is one – to – one in the domain $[2, \infty)$, so it is invertible.

To find the inverse: Let $y = f(x) \Rightarrow y = (x-2)^2 + 1$, for $x \geq 2$

$$\Rightarrow x = (y-2)^2 + 1 \Leftrightarrow x - 1 = (y-2)^2, \text{ for } y \geq 2$$

$$\Rightarrow y - 2 = \sqrt{x-1} \Rightarrow y = 2 + \sqrt{x-1}$$

$$\therefore f^{-1}(x) = 2 + \sqrt{x-1}$$

Domain of $f^{-1} = \{x: x \geq 1\}$ and range of $f^{-1} = \{y: y \geq 2\}$.

c) This function is one – to – one in the domain $(-\infty, -3]$, so it is invertible.

To find the inverse for $x \leq -3$

Let $y = f(x) = (x+3)^2$ then $x = (y+3)^2$, for $y \leq -3$

$$\Rightarrow y + 3 = \pm\sqrt{x} \Rightarrow y = -3 - \sqrt{x}, \text{ for } y \leq -3$$

$$\therefore f^{-1}(x) = -3 - \sqrt{x}$$

Domain of $f^{-1} = \{x: x \geq 0\}$ and range of $f^{-1} = \{y: y \leq -3\}$.

d) This function is not one – to – one in the domain $(-\infty, \infty)$ for example $(1, 6)$ and $(-5, 6)$ have the same y – coordinate. $\therefore f(x) = |x+2| + 3$ is not invertible in the domain $(-\infty, \infty)$.

e) This function is one – to – one in the domain $[-2, \infty)$, so it has inverse.

To find the inverse for $x \geq -2$

Unit Four

Mathematical Reasoning

- Mathematical logic is the art of reasoning.
- Mathematical reasoning is tool for organizing evidence in a systematic way through mathematical logic.
- Its reasoning is based on the form of the sequence of statements.

Statement and open statement

- **Statement (Proposition)**: is an assertive or declarative sentence which is either **true (T) or false (F) but not both**.
- **Open proposition** (open statement) is not a statement but is sentence which contains a variable and which can be converted into a statement by replacing the variable by the name of specific number or object.
- **Truth value**: The true and false value of the statement is truth value.

Note: Hope, wish, interest, beauty, intellegency command exclamatory interrogative and imperative sentences are not **logical statement** (not proposition) in mathematics.

Illustrative example

Identify whether proposition, an open proposition or neither

a) 4 is even number
b) 3 is less than 2
c) Long live Ethiopia
d) I wish all Ethiopian love their country
e) $4 + 10 = 12$
f) For some x, $x^2 - 4 = 0$
g) For all x, $x^2 + 1 > 0$
h) There exist x that satisfies: $x^2 + 5 = 0$
i) What do you feel about your Nationality.
j) Every student has the right to be engineer.
k) X is a city in Ethiopia.
l) $2x + 5 = 9$
m) You are good at mathematics

Solution:

) a, b, e, f, g j and h are proposition (statement)
(It can be said either true or false)

ii) c, d and m neither proposition nor open proposition (it can't be said either true or false)

iii) K and l are open proposition.

Fundamental logical connectives

As operating number by $(+,-,\times,\div)$ we operate statement by $(\neg,\vee,\wedge,\Rightarrow,\Leftrightarrow)$ which are called logical connectives (operator)

Name	Symbol	Meaning	How to write
• Negation	$\neg$	Not (It is false that)	$\neg P$
• Disjunction	$\vee$	Or, either	$P \vee q$
• Conjunction	$\wedge$	And, yet, But	$P \wedge q$
• Implication	$\Rightarrow$	If . . . then	$P \Rightarrow q$
• Bi-implication	$\Leftrightarrow$	If and only if	$P \Leftrightarrow q$

Rule. (Let P and q be a proposition)

i) **Rule of negation** $(\neg P)$:

$\neg P$ is true if and only if P is false

2) Find the negation of each of the following statement.

a) p: All rectangles are square b) P: $5 + 7 \leq 8$

Solution: a) $\neg P \equiv$ Note all rectangles are square

$\equiv$ It is false that all rectangle are square

$\equiv$ some rectangle are not square

b) $\neg P \equiv 5 + 7 > 8$

ii) **Rule of disjunction ($P \vee q$)**

Disjunction is the joining of two statement p,q by connective or denoted by $P \vee q$

- $(P \vee q)$ is false if and only if both p and q are false, but all other cases true.

i.e., $FVF \equiv F$, a but $TVF \equiv T$, $FVT \equiv T$, $TVT = T$

iii) **Rule of conjunction ($P \wedge q$)**

- $(P \wedge q)$ is true (T) if and only if both p and q true, but all other cases false.

i.e. $T \wedge T \equiv T$, $F \wedge T \equiv F$

$T \wedge F \equiv F$ $F \wedge F \equiv F$

iv) **Rule of implication ($P \Rightarrow q$)**

- $P \Rightarrow q$ is false if and only if p is true and q is false, but all other case true.

i.e., • $T \Rightarrow F \equiv F$, • $F \Rightarrow F \equiv T$

• $F \Rightarrow T \equiv T$ • $T \Rightarrow T \equiv T$

V) Rule of bi-implication (P⇔q)

- P⇔q is true if and only if both p and q have the same truth value.

i.e. • $F \Leftrightarrow F \equiv T$, • $F \Leftrightarrow T \equiv F$
• $T \Leftrightarrow T \equiv T$ • $T \Leftrightarrow F \equiv F$

3) Find the truth value of each of the following

a) $5 + 3 = 10$ or $2 + 9 = 11$
b) $6 - 4 = 8$ or $1 + 6 = 5$
c) $5 + 3 = 10$ and $2 + 9 = 11$
d) $6 - 4 = 8$ and $1 + 6 = 5$
e) If $5 + 3 = 10$ then $2 + 9 = 11$
f) If $6 - 4 = 8$ then $1 + 6 = 5$
g) $5 + 3 = 10$ if and only if $2 + 9 = 11$
h) $6 - 4 = 10$ if and only if $1 + 6 = 5$

Solution:

a) $F \vee T \equiv T$ e) $F \Rightarrow T \equiv T$
b) $F \vee F \equiv F$ f) $F \Rightarrow F \equiv T$
c) $F \wedge T \equiv F$ g) $F \Leftrightarrow T \equiv F$
d) $F \wedge F \equiv F$ h) $F \Leftrightarrow F \equiv T$

4) Let $P \equiv 4 < 3$, $q \equiv 5$ is an odd number $r \equiv 13$ is not prime number, then which of the following statements has truth value "**True**".

A) $q \Rightarrow (p \vee r)$ C) $\neg(P \vee r) \Rightarrow q$
B) $\neg r \Leftrightarrow (p \wedge q)$ D) $\neg P \wedge (q \Rightarrow r)$

Solution: the truth value of $P \equiv F$, $q \equiv T$ and $r \equiv F$

A) $q \Rightarrow (p \vee r)$
$T \Rightarrow (F \vee F)$
$T \Rightarrow F \equiv F$

C) $\neg(P \vee r) \Rightarrow q$
$\neg(F \vee F) \Rightarrow T$
$T \Rightarrow T \equiv T$

B) $\neg r \Leftrightarrow (p \wedge q)$
$T \Leftrightarrow (F \wedge T)$
$T \Leftrightarrow F \equiv F$

D) $\neg P \wedge (q \Rightarrow r)$
$T \wedge (T \Rightarrow F)$
$T \wedge F \equiv F$

Answer: C

) Let $P \equiv \sqrt{2}$ is an irrational number $q \equiv 2$ is an odd number then which of the following compound statement has the truth value "True"

A) $\neg p \vee q$ B) $p \Rightarrow q$ C) $\neg q \Leftrightarrow p$ D) $p \wedge q$

Solution: The truth value of $P \equiv T$, $q \equiv F$, $\neg p \equiv F$, $\neg q \equiv T$

A) $\neg p \vee q$ B) $p \Rightarrow q$ C) $\neg q \Leftrightarrow p$ D) $p \wedge q$
$F \vee F \equiv F$ $T \Rightarrow F \equiv F$ $T \Leftrightarrow T \equiv T$ $T \wedge F \equiv F$ Answer: C

Suppose proposition $P \Rightarrow \neg q$ is false (F), which of the following is true? … UEE 2004/2012.

A) $\neg q \vee (p \Rightarrow q)$ C) $(\neg q \vee p) \Leftrightarrow q$

B) $\neg q \vee (q \Rightarrow \neg p)$ D) $(p \wedge q) \Leftrightarrow \neg q$.

Solution: $p \Rightarrow \neg q \equiv F$ iff, $p \equiv T$, $\neg q \equiv F$ $q \equiv T$, $\neg p \equiv F$

A) $\neg q \wedge (p \Rightarrow q)$ C) $(\neg q \vee p) \Leftrightarrow q$

$F \wedge (T \Rightarrow T)$ $(F \vee T) \Rightarrow T$

$F \wedge T \equiv F$ $T \Leftrightarrow T \equiv T$

A) $\neg p \vee (q \Rightarrow \neg p)$ C) $(p \wedge p) \Leftrightarrow \neg q$

$F \vee (T \Rightarrow F)$ $(T \wedge T) \Rightarrow F$

$F \vee F \equiv F$ $T \Leftrightarrow F \equiv F$

Answer: C

If $(p \wedge \neg q) \Rightarrow \neg r$ is false, then the truth value of p,q and r respectively are _____

A) T,F,F B) F,T,T C) T,F,T D) F,T,T

Solution: $(p \wedge \neg q) \Rightarrow \neg r$ is false if and only if

$p \wedge \neg q \equiv T$ and $\neg r \equiv F$, so that

$p \equiv T$, $\neg q \equiv T$, $q \equiv F$ and $r \equiv T$

$\therefore P \equiv T$, $q \equiv F$ and $r \equiv T$

Answer: C

If the truth value of $(\neg p \Leftrightarrow \neg q) \Rightarrow (P \vee \neg r)$ is false then determine the truth value of

a) $q \Rightarrow p$ b) $\neg p \Leftrightarrow r$ c) $\neg q \wedge \neg r$ d) $(\neg p \Leftrightarrow r) \wedge p$

Solution: $(\neg p \Leftrightarrow \neg q) \Rightarrow (p \vee \neg r) \equiv F$ iff $p \vee \neg r \equiv F$ and $(\neg p \Leftrightarrow \neg q) \equiv T$

- From $p \vee \neg r \equiv F$, we have, $P \equiv F$, $\neg r \equiv F$ and $r \equiv T$
- From $(\neg P \Leftrightarrow \neg q) \equiv T$, we have $\neg q \equiv T$, $\neg p \equiv T$, $q \equiv F$

a) $q \Rightarrow p$ b) $\neg p \Leftrightarrow r$ c) $\neg q \wedge \neg r$ d) $(\neg q \Leftrightarrow r) \wedge p$

$F \Rightarrow F \equiv T$ $T \Leftrightarrow T \equiv T$ $T \wedge F \equiv F$ $(T \Leftrightarrow T) \wedge F \equiv F$

Suppose the value of $p \Rightarrow q$ is T. What can be said about the value of $(p \wedge q) \Leftrightarrow (p \vee q)$.

Solution: $p \Rightarrow q \equiv T$, we have three possibility

Case I. $(T \Rightarrow T)$ if $P = T$, $q = T$, then $(p \wedge q) \Leftrightarrow (p \vee q) \equiv (T \wedge T) \Leftrightarrow (T \vee T) \equiv T \Leftrightarrow T = T$.

Case II. $(F \Rightarrow F)$ if $P = F$, $q = F$, then

$(p \wedge q) \Leftrightarrow (p \vee q) \equiv (F \wedge F) \Leftrightarrow (F \vee F) \equiv F \Leftrightarrow F = T$.

Case III. $(F \Rightarrow T)$ if $P = F$, $q = T$, then

$(p \wedge q) \Leftrightarrow (p \vee q) \equiv (F \wedge T) \Leftrightarrow (F \vee T) \equiv F \Leftrightarrow T \equiv F$.

10) Suppose the value of $p \Leftrightarrow q$ is T what can be said about the value of: a) $p \Leftrightarrow \neg q$ b) $\neg q \Leftrightarrow q$

Solution: $p \Leftrightarrow q \equiv T$ iff both p and q have the same truth value, i.e $T \Leftrightarrow T$ or $F \Leftrightarrow F$

Case I. $(T \Leftrightarrow T)$ if $P = T, q = T$,

a) $p \Leftrightarrow \neg q \equiv T \Leftrightarrow F \equiv F$ b) $\neg p \Leftrightarrow q \equiv F \Leftrightarrow T \equiv F$

Case II. $(F \Leftrightarrow F)$ if $P = F, q = F$,

a) $p \Leftrightarrow \neg q \equiv F \Leftrightarrow T \equiv F$ b) $\neg p \Leftrightarrow q \equiv T \Leftrightarrow F \equiv T$

11) Let P, q, r have truth values F,F and T respectively, then which of the following have truth value "false"

A. $r \Leftrightarrow (\neg p \wedge \neg q)$ C. $\neg q \Rightarrow (p \vee \neg r)$

B. $\neg p \wedge (q \Rightarrow \neg r)$ D. $p \vee (\neg(q \wedge r))$

Solution: we have, $p \equiv F$, $q \equiv F$, and $r \equiv T$

A. $r \Leftrightarrow (\neg p \wedge \neg q)$ C. $\neg q \Rightarrow (p \vee \neg r)$

$T \Leftrightarrow (T \wedge T) \equiv T$ $T \Rightarrow (F \vee F) \equiv T \Rightarrow F \equiv F$

B. $\neg p \wedge (q \Rightarrow \neg r)$ D. $p \vee (\neg(q \wedge r))$

$T \wedge (F \Rightarrow F)$ $F \vee (\neg(F \wedge T)$

$T \wedge T \equiv T$ $F \vee T \equiv T$ **Answer: C**

12) If p,q and r are proposition such that $(p \Leftrightarrow r) \Rightarrow (\neg q \Leftrightarrow p)$ has truth value "false" then which of the following have truth value "false"

A. $\neg r \Rightarrow P$ B. q C. $\neg r$ D. $p \vee q$

Solution: we have $(p \Leftrightarrow r) \Rightarrow (\neg q \vee p) \equiv F$

Then $P \Leftrightarrow r \equiv T$ and $\neg q \vee p \equiv F$

- From $\neg q \vee p \equiv F$, we have $\neg q \equiv F$, $q \equiv T$, $P \equiv F$
- From $p \Leftrightarrow r \equiv T$, we have $r \equiv F$ since $P \equiv F$

A. $\neg r \Rightarrow p$ B. $q \equiv T$ C. $\neg r = T$ D. $p \vee q$

$T \Rightarrow F \equiv F$ $F \vee T \equiv T$ **Answer: A**

Equivalent proposition

Two compound statement **R** and **Q** are said to be equivalent if they have the same truth value for all possible cases (they have identical truth table)

Notation: If **R** and **Q** are equivalent, then denoted by $\mathbf{R \equiv Q}$

Each of the following pair are equivalent

- $P \Rightarrow q \equiv \neg p \vee q$
- $P \Rightarrow q \equiv \neg q \Rightarrow \neg p$ ← contra positive law
- $\neg(p \vee q) \equiv \neg p \wedge \neg q$ ← Demorgan's law

- $\neg(p \wedge q) \equiv \neg p \vee \neg q$ ← Demorgan's law
- $P \Rightarrow (q \wedge r) \equiv (p \Rightarrow q) \wedge (p \Rightarrow r)$
- $P \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)$
- **Algebra of proposition**

i) Idempotent law

a) $p \vee p \equiv p$ b) $p \wedge \neg p \equiv F$

ii) Complement law

a) $p \vee \neg p \equiv T$ b) $p \wedge \neg p \equiv F$

iii) Distributive law

a) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ b) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$

iv) Demorgan's law

a) $\neg(p \vee q) \equiv \neg p \wedge \neg q$ b) $\neg(p \wedge q) \equiv \neg p \vee \neg q$

v) Implication law

a) $p \Rightarrow q \equiv \neg p \vee q$ b) $p \Rightarrow (q \wedge r) \equiv \neg p \vee (q \wedge r)$

vi) Bi-implication law

a) $p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$

vii) Commutative law

a) $(p \wedge q) \equiv q \wedge p$ b) $(p \vee q) \equiv (q \vee p)$

Contradiction and Tautology

- **Tautology** A compound statement is said to be tautology if it is **true** for all possible combination.

  (In a truth table of a tautology, there will be only "T" in the last column.

- **Contradiction** A compound statement is said to be contradiction if it is **false** for all possible combination.

  (In a truth table of contradiction, there will be only "F" in the last column.

Note: If **$R \equiv Q$, then $R \Leftrightarrow Q$ is tautology.**

Illustrative example

13) Which of the following compound propositions are tautology, contradiction, or neither.

a) $(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$
b) $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$
c) $p \Rightarrow (p \wedge q)$
d) $p \vee (\neg(p \wedge q))$
e) $(p \wedge q) \wedge (\neg(p \vee q]$
f) $(p \Rightarrow \neg q) \wedge (p \Rightarrow q)$
g) $[p \Rightarrow (p \vee r)] \Leftrightarrow (p \Rightarrow q) \vee (p \Rightarrow r)$
h) $(\neg p \vee q) \Rightarrow (p \wedge \neg q)$
i) $\neg(p \wedge q) \Rightarrow (p \Rightarrow q)$
j) $[p \vee (\neg q \vee \neg p)] \Rightarrow q$

Solution: a) $p \Rightarrow q \equiv \neg p \vee q$

$\therefore (\neg p \vee q) \Leftrightarrow (\neg p \vee q) \equiv T$ for all possible case

$\therefore$ tautology

b) $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$

$(p \Rightarrow q) \Leftrightarrow (q \vee \neg q)$ Explanation: $\neg q \Rightarrow \neg p \equiv p \Rightarrow q = \neg p \vee q$

$(p \Rightarrow q) \Leftrightarrow (\neg p \vee q)$

$(p \Rightarrow q) \Leftrightarrow (p \Rightarrow q) \equiv T$ for all possible case

$\therefore$ It is tautology

c) $p \Rightarrow (p \wedge q) \quad = \neg p \vee (p \wedge q)$

$\equiv (\neg p \vee p) \wedge (\neg p \vee q)$ ← Distributive law

$\equiv (T) \wedge (\neg p \vee q)$

$\equiv \quad \neg p \vee q$

$\therefore p \Rightarrow (p \wedge q)$ is neither.

But $P \Rightarrow (p \vee q)$ is tautology.

d) $p \vee (\neg(p \wedge q)$ we can simplify as

$\equiv p \vee (\neg p \vee \neg q)$

$\equiv (p \vee \neg p) \vee \neg q$

$\equiv T \vee \neg q$

$\equiv T$ for all possible case.

$\therefore p \vee (\neg(p \wedge q)$ is tautology

p	q	$\neg(p \wedge q)$	$p \vee \neg(p \wedge q)$
T	T	F	T
T	F	T	T
F	T	T	T
F	F	T	T

• the 4th column is T

Explanation

e) $(p \wedge q) \wedge (\neg(p \vee q]$ simplified as using truth table

$\equiv (p \wedge q) \wedge (\neg p \wedge \neg q)$

$\equiv ((p \wedge q) \wedge \neg p) \wedge \neg q$

$\equiv (p \wedge \neg p) \wedge (q \wedge \neg q)$

$\equiv F \wedge F \equiv F$

p	q	$p \wedge q$	$\neg(p \vee q)$	$(p \wedge q) \wedge (\neg(p \vee q)$
T	T	T	F	F
T	F	F	F	F
F	T	F	F	F
F	F	F	T	F

$\therefore (p \wedge q) \wedge \neg(p \vee q)]$ is contradiction

a) $(p \Rightarrow \neg q) \wedge (p \Rightarrow q)$

By using truth table

P	Q	$\neg q$	$p \Rightarrow q$	$p \Rightarrow \neg q$	$(p \Rightarrow \neg q) \wedge (p \Rightarrow q)$
T	T	F	T	F	F
T	F	T	F	T	F
F	T	F	T	T	T
F	F	T	T	T	T

- From the last column (column 6), we see that $(P \Rightarrow \neg q) \wedge (p \Rightarrow q)$ is neither tautology nor contradiction.

b) $P \Rightarrow (q \vee r) \equiv (p \Rightarrow q) \vee (p \Rightarrow r)$

$\therefore p \Rightarrow (q \vee r) \Leftrightarrow (p \Rightarrow q) \vee (p \Rightarrow r)$ is tautology

i.e It is true for all possible combination.

c) $(\neg p \vee q) \Rightarrow (p \wedge \neg q)$, using truth table.

p	q	¬p	¬q	¬p∨q	p∧¬q	(¬p∨q) ∧(p∧¬q)
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	T	F	F
F	F	T	T	T	F	F

From the table we see that the last column is not all T or not all F

∴ neither tautology nor contradiction

i) $\neg(p \wedge q) \Rightarrow (p \Rightarrow q)$, using truth table

p	q	¬(p∧q)	(p⇒q)	¬(p∧q) ⇒(p⇒q)
T	T	F	T	T
T	F	T	F	F
F	T	T	T	T
F	F	T	T	T

From the table, last column is not all T

∴ ¬(p∧q) ⇒(p⇒q) is neither.

j) $[p \vee (\neg q \vee \neg p)] \Rightarrow q$ is simplified as

$[p \vee (\neg p \vee \neg q)] \Rightarrow q$

$[(p \vee \neg p) \vee \neg q)] \Rightarrow q$

$(T \vee \neg q) \Rightarrow q$

$T \Rightarrow q$

$\equiv q$

∴ It is neither

Explanation

Using table

p	q	¬ (p∧q)	p∨(¬(p∧q)) ⇒q
T	F	T	F
T	T	F	T
F	F	T	F
F	T	T	T

14) Which of the following is **not a tautology**? …EHEECE

A. $p \vee (q \Rightarrow \neg p)$ C. $(p \Rightarrow q) \Rightarrow \neg(p \wedge \neg q)$

B. $(p \wedge q) \Rightarrow (p \vee q)$ D. $p \Rightarrow (p \wedge q)$

Solution: A. $p \vee (q \Rightarrow \neg p) \equiv p \vee (\neg q \vee \neg p) \equiv p \vee (\neg p \vee \neg q)$

$\equiv (p \vee \neg p) \vee \neg q \equiv T \vee \neg q \equiv T$

B. $(p \wedge q) \Rightarrow (p \vee q) \equiv \neg (p \wedge q) \vee (p \vee q) \equiv (\neg p \vee \neg q) \vee (p \vee q)$

$\equiv (\neg p \vee p) \vee (\neg q \vee q) \equiv T \vee T \equiv T$

C. $(p \Rightarrow q) \Rightarrow \neg(p \wedge \neg q) \equiv (\neg p \vee q) \Rightarrow (\neg p \vee q) \equiv T$

$p \Rightarrow (p \wedge q)$ simplifies

$\neg p \vee (p \wedge q) \equiv (\neg p \vee p) \wedge (\neg p \vee q)$

$\equiv T \wedge (\neg p \vee q)$

$\equiv \neg p \vee q \equiv p \Rightarrow q$

s not tautology

∴ **Answer D**

Explanation

p	q	$p \Rightarrow (p \wedge q)$
T	T	T
T	F	F
F	T	T
F	F	T

erse and contrapositive

If $p \Rightarrow q$ is the implication (direct proposition) then

i) $q \Rightarrow p$ is called its converse

ii) $\neg q \Rightarrow \neg p$ is called its contrapositive

iii) $\neg p \Rightarrow \neg q$ is called its inverse

Illustrative Example

For each statement, give its converse and contrapositive

a) If $x = 3$ then $x^2 - 4 = 5$

b) If x is an integer, then 2x is an even integer

c) Either $4 + 2 \neq 8$ or $5 + 2 = 6$

***Solution*:** a) i) Converse: If $x^2 - 4 = 5$ then $x = 3$

ii) Contrapostive: If $x^2 - 4 \neq 5$ then $x \neq 3$.

b) i) Converse: If 2x is an even integer then x is an integer

ii) Contrapostive: If 2x is not an even integer then x is not an integer

c) Recall $\neg p \vee q \equiv p \Rightarrow q$

i) implication (direct) If $4 + 2 = 8$ then $5 + 2 = 6$

ii) Converse: If $5 + 2 = 6$ then $4 + 2 = 8$

iii) Contrapostive: If $5 + 2 \neq 6$ then $4 + 2 \neq 8$

Quantifiers

- A quantifiers is a specific collection of numbers, objects, or persons.
- A quantifiers change an open statements to statements by replacing variable (s)
- There are two important ways of changing an open statement in to statement with out replacing values for the variables or pronouns. This is done by attaching one of the two symbols, known as quantifiers before an open statement. These two symbol are

i) $\exists$, called the existential quantifiers

The **notation** $\exists x$ read as:

- *There exists x*
- *For at least one x*
- *For some x*
- $(\exists x)P(x) \equiv$ There exist x in universe that satisfies property P.

ii) $\forall$, called the universal quantifiers

The notation $\forall x$ read as:

- *For all x*
- *For each x*
- *For every x*

$(\forall x)P(x) \equiv$ For all x in given universe that satisfies property P.

Example a) Let: $P(x)$: $x^2 - 4 = 0$, u = R, then i) $(\exists x)P(x)$ is true

ii) $(\forall x)P(x)$ is false,

Illustrative example

16. Determine the truth value of each statements let the universe U = Real number (R)

a) $(\exists x)(4x - 3 = -2x + 1)$
b) $(\exists x)(x^2 + 2x + 3 = 0)$
c) $(\forall x)(x^2 + 2x + 3 > 0)$
d) $(\forall x)(x^2 + 1 > 1)$
e) $(\exists x)(x^2 = -4)$
f) $(\exists x)(x^2 = 3)$
g) $(\forall x)(x^2 + 4x + 3 = 0)$
h) $(\exists x)(x^2 + 4x + 3 = 0)$
i) $(\forall x)(x^2 + 2x + 3 \neq 0)$
j) $(\exists x)(x^2 > 0)$

Solution: U = R

a) Let $x = \frac{2}{3}$, then $\frac{8}{3} - 3 = \frac{-4}{3} + 1 \therefore (\exists x)(4x - 3 = -2x + 1)$ is True

b) $x \in R$, then for each x, $x^2 + 2x + 3 \neq 0$,
$\therefore (\exists x)(x^2 + 2x + 3 = 0)$ is F

c) For each real number x, $x^2 + 2x + 3 > 0$, $\therefore (\forall x)(x^2 + 2x + 3 > 0)$ is T

d) Let $x = 0$, $0^2 + 1 = 1$. $\therefore (\forall x)(x^2 + 1 > 1)$ is false.

e) No real number x, that satisfies, $x^2 = -4$
$\therefore (\exists x)(x^2 = -4)$ is false, But if $x \in$ complex number then $(\exists x)(x^2 = -4)$ is true.

f) Let $x = \sqrt{3}$, $\Rightarrow \sqrt{3}^2 = 3 \therefore (\exists x)(x^2 = 3)$ is True.

g) Let $x = 1 \Rightarrow 1^2 + 4(1) + 3 \neq 0 \therefore (\forall x)(x^2 + 4x + 3 = 0)$ is False

h) Let $x = -1$, so that $(-1)^2 + 4(-1) + 3 = 0$ $(\exists x)(x^2 + 4x + 3 = 0)$ is T

i) For each x, $x^2 + 2x + 3 \neq 0 \therefore (\forall x)(x^2 + 2x + 3 \neq 0)$ is True

j) Let $x = 0$ so that $0^2 = 0 \therefore (\forall x)(x^2 > 0)$ is False
But $(\exists x)(x^2 > 0)$ is true

17. Let U = The set of integers.
Let, $P(x) \equiv x$ is prime, $e(x) \equiv x$ is even and $R(x) \equiv x$ is odd.
Determine the truth value of each statement.

a) $(\exists x)[P(x) \wedge R(x)]$
b) $(\forall x)[P(x \wedge R(x)]$
c) $(\exists x)[P(x) \Rightarrow e(x)]$
d) $(\forall x)[P(x \Rightarrow e(x)]$
e) $\neg[(\exists x)(P(x) \Rightarrow e(x)]$
f) $\neg[(\forall x)(P(x) \Rightarrow e(x)]$
g) $(\forall x)[(P(x) \wedge R(x)) \Rightarrow e(x)]$
h) $(\forall x)[P(x) \vee R(x)]$

Solution:

a) Let $x = 3$, then $P(3) \wedge R(3)$ is true
$\therefore (\exists x)[P(x) \wedge R(x)]$ is true.

b) Let $x = 9$, then $P(9) \wedge R(9)$ is False
$\therefore (\forall x)[P(x \wedge R(x)]$ is False

c) Let $x = 2$, then $P(2) \Rightarrow e(2)$ is true.
$\therefore (\exists x)[P(x \Rightarrow e(x)]$ is true

d) Let $x = 3$, as counter example, then $P(3) \Rightarrow e(3)$ is false
therefore, $(\forall x)[P(x) \Rightarrow e(x)]$ is false

e) negation of true is false from (c), therefore,
$\neg[(\exists x)(P(x \Rightarrow e(x)]$ is false
Note: $\neg[(\exists x)(P(x \Rightarrow e(x)] \equiv (\forall x)(P(x) \wedge \neg e(x)]$

f) Negation of false is true from (d)
$\therefore [(\forall x)(P(x) \Rightarrow e(x)]$ is true
Note: $\neg[\forall x)(P(x) \Rightarrow e(x)] \equiv (\exists x)(P(x) \wedge \neg e(x)]$

g) Let $x = 3$, as counter example, then
$P(3) \wedge R(3) \Rightarrow e(3)$ is false
$\therefore (\forall x)[(P(x) \wedge R(x) \Rightarrow e(x)]$ is false

h) $(\forall x)[P(x) \vee R(x)]$ is true

Relation ship between quantifiers

i) $\neg(\exists x)P(x) \equiv (\forall x)\neg P(x)$.
ii) $\neg(\forall x)P(x) \equiv (\exists x)\neg P(x)$.
iii) $\neg[(\forall x)(P(x) \Rightarrow R(x))] \equiv (\exists x)(P(x) \wedge \neg R(x))$.
iv) $\neg[(\exists x)(P(x) \Rightarrow R(x)] \equiv (\forall x)(P(x) \wedge \neg R(x))$.

18. Give the negation of each statement and determine the truth values for each part.

a) $(\exists x)(4x - 2 = -3x + 12)$
b) $(\forall x)(x^2 + 1 = 0)$
c) $(\exists x)(x^2 + 1 \leq 0)$
d) $(\forall x)(x^2 > 0)$
e) $(\forall x)(x^2 - 1 > 0)$

Solution:

a) $\neg (\exists x)(4x-2= -3x+12) \equiv (\forall x)(4x - 2 \neq -3x + 12)$ is false

b) $\neg(\forall x)(x^2 + 1 = 0) \equiv (\exists x)(x^2 + 1 \neq 0)$ is false
c) $\neg(\exists x)(x^2 + 1 \leq 0) \equiv (\forall x)(x^2 + 1 > 0)$ is true
d) $\neg(\forall x)(x^2 > 0) \equiv (\exists x)(x^2 \leq 0)$ is true, let $x = 0$
e) $\neg(\forall x)(x^2 - 1 > 0) \equiv (\exists x)(x^2 - 1 \leq 0)$ is true, let $x = 0$

19. Suppose $e(x) \equiv x$ is even
$p(x) \equiv x$ is prime
$d(x) \equiv x$ is divisible by 2.
Which one of the following has the truth value "F" on the set of natural numbers? ..... UEE 2004/12
A) $(\exists x)[e(x) \wedge P(x)]$ C) $(\exists x)[e(x) \wedge \neg d(x)]$
B) $(\forall x)[e(x) \Rightarrow d(x)]$ D) $(\forall x)[e(x) \vee \neg d(x)]$
Solution: A) Let $x = 2$, then $e(2) \wedge p(2) \equiv$ True
B) True
C) $\neg(\forall(x)(e(x) \Rightarrow d(x))) \equiv (\exists x)(e(x) \wedge \neg d(x)) \equiv$ False
D) For each x, $e(x) \vee \neg d(x)$ is true.
Answer: C

20. Let $P(x)$: $x^2 + x$ is positive. Which of the following is equivalent to $\neg(\exists x)p(x)$? .....EHEECE 2002
A. $(\exists x) \neg (x^2 + x > 0)$ C) $(\exists x)(x^2 + x \leq 0)$
B) $(\forall x)(x^2 + x < 0)$ D) $(\forall x)(x^2 + x \leq 0)$
Solution: $\neg(\exists x)(x^2 + x > 0) \equiv (\forall x)(x^2 + x \leq 0)$

Answer: D

21. Let $p(x) \equiv x$ is positive, $Q(x) \equiv x > 5$ then the negation of $(\forall x)(Qx) \Rightarrow P(x)$ ...EHEECE.
A. There is x such that $x \leq 5$ and x is positive.
B. There is x such that $x > 5$ and x is not positive.
C. There is x such that $x \leq 5$ and x is not positive.
D. There is x such that $x > 5$ and x is positive.
Solution: $\neg(\forall x)(Q(x) \Rightarrow P(x)) \equiv (\exists x)(Q(x) \wedge \neg P(x))$
$\therefore$ Answer B

22. If $(P \wedge Q) \Rightarrow (\neg S \vee R)$ is false which of the following is true ....EHEECE.
A. $S \Rightarrow R$ B. $\neg p \vee \neg S$ C. $(p \wedge \neg Q) \Leftrightarrow R$ D. $P \wedge R$
Solution: $(P \wedge Q) \Rightarrow (\neg S \vee R) \equiv F$ so that
$\neg S \vee R \equiv F$ and $(P \wedge Q) \equiv T$
$R \equiv F$, $\neg S \equiv T$, and $P \equiv T$, $Q \equiv T$
Thus,
A) $S \Rightarrow R$ B) $\neg P \vee \neg S$ C) $(P \wedge \neg Q) \Leftrightarrow R$ D) $T \wedge F \equiv F$
$T \Rightarrow F \equiv T$ $F \vee F \equiv F$ $(T \wedge F) \Leftrightarrow F \equiv T$

Answer:

23. Which of the following propositions is equivalent $p \vee q$?...EHEECE 98/06.
A. $\neg p \Rightarrow q$ B. $\neg(\neg p \vee \neg q)$ C) $\neg(p \wedge q)$ D) $p \Rightarrow \neg q$

Solution: A) $\neg p \Rightarrow q \equiv \neg(\neg p) \vee q) \equiv p \vee q$ ∴ Answer A

If $\neg p \vee Q \Rightarrow (R \wedge \neg R)$ is true then which of the following is necessarily true? .....UEE 2003/11.

A. $R \Rightarrow Q$, B. $(Q \Rightarrow P) \wedge R$ C. $P \wedge \neg R$ D. $(Q \wedge R) \Rightarrow P$

Solution: $R \wedge \neg R \equiv F$, so that $\neg P \vee Q \equiv F$,

$\neg P \equiv F$, $P \equiv T$, $Q \equiv F$, since P is true

$\therefore (Q \wedge R) \Rightarrow P \equiv T$ ∴ Answer D

Which of the following is correct statement about the proposition $(p \Rightarrow q) \Rightarrow (\neg p \vee q)$? The proposition

A. is true only when q is true.
B. is false when q is false
C. is true for all possible combination of the truth value of p and q.
D. has the same truth value as that of q.

Solution: using truth table

p	q	$p \Rightarrow q$	$\neg p \vee q$	$(p \Rightarrow q) \Rightarrow (\neg p \vee q)$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	T
F	T	T	T	T
F	F	T	T	T

Thus, $(p \Rightarrow q) \Rightarrow (\neg p \vee q) \equiv T$ for all cases; ∴ Answer C

Quantifiers occurring in combination when two variables are involved.

Let P(x, y): $x + y = 6$ ⟵ an open statement you can change these open statement by attaching the following four cases.

i) $(\exists x)(\exists y)p(x, y)$ ≡ There is some x and some y that satisfies the relation P(x, y)

ii) $(\exists x)(\forall y)p(x, y)$ ≡ There is fixed x which stands to every y in the relation p(x, y)

iii) $(\forall x)(\exists y)p(x, y)$ ≡ For every x there is y such that p(x, y) holds.

iv) $(\forall x)(\forall y)p(x,y)$≡ Every x and every y satisfies the relation p(x, y)

Illustrative Example

Let universe, U = the set of real number

Let P(x, y): $x + y = 8$, then determine the truth value of the following.

a) $(\exists x)(\exists y)p(x, y)$ c) $(\forall x)(\exists y)p(x, y)$

b) $(\exists x)(\forall y)p(x, y)$ d) $(\forall x)(\forall y)p(x, y)$

Solution:

a) $(\exists x)(\exists y)p(x, y)$ means there is real number x and y that satisfies $x + y = 8$.
Let $x = 3$, and $y = 5$
$\therefore (\exists x)(\exists y)p(x, y)$ is true.

b) $(\exists x)(\forall y)p(x, y)$ means there is fixed x which standard for every y that satisfy $x + y = 8$ is false. Let $x = 2$, then $(\forall y)(2 + y = 8)$ is false

c) $(\forall x)(\exists y)p(x, y)$ means, for every x, we can find corresponding y value that satisfies $x + y = 8$ is true.
$x + y = 8$ is true
i.e. Let $x = 3$, then $y = 5$
Let $x = 0$, then $y = 8$
$\therefore (\forall x)(\exists y)p(x, y)$ is true

d) $(\forall x)(\forall y)p(x, y)$ is false, let $x = 1$, $y = 5 \Rightarrow 1 + 5 = 6 \neq 8$

27. Let $p(x, y)$: $x < y$ and $T(x, y)$: $y + x = y$ then determine the truth value of each of the following, if the universe is the set of real numbers:

a) $(\exists x)(\exists y)p(x, y)$
b) $(\exists x)(\forall y)p(x, y)$
c) $(\forall x)(\exists y)p(x, y)$
d) $(\exists x)(\exists x)T(x, y)$
e) $(\exists x)(\forall y)T(x, y)$
f) $(\forall x)(\exists y)T(x, y)$

Solution:

a) $(\exists x)(\exists y)(x < y)$ True

b) $(\exists x)(\forall y)p(x, y)$ means there is fixed x which is less than every real number y is false

c) $(\forall x)(\exists y)p(x, y)$ means, for every x, there is y such that $x < y$ is true.

d) $(\exists x)(\exists x)T(x, y)$ is true

e) $(\exists x)(\forall y)(y + x = y)$ is true
$(\exists x)(\forall y)T(x, y)$ means there is fixed x which is standard for all y, let $x = 0$.

f) $(\forall x)(\exists y)(y + x = y)$ is false
$(\forall x)(\exists y)T(x, y)$ means for every x, we can find y.
Let $x = 3 \Rightarrow y + 3. = y \Rightarrow 3 = 0$ false

28. Let U = the set of real number. Determine the truth value of each of the following.

A. $(\exists y)(\forall x)(x + y = 0)$
B. $(\forall x)(\exists y)(x + y = 0)$
C. $(\forall x \neq 0)(\exists y)(xy = 1)$
D. $(\exists x)(\forall y)(xy = 0)$
E) $(\exists x)(\forall y)(xy = y)$
F) $(\exists x)(\exists y)(x+y = 5$ and $x-y=1)$

Solution:

a) $(\exists y)(\forall x)(x + y = 0)$ is false

b) $(\forall x)(\exists y)(x + y = 0)$ is true for every x, there is y, such that $x + y = 0$

c) $(\forall x \neq 0)(\exists y)(xy = 1)$ is true

d) $(\exists x)(\forall y)(xy = 0)$ is true there is fixed x which is true for all y, let $x = 0$

e) $(\exists x)(\forall y)(xy=y)$ is true. Let $x = 1$, then it is true for all y

f) $(\exists x)(\exists y)(x+y=5$ and $x - y = 1)$ is true, let $x = 3$, $y = 2$

29. Which of the following is a true statement on the set of real number.

A) $(\forall x)(\exists y)(y = x + 5)$ C) $(\exists x)(\forall y)(x = y + 3)$

B) $(\exists y)(x^2 + 1 = 0)$ D) $(\forall x)(\forall y)\ (x \geq y) \wedge (y \neq x)$

Solution: $(\forall x)(\exists y)(y = x + 5)$ is true **Answer: A**

30. If x and y are non- negative integers which of the following is not rue? .... UEE 2004/12

A) $(\forall x)(\exists y)(y > x^2 - 1)$ C) $(\exists y)(\forall x)(y \leq x^2 - 1)$

B) $(\exists x)(\forall y)(y > x^2 - 1)$ D) $(\exists y)(\exists x)\ (y \leq x^2 - 1)$

Solution: A) Let $x = 2, \Rightarrow y > 4 - 1 \Rightarrow y > 3$

$\therefore (\forall x)(\exists y)(y > x^2 - 1)$ is true

B) $(\exists x)(\forall y)(y > x^2 - 1)$ means there is fixed x such that $y > x^2 - 1$ which is less than every y is false.

C) $(\exists y)(\forall x)(y \leq x^2 - 1)$ is true. There is fixed y which is less than every x such that $y \leq x^2 - 1$ is $y = -1$

D) $(\exists y)(\exists x)\ (y \leq x^2 - 1)$ is true

Arguments And Validity

- As introduction to this section, we consider the following example.

  i) If p is true and $p \Rightarrow q$ is true.
  What can be concluded about q?

  ii) If p and $p \wedge q$ is true,
  What can be concluded about q?

  i) $p \equiv T$, $p \Rightarrow q \equiv T$, $\therefore q \equiv T$

  ii) $p \equiv T$, $p \wedge q \equiv T$, $\therefore q \equiv T$

- In the above example, the statement, p, $p \Rightarrow q$ are called **premises or hypothesis** and q is called **conclusion.**

A logical deduction (argument form) is an assertion that a given statements, $P_1, P_2, P_3, \ldots P_n$ called **hypothesis** or **premises** yield another statement Q called the conclusion. Such logical deduction is denoted by $P_1, P_2, P_3, \ldots P_n \vdash Q$.

In the above example, the argument form is denoted by

i) p *the same as* $p, p \Rightarrow q \vdash q$ ii) $\dfrac{p \wedge q}{q}$ *the same as* $p \wedge q \vdash$

$$\frac{p \Rightarrow q}{q}$$

An argument from $P_1, P_2, P_3, \ldots P_n \vdash Q$ is said to be
i) Valid if Q is true, when ever all the premise $P_1, P_2, \ldots P_n$ are true.
ii) Invalid if Q is False, when ever all the premise $P_1, P_2, \ldots P_n$ are tru

Illustrative Example

30. Decide whether each of the following argument forms is valid invalid.

a) $\neg p \Rightarrow \neg q$

b) $p \Rightarrow q$

c) $p, \neg p \Rightarrow q \vdash q$

d) $p \Rightarrow \neg q, p, r \Rightarrow q \vdash \neg r$ $\dfrac{q}{p}$

e) $\neg q \Rightarrow \neg p, r \Rightarrow p, \neg q \vdash r$ $\dfrac{\neg q \Rightarrow r}{p}$

f) $(\neg r \wedge \neg s), (\neg s, \Rightarrow p) \Rightarrow r \vdash$

Solution: • Start from simple statement
• Premise is always true.

a) 1) $q \equiv T \leftarrow$ premise
2) $\neg p \Rightarrow \neg q \equiv T \leftarrow$ premise
3) $\neg q \equiv F$ and $\neg p \equiv F$ from (1) and (2)
4) $p \equiv T \leftarrow$ from 3.
$\therefore$ Argument form is in valid

b) By using truth table $p \Rightarrow q, \neg q \Rightarrow r \vdash p$

Row	p	q	r	$\neg p$	$p \Rightarrow q$	$\neg q \Rightarrow r$
1	T	T	T	F	T	T
2	T	T	F	F	T	T
3	T	F	T	T	F	T
4	T	F	F	T	F	F
5	F	T	T	F	T	T
6	F	T	F	F	T	T
7	F	F	T	T	T	T
8	F	F	F	T	T	F

the table in row 5, row 6 and row 7, the premise $p \Rightarrow q$, $\neg q \Rightarrow r$ are true. But the conclusion P is false.

∴ Argument form is in valid due to Row 5, Row 6 and Row 7.

By using truth table $p, \neg p \Rightarrow q \vdash q$

Row	p	q	$\neg p$	$\neg p \Rightarrow q$
	T	T	F	T
	T	F	F	T
	F	T	T	T
	F	F	T	F

From the table, the premise P, and $\neg p \Rightarrow q$ are true in the row 1 and row 2. But the conclusion q is false in row 2.

∴ Argument form is invalid

$p \Rightarrow \neg q, p, r \Rightarrow q \vdash \neg r$, start from simple statement

1. $p \equiv T$ ⟵ premise
2. $p \Rightarrow \neg q \equiv T$ ⟵ premise
3. $\neg q \equiv T$ ⟵ from (2), ∴ $q \equiv F$
4. $r \Rightarrow q \equiv T$ ⟵ premise
5. $q \equiv F, r \equiv F$ ⟵ from (4), definition ($\Rightarrow$)
6. $\neg r \equiv T$ ⟵ from (5)

∴ Argument form is valid

$\neg q \Rightarrow \neg p, r \Rightarrow p, \neg q \vdash r$

start from simple statement

1. $\neg q \equiv T$ ⟵ premise
2. $\neg q \Rightarrow \neg p \equiv T$ ⟵ premise
3. $\neg p \equiv T$, and $p \equiv F$ ⟵ from (2)
4. $r \Rightarrow p \equiv T$ ⟵ premise
5. $p \equiv F, r \equiv F$ ⟵ from (4), and definition ($\Rightarrow$)

∴ Argument form is invalid

$(\neg r \wedge \neg s), \neg(s \Rightarrow p) \Rightarrow r \vdash \neg p$

1. $\neg r \wedge \neg s \equiv T$ ⟵ premise
2. $\neg r \equiv T, r \equiv F$ and $\neg s \equiv T$ and $S \equiv F$
3. $(\neg s \Rightarrow p) \Rightarrow r) \equiv T$ ⟵ premise
4. $p \equiv F, \neg p \equiv T$

∴ Argument form is invalid

Prove the following are valid arguments by direct Method.

If the rain does not come, the crops are ruined and the people will starve. The crops are not ruined or the people will not starve. Therefore, the rain comes $\neg P \Rightarrow q$

Solution: p ≡ The rain comes

q ≡ The crops are ruined

r ≡ The people will starve

The argument forms can be write in symbols

$\neg p \Rightarrow (q \wedge r), \neg q \vee \neg r \vdash p$

1. $\neg q \vee \neg r \equiv T$ ⟵ premise
2. $\neg(q \wedge r) \equiv T$ ⟵ from (1) and Demorgan's law
3. $q \wedge r \equiv F$ ⟵ from (2) and rule of $(\neg)$
4. $\neg p \Rightarrow (q \wedge r) \equiv T$ ⟵ premise
5. $\neg p \equiv F$ ⟵ from (4) and rule of $(\Rightarrow)$
6. $p \equiv T$ ⟵ from (5)

∴ Argument form is valid

32. If the team is late, them it can not play the game. If the referee is here, then the team can play the game. The team is late. Therefore, the referee is not here

Solution: Let p ≡ The team is late

q ≡ It can play the game

r ≡ The referee is here

The argument form write as

$p \Rightarrow \neg q, r \Rightarrow q, p \vdash \neg r$

1. $p \equiv T$ ⟵ premise
2. $p \Rightarrow \neg q \equiv T$ ⟵ premise
3. $\neg q \equiv T$ ⟵ from (1) and (2)
4. $q \equiv F$ ⟵ from (3)
5. $r \Rightarrow q \equiv T$ ⟵ premise
6. $r \equiv F$ ⟵ from (5)
7. $\neg r \equiv T$ ⟵ from (1) and rule $(\neg)$

∴ Argument form is valid

Rule of Inference

Here are certain valid deductions called rules of inference

1. **Modes ponens:** $p, p \Rightarrow Q \vdash Q$
2. **Modes Tollens:** $\neg Q, p \Rightarrow Q \vdash \neg p$
3. **Principle of syllogism:** $p \Rightarrow Q, Q \Rightarrow R \vdash p \Rightarrow R$
4. **Principle of Detachment:** $p \wedge Q \vdash p, Q.$
5. **Modes Tollendo ponens:** $\neg p, p \vee Q \vdash Q$
6. **Modes ponendo Tollens:** $\neg(p \wedge Q), p \vdash \neg Q$
7. **Principle of equivalence:** $p \Leftrightarrow Q, p \vdash Q$

33. Investigate the validity of the following. If he works hard, he will be successful. If he is successful then he will be happy. Therefore hard work leads to happiness.

Solution: Let p ≡ he works hard

q ≡ he will be successful

r ≡ he will be happy.

Then, the argument form write as symbol $p \Rightarrow q, q \Rightarrow r \vdash p \Rightarrow r$

By the rule of inference (syllogism). It is valid

4. Decide the validity of each of the following

a) $\neg p \vee q, r \Rightarrow p, r \vdash q$

b) $p \Rightarrow q, \neg r \Rightarrow \neg q \vdash \neg r \Rightarrow \neg p$

c) $p \Rightarrow (q \vee r), \neg r, p \vdash q$

Solution: Start from the simple statement.

a)
1. $r \equiv T \longleftarrow$ premise
2. $r \Rightarrow p \equiv T \longleftarrow$ premise
3. $p \equiv T \longleftarrow$ from (2) and rule of $(\Rightarrow)$
4. $\neg p \vee q \equiv T \longleftarrow$ premise
5. $\neg p \equiv F \longleftarrow$ from (3) and then (4)
6. $q \equiv T \longleftarrow$ from (4) and rule of $(\vee)$

$\therefore$ Argument form is valid

b)
1. $p \Rightarrow q \equiv T \longleftarrow$ premise
2. $q \Rightarrow r \equiv T \longleftarrow$ premise and contrapoistive of $\neg r \Rightarrow \neg q$
3. $p \Rightarrow q, q \Rightarrow r \vdash p \Rightarrow r \longleftarrow$ principle of syllogism

$\therefore$ Argument form is valid

c)
1. $\neg r \equiv T \longleftarrow$ premise
2. $r \equiv F \longleftarrow$ from (1)
3. $p \equiv T \longleftarrow$ premise
4. $p \Rightarrow (q \vee r) \equiv T \longleftarrow$ premise
5. $p \equiv T, q \equiv T \longleftarrow$ from (3), (4)

5. If 6 is even, then 2 does not divide 7. either 5 is not prime or 2 divide 7. But 5 is prime. Therefore, 6 is not even.

Solution: Let $p \equiv 6$ is even

$q \equiv 2$ divides 7

$r \equiv 5$ is prime.

Hence, the argument form is

$p \Rightarrow \neg q, \neg r \vee q, r \vdash \neg p$

1. $r \equiv T \longleftarrow$ premise
2. $\neg r \equiv F \longleftarrow$ from (1) and rule $(\neg)$
3. $\neg r \vee q \equiv T \longleftarrow$ premise
4. $q \equiv T \longleftarrow$ from (3)
5. $p \Rightarrow \neg q \equiv T \longleftarrow$ premise
6. $p \equiv F \longleftarrow$ from (5) and rule $(\Rightarrow)$
7. $\neg p \equiv T$

$\therefore$ Argument form is Valid

36. Let p ≡ I pass the examination
q ≡ I study hard
Which of the following represents the argument "I pass the examination if I study hard. I don't study hard. Therefore, I pass the examination?" …. (EHEECE)

A. $p \Rightarrow q, \neg q \vdash p$
B. $p \Rightarrow q, q \vdash p$
c) $q \Rightarrow p, \neg q \vdash p$
d) $q \Rightarrow p, \neg q \vdash \neg p$

Solution: $q \Rightarrow p, \neg q \vdash p$ ∴ **Answer: C**

37. Consider the following argument form. Production is high if rain continues. Rain does not continue.
Therefore, either production is low or rain continues.
Let p ≡ production is low q ≡ rain continues
The following table is also given about p and q.

Row	P	q	¬p	¬q	¬q ⇒ q	q ⇒ ¬p	p ∨ q
1	T	T	F	F	T	F	T
2	T	F	F	T	T	T	T
3	F	T	T	F	T	T	T
4	F	F	T	T	F	T	F

Which of the following is necessarily true …. UEE
A. The argument form is valid due to row 2.
B. The argument form is valid due to row 2 and 3.
C. The argument form is invalid due to row 4.
D. The argument form is invalid due to row 1 and row 3.

Solution: The argument $q \Rightarrow \neg p, \neg q \vdash p \vee q$.
The premise $q \Rightarrow \neg p, \neg q$ are true in row 2 and row 3, but the conclusion $p \vee q$ is false in row 4
∴ Argument form is invalid due to row 4, **Answer: C**

Exercise

1. Let p ≡ F, q ≡ F, and r ≡ T respectively. Determine the truth value for each of the following.
   a) $(p \wedge \neg r) \Rightarrow \neg q$
   b) $p \Rightarrow (q \vee r)$
   c) $(p \wedge q) \Rightarrow \neg(q \vee r)$
   d) $[(p \vee q)] \Leftrightarrow [(\neg p) \wedge (\neg q)]$
   e) $[p \Rightarrow (q \vee r)] \Rightarrow [(p \wedge \neg q) \Rightarrow r]$
2. Find the negation for each of the following
   a) some students do not study their lesion
   b) $3 + 6 \leq 7$ or $4 - 1 = 5$
   c) If $5 < 3$ then $-3 < -5$
   d) $3 < 1$, if and only if $3 - 1$
   e) You will study or else you will pass your examination
   f) You will study if and only if you will pass your examination
   g) All Ethiopians are honest

3. For each statement, give its **inverse, converse, contrapositive**
   a) Either a number is not prime or it is odd
   b) If $5 < 3$ then $-3 < -5$
   c) If there are no cloud, then it is not raining
   d) Items are available in necessary for price not to be high.
   e) It is sufficient that every Ethiopia works hard in order for poverty to be eradicated.

Determine whether the following are tautologies, contradictions or neither.

a) $p \Rightarrow (p \wedge q)$ e) $[p \Rightarrow (q \vee r)] \Rightarrow [(p \wedge q) \Rightarrow r]$
b) $p \Rightarrow (p \vee q)$ f) $p \vee \neg(p \wedge q)$
c) $(p \vee q) \Rightarrow q$ g) $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$
d) $(p \wedge q) \Rightarrow q$ h) $[(p \Rightarrow q) \wedge (q \Rightarrow r)] \Rightarrow (\neg r \Rightarrow \neg p)$

Determine the validity of the following

i) a) $p \Rightarrow q, \neg q \vdash \neg p$ d) $\neg p, p \Leftrightarrow q \vdash \neg(p \wedge q)$
   b) $\neg p \Rightarrow q, p \vdash \neg p$ e) $p \Rightarrow \neg q, p, r \Rightarrow q \vdash \neg r$
   c) $p \Rightarrow \neg q, q \vdash \neg p$ f) $p \Rightarrow q, \neg r \Rightarrow \neg q \vdash p \Rightarrow r$

ii) a) If the rains does not come, the crops are ruined and the people will starve. The crops are not ruined or the people will not starve. Therefore the rain comes.
   b) If trees are cut soil erosion increase, soil erosion does not increase only if trees are planted. Therefore trees are cut.
   c) If it rains, I go to the cinema. I do not go to the cinema. Therefore, it does not rain.

Let $U = \{1, 2, 3, 4\}$
Let P(x): x is prime H(x): x is divisible by 2.
R(x): x is odd and prime Q(x): $x \leq 4$
Determine the truth – value of each of the following

a) $(\exists x)\, P(x)$
b) $(\exists x)[P(x) \wedge H(x)]$
c) $(\exists x)[P(x) \Rightarrow H(x)]$
d) $(\forall x)[R(x) \Rightarrow p(x)]$
e) $\neg[(\forall x)(P(x) \Rightarrow H(x)]$
f) $(\forall x)[P(x) \vee Q(x)] \Rightarrow (\exists x)[H(x) \Rightarrow R(x)]$

Simplify a) $\neg(p \Rightarrow q) \vee (p \Rightarrow q)$ c) $(\neg p \Rightarrow q) \wedge (p \vee \neg q)$
b) $(p \Leftrightarrow q) \wedge \neg(p \Leftrightarrow q)$

Statistics and probability

Statistics: is the science dealing with

- collecting data
- organizing data
- analysis and interpretation of quantitative data

Types of data

- Qualitative data
- Quantitative data

i) **Qualitative data:**

Not measurable, we can not expressed in number

Example: Color, ear lob attachment, sex, religion nationality, intere... beauty, honesty, intelligence, educational qualification....etc

ii) **Quantitative data:** It is measurable, we can expressed in number

Example: Age, weight, height, income, number of population, si... score in test.

iii) **Quantitative data:** is categorized in to two variable

These are:

a) Discrete variable: It is expressed or measured by whole number

Example: Number of student, Number of chair, Number of ho... Number of car

b) **Continuous variable:** It is expressed or measured ... fraction or decimal.

Example: height, weight, age, size, temperature, distance, income, ... in test

Frequency distribution method of data presentation

i) Discrete frequency distribution

List of all data values obtained with their respective frequencies

ii) Grouped frequency distribution

Used to present large amount of data in class.

Ungrouped data: These are data in their original form with out ... arranged or summarized

Grouped data:- data organized and summarized in freq... distribution are grouped data

The following are basic concept that used in frequency distribution

i) Cumulative frequency: is the sum of frequency of all the classes from the first to up to class.

Example: If f_1, f_2, f_3, f_4 are frequency of 4 classes, then the cumulative frequency of these classes are f_1, f_1+f_2, $f_1+f_2+f_3$ and $f_1 + f_2 + f_3 + f_4$ respectively

These is called "less than cumulative frequency.

If the process is done from the total frequency to back ward is called **more than frequency.**

ii) Class interval $= \dfrac{\text{(maximum value) - (minimum value)}}{\text{number of class required}}$

iii) Class limits is the lower and the upper elements in a given class

- Lower class limit is the first number in given class.
- Upper class limit is the second number in given class.

iv) Class mark $(x_c) = \dfrac{\text{(lower class limit)} + \text{(upper class limit)}}{2}$

v) class boundaries

Lower class boundary = (lower class limit) – (0.5)

Upper class boundary = (upper class limit) + (0.5)

Illustrative Example

1. The following are scores of 50 student in mathematics exam.

55	44	45	48	69	30	47	50	66	38	58	44	42
46	47	53	54	35	50	62	49	82	40	46	29	47
75	25	46	28	43	50	47	62	30	56	48	30	
49	39	53	70	46	32	61	48	35	60	51	42+	

Then

a) prepare a grouped frequency distribution using 6 classes
b) Find class interval
c) Determine the cumulative frequency distribution.
d) What is the upper class limit of 3rd class.
e) What is the lower class limit of 5th class
f) Find the cumulative frequency of 3rd class
g) How many students gets more than 54 marks
h) How many students gets less than 54 marks
i) Determine class boundaries
j) Determine class mark
k) Determine the correction factor

l) How many students gets more than 25

m) How many students gets more than 35

Solution: Here

b) class interval $= \dfrac{\text{(maximum value)} - \text{(minimum value)}}{\text{number of class required}}$

$$= \frac{82-25}{6} = 9.5 \approx 10$$

Score (x) (class limit)	Number of student (frequency)	Cumulative frequency		Class boundaries	Class mark (x_i)
		Less than cumulative frequency	More than cumulative frequency		
25-34	7	7	50	24.5-34.5	29.5
35-44	9	16	43	34.5-44.5	39.5
45-54	22	38	34	44.5-54.5	49.5
55-64	7	45	12	54.5-64.5	59.5
65-74	3	48	5	64.5-74.5	69.5
75-84	2	50	2	74.5-84.5	79.5

c) see the 3rd and 4th column

d) 54

e) 65

f) 38

g) 34 student get more than 54.

h) 38 student get less than 54.

i) see the 5th column.

j) see the 6th column.

k) correction factor $= \dfrac{35-34}{2} = 0.5$

l) 50 student gets more than 25.

m) 43 student gets more than 35.

Measures of Location
(Measures of central tendency)

It is typical representative of a set of data.

- The average of a given set of data is measure of central tendency (location)

There are types of averages in common use:

1. Mean 3. Mode 5. Decile
2. Median 4. Quartile 6. Percentile

i) $mean(\bar{x}) = \dfrac{x_1 + x_2 + x_3 + \ldots + x_n}{n} = \dfrac{\sum x_i}{n}$ ← mean of raw data

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$$= \frac{f_1x_1 + f_2x_2 + f_3x_3 + ... + f_nx_n}{f_1 + f_2 + f_3 + ... + f_n} = \frac{\sum f_ix_i}{f_i} \leftarrow \text{mean of discrete frequency distribution}$$

$$= \frac{f_1m_1 + f_2m_2 + f_3m_3 + ... + f_nm_n}{f_1 + f_2 + f_3 + ... + f_n} = \frac{\sum f_im_i}{f_i} \leftarrow \text{mean of grouped frequency distribution}$$

Where m is the class mark ($x_c = m$) ← class mark

ii) **Median** When observation are arranged in increasing or decreasing order of magnitude, the middle value is the median.

- **Median of Raw data is given by**
- $$M_d = \begin{cases} \left(\frac{n+1}{2}\right)^{th} item \text{ if } n \text{ is odd} \\ \frac{\left(\frac{n}{2}\right)^{th} item + \left(\frac{n}{2}+1\right)^{th} item}{2} \text{ item if n is even} \end{cases}$$

Where n is number of observation

- $$M_d = B_L + \left(\frac{\frac{n}{2} - cf_b}{f_c}\right) i \leftarrow$$ Median for grouped frequency distribution

Where, B_L = Lower class boundary of the median class

n = total frequency

Cf_b = cumulative frequency just before the median class

f_c = frequency of the median class.

i = the size of class interval.

Combined mean

If $\bar{x}_1, \bar{x}_2$ are mean of n_1, n_2 observation respectively then the combined mean ($\bar{x}$) is

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} \leftarrow \text{combined mean}$$

Illustrative example

2. One group of 50 student has a mean score 30 in exam. A second group of 100 student has a mean score of 40. find the mean of all the 150 student.

Solution: $n_1 = 50,\ \bar{x}_1 = 30,\quad n_2 = 100,\ \bar{x}_2 = 40$

$$\therefore\ \bar{x} = \frac{50\times 30 + 100\times 40}{50+100} = \frac{5500}{150} = 36.7$$

2. a) Find the mean and median of the number 8, 3, 5, 10, 12

Solution: $\bar{x} = \dfrac{8+3+5+10+12}{5} = \dfrac{35}{5} = 7.6 \longleftarrow$ mean

To find the median first a write in increasing order, 3, 5, 8, 10, 12
Since n = 6 is even number

$$\therefore M_d = \frac{8+10}{2} = \frac{18}{2} = 9 \longleftarrow \text{median}$$

b) The following frequency distribution tables,
Find i) the mean ($\bar{x}$) ii) the median (M_d)

Mark	3-5	6-8	9-11	12-14	15-17	18-20
Number of student frequ.	4	3	6	4	2	1

Solution: Here

- To find the mean first determine the class mark (x_c)
- Class mark (x_c) $= \dfrac{3+5}{2} = 4,\ \dfrac{6+8}{2} = 7,\ \dfrac{9+11}{2} = 10, 13, 16, 19$
- To find the class boundaries, use correction factor,

  which is $\dfrac{6-5}{2} = 0.5 \longleftarrow$ correction factor
  - $3 - 0.5 = 2.5$ and $5 + 0.5 = 5.5$
  - $fx_c \longleftarrow$ product of frequency and class mark.

  fx_c: $4 \times 4 = 16,\ 3 \times 7 = 21,\ 6 \times 10 = 60,$ .... Etc

Mark	Class	Class	Frequency	Cumulative

(class limit)	boundaries	mark(x_c)		frequency	
3-5	2.5-5.5	4	4	4	16
6-8	5.5-8.5	7	3	4+3 = 7	21
9-11	8.5-11.5	10	6	7+6=13	60
12-14	11.5-14.5	13	4	13+4 = 17	52
15-17	14.5-17.5	16	2	17+2=19	32
18-20	17.5-20.5	19	1	19+1 = 20	19
Total			$\sum f = 20$		$\sum fx_c = 200$

- $\sum f = 20$ ⟵ total frequency
- $\sum fx_c = 200$ ⟵ Multiplying x_c by its corresponding frequency and adding

i) $\therefore \bar{x} = \dfrac{\sum fx_c}{\sum f} = \dfrac{200}{20} = 10$ ⟵ mean

ii) To find the median of grouped frequency distribution use,

$$M_d = B_L + \left(\frac{\frac{n}{2} - cf_b}{f_c} \right) i$$

- Median class is $\left(\frac{n}{2}\right) = \left(\frac{20}{2}\right)^{th}$ item = 10^{th} item. It lies on 3^{rd} class (8.5 – 11.5)
- $B_L = 8.5$ ⟵ lower boundary of median class.
- $Cf_b = 7$ ⟵ cumulative frequency just before the median class
- $f_c = 6$ ⟵ frequency of the median class
- $i = 5.5 - 2.5 = 3$ ⟵ class interval width
- $n = 20$ ⟵ total frequency

$$\therefore M_d = 8.5 + \left(\frac{\frac{20}{2} - 7}{6} \right) 3 = 10 \longleftarrow \text{median}$$

c) The following frequency distribution tables

Find i) the mean
ii) the median iii) the mode

Daily income	6 – 10	11 – 15	16 – 20	21 – 25	26 – 30	31 – 35
Frequency	2	3	4	6	8	1

Solution: Here
First determine,

- class boundaries
- class mark
- cumulative frequency
- fx_c ⟵ (frequency)(class mark)

Class boundaries	5.5-10.5	10.5-15.5	15.5-20.5	20.5-25.5	25.5-30.5	30.5-35.5
Frequency	2	3	4	6	0	1
Class mark (x_c)	8	13	18	23	28	33
Cumulative frequency	2	5	9	15	15	16
fx_c	16	39	72	138	0	33

i) $$\therefore \bar{x} = \frac{\sum fx_c}{\sum f} = \frac{16+39+72+138+0+33}{2+3+4+6+0+1} = \frac{298}{16} = 18.6$$ ⟵ mean

ii) To find the median first determine the median class by

$$\left(\frac{n}{2}\right)^{th} = \left(\frac{16}{2}\right)^{th} \text{ item} = 8^{th} \text{ item}$$

Hence the median class lies on 3rd class. It is (15.5 – 20.5)
Thus,

- $B_L = 15.5$ ⟵ lower boundary of median class.
- $Cf_b = 5$ ⟵ cumulative frequency just before the median class
- $f_c = 4$ ⟵ frequency of the median class
- $i = 10.5 - 5.5 = 5$ ⟵ class interval width
- $n = 16$ ⟵ total frequency

$$\therefore M_d = 15.5 + \left(\frac{\frac{16}{2} - 5}{4} \right) 5 = 19.25 \longleftarrow \text{median}$$

Mode (m_0)

Definition: The mode is a value which occurs most frequently in a set of observation:

- The mode may not exist and even if it does not exist it may not be unique.

1. Find the mode of the following data.
   a) 2, 1, 0, 4, 3, 3, 2, 5, -8, 7, -2, 3
   b) 2, 5, 3, 1, 7, 8, 9, 2, 1, 4, 5, 3, 1, 2, 5, 4, 6, 0, 1, 2, 5
   c) -1, 4, 7, 6, 9, 1, 0, 5, 8

   Solution:

   a) mode is $\hat{x} = 3$
   b) modes are $\hat{x} = 1$, $\hat{x} = 2$, $\hat{x} = 5$. Here there are 3 modal value.
   c) mode does not exist.

Mode of grouped date

Mode of grouped data in given uniform class interval can be obtained by the formula

$$Mode = M_0 = B_L + \left(\frac{d_1}{d_1 + d_2} \right) i$$

Where: B_L = lower class boundary of Modal class
d_1 = the difference between the frequency of the modal class and the frequency of the preceding class (pre-modal class)
d_2 = the difference between the frequency of the modal class and the frequency of next class
i = size of the class interval

Illustrative Example

11) a) Find the mode.

Class Interval	Frequency
4 – 6	14
7 – 9	39

10 – 12	27
13 – 15	16

Solution: The modal class is the 2nd class because its frequency is the largest.

$B_L = 6.5, \quad d_1 = 39 - 14 = 25$

$d_2 = 39 - 27 = 12$

$i = 9 - 6 = 3$

$$\therefore m_0 = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i = 6.5 + \left(\frac{25}{25+12}\right) 3 = 8.52$$

b) Fnd the mode.

Marks	0 – 9	10 – 19	20 – 29	30 – 39	40 – 49
Frequency	10	16	25	18	15

Solution: The modal class is 3rd class because its frequency is the largest.

$\therefore B_L = 19.5, \quad d_1 = 25 - 16 = 9,$

$d_2 = 25 - 18 = 7, \quad i = 29 - 19 = 10$

$$\therefore m_0 = 19.5 + \left(\frac{9}{9+7}\right) 10 = 25.125$$

Quartiles, deciles and percentiles

- The median divides, the distribution into two equal parts.
  - The quartiles divides, the distribution in to four equal parts.
  - The deciles divides, the distribution into ten equal parts.
  - Percentiles in to 100 equal parts.

Note: Quartiles, deciles and percentiles measures of location

Quartiles, deciles and percentiles for un grouped data.

Quartiles are values that divides set of data into four equal parts. There are three quartiles, Q_1, Q_2 and Q_3.

To find quartiles for ungrouped data

- Arrange the data in increasing order
- If the number of observation is:

a) odd, $Q_k = \left(\frac{k(n+1)}{4}\right)^{th}$ **item b) Even** $Q_k = \frac{\left(\left(\frac{kn}{4}\right) + \left(\frac{kn}{4}+1\right)\right)}{2}$

Illustrative Example

12. Find Q_1, Q_2 and Q_3 for the following date. 3, 6, 10, 25, 19, 14, 17, 22, 26

Solution: Arranging in increasing order, we get 3, 6, 10, 14, 17, 19, 22, 25, 26

$$Q_1 = \left(\frac{1(9+1)}{4}\right)^{th} \text{ item} = (2.5)^{th} \text{ item}$$

here Q_1 lies half way between 2^{nd} and 3^{rd} item

$\therefore Q_1 = (2^{nd})$ item + ½(3^{rd} item - 2^{nd} item)

$= 6 + ½(10 - 6) = 6 + 2 = 8$

Or

$$Q_1 = \frac{6+10}{2} = 8$$

$$Q_2 = \left(\frac{2(9+1)}{4}\right)^{th} \text{ item} = 5^{th} \text{ item}$$

$\therefore Q_2 = 17 = \text{median}$

$$Q_3 = \left(\frac{3(9+1)}{4}\right)^{th} \text{ item} = 7.5^{th} \text{ item}$$

$$\therefore Q_3 = \frac{7^{th} \text{ item} + 8^{th} \text{ item}}{2} = \frac{22+25}{2} = 23.5$$

13. Find Q_1, Q_2 and Q_3

x	8	12	13	15	17	18	24
f	12	18	20	2	4	4	1

Solution: n = 61

$$Q_k = \left(\frac{k(n+1)}{4}\right)^{th} \text{ item} \Rightarrow Q_1 = \left(\frac{61+1}{4}\right)^{th} \text{ item} = (15.5)\text{th item}$$

$$= \frac{15^{th} \text{ item} + 16^{th} \text{ item}}{2} \quad = \frac{12+12}{2} = \frac{24}{2} = 12$$

$$Q_2 = \left(\frac{2(61+1)}{4}\right) = 31^{th} \text{ item} = 13 = \text{median}$$

$$Q_3 = \left(\frac{3(61+1)}{4}\right)^{th} \text{ item} = (46.5)^{th} \text{ item}$$

$$= \frac{46^{th} \ item + 47i^{th} \ item}{2} = \frac{13+13}{2} = 13$$

Deciles

Deciles are values that divide a set of data in to ten equal parts. There are $D_1, D_2, D_3, D_4, D_5, D_6, D_7, D_8, D_9$.

To find deciles for un grouped data

- Arrange the data in increasing order of magnitude
- If the number of observation is:

a) Odd, $D_i = \left(\frac{i(n+1)}{10}\right)^{th}$ item

b) even, $D_i = \left[\frac{\left(\frac{in}{10}\right)+\left(\frac{in}{10}+1\right)}{2}\right]^{th}$ item

Illustrative Example

14. Find D_2, D_3 and D_8 for each of the following data sets:

a) 18, 2, 4, 6, 10, 7, 9, and 11

Solution: Arranging in increasing order of magnitude, we get 2, 4, 6, 7, 9, 10, 11, 18

$$D_2 = \left(\frac{\frac{2(8)}{10}+\left(\frac{2(8)}{10}+1\right)}{2}\right)^{th} \text{ item} = (2.1)^{th} \text{ item}$$

Hence D_2 is between 2^{nd} and 3^{rd} item, i.e. $x_2 + 0.1(x_3 - x_2)$

$\therefore D_2 = 4 + 0.1(6-4) = 4 + 0.2 = 4.2$

$$D_3 = \left(\frac{\frac{3(8)}{10} + \left(\frac{3(8)}{10}+1\right)}{2}\right)^{th} \text{ item} = \left(\frac{2.4+3.4}{2}\right)^{th} \text{ item} = (2.9)^{th} \text{ item}$$

D_3 is between 2^{nd} item and 3^{rd} item i.e.

$x_2 + 0.9(x_3 - x_2)$ $\quad \therefore D_3 = 4 + 0.9(6-4) = 5.8$

$$D_8 = \left(\frac{\frac{8(8)}{10} + \left(\frac{8(8)}{10}+1\right)}{2}\right)^{th} \text{ item} = (6.9)^{th} \text{ item}$$

$\therefore D_8$ is between 6^{th} item and 7^{th} item

i.e. $x_6 + 0.9(x_7 - x_6)$

$\therefore D_8 = 10 + 0.9(11-10) = 10.9$

Percentiles

Data divided into 100 equal parts. There are $P_1, P_2, P_3 \ldots P_{99}$

Note: Percentiles are not the same as percentages.

To find percentiles

- Arrange the data in increasing order of value
- If the number of observation is

a) Odd, $P_t = \left(\frac{t(n+1)}{100}\right)^{th}$ item

b) Even, $P_t = \left(\frac{\left(\frac{tn}{100}\right) + \left(\frac{tn}{100}\right)+1}{2}\right)^{th}$ item

Illustrative example

15. Find P_{24}, P_{42} and P_{75} for the following data.

a) 23, 36, 40, 44, 29, 27, 19, 7, 3

$$P_{24} = \left(\frac{24(9+1)}{100}\right)ith \text{ item} = (2.4)^{th} \text{ item}$$

$\therefore P_{24}$ is between 2^{nd} and 3^{rd} item

i.e. $x_2 + 0.4(x_3 - x_2)$

$\therefore P_{24} = 7 + 0.4(19 - 7) = 11.8$

$$P_{42} = \left(\frac{42(9+1)}{100}\right)ith \text{ item} = 4.2 \text{ item}$$

Here, P_{42} lies between 4^{th} and 5^{th} item $= x_4 + 0.2\,(x_5 - x_4)$

$\therefore P_{42} = 23 + 0.2(27 - 23) = 23.8$

$$P_{75} = \left(\frac{75(9+1)}{100}\right)ith \text{ item} = (7.5)^{th} \text{ item}$$

Here, P_{75} lies between 7^{th} and 8^{th} item

i.e. $x_7 + 0.5(x_8 - x_7)$

$\therefore P_{75} = 36 + 0.5(40 - 36) = 38$

Quartiles, Deciles and percentiles for grouped data

Each formula for grouped data is also given as follow

1. Quartiles, $Q_k = B_L + \left[\frac{\frac{kn}{4} - cf_b}{f_c}\right]i$

Where, k = 1, 2, 3

B_L = lower class boundary of k^{th} quartile class

Cf_b = the cumulative frequency before the k^{th} quartile class

f_k = frequency in the k^{th} quartile class

i = the size of class

2. Deciles, $D_j = B_L + \left[\frac{\frac{jn}{10} - cf_b}{f_c}\right]i$

Where: f_c = frequency of j^{th} – deciles class

3. Percentiles, $P_t = B_L + \left[\dfrac{\frac{tn}{100} - cf_b}{f_c}\right] i$

Note, $Q_2 = D_5 = P_{50}$

16. Find Q_1, Q_3, D_4, D_8, P_{12}, P_{24}, P_{68} and P_{99} of each of the following data set.

Age	3 – 12	13 – 22	23 – 32	33 – 42	43 – 52
F	4	12	10	7	2

Solution: First, find the cum. frequency

Age	3 – 12	13 – 22	23 – 32	33 – 42	43 – 52
F	4	12	10	7	2
Cf	4	16	26	33	35

To find Q_1

$$Q_1 = \left(\frac{1\times 35}{4}\right)^{th} \text{ item} = (8.75) \text{ item}$$

It is found in 2nd class

$\therefore B_L = 12.5 \longleftarrow$ lower class boundary of 2nd class

$cf_b = 4 \longleftarrow$ cum.freq. before 2nd class of quartile

$f_c = 12 \longleftarrow$ freq. of 2nd class

$i = 22 - 12 = 10 \longleftarrow$ class size

$$\therefore Q_1 = 12.5 + \left[\frac{\frac{35}{4} - 4}{12}\right](10) = 16.45$$

To find Q_3

$$Q_3 = \left(\frac{3\times 35}{4}\right) = (26.25)^{th} \text{ item, it is found in } 4^{th} \text{ class}$$

$\therefore B_L = 32.5 \longleftarrow$ lower class boundary of 4th class

$cf_b = 26 \longleftarrow$ cum.freq. before 4th class of quartile

$f_c = 7 \longleftarrow$ freq. of 4th class

$i = 10$

$$\therefore Q_3 = 32.5 + \left[\frac{\frac{3\times 35}{4} - 26}{7}\right](10) = 32.857$$

To find D_4

$$D_4 = \left(\frac{4\times 35}{10}\right)^{th} \text{ item} = (14)^{th} \text{ item}$$

It is found in 2^{nd} class

$\therefore B_L = 12.5 \longleftarrow$ lower class boundary of 2^{nd} class

$cf_b = 4 \longleftarrow$ cum.freq. before 2^{nd} class of decile

$f_c = 12 \longleftarrow$ freq. of 2^{nd} class

$i = 10 \longleftarrow$ class size

$$\therefore D_4 = 12.5 + \left[\frac{\frac{4\times 35}{10} - 4}{12}\right](10) = 20.83$$

To find D_8

$$D_8 = \left(\frac{8\times 35}{10}\right)^{th} \text{ item} = (28)^{th} \text{ item}$$

It is found in 4^{th} class

$\therefore B_L = 32.5 \longleftarrow$ lower class boundary of 4^{th} class

$cf_b = 26 \longleftarrow$ cum.freq. before 4^{th} class of decile

$f_c = 7 \longleftarrow$ freq. of 4^{th} class

$i = 10 \longleftarrow$ class size

$$\therefore D_8 = 32.5 + \left[\frac{\frac{8\times 35}{10} - 26}{7}\right](10) = 35.35$$

To find P_{12}

$$P_{12} = \left(\frac{12 \times 35}{100}\right) \text{ item} = (4.2) \text{ item}$$

It is found, in 2^{nd} class

$\therefore B_L = 12.5 \longleftarrow$ lower class boundary of 2^{nd} class
$cf_b = 4 \longleftarrow$ cum.freq. before 2^{nd} class of percentile
$f_c = 12 \longleftarrow$ freq. of 2^{nd} class

$$\therefore P_{12} = 12.5 + \left[\frac{\frac{12 \times 35}{100} - 4}{12}\right](10) = 12.66$$

To find P_{24}

$$P_{24} = \left(\frac{24 \times 35}{100}\right)^{th} \text{ item} = (8.4)^{th} \text{ item}$$

It is found in 2^{nd} class

$\therefore B_L = 12.5$

$$\therefore P_{24} = 12.5 + \left[\frac{\frac{24 \times 35}{100} - 4}{12}\right](10) = 16.16$$

To find P_{68}

$$P_{68} = \left(\frac{68 \times 35}{100}\right)^{th} \text{ item} = (23.8)^{th} \text{ item}$$

It is found in 3^{rd} class

$\therefore B_L = 22.5 \longleftarrow$ lower class boundary of 3^{rd} class
$cf_b = 16 \longleftarrow$ cum.freq. before 3^{rd} class of percentile
$f_c = 10 \longleftarrow$ freq. of 3^{rd} class

$$\therefore P_{68} = 22.5 + \left[\frac{\frac{68 \times 35}{100} - 16}{10}\right](10) = 30.3$$

To find P_{99}

$$P_{99} = \left(\frac{99 \times 35}{100}\right)^{th} \text{ item} = (34.65) \text{ item}$$

It is found in the 5th class.

$\therefore B_1 = 42.5$ ← lower class boundary of 5th class

$cf_b = 33$ ← cum.freq. before 5th class of percentile

$f_c = 2$ ← freq. of 5th class

$$\therefore P_{99} = 42.5 + \left[\frac{\frac{99\times 35}{100} - 33}{2}\right](10) = 50.75$$

17. Find $D_4, D_8, D_9, P_{12}, P_{24}, P_{70}$ for each of the following data set

x	12	16	17	19	21	22	28
f	12	18	20	2	4	4	1

Solution: $\Sigma f = n = 61$

$$D_k = \left(\frac{k(n+1)}{10}\right)^{th} \text{ item}$$

$$\therefore D_4 = \left(\frac{4(61+1)}{10}\right)^{th} \text{ item} = (24.8)^{\text{th}} \text{ item}$$

$\therefore D_4 = x_{24} + 0.8(x_{25} - x_{24}) = 16 + (0.8)(16 - 16) = 16$

$$D_8 = \left(\frac{8(61+1)}{10}\right)^{th} \text{ item} = (49.6)^{\text{th}} \text{ item}$$

$= x_{49} + 0.6(x_{50} - x_{49}) = 17 + 0.6(17 - 17) = 17$

$$D_9 = \left(\frac{9(61+1)}{10}\right)^{th} \text{ item} = (55.8)^{\text{th}} \text{ item}$$

$= x_{55} + 0.8(x_{56} - x_{55}) = 21 + 0.8(21 - 21) = 21$

$$P_{12} = \left(\frac{12(61+1)}{100}\right)^{th} \text{ item} = (7.44)^{\text{th}} \text{ item}$$

$= x_7 + 0.44(x_8 - x_7) = 12 + 0.44(12 - 12) = 12$

$$P_{24} = \left(\frac{24(61+1)}{100}\right)^{th} \text{ item} = (14.88)^{\text{th}} \text{ item} = 16$$

Application of Quantiles

(Quartiles, deciles and percentiles are called quantiles).

Note: Quartiles ← Q_1 (lower quartile), Q_2 (middle quartile)

Q_3 (upper quartile

Q_1(lower) quartile $\equiv \left(\frac{1}{4}\right)^{th}$ of the values are less than or equal to Q_1

and $\left(\frac{3}{4}\right)^{th}$ of the values greater than it.

Q_3 (upper) quartile $\equiv \left(\frac{3}{4}\right)^{th}$ of the values are less than or equal to Q_3

and $\left(\frac{1}{4}\right)^{th}$ of the value are greater than it. $Q_2 = D_5 = P_{50}$

Illustrative example

17. Find Q_1,Q_2 and Q_3 of the following data. It is marks obtained in a mathematics exam (out of 60)

Mark	30–34	35–39	40–44	45–49	50–54	55–59
Number of student	7	12	3	5	3	6
Cumlative frequency	7	19	22	27	30	36

a) If students in the top 25% are to be awarded certificate of "Best in math". What is the minimum mark for a certificate

b) If student whose scores are in the bottom 25% of the marks are considered as failures, then what is the maximum failing mark?

c) Find the percentile of student who get more than 50

d) If student get more than 75 percentile find the minimum mark he/she get.

e) Find the pass mark if 70% of the student to pass.

Solution: To find Q_1

$$Q_1 = \left(\frac{1\times 36}{4}\right)^{th} \text{ item} = 9^{th} \text{ item, which falls in } 2^{nd} \text{ class } (35-39)$$

$B_L = 34.5$ ⟵ Lower class boundary of 2^{nd} class.

$Cf_b = 7$ ⟵ the cum. Frequence before 2^{nd} class.

$f_k = 12$ ⟵ the number of frequence in the 2^{nd} class.

$i = 5$ ⟵ the size of class interval.

$$\therefore Q_1 = 34.5 + \left(\frac{\frac{1\times 36}{4} - 7}{12}\right)(5) = 35.3$$

- Interpretation

$Q_1 = 35.3$ means $\left(\frac{1}{4}\right)^{th}$ or 25% of the student gets less than or equal to 35.3 and $\left(\frac{3}{4}\right)^{th}$ or 75% of the student scores greater than 35.3

To find Q_3

$$Q_3 = \left(\frac{3\times 36}{4}\right)^{th} \text{ item} = 27^{th} \text{ item which falls in } 4^{th} \text{ class } (45-49)$$

$\therefore B_L = 44.5 \longleftarrow$ lower class bourdry of 3^{rd} class.
$Cf_b = 22 \longleftarrow$ the cumfreq. Before 3^{rd} class.
$f_k = 5 \longleftarrow$ the number of frequency in the 3^{rd} class.
$i = 5$

$$\therefore Q_3 = 44.5 + \left[\frac{27-22}{5}\right]5 = 49.5$$

- Interpretation

$Q_3 = 49.5$ means $\left(\frac{3}{4}\right)^{th}$ or 75% student gets less than or equal to 49.5 and $\left(\frac{1}{4}\right)^{th}$ or 25% of the student score greater than 49.5

a) $Q_3 = 49.5 \longleftarrow$ minimum mark to award certificate "Best in math"
b) $Q_1 = 35.3 \longleftarrow$ maximum failing mark
c) Let the percentile be P_k, Hence $P_k > 50$

$$P_k = B_L + \left(\frac{\frac{k\times 36}{100} - cf_b}{f_c}\right)(i) \Rightarrow 50 + \left(\frac{\frac{k\times 36}{100} - 27}{3}\right)5 > 50$$

$\Rightarrow (0.12k - 9)5 > 0 \Rightarrow 0.6k - 45 > 0 \Rightarrow 0.6k > 45$

$\therefore k > \frac{45}{0.6} = 75 \qquad \therefore P_{75}$

d) More than 75 percentile means

$$\Rightarrow P_{25} = 34.5 + \left[\frac{9-7}{12}\right]5 = 35.3$$

35.3 ⟵ the minimum mark he/she get.

e) If 70% is to pass then 30% is to fail

Therefore, answer is the 30^{th} percentile which is P_{30}

To find P_{30}

$$P_{30} = \left(\frac{30 \times 36}{100}\right)^{th} \text{ item} = (10.8)^{th} \text{ item}$$

It is found in 2^{nd} class (35 – 39)

- $B_L = 34.5$ ⟵ lower boundary of 2^{nd} class.
- $Cf_b = 7$ ⟵ cumulative frequency just before 2^{nd} class if P_{30}
- $f_c = 12$ ⟵ frequency of the 2^{nd} class of P_{30}
- $i = 5$ ⟵ class width

$$\therefore \quad P_{30} = 34.5 + \left(\frac{10.8-7}{12}\right)5 = 36.08 \leftarrow \text{minimum pass mark}$$

Measures of dispersion

The scatter or spread of the items of a distribution is known as a dispersion or variation. It is the degree to which numerical data tend to spread about an average value.

Measure of centeral tendency is not enough to describe the nature of the data. It is therefore better to use both central tendency and dispersion.

Consider the following data set

Group	Data value	Total	Mean	Variance
A	5,5,5,5,5	25	5	0
B	3,6,4,6,6,	25	5	2
C	2,5,8,0,10	25	5	17

From the table the means are identical but variation in the given set of data

∴ Measure of dispersion, helps us to know the degree of uniformity and consistency of the series.

- If the difference between the item is large the dispression also large and vice versa.
- If the difference between is very small, measure of central tendency (Average) represents and describes the data adequately.

The common measure of dispersions are

* Range * Variance * Standard deviation

Range

The difference between the two extreme values Range = $x_{max} - x_{min}$

- It depends on extreme value

For group data

Range is the diffence between the upper class bounds of the highest class and the lower class boundary of the lowest class.

Illustrative example

18) Find the range for the following table

x	6 – 11	12 – 17	18 – 23
f	5	6	9

Range = 23.5 – 5.5 = 18

Variance and standard deviation

They are the most commonly used measure of dispersion.

Definition: Variance is the mean of the squared deviation of each item from the mean.

- Standard deviation = $\sqrt{\text{var}iance}$
- Variance $(S^2) = \dfrac{(x_1 - \bar{x})^2 + (x^2 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2}{n}$

$$\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}$$

- Standard deviation $S = \sqrt{\dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}}$

For frequency distribution

- $\text{Variance} = \dfrac{\sum_{i=1}^{n} f_i\left(x_i - \bar{x}\right)^2}{\sum_{i=1}^{n} f_i}$

Illustrative example

19) Find the variance and standard deviation of the population function

a) 1,2,6,3

b)

x	3	4	5	6	7
f	2	1	3	3	1

Solution: $\bar{x} = \dfrac{1+2+6+3}{4} = 3$

a)

x	$x - \bar{x}$	$\left(x - \bar{x}\right)^2$
1	$1-3=-2$	4
2	$2-3=-1$	1
3	$3-3=0$	0
6	$6-3=3$	9

$\sum\left(x-\bar{x}\right)^2 = 4+1+0+9 = 14$

$\therefore \text{Variance } \left(\delta^2\right) = \dfrac{\sum\left(x-\bar{x}\right)^2}{n} = \dfrac{14}{4} = 3.5$

$\therefore \text{Standard deviation } \left(\delta^2\right) = \sqrt{3.5}$

b) $\text{Mean } \left(\bar{x}\right) = \dfrac{2\times3+1\times4+3\times5+3\times6+1\times7}{2+1+3+3+1} = \dfrac{6+4+15+18+7}{10}$

$= \dfrac{50}{10} = 5$

X	f	$x - \bar{x}$	$(x - \bar{x})^2$	$f(x - \bar{x})^2$
3	2	3–5= –2	4	8
4	1	4–5= –1	1	1
5	3	5–5= 0	0	0

6	3	6–5= 1	1	3
7	1	7 – 5 = 2	4	4

$$\therefore \text{Variance} = \frac{\sum f\left(x-\bar{x}\right)^2}{\sum f} = \frac{8+1+0+3+4}{2+1+3+3+1} = \frac{16}{10}$$

$\therefore$ Standard deviation = $\sqrt{1.6}$

20) Find the range, variance and standard deviation for each of the following data.

Class	0 – 4	5 – 9	10 – 14	15 – 19	20 – 24
Frequency	3	4	1	2	1

Solution:

- Range 24.5 – (-0.5) = 25

To find variance

Class	Freq	Class mark (x_e	fx_c	$x_c - \bar{x}$	$(x_c - \bar{x})^2$	$f(x_c - \bar{x})$
0 – 4	3	2	6	– 7	49	147
5 – 9	4	7	28	–2	4	16
10 – 14	1	12	12	0	0	0
15 – 19	2	17	34	8	64	128
20 – 24	1	22	22	13	169	169

- Mean $\left(\bar{x}\right) = \frac{\sum fx_c}{\sum f} = \frac{102}{11} \approx 9$

- Variance $\left(\delta^2\right) = \frac{\sum f\left(x_c - \bar{x}\right)^2}{\sum f} = \frac{460}{11} = 41.7$

Standard deviation = $\sqrt{41.7} \approx 6.4$

21) The sum of eight numbers is 48 and the sum of their squares is 296

Find a) mean b) variance c) standard deviation

Solution: mean $\left(\bar{x}\right) = \frac{48}{8} = 6$

$$\text{Variance} = \bar{x^2} - (\bar{x})^2$$

Variance $= \frac{296}{8} - 6^2 = 37 - 36 = 1$ S.D $= \sqrt{1} = 1$

If the standard deviation of $x_1, x_2, x_3, \ldots x_n$ is 5, then find standard deviation of $3x_1 + 2, 3x_2 + 2, 3x_3 + 2 \ldots$

Solution: Recall grade lesson

Note, Adding constant to a data new variance = old variance

- Multiplying constant k to data
- New variance = k^2 times old variance

$\therefore$ New variance $= (3)^2(25)$

$\therefore$ New S.D $= \sqrt{(3^2)(25)} = (3)(5) = 15$

23. If marks obtained by 9 students in two test (out of 10)

Test I	10	9	8	7	6	5	4	3	2
Test II	7	5	9	4	7	5	7	4	6

a) determine the range, variance and standard deviation of each test.

b) Which test has highest variation.

c) Which test has more consistence score

d) Which test has lesser consistence score

Solution:

- Range of test I, $10 - 2 = 8$
- Range of test II, $9 - 4 = 5$
- Mean of test I =

$$\frac{10+9+8+7+6+5+4+3+2}{9} = \frac{54}{9} = 6$$

- Mean of test II $= \frac{7+5+9+4+7+5+7+4+6}{9} = 6$
- Variance of test I =

$$\frac{(10-6)^2 + (9-6)^2 + (8-6)^2 + \ldots + (2-6)^2}{9} = 6.7$$

- Standard dev. of test I $= \sqrt{6.7} = 2.58$
- Variance of test II $= \frac{(7-6)^2 + (5-6)^2 + \ldots + (6-6)^2}{9}$

$$= \frac{1+1+9+4+1+1+1+4+0}{9} \approx 2.44$$

$\therefore$ standard dev. of test II $= \sqrt{2.44} \approx 1.56$

Hence:

a)

	Range	Variance	S.D
Test I	8	6.7	2.58
Test II	5	2.44	1.56

b) Test I has highest variation.

c) Test II has more consistence score

- Lesser S.D is more consistence
- Greater S.D is less consistence.

d) Test I has lesser consistence score.

Permutations and combination

i) Fundamental principles of counting

Multiplication principle

If one event can occur in **m** ways and second event can occur **n** ways then both events can occur in **mn** ways.

Illustrative Example

24. There are 5 gates to go from A to B and 4 different gates to go from B to C. How many different ways to go from A to C

Solution:

There are $(5 \times 4) = 20$ ways

25. In how many different way can five books be arranged on shelf

Solution:

1st book has 5 choices
2nd book has 4 choices
3rd book has 3 choices
4th book has 2 choices
5th book has 1 choices

$\therefore$ The total number of arrangement

$5 \times 4 \times 3 \times 2 \times 1 = 120$ ways

26. In how many different ways can 6 people be seated in a row

Solution: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ ways

27. Abebe has 4 pair of trousers, 5 shirts and 2 pair of shoes. In how many different ways can he dress.

Solution: $4 \times 5 \times 2 = 40$ ways

28. In how many ways can a president, vice-president and secretary be elected from a committee of 12 people

Solution: There are 12 people to place in 3 positions. Thus

$12 \times 11 \times 10 = 1320$ ways

29. There are 6 girls and 7 boys for group work. In how many different ways can a girl-boy couple be formed for group work.

Solution: $6 \times 5 = 30$ ways

30. How many four-digit numbers can be formed from the digit 1,3,4,5,6 and 7 where the digit is used at most once (Repetitions are not allowed)

a) If the number must be odd
b) If the number must be even
c) If the number greater than 5000
d) If the number less than 4000.

Solution: Here

a) * selecting thousand places we can fill either of 6 digit
   * selecting hundreds places = 5 way
   * selecting tens places = 4 way
   * selecting the unit places = 2 way
   $\therefore 6 \times 5 \times 4 \times 2 = 240$ way
b) $6 \times 5 \times 4 \times 1 = 120$ way
c) $3 \times 5 \times 4 \times 3 = 180$ way
d) $2 \times 5 \times 4 \times 3 = 120$ way

31. From question #30, if repetition is allowed i.e. the digit used more than once then

a) $6 \times 6 \times 6 \times 4 = 864$ way
b) $6 \times 6 \times 6 \times 2 = 432$ way
c) $3 \times 6 \times 6 \times 6 = 648$ way
d) $2 \times 6 \times 6 \times 6 = 432$ way

Definition: Factorial Notation

For each positive integer n,

$n! = n(n-1)(n-2) \ldots\ldots 3.2.1.$ also $0! = 1$

Example: Simplify: a) $6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

b) $\frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1} = 336$

c) $5! + 3! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 + 3 \cdot 2 \cdot 1 = 126$

Permutation

A Permutation is an arrangement of objects.

The number of permutation of **n** different object taken r at a time is $P(n,r) = \dfrac{n!}{(n-r)!}$

- The permutation of n different objects taken r at a time means an arrangement of r out of the n-object with attention given to the order of arrangement.

Illustrative Example

32. Find a) All the permutation of A B C
b) All the permutation of A B C D
c) All the permutation of 1, 2, 3, 7

Solution: 1st place we can fill either of the 3 letter A, B or C
2nd place we can fill either of the two remaining
3rd place we can fill the remaining one letters.

$\therefore P(3,3) = \dfrac{3!}{(3-3)!} = \dfrac{3!}{0!} = \dfrac{3!}{1!} = \dfrac{3!}{1} = 3\times2\times1 = 6$ different arrangement

b) The 4 letters, A, B, C, D can be arranged as P(4, 4) =

$P(4,4) = \dfrac{4!}{(4-4)!} = \dfrac{4\times3\times2\times1}{1} = 24$ different arrangement

c) The digit 1, 2, 3, 7 can be arranged
- Thousand place we can fill either of the four digit.
- Hundred place we can fill either of the remaining three digit.
- Tens place we can fill either of the two digit
- Unit place we fill the remaining digit.

$\therefore 4\times3\times2\times1 = 24$ different 4 digit number we can make

$\therefore \dfrac{4!}{(4-4)!} = \dfrac{4!}{0!} = 4\times3\times2\times1 = 24$ different 4 digit # make

33. a) How many permutation are there of 3 people selected from a group of 10 people.
b) In how many ways can a first, a second, and third prize be awarded in class of 9 student.

c) A row of six seats in class room is to be filled by selecting individual from a group of ten students.
In how many ways can the seats

d) In how many ways can 7 books be arranged in shelf

e) There are 8 different colours how many ways of 3 different colour are possible

Solution: Here

a) $P(10,3) = \frac{10!}{(10-3)!} = \frac{10!}{7!} = 10 \times 9 \times 8 = 720$

b) $P(9,3) = \frac{9!}{(9-3)!} = \frac{9 \times 8 \times 7 \times 6!}{6!} = 504$

c) $P(10,6) = \frac{10!}{(10-6)!} = \frac{10!}{4!} = 10 \times 9 \times 8 \times 7 \times 6 \times 5 = 151{,}200$

d) $P(7,7) = \frac{7!}{(7-7)!} = 7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$ way

e) $P(8,3) = \frac{8!}{(8-3)!} = \frac{8!}{5!} = 8 \times 7 \times 6 = 336$ way

34. a) How many three digit numbers can be formed from 3, 4, 5, 6 and 7

i) If repetition allowed

ii) If each digit used at most once i.e. (repetition not allowed)

Solution: i) $5 \times 5 \times 5 = 125$

ii) $P(5, 3) = \frac{5!}{(5-3)!} = 5 \times 4 \times 3 = 60$

Permutation of some identical object

The number of permutation of n objects of each

n_1 are alike object of 1st kind,

n_2 are alike object of 2nd kind.

n_3 are alike object of kth kind, is

$\frac{n!}{n_1!n_2!\ldots n_k}$ where $n = n_1 + n_2 + \ldots + n_k$

Illustrative example

35. Find the number of arrangement (permutation) can be made from.

a) 4 black, 3 red, 2 white

b) 2 black, 2 red, 2 white

Solution: Here

a) $n = 4 + 3 + 2 = 9$

$$\therefore \frac{9!}{4!3!2!} = \frac{9\times8\times7\times6\times5}{3\times2\times2} = 630$$

b) $n = 2 + 2 + 2 = 6$

$$\therefore \frac{6!}{2!2!2!} = \frac{6\times5\times4\times3\times2\times1}{2\times2\times2} = 90$$

36. Find the number of arrangement (permutation can be made of the letters of the word

a) LAMP b) FOOD c) CALCULUS

Solution:

a) There are 4 different letters in word "LAMP"

$$\therefore {}^4P_4 = \frac{4!}{(4-4)!} = 4! = 4\times3\times2\times1 = 24$$

b) Again, there are 4 letters in "FOOD". 1F, 2'0',1D

$$\therefore \frac{4!}{1!2!1!} = \frac{4\times3\times2\times1}{2\times1} = 12 \text{ differ ways}$$

c) Although there are 8 letters in word "CALCULUS" 2C,1A,2L,2U,1S.

∴ The permutation (arrangement) will be

$$\frac{8!}{2!1!2!2!1!} = \frac{8\times7\times6\times5\times4\times3\times2\times1}{2\times1\times2\times1\times2\times1} = 5{,}040.$$

37. In how many ways can the letters of word :ARRANGE" be arranged? How many of these arrangements are there in which

a) the two R'S come together

b) the two R'S and the two A'S come together

Solution:

a) The number of arrangements in which the two RS come together can be obtained by treating the two R'S as one letter. Thus, total number of arrangement.

$$\therefore \frac{6!}{2!1!1!1!} = 360$$

b) $$\frac{5!}{1!1!1!1!1!} = 5! = 120$$

38. a) Find the number of arrangement (permutation) that can be made out of the letters of the word "MATHEMATICS". In how many of these permutations

b) Do all the vowels occur together

c) Do the word start with I

d) Begins with the two A

e) Begins with E and ends with C

Solution:

a) The word "MATHEMATICS" consists of 11 letters of which 2 are A_s,

$2\ M_s$

$2\ T_s$ and the rest all different

$\therefore$ The total number of arrangements (permutations) are

$$\frac{11!}{2!2!2!}$$

b) The word "MATHEMATICS" consists of 4 vowels A, A, E and I (two are similar)

To find the number of arrangements in which the four vowels occur together, forming one letter. Thus we are left with 7 letter of which

$$\left.\begin{array}{l} 2\ are\ M_s \\ 2\ are\ T_s \\ 1\ is\ H \\ 1\ is\ C \\ 1\ is\ S \end{array}\right\} \text{arranged as} = \frac{7!}{2!2!}$$

The 4 vowels arranged as $= \frac{4!}{2!}$

$\therefore$ Total number of arrangements are $\left(\frac{7!}{2!2!}\right)\left(\frac{4!}{2!}\right)$

c) $\frac{10!}{2!2!2!}$ d) $\frac{9!}{2!2!}$ e) $\frac{9!}{2!2!2!}$

39. a) In 3 different physics books, 4 different chemistry books and 2 different maths books are to be arranged on shelf and each type of a book are identical. In how many can these books be arranged

Solution: $\frac{9!}{3!4!2!}$ ..... Explanation $3 + 4 + 2 = 9$

40. How many different number of five digits can be made using the digit 4, 4, 2, 2, 2

Solution: There are two 4S

Three, 2S

Hence the number of different numbers can be formed

$$\frac{5!}{2!3!} = \frac{1\times2\times3\times4\times5}{1\times2\times1\times2\times3} = 10$$

Circular permutation

A circular permutation is an arrangement of element in a circular pattern.

- In circular arrangement, there is no "first place" so that fix the position of one object
- A circular arrangement of **n** object is (n – 1)! different way

41. **Example:** In how many ways can 6 people be seated around circular table?

Solution: Use (n – 1)!

$\therefore (6 - 1)! = 5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$ way

Combination

A combination is a selection of objects with out considering the order

- **AB** and **BA** represent the same combination

The number of combination of **n** object taken **r** at a time is denoted by $C(n, r) = \binom{n}{r} = \frac{n!}{(n-r)!r!}$

Illustrative Example

42. Compute each of the following
    a) C(8, 5) b) C(7, 3) c) C(n, n)

    Solution: Here

    a) $C(8,5) = \frac{8!}{(8-5)!5!} = \frac{8!}{3!5!} = \frac{8\times7\times6}{3\times2} = 56$

    b) $C(7,3) = \frac{7!}{(7-3)!3!} = \frac{7\times6\times5}{3\times2} = 35$

    c) $C(n,n) = \frac{n!}{(n-n)!n!} = 1$

43. If C(n, 3) = C(n, 5), find n

    Solution: since C(n, r) = C(n, n-r)

    Thus, C(n, 3) = C(n, n-3) $\Rightarrow$ C(n, n-3) = C(n, 5)

    $\therefore$ n – 3 = 5 $\therefore$ n = 8

44. If C(n, 15) = C(n, 11), find, n

    Solution: C(n, 15) = C(n, n – 15)

    $\Rightarrow$ C(n, n-15) = C(n,11)

    $\therefore$ n – 15 = 11

    $\therefore$ n = 26

45. If C(18, r) = C(18, r + 2), then find r

    Solution: C(18, r) = C(18, 18 – r)

    $\Rightarrow$ C(18, 18-r) = C(18, r +2)

    $\Rightarrow$ 18 – r = r + 2 $\Rightarrow$ 16 = 2r $\therefore$ r = 8

46. a) In how many way can 2 books be selected out of 10 different books?

    b) In how many ways can a committee of 3 be selected from 8 people

    c) In how many ways can three or more people be selected out of 5.

    Solution: Here

a) $C(10,2) = \frac{10!}{8!2!} = \frac{10\times 9}{2} = 45$

b) $C(8,3) = \frac{8!}{(8-3)!3!} = \frac{8\times 7\times 6}{3\times 2} = 56$

c) $C(5,3) + C(5,4) + C(5,5) = 10 + 5 + 1 = 16$

47. In how many ways can 3 men and 2 women be selected from 12 men and 8 women?

Solution: Out of 12 men select 3 men
Out of 8 women select 2 women.

$$\therefore C(12,3)\cdot C(8,2) = \left(\frac{12!}{(12-3)!3!}\right)\cdot\left(\frac{8!}{(8-2)!2!}\right) = 6160$$

48. A number of ways of selecting a committee of 3 boys and girls from a group of 5 boys and 8 girls from a group of 5 boys and 8 girls is ---------- (Entrance exam)

A) 560 B) 128 C) 40 D) 256

Solution: $C(5,3)\cdot C(8,5) = (10)(56) = 560$ **Answer:**

49. A committee of 3 members has to be formed from 5 men and 6 women. In how many ways can this be done when the committee consists of

a) 2 women and 1 men
b) all men
c) at least 2 men
d) all women
e) at most 2 men

Solution:

a) 2 women from 6 women can be selected $\binom{6}{2}$.

1 men from 5 men can be selected $\binom{5}{1}$

$$\therefore \binom{6}{2}\binom{5}{1} = \left(\frac{6!}{(6-2)!2!}\right)\left(\frac{5!}{(5-1)!1!}\right) = (15)(5) = 75 \text{ ways}$$

b) 3 men from 5 men selected is $C\binom{5}{3} = 10$

c) at least 2 men means:

2 men and 1 women or 3 men and no women

$$\therefore C\binom{5}{2}\cdot\binom{6}{1}+\binom{5}{3}\cdot\binom{6}{0}=(10)(6)+(10)(1)=70$$

d) All women means 3 of them are women.

$$\therefore \binom{6}{3}\cdot\binom{5}{0}=20 \text{ ways}$$

e) at most 2 men: 2 men or 1 men

2 men and 1 women or 1 men and 2 women

$$\therefore \binom{5}{2}\cdot\binom{6}{1}+\binom{5}{1}\cdot\binom{6}{2}=(10)(6)+(5)(15)=135$$

50. A set contains 5 elements. How many of its subset has:

a) exactly 3 element

b) at least 3 element

c) at most 3 element

Solution:

a) $$C\binom{5}{3}=\frac{5!}{2!3!}=\frac{5\times 4}{2}=10$$

b) at least 3 element mean that (3, 4 or 5 element)

$$\therefore \binom{5}{3}+\binom{5}{2}+\binom{5}{1}+\binom{5}{0}=10+10+5+1=26$$

51. A box contains 5 white, 2 Red and 3 Black balls. If three balls are selected at random. Find the number of ways such that

a) One ball is black

b) 3 of them white

c) exactly 2 white

d) 1 white, 1 red and 3 black

Solution: Total ball: 5 + 2 + 3 = 10ball

a) $$\binom{3}{1}\binom{7}{2}=(3)(21)=63$$

b) $$\binom{5}{3}\binom{5}{0}=10$$

c) $$\binom{5}{2}\binom{5}{1}=(10)(5)=50$$

d) $$\binom{5}{1}\binom{2}{1}\binom{3}{3}=5\cdot(2)\cdot 1=10$$

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Binomial theorem

Binomial theorem for positive integers if n is a positive integer, then

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$$

$$= \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \ldots + \binom{n}{n} b^n$$

Illustrative example

52. Expand: $(2x^2 + 3)^4$

Solution:

$$(2x^2+3)^4 = \binom{4}{0}(2x^2)^4 + \binom{4}{1}(2x^2)^3(3) + \binom{4}{2}(2x^2)^2(3)^2 + \binom{4}{3}(2x^2)$$

$$= 16x^8 + 96x^6 + 216x^4 + 216x^2 + 81$$

Example: $\left(\sqrt{x} - 2y\right)^5$

$$\left(\sqrt{x} - 2y\right)^5 = \binom{5}{0}\sqrt{x}^5 + \binom{5}{1}\sqrt{x}^4(-2y)^1 + \binom{5}{2}\sqrt{x}^3(-2y)$$

$$+ \binom{5}{3}\sqrt{x}^2(-2y)^3 + \binom{5}{4}\sqrt{x}(-2y)^4 + \binom{5}{5}(-2y)^5$$

$$= x^{\frac{5}{2}} - 10x^2y + 40x^{\frac{3}{2}}y^2 - 80xy^3 + 80\sqrt{x}y^4 - 32y^5$$

53. Find the 4th term in the expansion of $(2x^3 - 3y^2)^5$

Solution: with $a = 2x^3$, and $b = -3y^2$, and using the binomial theo

with k = 3 and n = 5 we have $\binom{5}{3}(2x^3)^2(-3y^2)^3 = -1080x^$

$\therefore$ The fourth term is $-1080x^6y^6$

54. Find the indicated term without expanding

a) $(3x - y)^{10}$, eighth term

b) $(x + 2y)^{12}$, the fourth term

c) $\left(\frac{3}{x} - \frac{x}{3}\right)^{13}$, the se

term

Solution: a) a = 3x, b = -y, k = 7, and n = 10, we have

$$\binom{10}{7}(3x)^3(-y)^7 = -3240x^3y^7$$

b) a = x, b = 2y, k = 3, and n = 12 $\therefore$

$$\binom{12}{3}x^9(2y)^3 = -960x^7y^3$$

c) $a = 3x^{-1}$, $b = \frac{1}{3}x$, k = 6 and n = 1 $\therefore$

$$\binom{13}{6}(3x^{-1})^7\left(\frac{1}{3}x\right)^6 = \binom{13}{6}(3)x$$

55. Find the coefficient of $x^5 y^3$ in the expansion of $\left(\frac{5}{2}x - \frac{2}{5}y\right)^8$

Solution: $\binom{8}{k}\left(\frac{5}{2}x\right)^{8-k}\left(\frac{-2}{5}y\right)^k = x^5y^3$

$\therefore$ k = 3

$$\therefore \binom{8}{k}\left(\frac{5}{2}x\right)^5\left(\frac{-2}{5}y\right)^3 \quad \therefore \binom{8}{3}\left(\frac{5}{2}x\right)^5\left(\frac{-2}{5}y\right)^3$$

$$\therefore (56)\left(\frac{5^5}{2^5}\right)\left(\frac{(-2)^3}{5^3}\right)x^5y^3 = -350$$

56. Find whether there is any term-containing x^9 in the expansion of $\left(2x^2 - \frac{1}{x}\right)^{20}$

Solution: $\binom{20}{k}(2x^2)^{20-k}\cdot(-x^{-1})^k = (\ )x^9$

$$\Leftrightarrow x^{40-2k-k} = x^9$$

$$40 - 3k = 9 \Leftrightarrow k = \frac{31}{3}$$

Which is not possible because r must be a positive integer. Therefore there is no term containing x^9 in the given expansion.

57. Find the term independent (constant) of x in the expansion of each of the following.

a) $\left(x^2 - \frac{1}{x}\right)^9$ b) $\left[2x^3 - \frac{1}{x}\right]^{12}$ c) $\left(x - \frac{1}{2x}\right)^{10}$

Here, we make the exponent of x zero

Solution: $\binom{9}{k}\left(x^2\right)^{9-k} \cdot \left(-x^{-1}\right)^k$

$\Rightarrow 18 - 2k - k = 0 \Rightarrow 18 - 3k = 0$

$k = 6$

$\therefore \binom{9}{6}\left(x^2\right)^3\left(-x^{-1}\right)^6 = 84$

b) By making the exponent of x zero

$\binom{12}{k}\left(2x^3\right)^{12-k} \cdot \left(-x^{-1}\right)^k$

$\Rightarrow x^{36-3k-k} \cdot x = x^0 \Rightarrow 36 - 4k = 0 \Rightarrow 4k = 36$

$\therefore k = 9$

$\therefore \binom{12}{9}\left(2x^3\right)^3 \cdot (-1)^9 x^{-9} = -1760$

c) $\left(x - \frac{1}{2x}\right)^{10}$

$\binom{10}{k} x^{10-k} \cdot \left(-\frac{1}{2}x^{-1}\right)^k$

$\binom{10}{k} x^{10-k} \cdot x^{-k}\left(-\frac{1}{2}\right)^k$

By setting exponent of x zero we get

$10 - 2k = 0$

$k = 5$

ally exclusive events:

E_1 and E_2 are mutually exclusive if and only if they have out comes, i.e. $E_1 \cap E_2 = \varnothing$

ı Rolling of die

= getting even number

= getting prime number

= {2, 4, 6}

= {2, 3, 5}

$\cap E_2$ = {2}

nd E_2 are not mutually exclusive event.

tossing of coin once.

and {T} are mutually exclusive event.

$\cap E_2 = \varnothing$ then E_1 and E_2 are mutually exclusive

$\cap E_2 \neq \varnothing$ then E_1 and E_2 are not mutually exclusive

ndent Events: Two event E_1 and E_2 are said to be dent events if the occurrence of one does not influence ırrence of the other.

le: In tossing of two coin, the event getting head {H} irst coin and the event getting head {H} on the second independent.

are independent event then $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

ent Event: Two events said to be dependent if the ce of one affect the occurrence of the other.

: A box contain 4 Red and 6 black ball. If two balls are what is the probability that

th are black (with out replacement)

th are red: (with out replacement)

d then black

a) The two event (drawing single black ball) are t

the first draw, $P(\text{black}) = \frac{6}{10}$

the 2nd draw, given that the first draw was black

$\text{lack}) = \frac{5}{9}$

Probability

Probability is measure of uncertainty involved in the happening of event so that definite value may be assigned to it.

Definition:

i) **Random experiment:** any happening whose result is uncertain.
Example: Tossing of coin, Rolling of die.

ii) **Out comes:** The possible result of the experiment.

iii) **Sample space** is the set of all possible out comes of an experiment.
Example: S = {1,2,3,4,5,6} ⟵ sample space in rolling of die.

iv) **Events:** An event is any subset of sample space.
Example: In tossing of coin once, {H,T} is event

Types of Events

a. **Simple event** containing exactly one sample point.

b. **Compound event** is an event that has more than one sample point
Example: tossing of two coin {HH, HT, TH, TT}

c. **Complement** of an event, E is the collection of all out comes in the sample space that are not in E, denoted by E'
Example: In Rolling of die
Find a) the event E getting 6 b) E'
Solution: Sample space S = {1,2,3,4,5,6}
a) E = {6} ⟵ occurrence of event
b) E' = {1,2,3,4,5} ⟵ complement of event (Non occurrence of event)

d) **Exhaustive Event:** Events are said to be exhaustive if at least one of them must necessarily occur.
Example: In rolling of die:
{1}, {2}, ..., {6}, {1, 2}, {3, 6}, {1,2,3,4,5,6}
are exhaustive event.
Example: In tossing of coin:
{H}, {T}, {HT} are exhaustive event

Let $E_1, E_2, \ldots E_n$ forms set of exhaustive event then
$E_1 \cup E_2 \cup E_3 \cup \ldots\ldots E_n$ = subset.

e) Mutually exclusive events:

Two events E_1 and E_2 are mutually exclusive if and only if they have no common out comes, i.e. $E_1 \cap E_2 = \varnothing$

Example: In Rolling of die

Let E_1 = getting even number

Let E_2 = getting prime number

$\Rightarrow E_1 = \{2, 4, 6\}$

$\Rightarrow E_2 = \{2, 3, 5\}$

Thus $E_1 \cap E_2 = \{2\}$

$\therefore E_1$ and E_2 are not mutually exclusive event.

Example: In tossing of coin once.

: {H} and {T} are mutually exclusive event.

Note: i) If $E_1 \cap E_2 = \varnothing$ then E_1 and E_2 are mutually exclusive

ii) If $E_1 \cap E_2 \neq \varnothing$ then E_1 and E_2 are not mutually exclusive

f) In dependent Events: Two event E_1 and E_2 are said to be independent events if the occurrence of one does not influence the occurrence of the other.

Example: In tossing of two coin, the event getting head {H} on the first coin and the event getting head {H} on the second coin are independent.

* If E_1 and E_2 are independent event then $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$

g) Dependent Event: Two events said to be dependent if the occurrence of one affect the occurrence of the other.

Example: A box contain 4 Red and 6 black ball. If two balls are drawing what is the probability that

a) both are black (with out replacement)

b) both are red: (with out replacement)

c) Red then black

Solution: a) The two event (drawing single black ball) are dependent

- on the first draw, $P(\text{black}) = \frac{6}{10}$
- on the 2^{nd} draw, given that the first draw was black

$$P(\text{black}) = \frac{5}{9}$$

$\therefore$ P(black then black) $= \frac{6}{10} \cdot \frac{5}{9} = \frac{30}{90} = \frac{1}{3}$

b) Probability of getting Red, in the first draw P(Red) $= \frac{4}{10}$

- Probability of getting Red, in the 2nd draw, given that the first draw was Red, P(Red) $= \frac{3}{9}$

$\therefore$ P(Red then Red) $= \frac{4}{10} \cdot \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$

c) P(Red then Black) $= \frac{4}{10} \cdot \frac{6}{9} = \frac{24}{90} = \frac{4}{15}$

d) P(Black then Red) $= \frac{6}{10} \cdot \frac{4}{9} = \frac{24}{90} = \frac{4}{15}$

Equally likely event: Each element (event) has equal chance to select.

Probability of an Event

Probability can be measured by three different approaches

i) The classical (mathematical) approach
ii) The empirical (relative frequency) approach
iii) The axiomatic approach

i) The classical approach

Definition: If all the out comes of random experiment are equally likely out comes, then the probability of an event E is

$$P(E) = \frac{n(E)}{n(S)}$$

Example: A fair die is tossed once. Find the probability that
a) E_1: An odd number appears.
b) E_2: An even number appears

Solution:

a) $S = \{1, 2, 3, 4, 5, 6\}$, $n(S) = 6$
$E_1 = \{1, 3, 5\}$, $n(E_1) = 3$

$$\therefore P(E_1) = \frac{n(E_1)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

b) $n(S) = 6$, $E_2 = \{2, 4, 6\}$, $n(E_2) = 3$

$$\therefore P(E_2) = \frac{n(E_2)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

ii) The empirical approach

$$P(E) = \frac{frequency\ of\ E}{total\ number\ of\ observation}$$

Example 1: The following table gives a distribution of daily income.

Income	18-24	25-31	32-37	38-44	45-51
No. of work	5	40	15	23	17

Find the probability that:

a) their income less than or equal to 31
b) their income 32 or above
c) their income either between 32 -27 or 18 -24

Solution: Total number of observation = 5+40+15+23+17= 100

a) $p(income \le 31) = \frac{5+40}{100} = \frac{45}{100} = \frac{9}{20}$

b) $p(Income \ge 32) = \frac{15+23+17}{100} = \frac{55}{100} = \frac{11}{20}$

c) $p(Income\ 32-37\ or\ 18-24) = \frac{15+5}{100} = \frac{20}{100} = \frac{1}{5}$

2. 100 bulb were found to be defective in lot of 800 bulb. Find the probability that a bulb selected from the lot is defective is

$$p(Def) = \frac{f_D}{800} = \frac{100}{800} = \frac{1}{8}$$

ii. The axiomatic approach

Let S be the sample space of an experiment and E an event. Then **probability** P (E) of **E** is given by

$$P(E) = \frac{n(E)}{n(S)}$$

Since E is a subset of S, we see that

$$0 \le n(E) \le n(S)$$

$$0 \le \frac{n(E)}{n(S)} \le \frac{n(S)}{n(S)} \longleftarrow \text{Dividing both side by n(S)}$$

$$0 \le \frac{n(E)}{n(S)} \le 1$$

$$0 \le \frac{n(E)}{n(S)} \le 1$$

$0 \le p(E) \le 1 \longleftarrow$ axiom property

Note: i) if p(E) = 0 then E = ∅. (Impossible event)
ii) if p(E) = 1, then E = S (Certain event)
iii) p(not E) = 1 – P(E) = p(E') where E' = S E

Odds in favour of and odds against an event

If p(E) = m and p(E') = n then the ratio:

i) $\frac{p(E)}{p(E')} = \frac{m}{n} \rightarrow$ odd in favor of Event E (ratio of succe... failures)

ii) $\frac{p(E')}{p(E)} = \frac{n}{m} \rightarrow$ odd against an event E

Note: p (E') = 1 – p(E) and (E') = (not E)
n(S) = n(E) + n(E') = m + n

Illustrative Example

52. The odd infavour of an event are 4:7. Find probability ... occurrence (P(E))

Solution: $\frac{p(E)}{p(E')} = \frac{4}{7} \Rightarrow \frac{p(E)}{1-p(E)} = \frac{4}{7}$

$\Rightarrow 7P(E) = 4 - 4p(E) \Rightarrow 11p(E) = 4$

$\therefore p(E) = \frac{4}{11}$ and $p(E') = 1 - \frac{4}{11} = \frac{7}{11}$

54. The odd against certain event are 2:3. Find
a) the probability of its occurrence
b) the probability of non-occurrence.

Solution: a) $\frac{p(E')}{p(E)} = \frac{2}{3} \Rightarrow \frac{1-p(E)}{p(E)} = \frac{2}{3} \Rightarrow 3 - 3p(E) = 2p(E)$

$\therefore 5\,P(E) = 3$

$\therefore p(E) = \frac{3}{5}$

b) $p(E') = 1 - p(E) = 1 - \frac{3}{5} = \frac{2}{5}$

55. A box containing 4 green, 5 black and 6 white marbles. One marble is drawn at random.

i) Find the probability:

a) green marble.

b) not green marble

ii) Find c) the odds in favor of drawing green marble

d) the odds against drawing green marble

e) the odds against drawing white marble

Solution: Here

a) $p(\text{green}) = \frac{4}{4+5+6} = \frac{4}{15}$

b) $p(\text{not green}) = 1 - p(\text{green}) = 1 - \frac{4}{15} = \frac{11}{15}$

ii) c) $\frac{p(E)}{p(E')} = \frac{p(green)}{p(not\ green)} = \frac{\frac{4}{15}}{\frac{11}{15}} = 4:11$

d) $\frac{p(E')}{p(E)} = \frac{p(not\ green)}{p(green)} = \frac{\frac{11}{15}}{\frac{4}{15}} = 11:4$

e) $\frac{p(E')}{p(E)} = \frac{p(not\ white)}{p(white)} = \frac{1-p(white)}{p(white)} = \frac{1-\frac{6}{15}}{\frac{6}{15}} = \frac{\frac{9}{15}}{\frac{6}{15}} = 9:6$

56. If p(E) = 0.6, Find

a) The odd in favor of event E.

b) The odd against the event E.

Solution: Let = O(E) odd infavor of E

a) $O(E) = \frac{P(E)}{P(E')} = \frac{0.6}{1-0.6} = \frac{0.6}{0.4} = \frac{3}{2}$ ← odd infavor of E.

b) $O(E') = \frac{P(E')}{P(E')} = \frac{1-0.6}{0.6} = \frac{0.4}{0.6} = \frac{4}{6} = \frac{2}{3}$ ← odd against E.

Rules of probability

I. Addition rule

If E_1 and E_2 are any two events then

i) $P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

ii) $P(E_1 \cup E_2) = P(E_1) + P(E_2)$, if E_1 and E_2 are mutually exclusive i.e. $E_1 \cap E_2 = \varnothing$

Note: $P(E_1 \text{ or } E_2) = P(E_1 \cup E_2)$

II. Multiplication Rule

If E_1 and E_2 are any two events, the probability that both events occur denoted by $P(E_1 \text{ and } E_2) = P(E_1 \cap E_2) = P(E_1E_2)$ is given by

i) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2)$ If E_1 and E_2 are independent event

ii) $P(E_1 \cap E_2) = P(E_1) \cdot P(E_2|E_1)$ if E_1 and E_2 are dependent event (conditional probability)

$= P(E_2) \cdot P(E_1|E_2)$

Condional Probability: When occurrence of one event depends on the occurrence of another event, we say the second event is conditioned by the first event.

This is called **conditional probability.**

i) The conditional probability of event E_1 given that E_2 has already occurred is defined by $P(E_1|E_2) = \frac{P(E_1 n E_2)}{P(E_2)}$

ii) If event E_2, given that E_1 has already occurred is defined by $P(E_2|E_1) = \frac{P(E_1 n E_2)}{P(E_1)}$

Note: The vertical bar is read "given"

Illustrative Example

57. A box containing 6 red ball, 4 white ball and 5 blue balls. One ball is drawn at random. Find the probability that is:

a) red or white
b) white or blue
c) neither red nor blue
d) not blue
e) odd in favor or red or white
f) odd against white or blue

Solution: Let R = Red, B = Blue, W = White by using addition rule

a) $P(R \text{ or } W) = P(R) + P(W) = \frac{6}{15} + \frac{4}{15} = \frac{10}{15} = \frac{2}{3}$

b) $P(W \text{ or } B) = P(W) + P(B) = \frac{4}{15} + \frac{5}{15} = \frac{9}{15} = \frac{3}{5}$

c) $P(\text{neither red nor blue}) = 1 - P(R \text{ or } W)$

$$= 1 - \left(\frac{6}{15} + \frac{4}{15}\right) = 1 - \frac{2}{3} = \frac{1}{3}$$

d) $P(\text{not } B) = 1 - P(B) = 1 - \frac{5}{15} = \frac{10}{15} = \frac{2}{3}$

e) $O(R \text{ or } W) = \frac{P(R) + P(W)}{1 - (P(R) + P(W))} = \frac{\frac{2}{3}}{1 - \frac{2}{3}} = 2:1$

f) Exercise left for the student

58. A number is selected at random from the set of integers from 1 to 20 inclusive. Find the probability that the number selected is:

a) Multiple of 2 or 3 b) multiple of 2 and 3

Solution: Let $S = \{1, 2, 3, \ldots.20\}$ $\therefore n(S) = 20$

Let E_1 = selecting multiple of 2,

E_2 = event selecting multiple of 3.

Then, $E_1 = \{2,4,6,8,10,12,14,16,18,20\}$, $n(E_1) = 10$

$E_2 = \{3, 6,9,12,15,18\}$, $n(E_2) = 6$

Since, $E_1 \cap E_2 = \{6, 12, 18\}$

$\therefore$ E_1 and E_2 are not mutually exclusive

a) $P(E_1 \text{ or } E_2) = P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$

$$= \frac{10}{20} + \frac{6}{20} - \frac{3}{20} = \frac{13}{20}$$

b) $P(E_1 \text{ and } E_2) = P(E_1 \cap E_2) = P(E_1) \cdot P(E_2|E_1)$

$= P(E_1) \cdot P(E_2)$ ⟵ Since E_1 and E_2 are independent event ($P(E_2|E_1) = P(E_2)$)

$$= \left(\frac{10}{20}\right) \cdot \frac{6}{20} = \frac{3}{20}$$

59. A die and coin is tossed together. Find thee probability of
a) six and head
b) odd number and head
c) 3 and tail
d) even number and tail

Solution: Tossing die = {1, 2, 3, 4, 5 , 6}
Tossing coin = {H, T}
Tossing coin and die together is $6 \times 2 = 12$ out come
$\therefore S = \{1H, 2H, 3H, 4H, 5H, 6H, 1T, 2T, 3T, 4T, 5T, 6T\}$

Thus,

a) $p(\text{six and head}) = p(\text{six}) \cdot p(\text{head}) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$. Event = {6H}

b) $p(\text{odd and head}) = p(\text{odd}) \cdot p(\text{head}) = \frac{3}{6} \cdot \frac{1}{2} = \frac{1}{4}$. Event = {1H,3H,5H}

c) $p(3 \text{ and tail}) = p(\text{getting } 3) \cdot p(\text{tail}) = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12}$. Event = {3T}

d) $p(\text{even and tail}) = p(\text{even}) \cdot p(\text{tail}) = \frac{3}{6} \cdot \frac{1}{2} = \frac{1}{4}$. Event = {2T,4T,6T}

60) A box contains 4 white and 3 red balls. If two balls are drawn after the other with out replacement. Find the probability that
a) both are red.
b) both are white
c) the first ball is red and the second is white.
d) the first ball is white and the second is red
e) both are different color

Solution: Let R = Red and W = White, then

a) $P(R \text{ and } R) = \frac{3}{7} \cdot \frac{3-1}{7-1} = \frac{3}{7} \cdot \frac{2}{6} = \frac{1}{7} \longrightarrow$ (Explanation: $\frac{3-1}{7-1}$ is one red ball is removed from 1st draw)

b) $P(W \text{ and } W) = \frac{4}{7} \cdot \frac{4-1}{7-1} = \frac{4}{7} \cdot \frac{3}{6} = \frac{2}{7} \longrightarrow \frac{4-1}{7-1}$ is one white ball is removed on 1st draw

c) $P(R \text{ and then } W) = \frac{3}{7}\cdot\frac{4}{7-1} = \frac{3}{7}\cdot\frac{4}{6} = \frac{2}{7} \longrightarrow$ 7-1 is Red is removed on the 1st draw

d) $P(W \text{ and then } R) = P(W)\cdot P(R|W)$

$= \frac{4}{7}\cdot\frac{3}{7-1} = \frac{4}{7}\cdot\frac{3}{6} = \frac{2}{7} \longrightarrow$ 7-1 is White is removed on the 1st draw

e) $P(RW \text{ or } WR) = P(RW) + P(WR) =$

$= P(R)\cdot P(W|R) + P(W)\cdot P(R|W) = \frac{3}{7}\cdot\frac{4}{6} + \frac{4}{7}\cdot\frac{3}{6} = \frac{4}{7}$

Method II, a) $P(RR) = \frac{C(3,2)}{C(7,2)} = \frac{1}{7}$, b) $P(WW) = \frac{C(4,2)}{C(7,2)} = \frac{2}{7}$,

51) A box contains 4 red, 3 white and 5 blue balls. Three balls are drawn one after the other. Find the probability of getting:

a) all 3 are red. Drawing with i) replacement
ii) out replacement

b) all 3 are white.
i) With replacement ii) with out replacement

c) all the balls are of different colour.

d) the balls are drawn in the order red, white, blue. Each ball is
i) replaced ii) not replaced

Solution: Let Red = R_1, R_2, R_3 (1st, draw 2nd draw, 3rd draw of red ball respectively W = W_1, W_2, W_3 (1st, 2nd, 3rd draw of white ball respectively.

Then

a) i) If each ball is replaced, then R_1, R_2 and R_3 are in dependent event

$$\therefore P(R_1\, R_2\, R_3) = P(R_1).P(R_2).P(R_3) = \frac{4}{12}\cdot\frac{4}{12}\cdot\frac{4}{12} = \frac{1}{27}$$

ii) If ball is not replaced, then R_1, R_2 and R_3 are dependent event

$$\therefore P(R_1\, R_2\, R_3) = P(R_1).P(R_2|R_1).P(R_3|R_1\, R_2) = \frac{4}{12}\cdot\frac{3}{11}\cdot\frac{2}{10} = \frac{1}{55}$$

b) i) $P(W_1W_2W_3) = P(W_1)\cdot P(W_2)\cdot$

$P(W_3) = \frac{3}{12}\cdot\frac{3}{12}\cdot\frac{3}{12} = \frac{1}{64}$ ← (each ball is replaced i.e. independent)

ii) $P(W_1W_2W_3) = P(W_1)\cdot P(W_2|W_1)\cdot P(W_3|W_1W_2)$ ← (No replaced dependent)

$= \frac{3}{12}\cdot\frac{2}{11}\cdot\frac{1}{10} = \frac{1}{220} \rightarrow P(W_1) = \frac{3}{12}$

$P(W_2|W_1) = \frac{3-1}{12-1} = \frac{2}{11}$

Note: We can use: $P(W_1W_2W_3) = \frac{C(3,3)}{C(12,3)}$

c) P(1 of each colour is draw)

$= \frac{C(4,1)\cdot C(3,1)\cdot C(5,1)}{C(12,3)} = \frac{4\times 3\times 5}{210} = \frac{3}{11}$

d) i) If each ball is replaced then R,W,B are independent

$\therefore P(RWB) = P(R)\cdot P(W)\cdot P(B) = \frac{4}{12}\cdot\frac{3}{12}\cdot\frac{5}{12} = \frac{5}{144}$

ii) If each ball is not replaced then events are dependent.

$\therefore P(RWB) = P(R)\cdot P(W|R)\cdot P(B|RW) = \frac{4}{12}\cdot\frac{3}{11}\cdot\frac{5}{10} = \frac{1}{22}$

62) From a deck of 52 playing cards, one card is drawn. Find probability that it is

a) either king or Queen
b) either Queen or Jack.
c) ether Queen or Red.
d) either king or black
e) either heart or king
f) either an Ace or He

***Solution*: Here:**

A standard deck of 52 playing cards made up as shown.

Black (B)	Spade	A ♠	K ♠	Q ♠	J ♠	10 ♠	9 ♠	8 ♠	7 ♠	6 ♠	5 ♠	4 ♠	3 ♠	2 ♠
	Clube	A ♣	K ♣	Q ♣	J ♣	10 ♣	9 ♣	8 ♣	7 ♣	6 ♣	5 ♣	4 ♣	3 ♣	2 ♣
Red (R)	Heart (H)	A ♥	K ♥	Q ♥	J ♥	10 ♥	9 ♥	8 ♥	7 ♥	6 ♥	5 ♥	4 ♥	3 ♥	2 ♥
	Diamond	A ♦	K ♦	Q ♦	J ♦	10 ♦	9 ♦	8 ♦	7 ♦	6 ♦	5 ♦	4 ♦	3 ♦	2 ♦

a) $P(K \text{ or } Q) = P(K) + P(Q) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$

b) $P(Q \text{ or } J) = P(Q) + P(J) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$

c) $P(Q \text{ or } R) = P(Q) + P(R) - P(QnR) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{7}{13}$

d) $P(K \text{ or } B) = P(K) + P(B) - P(KnB) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{7}{13}$

e) $P(H \text{ or } K) = P(H) + P(K) - P(HnK) = \frac{13}{52} + \frac{4}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13}$

f) $P(A \text{ or } H) = P(A) + P(H) - P(AnH) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{4}{13}$

63) Two cards are drawn from pack of 52 cards. Find the probability that

a) all are Jack if the first card was replaced

b) All are king if the first card was not replaced

c) The first is an Ace and the second is king if the first cad was not replaced

***Solution:* Here**

a) $P(J \text{ and } J) = P(J) \cdot P(J) = \frac{4}{52} \cdot \frac{4}{52} = \frac{1}{169}$ ← the 1st king was replaced

b) $P(K \text{ and } K) = \frac{4}{52} \cdot \frac{4-1}{52-1} = \frac{4}{52} \cdot \frac{3}{51} = \frac{1}{221}$ ← the 1st King was not replaced

c) $P(AK) = P(A) \cdot P(K|A) = \frac{4}{52} \cdot \frac{4}{52-1} = \frac{4}{52} \cdot \frac{4}{51} = \frac{4}{663}$

64) Two balls are selected from a box of containing 7 red and 5 blue balls one after the other. Find the probability that

a) the second ball selected is blue given that the first was red ball.

b) The second ball selected is red ball given that the first was blue ball.

c) The second ball selected is blue given that the first was blue ball

d) The second ball is red knowing that the first ball is red

Solution: $P(\text{Red then blue}) = P(R) \cdot P(B|R) = \frac{7}{12} \cdot \frac{5}{11}$

a) $\therefore P(B|R) = \frac{5}{11}$ ← given that Red has already occurred

b) $P(\text{Band then Red}) = P(B) \cdot P(R|B) = \frac{5}{12} \cdot \frac{7}{11}$

$\therefore P(R|B) = \frac{7}{11}$ ← given that the blue ball has already occurred.

c) $P(\text{blue and then blue}) = P(B_1) \cdot P(B_2|B_1) = \frac{5}{12} \cdot \frac{4}{11}$

$\therefore P(B_2|B_1) = \frac{4}{11}$

d) $P(\text{Red then Red}) \; P(R_1) \cdot P(R_2|R_1) = \frac{7}{12} \cdot \frac{6}{11}$ $\therefore P(R_2|R_1) = \frac{6}{11}$

65) From a pack of 52 playing cards, three cards are drawn one after the other with out replacement. Find the probability that King, Queen and Jack will be obtained respectively?

Solution: $P(KQJ) = P(K) \cdot P(Q|K) \cdot P(J|QK)$

$$= \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} = \frac{8}{16575}$$

If A and B are independent event and $P(A) = \frac{1}{3}$, $P(B) = \frac{3}{4}$ find P(AUB)

Solution:

$P(AUB) = P(A) + P(B) - P(AnB) = P(A) + P(B) - (P(A) \cdot P(B))$

$$= \frac{1}{3} + \frac{3}{4} - \left(\frac{1}{3} \cdot \frac{3}{4}\right) = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

If a pair of dice is thrown. Find the probability that the sum of the top number is

a) 7 b) greater than 10, c) 11 d) either 6 or 10
e) 5 f) neither 6 nor 10 g) total of 7 or 11

Solution: S = {(1,1),(1,2),(1,3),(1,4),(1,5)(1,6),(2,1). . . (5,6),(6,6)

$\therefore n(S) = 36$

a) let E = Event that the sum is 7

Thus, E = {(2,5),(5,2,(1,6)(6,1),(3,4),(4,3)}, Then n(E) = 6

$$\therefore P(E) = \frac{n(E)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

b) Let E = Event sum greater than 10 i.e, 11 or 12

$\therefore$ E = {(5,6),(6,5),(6,6)}, n(E) = 3

$$\therefore P(E) = \frac{3}{36} = \frac{1}{12}$$

c) Let E = Event sum is 11

$\therefore$ E = {(5,6),(6,5)}, n(E) = 2

$$\therefore P(E) = \frac{2}{36} = \frac{1}{18}$$

d) Let E = Event sum 6 or 10.

Sum 6 = {(1,5),(5,1),(2,4),(4,2),(3,3),}

Sum 10 = {(4,6),(6,4),(5,5)}

$\therefore$ E = {(1,5),(5,1),(2,4),(4,2),(3,3),(4,6),(6,4),(5,5)}

$$\therefore P(E) = \frac{8}{36} = \frac{2}{9}$$

e) Let E = Event sum 5, $\therefore$ E = {(1,4),(4,1),(2,3),(3,2)}

$$\therefore P(E) = \frac{5}{36}$$

f) P(neither 6 nor 10) = 1 – p (sum either 6 or 10) = $1-\frac{2}{9}=\frac{7}{9}$

g) Event E = {(1,6),(6,1),(2,5),(5,2),(3,4),(4,3),(5,6),(6,5)}

$\therefore P(E)=\frac{6}{36}+\frac{2}{36}=\frac{8}{36}=\frac{2}{9}$

68) If $P(A)=\frac{2}{5}, P(B)=\frac{1}{3}, P(A\cup B)=\frac{1}{2}$

Find a) P(A|B) b) P(B|A)

Solution:

a) $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$\frac{1}{2}=\frac{2}{5}+\frac{1}{3}-P(A\cap B)\Rightarrow P(A\cap B)=\frac{7}{30}$, but

$P(AnB)=P(B)\cdot P(A|B)$

$\therefore P(A|B)=\frac{P(A\cap B)}{P(A)}=\frac{\frac{7}{30}}{\frac{2}{5}}=\frac{7}{12}$

69) If $P(A)=\frac{1}{2}, P(B)=\frac{1}{3}, P(A\cap B)=\frac{1}{4}$ Find

a) P(AUB) b) P(A|B) c) P(B|A)

Solution: Here

a) $P(AUB)=P(A)+P(B)-P(AnB)=\frac{1}{2}+\frac{1}{3}-\frac{1}{4}=\frac{7}{12}$

b) $P(A|B)=\frac{P(AnB)}{P(B)}=\frac{\frac{1}{4}}{\frac{1}{3}}=\frac{3}{4}$ c)

Probability and Combination analysis

70) A box contains 5 red, 3 white and 4 blue balls. If 3 balls are selected at random. Find the probability that

a) all are red
b) all are white
c) at least one ball is white
d) all the balls are different color
e) two are red and one is blue

Solution: Here

a) all are red means that 3 of them is red from 5 red

$$\therefore \text{P(all red)} = \frac{C(5,3)}{C(12,3)} = \frac{10}{210} = \frac{1}{21}$$

b) $$\text{P(3 white)} = \frac{C(3,3)}{C(12,3)} = \frac{1}{210}$$

c) P(at least 1 white) = P(1w or 2 white or 3 white)

$$= \frac{C(3,1)C(9,2) + C(3,2)C(9,1) + C(3,3)C(9,0)}{C(12,3)}$$

$$= \frac{(3)(36) + (3)(9) + 1}{210} = \frac{136}{210}$$

d) $$\text{P(all are different color)} = \frac{C(5,1)\cdot C(3,1)\cdot C(4,1)}{C(12,3)} = \frac{1}{7}$$

e) $$\text{P(2 red and 1 blue)} = \frac{C(5,1)\cdot C(4,1)}{C(12,3)} = \frac{(10)\times 4}{210} = \frac{4}{21}$$

71) Out of group of 5 girls and 4 boys, if three of them are selected for group work leader. What is the probability that

a) all are girls b) all are boys c) at least one girl

Solution:

a) $$\text{P(all girls)} = \frac{C(5,3)\cdot(4,0)}{C(9,3)} = \frac{10}{84} = \frac{5}{42}$$

b) $$\text{P(all boys)} = \frac{C(4,3)\cdot(5,0)}{C(9,3)} = \frac{4}{84} = \frac{1}{21}$$

c) $$\text{P(at least one girl)} = 1 - \text{P(all boys)} = 1 - \frac{1}{21} = \frac{20}{21}$$

72. In pack of ten books, three are geometry books while the rest are book on algebra. If three books are drawn at random, what is the probability that one of them is geometry book and the others are book on algebra

A. $\frac{21}{40}$ B. $\frac{21}{100}$ C. $\frac{2}{21}$ D. $\frac{7}{10}$

Solution:

$$\frac{C(3\,1)C(7,2)}{C(10,3)} = \frac{(3)(21)}{120} = \frac{21}{40}$$

Answer:

73. Ajar contains 3 red and 5 green balls identical in all aspects except colour. If two balls are drawn at random, then the probability that the two balls drawn are not green is

A. $\frac{9}{14}$ B. $\frac{15}{28}$ C. $\frac{3}{28}$ D. $\frac{5}{14}$

Solution: P(not green) = 1 – p(green)

$$= 1 - \frac{C(5,2)C(3,0)}{C(8,2)} = 1 - \frac{10}{28} = \frac{18}{28} = \frac{9}{14}$$

Answer:

74. A box contains 8 items of which 3 are defective. If a man selects two items are drawn from the box together, what is the probability that

a) both items are defective b) both are non-defective
c) one item is defective d) at least one is defective
e) at most one is defective

Solution: from 8 item $\begin{cases} 3 \text{ defective item} \\ 5 \text{ non – defective item} \end{cases}$

a) 2 defective items out of 3 defective items = C(3, 2)

$$\therefore p(\textit{both defective}) = \frac{C(3,2)}{C(8,2)} = \frac{3}{28}$$

b) 2 non – defective out of 5 non-defective

$$\therefore p(\textit{both non defective}) = \frac{C(5,2)}{C(8,2)} = \frac{10}{28}$$

c) One item defective means
The other one is non-defective

$$\therefore p(1 = \textit{defective}) = \frac{\binom{3}{1}\binom{5}{1}}{\binom{8}{2}} = \frac{(3)(5)}{28} = \frac{15}{28}$$

d) $p(x \geq 1) = p(x = 1) + p(x = 2)$

$$= \frac{C(3,1)C(5,1)+C(3,2)}{C(8,2)} = \frac{15}{28}+\frac{3}{28} = \frac{18}{28} = \frac{9}{14}$$

e) p(at most one defective) = p(no defective) + p(1-defective)

$$= \frac{C(5,2)C(3,0)+C(5,1)C(3,1)}{C(8,2)} = \frac{10+15}{28} = \frac{25}{28}$$

75. A bag contains 3 fresh and 5 spoiled oranges of equal size. If three oranges are to be drawn at random from the bag, what is the probability that at least one of the oranges is fresh?

A. $\frac{15}{28}$ B. $\frac{15}{56}$ C. $\frac{45}{56}$ D. $\frac{23}{28}$

Solution: $p\begin{pmatrix}\text{at least one of}\\ \text{the orange fresh}\end{pmatrix} = p(1\text{fresh}) + p(2\text{fresh}) + p(3\text{fresh})$

$$= \frac{C(3,1)C(5,2)+C(3,2)C(5,1)+C(3,3)}{C(8,3)}$$

$$= \frac{30+15+1}{56} = \frac{46}{56} = \frac{23}{28}$$

Answer: C

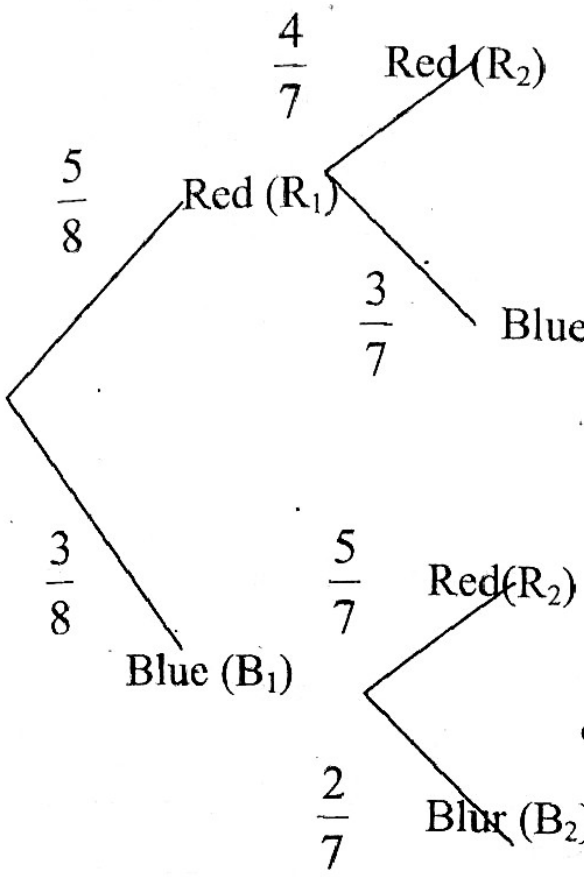
Calculating probability using tree diagram

When we have two or more events the probability of any out come we can obtain in simple ways from tree diagram.

76. A box contains 5 red balls, 3 blue balls. Two balls are drawn one after the other with out replacement. Find the probability that:

a) both are red
b). both are blue
c) not the same colour
d) the 2nd ball is blue given that the first ball is ed.
e) the 2nd ball is red given that the first ball is blue
f) 2nd ball is red given that 1st ball is red
g) 2nd ball is blue given that 1st ball is blue.

Solution: Let R_1 = red on 1st draw and B_1 = blue on 1st draw.
R_2 = red on 2nd draw B_2 = blue on 2nd draw.



• a) $P(R_1R_2) = P(R_1)\cdot P(R_2|R_1)$

$= \frac{5}{8}\cdot\frac{4}{7} = \frac{5}{14}$

• b) $P(B_1B_2) = P(B_1).P(B_2|B_1)$

$= \frac{3}{8}\cdot\frac{2}{7} = \frac{3}{28}$

• c) P(not the same colour)

= P(Red then blue or blue then red)

$= P(RB) + P(BR) = P(R)\cdot P(B|R) + P(B)P(R|B)$

$= \frac{5}{8}\cdot\frac{3}{7} + \frac{3}{8}\cdot\frac{5}{7} = \frac{15}{28}$

Or P(not the same colour)

$= 1 - P(\text{same colour}) = 1 - P(\text{both red}) - P(\text{both blue})$

$= 1 - \frac{5}{14} - \frac{3}{28} = \frac{15}{28}$

d) $P(B|R) = \frac{3}{7}$ ← probability that 2nd ball is blue

Given that 1st ball is red.

e) $P(R|B) = \frac{5}{7}$ ← probability that 2nd ball is red.

f) $P(R_2|R_1) = \frac{4}{7}$ ← probability that 2nd ball is red give that

ball is red

g) $P(B_2|B_1) = \frac{2}{7}$

Repeated Trials

Let p be the probability of success of an event
q be the probability of its failure.
n be repeated the trail n times
Now out of n trail r successes, n – r failure.
∴ P(r successes) = $c(n, r)\, p^r \cdot q^{n-r}$

The probability of getting two head in 5 to sses of coin.

Solution: $P(2) = C(5, 2)\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 = \frac{5}{16}$

78. The probability of getting 6 correct answer in 10 question each question has 4 multiple choice.

Solution: $p = \frac{1}{4},\ q = \frac{3}{4}\ \therefore\ p(6) = c(10,6)\left(\frac{1}{4}\right)^6\left(\frac{3}{4}\right)^4$

Solved Problem

79) If P(x) = 0.4, P(y) = 0.6 and P(x and y) = 0.15 then

a) Are the events x and y dependent or independent.

b) find i) P(x|y) ii) P(y|x)

Solution: Here

a) P(x and y) = P(x) . P(y|x) ⇒ 0.15 = (0.4) P(y|x)

$\therefore$ P(y|x) = $\frac{0.15}{0.4} = 0.375$

This shows it is dependent because

P(y|x) ≠ P(y) ⇒ (0.375 ≠ 0.6)

b) P(x and y) = p(y) .P(x|y)

0.15 = (0.6) P(x|y)

∴ P(x|y) = 0.25

78) There are 3 red and 9 green ball in a box. What is the probability of drawing at random with out replacement green then another green.

A) $\frac{9}{16}$ B) $\frac{6}{11}$ C) $\frac{9}{12}$ D) $\frac{1}{4}$

Solution: P(green then green) = $\frac{9}{12} \cdot \frac{8}{11} = \frac{6}{11}$

81) How many possible 3 member committees can be selected from an organization that contains 10 members
A. 120 B) 360 C) 240 D) 720

Solution: $C(10, 3) = \frac{10!}{(10-3)!3!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120$, **Answer: A**

82) One box contains 2 red ball and 4 blue marbles. A second box contains 3 red ball and 2 blue ball. What is the probability that drawing one ball from each box will pick 2 red ball.

A) $\frac{3}{10}$ B) $\frac{1}{5}$ C) $\frac{2}{5}$ D) $\frac{1}{2}$

Solution: Let R_1 = event "Red" ball from 1st box
R_2 = event "Red" ball from 2nd box

Then $P(R_1) = \frac{2}{2+4} = \frac{2}{6}$, *and* $P(R_2) = \frac{3}{3+2} = \frac{3}{5}$

Since R_1 and R_2 are in different box, there fore they are independent

$\therefore P(R_1 \cdot R_2) = P(R_1) \cdot P(R_2) = \frac{2}{6} \cdot \frac{3}{5} = \frac{1}{5}$ **Answer: B**

83) A black ball and 4 red balls are in a box. If two balls are drawn with out replacement, what is the probability of getting.
a) black ball on the first draw and red ball on the second.
b) two red ball
c) second ball is black given that the first ball was red.

Solution: Let B = event black and R = Red

Then a) $P(BR) = P(B) \cdot P(R|B) = \frac{1}{5} \cdot \frac{4}{4} = \frac{1}{5}$

b) $P(R_1 \text{ and } R_2) = P(R_1) \cdot P(R_2|R_1) = \frac{4}{5} \cdot \frac{3}{4} = \frac{3}{5}$

c) we know that $P(RB) = P(R) \cdot (B|R) = \frac{4}{5} \cdot \frac{1}{4}$

$\therefore P(B|R) = \frac{1}{4} \leftarrow$ required probability.

A probability that a student to pass mathematics exam is $\frac{2}{3}$ and probability that he passes both mathematics and an English exam is $\frac{14}{45}$.

The probability that he pass at least one exam is $\frac{4}{5}$. What is the probability that he pass the English exam is _____

A) $\frac{7}{15}$ B) $\frac{44}{45}$ C) $\frac{4}{9}$ D) $\frac{8}{15}$

Solution: P (math) = $\frac{2}{3}$; P(math $\cap$ Eng) = $\frac{14}{45}$.

P (math $\cup$ Eng) = $\frac{4}{5}$, Now we find P(Eng)

$\therefore$ P (math $\cup$ Eng) = P(math) + P(Eng) – P(math $\cap$ Eng)

$$\frac{4}{5} = \frac{2}{3} + P(Eng) - \frac{14}{45}$$

$$\therefore P(Eng) = \frac{4}{5} - \frac{2}{3} + \frac{14}{45} = \frac{36 - 30 + 14}{45} = \frac{4}{9}$$

85) A school has three class room for grade 11, namely, 11A, 11B and 11C.
The number of students in these class rooms is 28, 20 and 22 respectively. All student took an exam and the average score of student of 11A, 11B and 11C is 60,70 and 70 respectively. What is the average score in this examination for all grade 11 students …UEE 2004/12

A) 66 B) 66.67 C. 65 D) 65.67

Solution: $\overline{X} = \frac{60 \times 28 + 20 \times 70 + 22 \times 70}{28 + 20 + 22} = \frac{4620}{70} = 66$ **Ans.: A**

86) A three – digit library identification card is to be printed from the numbers 0,1,2,3,4,5 in such a way that the first is non – zero and no number is to be repeated. How many such cards can be printed? … UEE 2004/12

A) 100 B) 120 C) 150 D) 180

Answer: A

Solution: 5×5×4 = 100,

87) A student needs to select 3 books from 3 different mathematics, 3 different physics and 1 history book. What is the probability that one of them is mathematics and the other two are either physics or history books . . . UEE 2004/12

A) $\frac{3}{35}$ B) $\frac{9}{35}$ C) $\frac{15}{35}$ D) $\frac{18}{35}$

Solution: $\frac{C(3,1)C(4,2)}{C(7,3)} = \frac{3\times 6}{35} = \frac{18}{35}$ **Answer: D**

88) Items produced by certain company are subjected to two kinds of defects D_1 and D_2. out of total production, if 5% have defect D_1, 2% have Defect D_2 and 2% have both defect S, then what is the probability for an item to have defect D_2, given that it has defect D_1

A) 0.2 B) 0.05 C) 0.1 D) 0.4

Solution: $P(D_1 \text{ and } D_2) = 0.02$, $P(D_1) = 0.05$

$\Rightarrow P(D_1 \text{ and } D_2) = P(D_1).P(D_2|D_1) \Rightarrow 0.02 = 0.05\ P(D_2|D_1)$

$\Rightarrow P(D_2|D_1) = \frac{0.02}{0.05} = \frac{2}{5} = 0.4$, **Answer: D**

Supplementary exercise

1. Find the mean, mode(s) and median of the following data 1, 9, 4, 3, 10, 2, 8, 5.
2. Find the mean, mode(s) and median of the following distribution

v	-1	0	2	3	5
f	5	7	4	1	3

3. Given 1, 2, x, 5, y, 8. Find the values of x and y if the mode of the resulting numbers is 5 and the mean is 4.
4. What number should be included in the following data so that the median is 5.2? 4, 7, 11, 5, 3
5. If the mean of a, b, c, d is k then what is
   a) a + d, b + d, c + d, 2d b) ac, bc, c^2, dc
6. Find the median, lower and upper quartiles from the following table

Marks below	10	20	30	40	50	60	70	80
Number of student	15	35	60	84	94	127	198	240

7. Find the median from the following table

V	5	7	9	11	13	15	17	19
F	1	2	7	9	11	8	5	4

8. Calculate the mean and median from the following table

Class Interval	Frequency
6.5 – 7.5	5
7.5 – 8.5	12
8.5 – 9.5	25
9.5 – 10.5	48
10.5 – 11.5	32
11.5 – 12.5	6
12.5 – 13.5	1

9. In how many different ways can 5 student be seated in a row of 5 chair?
10. How many different ways are there to select 4 different players from 10 players on a team to play four table tennis matches, where the matches are ordered.
11. How many three digit numerals can be written by using only the digits 4, 5, 6, 7, 8, and 9 if no digit is repeated in the same numeral.
12. How many integers can be represented using the digit 3, 5, 7, 9 with each digit used at most once in a numeral?
13. A map of four countries is to be colored with a different color for each country. If six color are available, in how many different ways can the map be colored.
14. Two lottery tickets are drawn from 30 for first and second prizes. Find the number of ways of doing this.
15. How many ways can a football team schedule 3 holiday game with 3 teams if they are all available on any of 5 possible dates?
16. In how many different ways can 30 aks, 4 pins and 2 apples be arranged along a property line if one does not distinguish between trees of the kind?
17. How many ways can 8 people be assigned to 2 triple 2 double rooms?
18. If $P(n, 4) = 12.P(n, 2)$ find n
19. From a group of 4 men and 5 women, how many committees of size 3 are possible
    a) with no restriction
    b) with 2 men and a women
    c) with 2 men and a women if a certain man must be on the committee
20. Find the value of r if $C(18, r) = C(18, r + 2)$
21. If $C(n, 3) = C(n, 5)$, find the value of $C(2n, 2)$

22. How many signals can be made with 5 different flags by taking them any number at a time
23. There are 3 copies each of 4 different books. In how many ways can they be arranged on a shelf?
24. In how many ways can 2 or more ties be selected out of 8 ties
25. If k is a real number and if the middle term in the expansion of $\left(\frac{k}{2}+2\right)^8$ is 1120, find k
26. Find the term independent of x in

i) $\left(2x+\frac{1}{3x^2}\right)^9$ ii) $\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ iii) $\left(\frac{4}{3}x^2-\frac{3}{2x}\right)^9$

27. Find the coefficient of x^{18} in the expansion of $(ax^4-bx)^9$
28. If the coefficient of x^2 and x^3 in the expansion of $(3+kx)^9$
29. In the expansion of $(1+x)^{2n+1}$, the coefficient of x^r and x^{r-1} are equal. Find r
30. Find the value of r if the coefficient of $(2r+4)$th term and $(r-2)$th term in the expansion of $(1+x)^{18}$
31. Determine the probability that in a family of 5 children there will be at least 2 boys and 1 girl
32. find the probability of throwing a total of 7 in single throw with two dice.
33. what is the probability of throwing a total of 8 or 11 in single throw with two dice
34. A ball is drawn at random from a box containing 10 red, 30 white, 20 blue and 15 orange marbles. Find the probability that it is
    a) orange or red c) not blue
    b) not red or blue d) white e) red, white or blue
35. Determine the probability of three 6's in 5 tosses of fair die (Hint, P(666 $\bar{6}$ $\bar{6}$) or P(66 $\bar{6}$ 6 $\bar{6}$) or (6 $\bar{6}$ 66 $\bar{6}$) ...
36. A probability that a man will be alive in 25 years is $\frac{3}{5}$ and the probability that his wife will be alive in 25 years is $\frac{2}{3}$. Find the probability that
    a) both will be alive b) only the man will be alive
    c) only the wife will be alive d) at least one will be alive

Unit Six

Matrices and Determinants

A matrix is rectangular array (or arrangement) of numbers in rows and column.

Here are the general 3×4 matrix and

matrix $A_{3\times 4} \longrightarrow$ 3 the number of Row
4 the number of column

Row by Column
→ by ↓

$$A=\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{bmatrix}$$

- Each entry in a matrix is called element of matrix
- Elements in the **horizontal line** are called rows.
- Elements in the **vertical line** are called column.

- Element is identified by its row and column number. We write in subscript notation like
  a_{ij} called **ij**th element or entry of given matrix.
- a_{ij} indicates element in **i**th row and **j**th column.
- The number of rows and the number of columns are called **the size or, order dimension of matrix.**
- The dimension of the above matrix are 3×4 (read three by four)
- Write the number of rows it first and the number of column second.

Illustrative example

1. If $A=\begin{bmatrix} 3 & 6 & -2 & 4 & 1 \\ 7 & 0 & 5 & -6 & 8 \\ 11 & 9 & -4 & -3 & 2 \\ -8 & -1 & -7 & \frac{1}{2} & -5 \end{bmatrix}$, then

   a) Find the order of matrix A
   b) Find he value of a_{13}, a_{31}, a_{24}, a_{42}, a_{21}, a_{25} a_{52}, a_{36}
   c) Lable these element of A with variable a and subscripts.
      i) -7 ii) -3 iii) 2 iv) -6

4 x 3

Solution: a) $4 \times 5 \longrightarrow$ It has 4 rows and 5 column

b) $a_{13} = -2$, $a_{31} = 11 \longleftarrow 3^{rd}$ row and 1^{st} column

$a_{24} = -6$, $a_{42} = -1$, $a_{21} = 7 \longleftarrow 2^{nd}$ row 1^{st} column

$a_{25} = 8$, a_{52} = does not exist

a_{36} = does not exist

c) i) $a_{43} = -7$ ii) $a_{34} = -3$, iii) $a_{35} = 2$ iv) $a_{24} = -6$

- **Type of matrices**

These are row matrix, column matrix, square matrix, diagonal matrix, zero matrix, scalar matrix, identity matrix, and triangular matrix.

I. **Row matrix.** A matrix having exactly single row (3 5 7)

II. **Column Matrix.** A matrix having exactly single column $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$

III. **Square matrix.** A matrix with the same number of rows and column is called a square matrix

$$\begin{pmatrix} 3 & 4 \\ 7 & 1 \end{pmatrix}_{2\times2}$$

IV. **Diagonal matrix.** A square matrix in which all non diagonal elements are 0 is called a diagonal matrix.

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

V. **Zero matrix.** A matrix in which each element is zero is called zero matrix or null matrix for example.

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}_{2\times2} \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}_{3\times2}$$

VI. **Scalar Matrix.** A diagonal matrix in which all the diagonal elements are equal

Example $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ are scalar matrix